

54. 16-bit A/D Converter (ADC16H)

This is the ADC_B version of the ADC16H peripheral module. ADC_B is referred to as ADC16H in this chapter.

54.1 Overview

This MCU contains two units of Noise-Shaping SAR-type A/D converters (ADC16H) that is the hybrid architecture with features of SAR-type and Delta-Sigma modulation-type. The A/D converter unit 0 (ADC0) can select up to 15 channels of analog inputs. The A/D converter unit 1 (ADC1) can select up to 15 channels of analog inputs. The temperature sensor, internal reference voltage, VBATT 1/6 voltage monitor output, and D/A converters can be A/D-converted by A/D converter unit 0 or unit 1. A/D conversion data can be selected from 16-bit, 14-bit, 12-bit, and 10-bit length data formats.

The ADC16H has the following features:

- Resolution: Up to 16-bit
- Fast conversion: Up to 6.25 Msps (0.16 μ s per channel) (when A/D conversion clock ADCLK = 50 MHz)
- Input channels: Up to 23 analog input channels
- Support single-ended input or differential inputs
- Self-calibration function
- Built-in channel-dedicated sample-and-hold circuit (S&H)

Table 54.1 shows the ADC16H specifications, Table 54.2 shows the correspondence between ADC16H functions and ADC16H operation mode, and Table 54.3 shows the ADC16H I/O pins.

Figure 54.1 shows the block diagram of ADC16H. Figure 54.2 shows the analog channel structure for A/D converter unit 0. Figure 54.3 shows the analog channel structure for A/D converter unit 1.

Table 54.5 shows the ADC16H channel configuration of the extended analog function.

Table 54.1 ADC16H specifications (1 of 3)

Item	Description
Number of units	Two units (unit 0 and unit 1)
Input channels	• Support single-ended input or differential inputs • Up to 23 analog input channels – A/D converter unit 0: Up to 15 analog input channels at Single-ended input, 6 analog input channels (3 pairs) of which support Differential inputs. – A/D converter unit 1: Up to 15 analog input channels at Single-ended input, 6 analog input channels (3 pairs) of which support Differential inputs. – 7 analog input channels are shared by A/D converter unit 0 and unit 1
Extended analog function	Self-diagnosis, temperature sensor, internal reference voltage, D/A converter (DA0, DA1), VBATT 1/6 voltage monitor output
A/D conversion method	• Successive approximation register method in SAR mode • Noise-Shaping Successive approximation register method in Oversampling mode and Hybrid mode
Resolution of A/D converter	• Up to 12-bits in SAR mode • Up to 16-bits in Oversampling mode or Hybrid mode
Conversion time	0.16 μ s per channel (when A/D conversion clock ADCLK = 50 MHz).
A/D conversion clock	The A/D conversion clock (ADCLK) can be set by selecting the clock source and the division ratio as follows: • The clock source: the ADC clock ADCCLK*¹, the peripheral module clock PCLKA*¹, the GPT clock GPTCLK*¹. • The division ratio: 1/2/3/4/5 A/D conversion clock (ADCLK) can operate between 25 MHz at a minimum and 60 MHz at a maximum.
A/D conversion data	• A/D conversion results are stored in data register or FIFO. • A/D conversion results are available in 16-, 14-, 12-, and 10-bit data formats.

Table 54.1 ADC16H specifications (2 of 3)

Item	Description																											
Operation mode	• SAR mode – A/D converter samples the signal source once, and convert by Successive Approximation Register method – Fast A/D conversion – Support up to 8 channels per one scan group – Support single-ended input and differential input • Oversampling mode – A/D converter oversamples the signal source, and converts analog-to-digital by Noise Shaping Successive Approximation Register method – High-accuracy A/D conversion – Support up to 8 channels per one scan group – Support single-ended input and differential input • Hybrid mode – A/D converter oversamples the signal source, and converts analog-to-digital by Noise Shaping Successive Approximation Register method – High-accuracy A/D conversion – High data rate in continuous scanning operation – Support up to 4 channels per one scan group – Support single-ended input and differential input																											
Scan mode	• Single scan mode – Assign any selected analog input or analog channel of the extended analog function to any scan group, and convert the selected analog input only once per scan group for A/D conversion. – By selecting the scan start conditions for each scan group individually, A/D conversion for each scan group can be started at different times. • Continuous scan mode – Assign any selected analog input or analog channel of the extended analog function to any scan group and repeat A/D conversion in scan group units. • Background continuous scan mode – Assign any selected analog input or analog channel of the extended analog function to any scan group and repeat A/D conversion in scan group units. – The A/D conversion is performed in the background and, if the A/D conversion start trigger is entered, the A/D conversion data at that point is acquired. • One-channel continuous scan mode – Repeat A/D conversion of a single analog input or analog channel of an extended analog function. – If the A/D conversion start trigger is entered, the channel to be converted can be switched. – This conversion mode is suitable for use cases where a change to the A/D conversion channel is required according to an external factor. • Fixed-channel continuous scan mode – First scan a fixed number of fixed channels, then scan the non-fixed channels assigned to the scan group in sequence, and repeat these operations. – Enabling adjustment of the A/D conversion frequency and A/D converts more channels than in normal Hybrid mode.																											
Combination of operation mode and scan mode	<table><tr><th rowspan="2">Scan mode</th><th colspan="3">Operation mode of A/D converter</th></tr><tr><th>SAR mode</th><th>Oversampling mode</th><th>Hybrid mode</th></tr><tr><td>Single scan mode</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Continuous scan mode</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Background continuous scan mode</td><td>—</td><td>—</td><td>✓</td></tr><tr><td>One-channel continuous scan mode</td><td>—</td><td>✓</td><td>—</td></tr><tr><td>Fixed-channel continuous scan mode</td><td>—</td><td>—</td><td>✓</td></tr></table> Note: ✓ : available, —: not available	Scan mode	Operation mode of A/D converter			SAR mode	Oversampling mode	Hybrid mode	Single scan mode	✓	✓	✓	Continuous scan mode	✓	✓	✓	Background continuous scan mode	—	—	✓	One-channel continuous scan mode	—	✓	—	Fixed-channel continuous scan mode	—	—	✓
Scan mode	Operation mode of A/D converter																											
	SAR mode	Oversampling mode	Hybrid mode																									
Single scan mode	✓	✓	✓																									
Continuous scan mode	✓	✓	✓																									
Background continuous scan mode	—	—	✓																									
One-channel continuous scan mode	—	✓	—																									
Fixed-channel continuous scan mode	—	—	✓																									
Conditions for A/D conversion start	• Software trigger (for simultaneous activation of scan group: max. 9 triggers) • Software trigger (for individual scan group activation: up to 9 triggers) • Trigger from Event Link Controller: 6 triggers • Triggers from GPT: 28 triggers • External trigger input: 2 triggers (ADTRGn input (n = 0, 1))																											

Table 54.1 ADC16H specifications (3 of 3)

Item	Description
Functions	- Virtual Channel function (33 virtual channels) - Scan Group function (up to 9 scan groups) - Variable sampling time (select from 16 tables per virtual channel) - Channel-dedicated sample-and-hold circuit (SH) (3 SH units for A/D converter unit 0, 3 SH units for A/D converter unit 1) - Self-diagnosis function for A/D converter - Selectable A/D-converted value addition mode or average mode - Analog input disconnection detection assist function (discharge function/precharge function) - Selectable the data format from 16- / 14- / 12- / 10-bit. - Selectable the signed-data or the unsigned-data format - Limiter Clip function (up to 8 tables) - Compare Match function (up to 8 tables) - Self-calibration function - User's Gain adjustment function - User's Offset adjustment function - Built-in FIFO (8 stages per each scan group) - Built-in digital filter (support only Oversampling mode and Hybrid mode) - Multiple A/D converters unit-to-unit synchronous operation function
Interrupt sources	- A/D scan end interrupt - Generates the interrupt requests and the ELC events at the end of A/D scan operation for the scan group i (ADC_ADli (i = 0 to 8)). The interrupt requests are independent for each scan group. - FIFO data read request interrupt - Generates the interrupt requests when the number of vacant stages of FIFO for the scan group i becomes less than or equal to the specified value (ADC_FIFOREQi (i = 0 to 8)). The interrupt requests are independent for each scan group. - FIFO data overflow interrupt - Generates the interrupt request when the overflow occurs in any of the FIFO for the scan groups 0 to 8 (ADC_FIFOOVF). - Limiter clip interrupt - Generates the interrupt request when the limiter clip using the limiter tables 0 to 7 is occurred for A/D conversion results (ADC_LIMCLPI). - Compare match interrupt - Generates the interrupt requests when the compare match using the compare match tables j are occurred for A/D conversion results (ADC_CMPIj (j = 0 to 3)). The interrupt requests are independent for each compare match table. - Composite compare match interrupt - Generates the interrupt requests and the ELC events when the compare match of the composite condition using compare match tables 0 to 7 occurs (ADC_CCMPMm (m = 0, 1), ADC_CCMPUMm (m = 0, 1)). - A/D converter error interrupt - Generates the interrupt requests when the operational error is detected in A/D converter Unit j (ADC_ERRj (j = 0, 1)). - A/D conversion overflow interrupt - Generates the interrupt request when the A/D conversion result overflow is occurred (ADC_RESOVFj (j = 0, 1)). The interrupt requests are independent for A/D converter unit j. - A/D converter calibration end interrupt - Generates the interrupt requests at the end of calibration operation of A/D converter unit j (ADC_CALENDj (j = 0, 1)).
ELC interface	- Trigger Input - Scan can be started by the trigger from the ELC. - Event Generation - An event is generated at the end of each scan operation for the scan group 0 to 8. - An event is generated when Composite Compare Match is occurred.
Reference voltage	- Unit 0: - VREFH0 is the analog reference voltage. - VREFL0 is the analog reference ground. - Unit 1: - VREFH is the analog reference voltage. - VREFL is the analog reference ground.
Module-stop function	Module-stop state can be set to reduce power consumption.*2

Note 1. For details, see [section 9, Clock Generation Circuit](#).Note 2. For details, see [section 11, Low Power Mode](#).

Table 54.2 Association between ADC16H functions and ADC16H operation mode (1 of 2)

Function		SAR mode		Oversampling mode			Hybrid mode			
		Single scan mode	Continuous scan mode	Single scan mode	Continuous scan mode	One-channel continuous scan mode	Single scan mode	Continuous scan mode	Background continuous scan mode	Fixed-channel continuous scan mode
Analog input	Single-ended input	✓	✓	✓	✓	—	✓	✓	✓	—
	Differential input	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Channel-dedicated sample-and-hold circuit	✓	—	—	—	—	✓	✓	✓	✓ ^{*3}
Diagnosis function/extended analog function	Disconnection detection assist function	✓	✓	✓	✓	✓	✓	✓	—	—
	Self-diagnosis	✓	✓	✓	—	—	✓	—	—	—
	Disconnection detection assist function of channel-dedicated sample-and-hold circuit	✓	—	—	—	—	—	—	—	—
	Self-diagnosis of channel-dedicated sample-and-hold circuit	✓	—	—	—	—	—	—	—	—
	Internal reference voltage	✓	✓	✓	✓	✓	✓	✓	✓	✓
	VBATT 1/6 voltage monitor output	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Temperature sensor	✓	✓	✓	✓	✓	✓	✓	✓	✓
	D/A converter (DA0, DA1)	✓	✓	✓	✓	✓	✓	✓	✓	✓
Scanning operation	Group priority operation	✓ ^{*2}	✓ ^{*2}	—	—	—	—	—	—	—
	Multiple A/D converter synchronous operation	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 54.2 Association between ADC16H functions and ADC16H operation mode (2 of 2)

Function		SAR mode		Oversampling mode			Hybrid mode			
		Single scan mode	Continuous scan mode	Single scan mode	Continuous scan mode	One-channel continuous scan mode	Single scan mode	Continuous scan mode	Background continuous scan mode	Fixed-channel continuous scan mode
Digital calculation	Digital filter function	—	—	✓*1	✓*1	✓*1	✓*1	✓*1	✓*1	✓*1
	User's gain/offset adjustment function	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Addition/average function	✓	✓	✓	✓	✓	✓	✓	✓	—
	Limiter clip function	✓	✓	✓	✓	✓	✓	✓	✓	✓
	Compare match function	✓	✓	✓	✓	✓	✓	✓	✓	✓
FIFO function		✓	✓	✓	✓	✓	✓	✓	✓	✓

Note 1. Use of the digital filter function is required when in Oversampling mode and Hybrid mode.

Note 2. When using the channel-dedicated sample-and-hold circuit in group priority operation, it is available only for the highest priority scan group.

Note 3. Only fixed channels are supported when the number of fixed channels is set to 3 CH and all fixed channels are set to channel-dedicated sample-and-hold circuit. Non-fixed channels cannot use channel-dedicated sample-and-hold circuit.

Table 54.3 ADC16H I/O pins

Pin name	I/O	Function
AVCC0	Input	Analog block power supply pin for unit 0 and unit 1
AVSS0	Input	Analog block power supply ground pin for unit 0 and unit 1
VREFH0	Input	Analog reference voltage supply pin for unit 0
VREFL0	Input	Analog reference ground pin for unit 0
VREFH1	Input	Analog reference voltage supply pin for unit 1
VREFL1	Input	Analog reference ground pin for unit 1
AN000 to AN022	Input	Analog input pins
SHIN0P to SHIN2P, SHIN4P to SHIN6P	Input	Analog input pins for channel-dedicated sample-and-hold circuit - For single-ended input: Connect to signal source - For differential input: Functions as the non-inverting inputs (+) terminal for differential input
SHIN0N to SHIN2N, SHIN4N to SHIN6N	Input	Analog input pins for channel-dedicated sample-and-hold circuit - For single-ended input: Connect to signal source - For differential input: Functions as the inverting inputs (-) terminal for differential input
ADTRG0, ADTRG1	Input	External trigger input pin for starting A/D conversion, active-low.
ADST0, ADST1	Output	A/D conversion start
AD0FLAG1, AD1FLAG1	Output	A/D conversion end
ADSYNC	Output	Synchronization signal between units

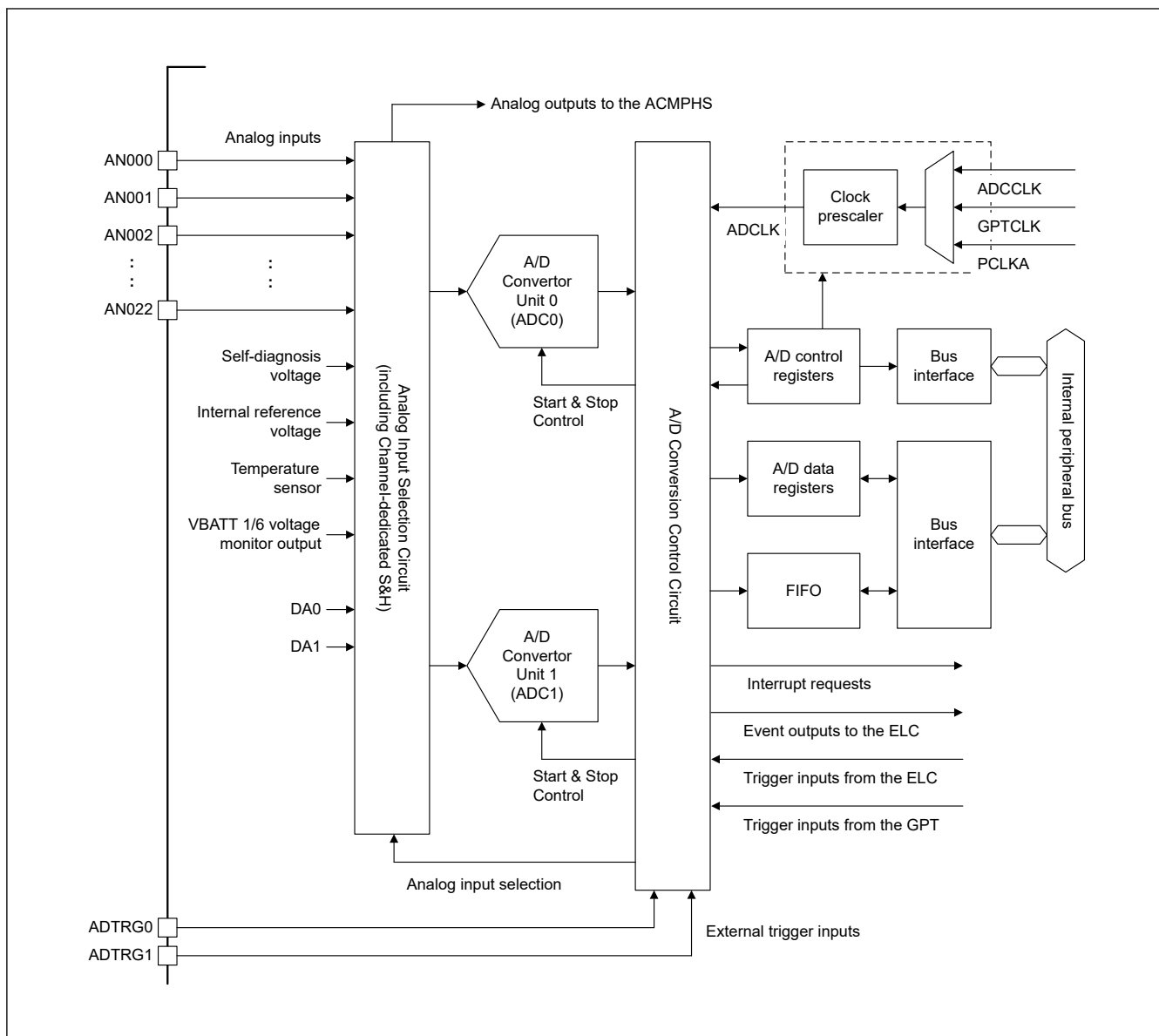


Figure 54.1 Block diagram of ADC16H

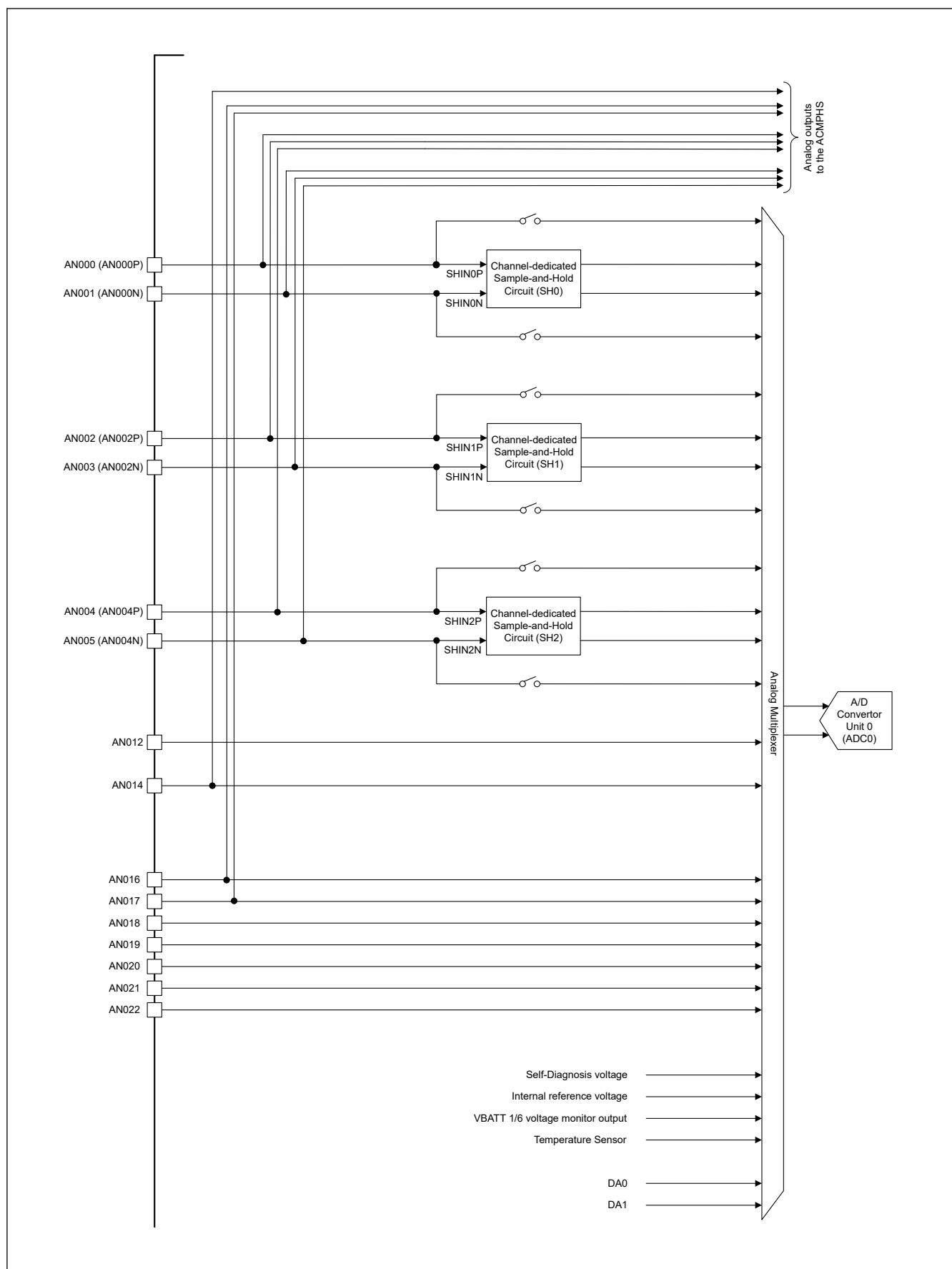


Figure 54.2 Analog channel structure for A/D converter unit 0

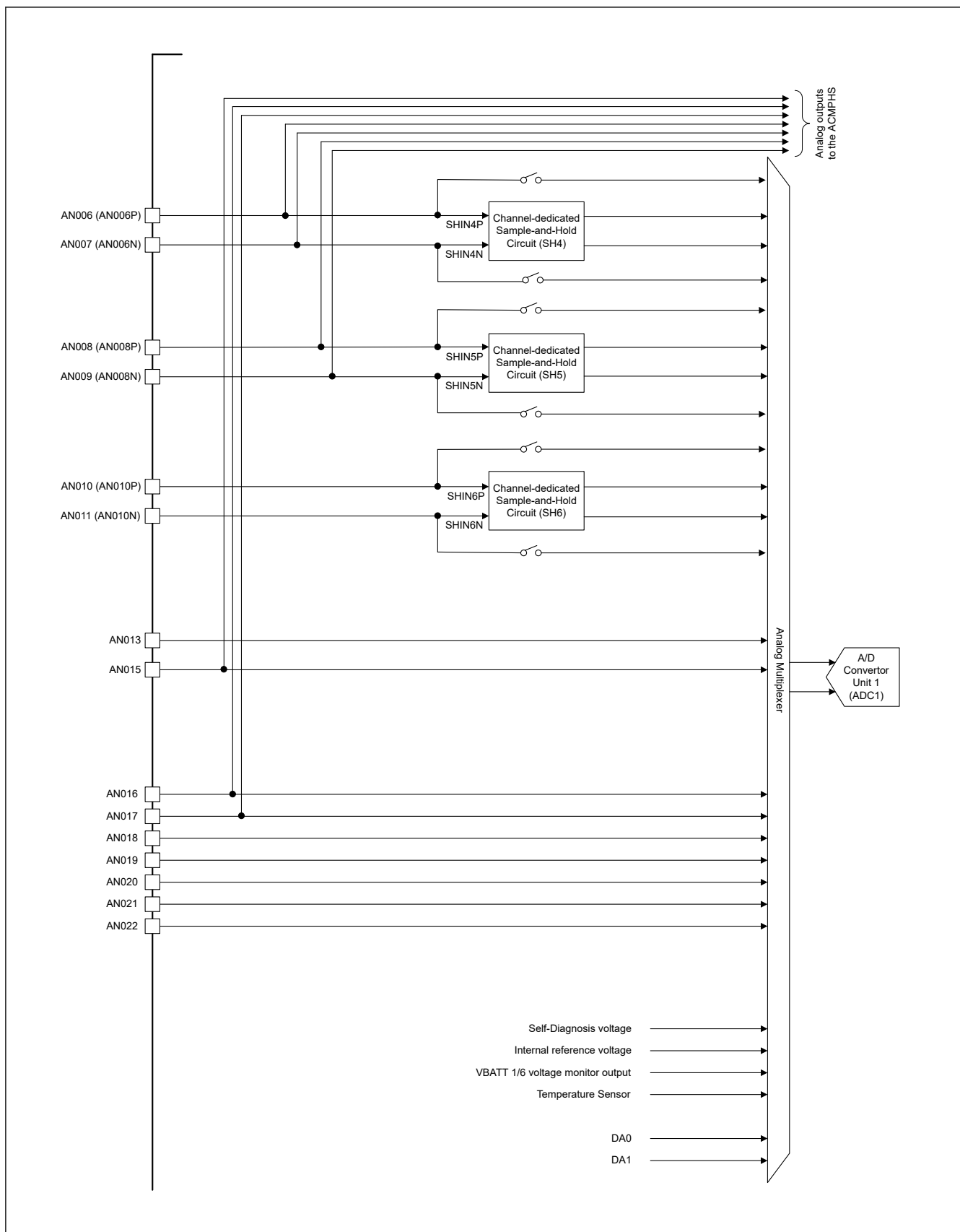


Figure 54.3 Analog channel structure for A/D converter unit 1

Table 54.4 shows the ADC16H channel configuration of the analog inputs.

Table 54.4 ADC16H channel configuration of the analog inputs

Analog channel number	Source of analog signal		A/D converter unit 0		A/D converter unit 1	
	Analog input	Channel-dedicated Sample-and-hold Circuit (S&H)* ²	Single-ended input	Differential input* ³	Single-ended input	Differential input* ³
0	AN000 (AN000P)	SHIN0	✓	✓	—	—
1	AN001 (AN000N)		✓		—	
2	AN002 (AN002P)	SHIN1	✓	✓	—	—
3	AN003 (AN002N)		✓		—	
4	AN004 (AN004P)	SHIN2	✓	✓	—	—
5	AN005 (AN004N)		✓		—	
6	AN006 (AN006P)	SHIN4	—	—	✓	✓
7	AN007 (AN006N)		—		✓	
8	AN008 (AN008P)	SHIN5	—	—	✓	✓
9	AN009 (AN008N)		—		✓	
10	AN010 (AN010P)	SHIN6	—	—	✓	✓
11	AN011 (AN010N)		—		✓	
12	AN012	—	✓	—	—	—
13	AN013	—	—	—	✓	—
14	AN014	—	✓	—	—	—
15	AN015	—	—	—	✓	—
16	AN016* ¹	—	✓	—	✓	—
17	AN017* ¹	—	✓	—	✓	—
18	AN018* ¹	—	✓	—	✓	—
19	AN019* ¹	—	✓	—	✓	—
20	AN020* ¹	—	✓	—	✓	—
21	AN021* ¹	—	✓	—	✓	—
22	AN022* ¹	—	✓	—	✓	—

Note 1. Do not perform A/D conversions of the same signal source from both A/D converter unit 0 and unit 1.

Note 2. When the channel-dedicated sample-and-hold circuit is used in single-ended input, the SHINxP (non-inverting input (+)) channel and the SHINxN (inverting input (−)) channel must not be assigned to the same scan group (x = 0 to 2, 4 to 6).

Note 3. When performing A/D conversion with differential-input, set the even-numbered channel as the A/D conversion target. For differential input, the even-numbered channels function as non-inverting inputs (+), and the odd-numbered channels function as inverting inputs (−).

Table 54.5 ADC16H channel configuration of the extended analog function (1 of 2)

Analog channel number	Source of analog signal	A/D converter unit 0		A/D converter unit 1	
		Single-ended input	Differential input	Single-ended input	Differential input
96	Self-diagnosis A/D unit 0* ²	—	✓	—	—
97	Self-diagnosis A/D unit 1* ²	—	—	—	✓
100	Temperature sensor* ¹	✓	—	✓	—
101	Internal reference voltage* ¹	✓	—	✓	—
102	VBATT 1/6 voltage monitor output* ¹	✓	—	✓	—
104	D/A converter channel 0* ¹	✓	—	✓	—
105	D/A converter channel 1* ¹	✓	—	✓	—

Table 54.5 ADC16H channel configuration of the extended analog function (2 of 2)

Analog channel number	Source of analog signal	A/D converter unit 0		A/D converter unit 1	
		Single-ended input	Differential input	Single-ended input	Differential input
112	Self-diagnosis S&H unit 0 ^{*2}	—	✓	—	—
113	Self-diagnosis S&H unit 1 ^{*2}	—	✓	—	—
114	Self-diagnosis S&H unit 2 ^{*2}	—	✓	—	—
116	Self-diagnosis S&H unit 4 ^{*2}	—	—	—	✓
117	Self-diagnosis S&H unit 5 ^{*2}	—	—	—	✓
118	Self-diagnosis S&H unit 6 ^{*2}	—	—	—	✓

Note 1. Do not perform A/D conversions of the same signal source from both A/D converter unit 0 and unit 1.

Note 2. The self-diagnosis function is only supported with differential input.

54.2 Register Descriptions

54.2.1 System

54.2.1.1 ADCLKENR : A/D Conversion Clock Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLKEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLKEN	ADCLK Operating Enable 0: Stop ADCLK 1: Supply ADCLK	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

The ADCLKENR controls the supply of the A/D conversion clock (ADCLK).

54.2.1.2 ADCLKSR : A/D Conversion Clock Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLKSR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLKSR	ADCLK status 0: ADCLK is stopped 1: ADCLK is in supply	R
31:1	—	These bits are read as 0.	R

The ADCLKSR register indicates the supply status of the A/D conversion clock (ADCLK).

54.2.1.3 ADCLKCR : A/D Conversion Clock Control Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DIVR[2:0]	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLKSEL[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CLKSEL[1:0]	ADCLK Clock Source Select 0 0: ADC Clock (ADCCLK) 0 1: GPT Clock (GPTCLK) 1 0: Peripheral Module Clock A (PCLKA) 1 1: Setting prohibited	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
18:16	DIVR[2:0]	Clock Division Ratio Select 0 0 0: 1/1 0 0 1: 1/2 0 1 0: 1/3 0 1 1: 1/4 1 0 0: 1/5 Others: Setting prohibited	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

The ADCLKCR register selects the frequency of A/D conversion clock (ADCLK). The frequency of ADCLK should be set within the range specified in the [section 70, Electrical Characteristics](#).

CLKSEL[1:0] bits (ADCLK Clock Source Select)

CLKSEL[1:0] bits select the clock source of A/D conversion clock (ADCLK).

DIVR[2:0] bits (Clock Division Ratio Select)

DIVR[2:0] bits select the division ratio for the clock source selected by CLKSEL[1:0] bits. The clock divided by the clock source selected by CLKSEL[1:0] bits at the ratio set by DIVR[2:0] bits becomes the A/D conversion operating clock (ADCLK).

54.2.1.4 ADSYCR : A/D Converter Synchronous Operation Control Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x00C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADSY DIS1	ADSY DIS0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	ADSYCYC[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1

Bit	Symbol	Function	R/W
7:0	ADSYCYC[7:0]	Synchronous Operation Period Cycle These bits should be set in the range from 5 to 255. The settings of 0 to 4 are prohibited.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
16	ADSYDIS0	ADC0 Synchronous Operation Select 0: Enable ADC0 synchronous operation 1: Disable ADC0 synchronous operation	R/W
17	ADSYDIS1	ADC1 Synchronous Operation Select 0: Enable ADC1 synchronous operation 1: Disable ADC1 synchronous operation	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

The ADSYCR register controls the synchronous operation function.

ADSYCYC[7:0] bits (Synchronous Operation Period Cycle)

These bits are selected the synchronous operation period cycles.

These bits set in the range from 5 to 255 regardless of whether the synchronous operation function is used or not. The settings from 0 to 4 are prohibited.

ADSYDIS0 bit (ADC0 Synchronous Operation Select)

ADSYDIS0 bit selects enable/disable the synchronous operation of ADC0. If the synchronous operation is enabled, the ADC0 operates synchronously with other A/D converter that has synchronous operation enabled.

ADSYDIS1 bit (ADC1 Synchronous Operation Select)

ADSYDIS1 bit selects enables/disables to the synchronous operation of ADC1. If the synchronous operation is enabled, the ADC1 operates synchronously with other A/D converter that has synchronous operation enabled.

54.2.1.5 ADUSLPCR0 : A/D Converter Unit Sleep Control Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADUS LP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADUSLP	Sleep setting of A/D converter unit 0 0: Set A/D converter to Normal operation mode 1: Set A/D converter to Sleep operation mode	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

This register is used to control the A/D converter unit 0 Sleep mode.

54.2.1.6 ADUSLPCR1 : A/D Converter Unit Sleep Control Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADUS LP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADUSLP	Sleep setting of A/D converter unit 1 0: Set A/D converter to Normal operation mode 1: Set A/D converter to Sleep operation mode	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

This register is used to control the A/D converter unit 1 Sleep mode.

54.2.1.7 ADERINTCR : A/D Conversion Error Interrupt Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADEIE 1	ADEIE 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADEIE0	ADC0 A/D Conversion Error Interrupt Enable 0: Disable A/D conversion error interrupt for ADC0 1: Enable A/D conversion error interrupt for ADC0	R/W
1	ADEIE1	ADC1 A/D Conversion Error Interrupt Enable 0: Disable A/D conversion error interrupt for ADC1 1: Enable A/D conversion error interrupt for ADC1	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

The ADERINTCR register controls the enable/disable to the A/D conversion error interrupt.

54.2.1.8 ADOVFINTCR : A/D Conversion Overflow Interrupt Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADOV FIE1	ADOV FIE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADOVFIE0	ADC0 A/D Conversion Overflow Interrupt Enable 0: Disable A/D conversion overflow interrupt for ADC0 1: Enable A/D conversion overflow interrupt for ADC0	R/W
1	ADOVFIE1	ADC1 A/D Conversion Overflow Interrupt Enable 0: Disable A/D conversion overflow interrupt for ADC1 1: Enable A/D conversion overflow interrupt for ADC1	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

The ADOVFINTCR register controls the enable/disable to the A/D conversion overflow interrupt.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x028

[illegible]

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
16	CALENDIE0	ADC0 Calibration End Interrupt Enable 0: Disable Calibration End Interrupt for ADC0 1: Enable Calibration End Interrupt for ADC0	R/W
17	CALENDIE1	ADC1 Calibration End Interrupt Enable 0: Disable Calibration End Interrupt for ADC1 1: Enable Calibration End Interrupt for ADC1	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

54.2.1.10 ADMDR : A/D Converter Mode Selection Register

Bit position:	31																11				8				3				0	
Bit field:	—																ADMD1[3:0]				—				ADMD0[3:0]					
Value after reset:	0																0				0				0				0	

Bit	Symbol	Function	R/W
3:0	ADMD0[3:0]	ADC0 Mode Selection 0x0: SAR mode – Single scan mode 0x1: SAR mode – Continuous scan mode 0x4: Oversampling mode – Single scan mode 0x5: Oversampling mode – Continuous scan mode 0x6: Oversampling mode – One-channel continuous scan mode 0x8: Hybrid mode – Single scan mode 0x9: Hybrid mode – Continuous scan mode 0xA: Hybrid mode – Background continuous scan mode 0xB: Hybrid mode – Fixed-channel continuous scan mode Others: Setting prohibited	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
11:8	ADMD1[3:0]	ADC1 Mode Selection 0x0: SAR mode – Single scan mode 0x1: SAR mode – Continuous scan mode 0x4: Oversampling mode – Single scan mode 0x5: Oversampling mode – Continuous scan mode 0x6: Oversampling mode – One-channel continuous scan mode 0x8: Hybrid mode – Single scan mode 0x9: Hybrid mode – Continuous scan mode 0xA: Hybrid mode – Background continuous scan mode 0xB: Hybrid mode – Fixed-channel continuous scan mode Others: Setting prohibited	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

The ADMDR register selects the operation mode and the scan mode for each A/D converter.

54.2.2 Scan Group

54.2.2.1 ADGSPCR : A/D Group Scan Priority Control Register

Base address: ADC_B = 0x4033_8000
 ADC_B_NS = 0x5033_8000

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	GRP1	LGRR S1	RSCN 1	PGS1	—	—	—	—	GRP0	LGRR S0	RSCN 0	PGS0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PGS0	ADC0 Group Priority Control Setting 0: ADC0 operation is without group priority control 1: ADC0 operation is with group priority control in SAR mode. Setting prohibited when other than SAR mode.	R/W
1	RSCN0	ADC0 Group Priority Control Setting 2 0: Set when PGS0 is set to 0. 1: Set when PGS0 is set to 1.	R/W
2	LGRRS0	ADC0 Group Priority Control Setting 3 0: Set when PGS0 is set to 0. 1: Set when PGS0 is set to 1.	R/W
3	GRP0	ADC0 Group Priority Control Setting 4 0: Set the following case: - When PGS0 is set to 0 - When PGS0 is set to 1 and ADC0 is SAR mode – Single scan mode. 1: Set when PGS0 is set to 1 and ADC0 is SAR mode – Continuous scan mode.	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	PGS1	ADC1 Group Priority Control Setting 0: ADC1 operation is without group priority control 1: ADC1 operation is with group priority control in SAR mode. Setting prohibited when other than SAR mode.	R/W
9	RSCN1	ADC1 Group Priority Control Setting 2 0: Set when PGS1 is set to 0. 1: Set when PGS1 is set to 1.	R/W

Bit	Symbol	Function	R/W
10	LGRRS1	ADC1 Group Priority Control Setting 3 0: Set when PGS1 is set to 0. 1: Set when PGS1 is set to 1.	R/W
11	GRP1	ADC1 Group Priority Control Setting 4 0: Set the following case: - When PGS1 is set to 0 - When PGS1 is set to 1 and ADC1 is SAR mode – Single scan mode. 1: Set when PGS1 is set to 1 and ADC1 is SAR mode – Continuous scan mode.	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

The ADGSPCR register controls the group priority operation to each A/D converter. For details on setting this register, see [section 54.3.18. Group Priority Operation](#).

54.2.2.2 ADSGER : Scan Group Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x048

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SGRE 8	SGRE 7	SGRE 6	SGRE 5	SGRE 4	SGRE 3	SGRE 2	SGRE 1	SGRE 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	SGRE0 to SGRE8	Scan Group n Enable The suffix number of each bit symbol corresponds to the scan group number n. 0: Disable the scan group n 1: Enable the scan group n	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

The ADSGER register selects the enable/disable to each scan group.

SGREn bit (Scan Group n Enable) (n = 0 to 8)

The SGREn bit selects the enable/disable to the scan group n. When SGREn is set to 1, A/D conversion with the scan group n is available. When SGREn is set to 0, A/D conversion with the scan group n is not available, even if the trigger corresponding to the scan group n is input.

54.2.2.3 ADSGCR0 : Scan Group Control Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x04C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	SGADS3[1:0]		—	—	—	—	—	—	SGADS2[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SGADS1[1:0]		—	—	—	—	—	—	SGADS0[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SGADS0[1:0]	Scan Group 0 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	SGADS1[1:0]	Scan Group 1 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
17:16	SGADS2[1:0]	Scan Group 2 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
25:24	SGADS3[1:0]	Scan Group 3 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSGCR0 register selects which A/D converter is used for each scan group.

54.2.2.4 ADSGCR1 : Scan Group Control Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	SGADS7[1:0]		—	—	—	—	—	—	SGADS6[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SGADS5[1:0]		—	—	—	—	—	—	SGADS4[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SGADS4[1:0]	Scan Group 4 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	SGADS5[1:0]	Scan Group 5 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
17:16	SGADS6[1:0]	Scan Group 6 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
25:24	SGADS7[1:0]	Scan Group 7 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSGCR1 register selects which A/D converter is used for each scan group.

54.2.2.5 ADSGCR2 : Scan Group Control Register 2

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x054

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SGADS8[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	SGADS8[1:0]	Scan Group 8 A/D Converter Selection 0 0: Select ADC0 0 1: Select ADC1 Others: Setting prohibited	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

The ADSGCR2 register selects which A/D converter is used for each scan group.

54.2.2.6 ADINTCR : Scan End Interrupt Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x05C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADIE8	ADIE7	ADIE6	ADIE5	ADIE4	ADIE3	ADIE2	ADIE1	ADIE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	ADIE0 to ADIE8	Scan Group n Scan End Interrupt Enable The suffix number of each bit symbol corresponds to the scan group number n. 0: Disable scan end interrupt 1: Enable scan end interrupt	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

The ADINTCR register enables/disables the scan end interrupt.

54.2.2.7 ADTRGEXTn : External Trigger Enable Register n (n = 0 to 8)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x0C0 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TRGE XT1	TRGE XT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TRGEXT0	External Trigger Input 0 (ADTRG0) Enable 0: Disable ADTRG0 1: Enable ADTRG0	R/W
1	TRGEXT1	External Trigger Input 1 (ADTRG1) Enable 0: Disable ADTRG1 1: Enable ADTRG1	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGEXTn register enables/disables the External Trigger input 0 and 1 (ADTRG0 and ADTRG1) as the starting conditions for A/D conversion of scan group n.

54.2.2.8 ADTRGELCn : ELC Trigger Enable Register n (n = 0 to 8)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x0C4 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	TRGE LC5	TRGE LC4	TRGE LC3	TRGE LC2	TRGE LC1	TRGE LC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
5:0	TRGELC0 to TRGELC5	ELC Trigger m Enable The suffix number of each bit symbol corresponds to the ELC trigger number m. 0: Disable ELC Trigger m 1: Enable ELC Trigger m	R/W
31:6	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGELCn register enables/disables the ELC Trigger m as the starting conditions for A/D conversion of scan group n.

54.2.2.9 ADTRGGPTn : GPT Trigger Enable Register n (n = 0 to 8)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x0C8 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	TRGG PTB13	TRGG PTB12	TRGG PTB11	TRGG PTB10	TRGG PTB9	TRGG PTB8	TRGG PTB7	TRGG PTB6	TRGG PTB5	TRGG PTB4	TRGG PTB3	TRGG PTB2	TRGG PTB1	TRGG PTB0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	TRGG PTA13	TRGG PTA12	TRGG PTA11	TRGG PTA10	TRGG PTA9	TRGG PTA8	TRGG PTA7	TRGG PTA6	TRGG PTA5	TRGG PTA4	TRGG PTA3	TRGG PTA2	TRGG PTA1	TRGG PTA0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
13:0	TRGGPTA0 to TRGGPTA13	GPT channel m A/D Conversion Starting Request A Enable The suffix number of each bit symbol corresponds to the GPT channel number m. 0: Disable the A/D conversion starting request A from GPT channel m 1: Enable the A/D conversion starting request A from GPT channel m	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
29:16	TRGGPTB0 to TRGGPTB13	GPT channel m A/D Conversion Starting Request B Enable The suffix number of each bit symbol corresponds to the GPT channel number m. 0: Disable the A/D conversion starting request B from GPT channel m 1: Enable the A/D conversion starting request B from GPT channel m	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGGPTn register enables/disables the A/D conversion starting request A/B from GPT channel m as the starting conditions for A/D conversion of scan group n.

54.2.2.10 ADTRGENR : A/D Conversion Start Trigger Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	STTR GEN8	STTR GEN7	STTR GEN6	STTR GEN5	STTR GEN4	STTR GEN3	STTR GEN2	STTR GEN1	STTR GEN0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	STTRGEN0 to STTRGEN8	Scan Group n A/D Conversion Start Trigger Enable The suffix number of each bit symbol corresponds to the scan group number n. 0: Disable the A/D conversion start trigger 1: Enable the A/D conversion start trigger	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGENR register enables/disables the trigger from the peripheral modules for starting the A/D conversion of Scan group n. The triggers for each scan group are selected in the ADTRGEXTn, ADTRGELCn, and ADTRGGPTn registers.

STTRGENn bit (Scan Group n A/D Conversion Start Trigger Enable)

The STTRGENn bit selects whether the trigger inputs from the peripheral module, which are selected in the ADTRGEXTn, ADTRGELCn, and ADTRGGPTn registers, is used for the A/D conversion start condition.

When the STTRGENn bit is set to 1, the A/D conversion of scan group n can be started by trigger inputs from the peripheral module.

When the STTRGENn bit is set to 0, the A/D conversion of scan group n cannot be started by trigger inputs from the peripheral module. In this setting, the A/D conversion of scan group n can be only started by writing to the ADSYSTR or ADSTRn registers.

54.2.2.11 ADTRGDLR0 : A/D Conversion Start Trigger Delay Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x1C0

Bit position:	31	23	16	7	0
Bit field:	— — — — — — — —	TRGDLY1[7:0]	— — — — — — — —	TRGDLY0[7:0]	
Value after reset:	0 0				

Bit	Symbol	Function	R/W
7:0	TRGDLY0[7:0]	Scan Group 0 Trigger Input Delay Configuration	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	TRGDLY1[7:0]	Scan Group 1 Trigger Input Delay Configuration	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGDLR0 register controls the input delay added to the External trigger inputs, ELC triggers and GPT triggers which are the starting conditions for the A/D conversion of each scan group. The input delay value is given as the register setting value × the cycle of the A/D conversion clock (ADCLK).

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x1C4

[illegible]

Bit	Symbol	Function	R/W
7:0	TRGDLY2[7:0]	Scan Group 2 Trigger Input Delay Configuration	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	TRGDLY3[7:0]	Scan Group 3 Trigger Input Delay Configuration	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGDLR1 register controls the input delay added to the External trigger inputs, ELC triggers and GPT triggers which are the starting conditions for the A/D conversion of each scan group. The input delay value is given as the register setting value $\times$ the cycle of the A/D conversion clock (ADCLK).

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000
Offset address: 0x1C8

Bit position:	31								23								16								7								0							
Bit field:									TRGDLY5[7:0]																TRGDLY4[7:0]															
Value after reset:	0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0							

Bit	Symbol	Function	R/W
7:0	TRGDLY4[7:0]	Scan Group 4 Trigger Input Delay Configuration	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	TRGDLY5[7:0]	Scan Group 5 Trigger Input Delay Configuration	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGDLR2 register controls the input delay added to the External trigger inputs, ELC triggers and GPT triggers which are the starting conditions for the A/D conversion of each scan group. The input delay value is given as the register setting value $\times$ the cycle of the A/D conversion clock (ADCLK).

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000
Offset address: 0x1CC

Bit position:	31								23								16								7								0							
Bit field:	— — — — — — — —								TRGDLY7[7:0]								— — — — — — — —								TRGDLY6[7:0]															
Value after reset:	0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0								0 0 0 0 0 0 0 0							

Bit	Symbol	Function	R/W
7:0	TRGDLY6[7:0]	Scan Group 6 Trigger Input Delay Configuration	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
23:16	TRGDLY7[7:0]	Scan Group 7 Trigger Input Delay Configuration	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGDLR3 register controls the input delay added to the External trigger inputs, ELC triggers and GPT triggers which are the starting conditions for the A/D conversion of each scan group. The input delay value is given as the register setting value $\times$ the cycle of the A/D conversion clock (ADCLK).

54.2.2.15 ADTRGDLR4 : A/D Conversion Start Trigger Delay Register 4

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x1D0

Bit position: 31 7 0

[illegible]

Value after reset: 0

Bit	Symbol	Function	R/W
7:0	TRGDLY8[7:0]	Scan Group 8 Trigger Input Delay Configuration	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

The ADTRGDLR4 register controls the input delay added to the External trigger inputs, ELC triggers and GPT triggers which are the starting conditions for the A/D conversion of each scan group. The input delay value is given as the register setting value $\times$ the cycle of the A/D conversion clock (ADCLK).

54.2.2.16 ADSWNR0 : A/D conversion times per scan Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
---------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Bit field:	—	—	—	—	—	SWNOFIX[2:0]	—	—	—	—	—	—	SWFIX[1:0]
------------	---	---	---	---	---	--------------	---	---	---	---	---	---	------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
---------------	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Bit field:	—	—	—	—	BCSSTSL[3:0]	—	—	—	—	—	SWNUM[2:0]
------------	---	---	---	---	--------------	---	---	---	---	---	------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
2:0	SWNUM[2:0]	Sweep Conversion Channel Number per scan Specifies the scan period of fixed channel scan mode. 0x0: 1 times 0x1: 2 times ⋮ 0x7: 8 times	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R

Bit	Symbol	Function	R/W
11:8	BCSSTSL[3:0]	Blank Channel Sampling State Table Select Sampling state table selection bit for blank channel. Select the analog input sampling time setting table. The sampling state value of the selected table is used. 0x0: Select sampling state table 0 (ADSSTR0.SST0[9:0]) 0x1: Select sampling state table 1 (ADSSTR0.SST1[9:0]) 0x2: Select sampling state table 2 (ADSSTR1.SST2[9:0]) 0x3: Select sampling state table 3 (ADSSTR1.SST3[9:0]) ⋮ 0xF: Select sampling state table 15 (ADSSTR7.SST15[9:0])	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R
17:16	SWFIX[1:0]	Fixed channel Number Specifies the fixed channel number. 0x0: 1 fixed channel 0x1: 2 fixed channels 0x2: 3 fixed channels 0x3: Setting prohibited	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R
26:24	SWNOFIX[2:0]	Non-Fixed Channel Number Specifies the non-fixed channel number. 0x1: 1 channels Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R

The ADSWNR0 register sets the A/D conversion period for the fixed channel scan of A/D converter Unit 0.

SWNUM[2:0] bits (Sweep Conversion Channel Number per scan)

The SWNUM[2:0] bits specify the AD conversion cycle for fixed channel scan of A/D converter Unit 0.

The A/D conversion cycle is the value set by SWNUM[2:0] + 1.

BCSSTSL[3:0] bits (Blank Channel Sampling State Table Select)

The BCSSTSL[3:0] bits set the number of sampling states on blank channels that do not get the A/D conversion data.

SWFIX[1:0] bits (Fixed channel Number)

The SWFIX[1:0] bits set the number of fixed channels that are converted channels sweeping through each scan.

SWNOFIX[2:0] bits (Non-Fixed Channel Number)

The SWNOFIX[2:0] bits set the number of non-fixed channels that are converted while switching channels in each scan.

Note: This register is valid only when fixed channel continuous scan is set (ADMD0[3:0] = 0xB).

54.2.2.17 ADSWNR1 : A/D conversion times per scan Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x064

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	SWNOFIX[2:0]			—	—	—	—	—	—	SWFIX[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	BCSSTSL[3:0]			—	—	—	—	—	—	SWNUM[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SWNUM[2:0]	Sweep Conversion Channel Number per scan Specifies the scan period of fixed channel scan mode. 0x0: 1 times 0x1: 2 times ⋮ 0x7: 8 times	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R
11:8	BCSSTSL[3:0]	Blank Channel Sampling State Table Select Sampling state table selection bit for blank channel. Select the analog input sampling time setting table. The sampling state value of the selected table is used. 0x0: Select sampling state table 0 (ADSSTR0.SST0[9:0]) 0x1: Select sampling state table 1 (ADSSTR0.SST1[9:0]) 0x2: Select sampling state table 2 (ADSSTR1.SST2[9:0]) 0x3: Select sampling state table 3 (ADSSTR1.SST3[9:0]) ⋮ 0xF: Select sampling state table 15 (ADSSTR7.SST15[9:0])	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R
17:16	SWFIX[1:0]	Fixed channel Number Specifies the fixed channel number. 0x0: 1 fixed channel 0x1: 2 fixed channels 0x2: 3 fixed channels 0x3: Setting prohibited	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R
26:24	SWNOFIX[2:0]	Non-Fixed Channel Number Specifies the non-fixed channel number. 0x1: 1 channels Others: Setting prohibited	R/W
31:27	—	These bits are read as 0. The write value should be 0.	R

The ADSWNR1 register sets the A/D conversion period for the fixed channel scan of A/D converter Unit 1.

SWNUM[2:0] bits (Sweep Conversion Channel Number per scan)

The SWNUM[2:0] bits specify the AD conversion cycle for fixed channel scan of A/D converter Unit 1.

The A/D conversion cycle is the value set by SWNUM[2:0] + 1.

BCSSTSL[3:0] bits (Blank Channel Sampling State Table Select)

The BCSSTSL[3:0] bits set the number of sampling states on blank channels that do not get the A/D conversion data.

SWFIX[1:0] bits (Fixed channel Number)

The SWFIX[1:0] bits set the number of fixed channels that are converted channels sweeping through each scan.

SWNOFIX[2:0] bits (Non-Fixed Channel Number)

The SWNOFIX[2:0] bits set the number of non-fixed channels that are converted while switching channels in each scan.

Note: This register is valid only when fixed channel continuous scan is set (ADMD1[3:0] = 0xB).

54.2.2.18 ADDECCR : One-channel Continuous Scan Mode Decimation Control Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	DCIM1[4:0]				—	—	—	DCIM0[4:0]				—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	DCIM0[4:0]	One-channel continuous scan mode output decimation setting for A/D converter unit 0 0x0: No decimation 0x3: Output data once for every 4 conversions (thinned 3 times) 0x7: Output data once for every 8 conversions (thinned out 7 times) Others: Setting prohibited	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
12:8	DCIM1[4:0]	One-channel continuous scan mode output decimation setting for A/D converter unit 1 0x0: No decimation 0x3: Output data once for every 4 conversions (thinned 3 times) 0x7: Output data once for every 8 conversions (thinned out 7 times) Others: Setting prohibited	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

ADDECCR register controls the decimation of data output in the one-channel continuous scan mode.

54.2.2.19 ADACMDR : A/D Converter Accuracy Mode Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x084

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADHA CMD1	—	—	—	—	—	—	—	ADHA CMD0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADHACMD0	A/D converter unit 0 high accuracy mode setting 0: High accuracy mode of A/D converter is disabled 1: High accuracy mode of A/D converter is enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	ADHACMD1	A/D converter unit 1 high accuracy mode setting 0: High accuracy mode of A/D converter is disabled 1: High accuracy mode of A/D converter is enabled	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

The ADACMDR register controls the accuracy mode of the A/D converter.

ADHACMDx (x = 0, 1) bit (A/D Converter Accuracy Mode Setting Register)

The ADHACMDx (x = 0, 1) bit selects the accuracy mode of the A/D converter.

Setting this bit to 1 enables high accuracy mode of the A/D converter. Enabling the high accuracy mode provides higher accuracy A/D conversion results than when the high accuracy mode is disabled. For details on analog characteristics, see [section 70, Electrical Characteristics](#).

When setting this bit to 1 (enable high accuracy mode), some register setting restrictions apply to this function. For details, see .

54.2.3 Virtual Channel

54.2.3.1 ADCHCRn : A/D Conversion Channel Configuration Register n (n = 0 to 32)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x600 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	SSTSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AINM D	CNVCS[6:0]							—	—	—	SGSEL[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	SGSEL[4:0]	Scan Group Selection 0x00: Not select any scan group 0x01: Select the scan group 0 0x02: Select the scan group 1 0x03: Select the scan group 2 0x04: Select the scan group 3 ⋮ 0x09: Select the scan group 8 Others: Setting prohibited	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
14:8	CNVCS[6:0]	A/D Conversion Channel Selection 0x00: AN000 0x01: AN001 ⋮ 0x16: AN022 0x60: Self-diagnosis A/D unit 0 ^{*1} 0x61: Self-diagnosis A/D unit 1 ^{*1} 0x64: Temperature sensor 0x65: Internal reference voltage 0x66: VBATT 1/6 voltage monitor output 0x68: D/A converter channel 0 0x69: D/A converter channel 1 0x70: Self-diagnosis Sample-and-hold Circuit Unit 0 ^{*1} 0x71: Self-diagnosis Sample-and-hold Circuit Unit 1 ^{*1} 0x72: Self-diagnosis Sample-and-hold Circuit Unit 2 ^{*1} 0x74: Self-diagnosis Sample-and-hold Circuit Unit 4 ^{*1} 0x75: Self-diagnosis Sample-and-hold Circuit Unit 5 ^{*1} 0x76: Self-diagnosis Sample-and-hold Circuit Unit 6 ^{*1} Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
15	AINMD	Analog Input mode selection*1,*2 0: Single-ended input 1: Differential input	R/W
19:16	SSTSEL[3:0]	Sampling State Table Selection 0x0: Select the sampling state table 0 0x1: Select the sampling state table 1 0x2: Select the sampling state table 2 0x3: Select the sampling state table 3 ⋮ 0xF: Select the sampling state table 15	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Select Differential input when self-diagnosis is selected.

Note 2. When using channel-dedicated sample-and-hold circuit, ADSHCRm.SHMDn (m = 0, 1, n = 0 to 2, 4 to 6) and corresponding ADCHCRn.AINMD (n = 0 to 32) setting must be matched.

The ADCHCRn register selects the Virtual channel configuration for A/D conversion.

SGSEL[4:0] bits (Scan Group Selection)

The SGSEL[4:0] bits select which scan group to assign the selected channel in CNVCS[6:0]. If SGSEL[4:0] is set to 0x00, the channel selected by CNVCS[6:0] is not available for A/D conversion. If SGSEL[4:0] is set to other than 0x00, the channel selected by CNVCS[6:0] is available for A/D conversion in the scan operation of the scan group corresponding to the setting value.

Up to 8 A/D conversion channels can be assigned per scan group in the SAR mode or Oversampling mode; Up to 4 A/D conversion channels can be assigned per scan group in the Hybrid mode. Do not assign more than specified number of A/D conversion channels per scan group.

CNVCS[6:0] bits (A/D Conversion Channel Selection)

The CNVCS[6:0] bits select the analog signal source for the A/D conversion. The CNVCS[6:0] bits should be set for the channel number corresponding to the Analog Input or the Extended Analog function. For A/D conversion of the analog signal source in Differential input mode, the CNVCS[6:0] bits should be set the even-numbered channel; When the odd-numbered channel is set in Differential input mode, A/D conversion is not guaranteed.

To A/D convert the channel selected by the CNVCS[6:0] bits, it should be assigned to one of the scan groups by the SGSEL[4:0] bits.

AINMD bits (Analog Input mode selection)

The AINMD bit selects the input method for A/D conversion of the analog signal source selected by the CNVCS[6:0] bits. When the AINMD bit is set to 0, A/D conversion is performed in single-ended input. When the AINMD bit is set to 1, A/D conversion is performed in differential input.

If the self-diagnosis channel is selected with the CNVCS[6:0] bits, a differential input should be selected.

SSTSEL[3:0] bits (Sampling State Table Selection)

The SSTSEL[3:0] bits select the sampling state table for A/D conversion of the analog signal source selected by the CNVCS[6:0] bits. The number of sampling states is the value set in the selected sampling state table.

54.2.3.2 ADDOPCRAn : A/D Conversion Data Operation Control A Register n (n = 0 to 32)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x604 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	OFSETSEL[3:0]				—	—	—	—	GAINSEL[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DFSEL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	DFSEL[2:0]	Digital Filter Selection *1 *2 0x0: Not use the digital filter 0x1: Use the 1st digital filter 0x2: Use the 2nd digital filter 0x3: Use the 3rd digital filter 0x4: Use the 4th digital filter Others: Setting prohibited	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
19:16	GAINSEL[3:0]	User Gain Table Selection 0x0: Not use the User Gain Table 0x1: Use the User Gain Table 0 0x2: Use the User Gain Table 1 0x3: Use the User Gain Table 2 ⋮ 0x8: Use the User Gain Table 7 Others: Setting prohibited	R/W
23:20	—	These bits are read as 0. The write value should be 0.	R/W
27:24	OFSETSEL[3:0]	User Offset Table Selection 0x0: Not use the user offset table 0x1: Use the user offset table 0 0x2: Use the user offset table 1 0x3: Use the user offset table 2 ⋮ 0x8: Use the user offset table 7 Others: Setting prohibited	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Whenever A/D conversion is performed in Oversampling mode or Hybrid mode, the digital filter must be used.

Note 2. Whenever A/D conversion is performed in SAR mode, the digital filter must not be used.

The ADDOPCRAn register is one of the registers that selects the data calculation function for the A/D conversion data of virtual channel n.

DFSEL[2:0] bits (Digital Filter Selection)

The DFSEL[2:0] bits select the digital filter for inputting A/D converted data of virtual channel n. When 000b is set in the DFSEL[2:0] bits, the digital filter is not used. When 001b, 010b, 011b, or 100b is set in the DFSEL[2:0] bits, the digital filter corresponding to the setting of the DFSEL[2:0] bits is selected. The characteristics of the digital filters are specified by ADDFSRm (m = 0, 1) register.

For SAR mode, select 000b for the DFSEL[2:0] bits. For Oversampling mode or Hybrid mode, set the DFSEL[2:0] bits to 001b, 010b, 011b, or 100b.

GAINSEL[3:0] bits (User Gain Table Selection)

The GAINSEL[3:0] bits select the User Gain Table for adjusting the A/D conversion data of virtual channel n.

When the GAINSEL[3:0] bits are set to 0x0, no gain adjustment is performed. When the GAINSEL[3:0] bits are set to any of 0x1 to 0x8, the gain adjustment is performed according to the User Gain Table selected by the GAINSEL[3:0] bits.

OFSETSEL[3:0] bits (User Offset Table Selection)

The OFSETSEL[3:0] bits select the user offset table for adjusting the A/D conversion data of virtual channel n. When the OFSETSEL[3:0] bits are set to 0x0, no offset adjustment is performed. When the OFSETSEL[3:0] bits are set to any of 0x1 to 0x8, the offset adjustment is performed according to the User Offset Table selected by the OFSETSEL[3:0] bits.

54.2.3.3 ADDOPCRBn : A/D Conversion Data Operation Control B Register n (n = 0 to 32)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x608 + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	CMPT BLE7	CMPT BLE6	CMPT BLE5	CMPT BLE4	CMPT BLE3	CMPT BLE2	CMPT BLE1	CMPT BLE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ADC[3:0]				—	—	—	—	—	—	AVEMD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	AVEMD[1:0]	Addition/Averaging Mode Selection 0 0: Not use Addition/Averaging mode 0 1: Addition mode 1 0: Averaging mode 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
11:8	ADC[3:0]	Addition/Averaging Times Selection 0x0: 1-time conversion (no addition, same as normal conversion) 0x1: 2-time conversion (1 addition) 0x3: 4-time conversion (3 additions) 0x4: 8-time conversion (7 additions) 0x5: 16-time conversion (15 additions) 0x6: 32-time conversion (31 additions) 0x7: 64-time conversion (63 additions) 0x8: 128-time conversion (127 additions) 0x9: 256-time conversion (255 additions) 0xA: 512-time conversion (511 additions) 0xB: 1024-time conversion (1023 additions) Others: Setting prohibited	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
23:16	CMPTBLE0 to CMPTBLE7	Compare Match Enable The suffix number of each bit symbol corresponds to the compare match table number m. 0: Disable the compare match with the compare match table m 1: Enable the compare match with the compare match table m	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADDOPCRBn register is one of the registers that selects the data calculation function for the A/D conversion data of virtual channel n.

AVEMD[1:0] bits (Addition/Averaging Mode Selection)

The AVEMD[1:0] bits select the addition mode or the averaging mode for the A/D conversion of virtual channel n. When the AVEMD[1:0] bits are set to 00b, the addition/averaging operation is not performed. When the AVEMD[1:0] bits are set to 01b or 10b, the addition/averaging operation is performed according to the setting value of the ADC[3:0] bits.

ADC[3:0] bits (Addition/Averaging Times Selection)

The ADC[3:0] bits select the number of times of addition/averaging for the A/D conversion of virtual channel n. If the addition mode or the averaging mode is selected in the AVEMD[1:0] bits, the A/D conversion is performed a number of times according to the setting value of the ADC[3:0] bits.

CMPTBLEm bits (Compare Match Enable) (m = 0 to 7)

The CMPTBLEm bits enable/disable the compare match with the compare match table m for the A/D conversion data of virtual channel n.

54.2.3.4 ADDOPCRCn : A/D Conversion Data Operation Control C Register n (n = 0 to 32)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x60C + 0x10 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	SIGNSEL	—	—	ADPRC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	LIMTBLS[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	LIMTBLS[3:0]	Limiter Clip Table Selection 0x0: Not use the Limiter Clip Table 0x1: Use the Limiter Clip Table 0 0x2: Use the Limiter Clip Table 1 0x3: Use the Limiter Clip Table 2 : 0x8: Use the Limiter Clip Table 7 Others: Setting prohibited	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
17:16	ADPRC[1:0]	A/D Conversion Data Format Selection 0 0: Store the A/D conversion result as 16-bit data format 0 1: Store the A/D conversion result as 14-bit data format 1 0: Store the A/D conversion result as 12-bit data format 1 1: Store the A/D conversion result as 10-bit data format	R/W
19:18	—	These bits are read as 0. The write value should be 0.	R/W
20	SIGNSEL	A/D Conversion Data Signed/Un-signed Selection 0: Signed data format 1: Un-signed data format	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

The ADDOPCRCn register is one of the registers that selects the data calculation function for the A/D conversion data of virtual channel n.

LIMTBLS[3:0] bits (Limiter Clip Table Selection)

The LIMTBLS[3:0] bits select the limiter clip table for clipping the A/D conversion data of virtual channel n. When the LIMTBLS[3:0] bits are set to 0x0, the A/D conversion data is not clipped. When the LIMTBLS[3:0] bits are set to any of 0x1 to 0x8, the A/D conversion data is clipped according to the limiter clip table selected by the LIMTBLS[3:0] bits.

ADPRC[1:0] bits (A/D Conversion Data Format Selection)

The ADPRC[1:0] bits select the data length of A/D conversion data. The A/D conversion data of virtual channel n is stored to the A/D Data Register or the FIFO with selected data length.

SIGNSEL bit (A/D Conversion Data Signed/Un-signed Selection)

The SIGNSEL bit selects the Signed/Unsigned data format of the A/D conversion data. The A/D conversion data of the virtual channel n is stored to the A/D Data Register or the FIFO with selected data format.

Select the Signed data format (SIGNSEL = 0) whenever the A/D conversion of the self-diagnosis channel is performed. Select the Unsigned data format (SIGNSEL = 1) whenever the A/D conversion in SAR mode, unless to convert the self-diagnosis channel, is performed. The signed data format or the unsigned data format is available for the A/D conversion in Oversampling mode or Hybrid mode unless converting to the self-diagnosis channel.

54.2.4 A/D Conversion Configuration**54.2.4.1 AD SGDCRn : Scan Group Diagnosis Function Control Register n (n = 0 to 8)**

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x200 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ADNDIS[7:0]								—	—	ADDIS N	ADDIS P	—	—	—	ADDIS EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DIAGVAL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	DIAGVAL[2:0]	Self-diagnosis Mode Selection 0 0 0: Set when any self-diagnosis channel is not included. Setting prohibited when any self-diagnosis channels are included. 1 0 0: Self-diagnosis mode 1 1 0 1: Self-diagnosis mode 2 1 1 0: Self-diagnosis mode 3 Others: Setting prohibited	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	ADDISEN	Disconnection Detection Assist Enable 0: Disable the disconnection detection assist function 1: Enable the disconnection detection assist function	R/W
19:17	—	These bits are read as 0. The write value should be 0.	R/W
20	ADDISP	Disconnection Detection Assist Mode Selection (for the even-numbered analog channel) 0: Discharge 1: Precharge	R/W
21	ADDISN	Disconnection Detection Assist Mode Selection (for the odd-numbered analog channel) 0: Discharge 1: Precharge	R/W

Bit	Symbol	Function	R/W
23:22	—	These bits are read as 0. The write value should be 0.	R/W
31:24	ADNDIS[7:0]	Disconnection Detection Assist Period 0x0: Setting prohibited when the disconnection detection assist function is enabled 0x1: Setting prohibited 0x2: Setting prohibited 0x3: Setting prohibited Others: The number of states for the discharge or precharge period. The number of states actually valid in the operation is the value of the register setting minus one.	R/W

The AD SGDCRn register controls the diagnosis function at the scanning operation of scan group n.

DIAGVAL[2:0] bits (Self-diagnosis Mode Selection)

The DIAGVAL[2:0] bits select the self-diagnosis mode for A/D converter.

The DIAGVAL[2:0] bits must be set to 000b when the scan group n does not include any virtual channels that selected the self-diagnosis channel. The DIAGVAL[2:0] bits must be set to 100b, 101b or 110b when the scan group n is including the virtual channel that selected the self-diagnosis channel.

ADDISEN bit (Disconnection Detection Assist Enable)

The ADDISEN bit enables/disables the disconnection detection assist function. When the ADDISEN bit is set to 0, the disconnection detection assist operation is not performed. When the ADDISEN bit is set to 1, the disconnection detection assist operation according to the setting of the ADDISP bit and the ADDISN bit is performed at the scanning operation of scan group n.

ADDISP bit / ADDISN bit (Disconnection Detection Assist Mode Selection)

The ADDISP bit and the ADDISN bit select the discharge/precharge operation for the disconnection detection.

The ADDISP bit selects the discharge/precharge operation for the even-numbered analog channel.

The ADDISN bit selects the discharge/precharge operation for the odd-numbered analog channel.

ADNDIS[7:0] bits (Disconnection Detection Assist Period)

The ADNDIS[7:0] bits select the discharge/precharge period. When the disconnection detection assist function is disabled (ADDISEN = 0), the ADNDIS[7:0] bits should be set to 0x0. When the disconnection detection assist function is enabled (ADDISEN = 1), the ADNDIS[7:0] bits should be set the value within 0x3 to 0xFF. The number of states actually valid in the operation is the value of the register setting minus one. For example, if ADNDIS[7:0] = 0x05 is set, the discharge/precharge period is 4 states.

54.2.4.2 ADSSTR0 : Sampling State Table Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x240

Bit position:	31	25	16	9	0
Bit field:	— — — — —	SST1[9:0]	— — — — —	SST0[9:0]	
Value after reset:	0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0	

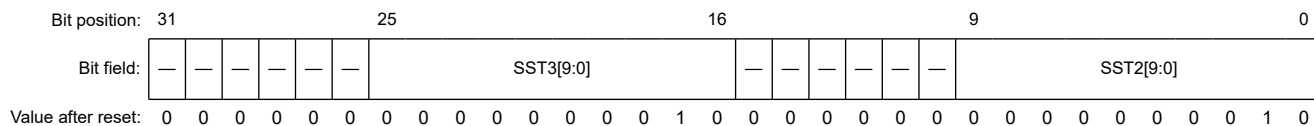
Bit	Symbol	Function	R/W
9:0	SST0[9:0]	Sampling State Table 0 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST1[9:0]	Sampling State Table 1 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR0 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.3 ADSSTR1 : Sampling State Table Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x244



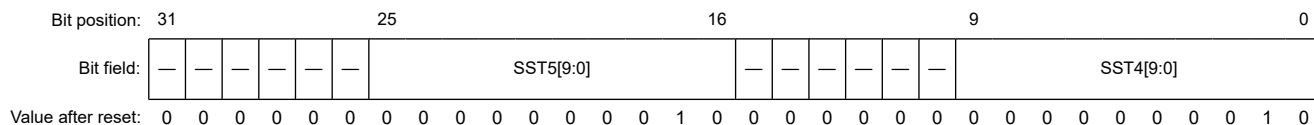
Bit	Symbol	Function	R/W
9:0	SST2[9:0]	Sampling State Table 2 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST3[9:0]	Sampling State Table 3 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR1 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.4 ADSSTR2 : Sampling State Table Register 2

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x248



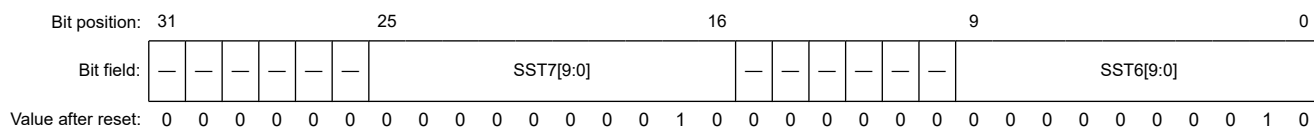
Bit	Symbol	Function	R/W
9:0	SST4[9:0]	Sampling State Table 4 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST5[9:0]	Sampling State Table 5 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR2 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.5 ADSSTR3 : Sampling State Table Register 3

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x24C



Bit	Symbol	Function	R/W
9:0	SST6[9:0]	Sampling State Table 6 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST7[9:0]	Sampling State Table 7 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR3 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.6 ADSSTR4 : Sampling State Table Register 4

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x250

Bit position:	31	25	16	9	0
Bit field:	—	—	—	—	—
Value after reset:	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	SST8[9:0]	Sampling State Table 8 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST9[9:0]	Sampling State Table 9 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR4 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.7 ADSSTR5 : Sampling State Table Register 5

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x254

Bit position:	31	25	16	9	0
Bit field:	—	—	—	—	—
Value after reset:	0	0	0	0	0

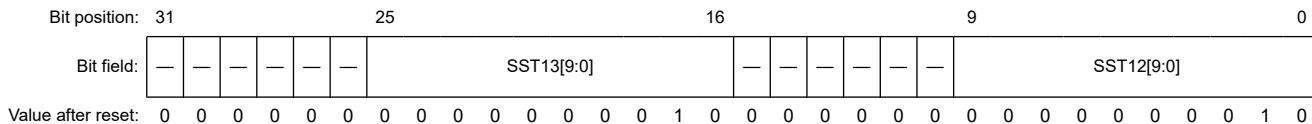
Bit	Symbol	Function	R/W
9:0	SST10[9:0]	Sampling State Table 10 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST11[9:0]	Sampling State Table 11 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR5 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.8 ADSSTR6 : Sampling State Table Register 6

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x258



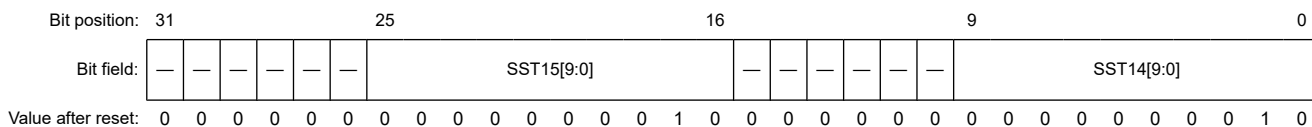
Bit	Symbol	Function	R/W
9:0	SST12[9:0]	Sampling State Table 12 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST13[9:0]	Sampling State Table 13 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR6 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.9 ADSSTR7 : Sampling State Table Register 7

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x25C



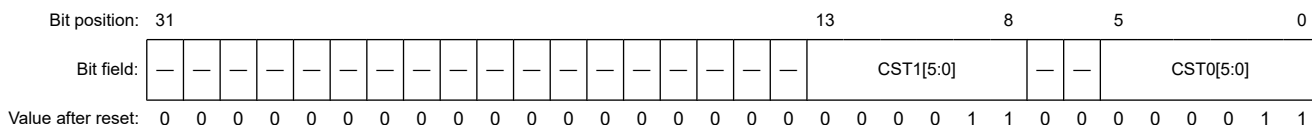
Bit	Symbol	Function	R/W
9:0	SST14[9:0]	Sampling State Table 14 These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	SST15[9:0]	Sampling State Table 15 These bits set the sampling time in the range from 2 to 1023 states.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADSSTR7 register selects the sampling time at the A/D conversion that is using each sampling state table.

54.2.4.10 ADCNVSTR : A/D Conversion State Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x260



Bit	Symbol	Function	R/W
5:0	CST0[5:0]	A/D Converter Unit 0 (ADC0) A/D successive approximation time configuration These bits set the successive approximation time in the range from 3 to 63 states.	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
13:8	CST1[5:0]	A/D Converter Unit 1 (ADC1) A/D successive approximation time configuration These bits set the successive approximation time in the range from 3 to 63 states.	R/W
31:14	—	These bits are read as 0. The write value should be 0.	R/W

The ADCNVSTR register specifies the A/D successive approximation time, which required by the A/D converter Unit 0 or Unit 1 to convert the sampled analog input to the digital value, in the number of cycles based on A/D conversion clock (ADCLK). The A/D successive approximation time does not include the sampling time of the analog input.

The A/D successive approximation time must be set to meet the value specified in [section 70, Electrical Characteristics](#). If the successive approximation time is set to a value over the range specified for the electrical characteristics, the A/D conversion result is not guaranteed. In addition, the A/D successive approximation time for each A/D converter should be set to be the same. (Set the same value to CST0[5:0] bits and CST1[5:0] bits).

54.2.5 S&H and Others

54.2.5.1 ADShCR0 : Channel-dedicated Sample-and-hold Circuit Control Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x280

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SHMD 2	SHMD 1	SHMD 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SHEN 2	SHEN 1	SHEN 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SHEN0	Channel-dedicated Sample-and-hold Circuit Unit 0 Select 0: Bypass the circuit unit 0 1: Use the circuit unit 0	R/W
1	SHEN1	Channel-dedicated Sample-and-hold Circuit Unit 1 Select 0: Bypass the circuit unit 1 1: Use the circuit unit 1	R/W
2	SHEN2	Channel-dedicated Sample-and-hold Circuit Unit 2 Select 0: Bypass the circuit unit 2 1: Use the circuit unit 2	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	SHMD0*1	Channel-dedicated Sample-and-hold Circuit Unit 0 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W
17	SHMD1*1	Channel-dedicated Sample-and-hold Circuit Unit 1 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W
18	SHMD2*1	Channel-dedicated Sample-and-hold Circuit Unit 2 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W

Bit	Symbol	Function	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. When using channel-dedicated sample-and-hold circuit, ADShCRm.SHMDn (m = 0, 1, n = 0 to 2, 4 to 6) and corresponding ADChCRn.AINMD (n = 0 to 32) setting must be matched.

The ADShCR0 register controls the channel-dedicated sample-and-hold circuits (unit 0, unit 1 and unit 2) connected to the A/D converter unit 0 (ADC0).

SHENn bit (Channel-dedicated Sample-and-hold Circuit Bypass Select) (n = 0 to 2)

The SHENn bit selects whether to use or bypass the channel-dedicated sample-and-hold circuits unit n.

When the SHENn bit is set to 0, the channel-dedicated sample-and-hold circuits unit n is disabled. If A/D conversion of analog input channel which is connected to the channel-dedicated sample-and-hold circuit unit n is performed, the channel-dedicated sample-and-hold circuit is not used and is bypassed.

When the SHENn bit is set to 1, the channel-dedicated sample-and-hold circuits unit n is enabled. In this case, A/D conversion of analog input channel to which the channel-dedicated sample-and-hold circuit unit n is performed with the channel-dedicated sample-and-hold circuit used.

SHMDn bit (Channel-dedicated Sample-and-hold Circuit Input Mode Select) (n = 0 to 2)

The SHMDn bit selects the input mode of the channel-dedicated sample-and-hold circuits unit n.

When the SHMDn bit is set to 0, the channel-dedicated sample-and-hold circuits unit n is in Single-ended input mode.

When the SHMDn bit is set to 1, the channel-dedicated sample-and-hold circuits unit n is in Differential input mode.

54.2.5.2 ADShCR1 : Channel-dedicated Sample-and-hold Circuit Control Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x28C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SHMD 6	SHMD 5	SHMD 4
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SHEN 6	SHEN 5	SHEN 4
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SHEN4	Channel-dedicated Sample-and-hold Circuit Unit 4 Select 0: Bypass the circuit unit 4 1: Use the circuit unit 4	R/W
1	SHEN5	Channel-dedicated Sample-and-hold Circuit Unit 5 Select 0: Bypass the circuit unit 5 1: Use the circuit unit 5	R/W
2	SHEN6	Channel-dedicated Sample-and-hold Circuit Unit 6 Select 0: Bypass the circuit unit 6 1: Use the circuit unit 6	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	SHMD4 ^{*1}	Channel-dedicated Sample-and-hold Circuit Unit 4 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W
17	SHMD5 ^{*1}	Channel-dedicated Sample-and-hold Circuit Unit 5 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W

Bit	Symbol	Function	R/W
18	SHMD6*1	Channel-dedicated Sample-and-hold Circuit Unit 6 Input Mode Select 0: Single-ended Input mode 1: Differential Input mode	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. When using channel-dedicated sample-and-hold circuit, ADSHCRm.SHMDn (m = 0, 1, n = 0 to 2, 4 to 6) and corresponding ADCHCRn.AINMD (n = 0 to 32) setting must be matched.

The ADSHCR1 register controls the channel-dedicated sample-and-hold circuits (unit 4, unit 5 and unit 6) connected to the A/D converter unit 1 (ADC1).

SHENn bit (Channel-dedicated Sample-and-hold Circuit Bypass Select) (n = 4 to 6)

The SHENn bit selects whether to use or bypass the channel-dedicated sample-and-hold circuits unit n.

When the SHENn bit is set to 0, the channel-dedicated sample-and-hold circuits unit n is disabled. If A/D conversion of analog input channel which is connected to the channel-dedicated sample-and-hold circuit unit n is performed, the channel-dedicated sample-and-hold circuit is not used and is bypassed.

When the SHENn bit is set to 1, the channel-dedicated sample-and-hold circuits unit n is enabled. In this case, A/D conversion of analog input channel to which the channel-dedicated sample-and-hold circuit unit n is performed with the channel-dedicated sample-and-hold circuit used.

SHMDn bit (Channel-dedicated Sample-and-hold Circuit Input Mode Select) (n = 4 to 6)

The SHMDn bit selects the input mode of the channel-dedicated sample-and-hold circuits unit n.

When the SHMDn bit is set to 0, the channel-dedicated sample-and-hold circuits unit n is in Single-ended input mode.

When the SHMDn bit is set to 1, the channel-dedicated sample-and-hold circuits unit n is in Differential input mode.

54.2.5.3 ADSDCR0 : Channel-dedicated Sample-and-hold Circuit Diagnosis Control Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x284

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	SH0DIS	—	—	—	SH0DISEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SH0DISEN	A/D converter unit 0 channel-dedicated sample-and-hold circuit disconnection detection assist enable 0: Disconnection detection assist is disabled 1: Disconnection detection assist is enabled	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	SH0DIS	A/D converter unit 0 channel-dedicated sample-and-hold circuit disconnection detection assist selection 0: Discharge 1: Precharge	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: For more details about restrictions, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit.](#)

54.2.5.6 ADShSTR1 : Channel-dedicated Sample-and-hold Circuit State Register 1

Offset address: 0x294

Bit	Symbol	Function	R/W
7:0	SHSST[7:0]	Channel-dedicated Sample-and-hold Circuit Unit 4 to 6 Sampling Time Setting These bits set the sampling time in the range from 4 to 255 states.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
18:16	SHHST[2:0]	Channel-dedicated Sample-and-hold Circuit Unit 4 to 6 Hold Mode Switching Time Setting These bits set the hold mode switching time in the range from 2 to 7 states.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

54.2.5.7 ADHSBPCR : Channel-dedicated Sample-and-hold Circuit Single-ended Input Mode Bypass Control Register

Offset address: 0x310

Bit	Symbol	Function	R/W
0	SHSBP0	Channel-dedicated sample-and-hold circuit unit 0 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN0 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W

Bit	Symbol	Function	R/W
1	SHSBP1	Channel-dedicated sample-and-hold circuit unit 1 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN1 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
2	SHSBP2	Channel-dedicated sample-and-hold circuit unit 2 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN2 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
3	—	This bit is read as 0. The write value should be 0.	R
4	SHSBP4	Channel-dedicated sample-and-hold circuit unit 4 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR1.SHEN4 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
5	SHSBP5	Channel-dedicated sample-and-hold circuit unit 5 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR1.SHEN5 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
6	SHSBP6	Channel-dedicated sample-and-hold circuit unit 6 single-ended input bypass selection (even-numbered analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR1.SHEN6 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R
16	SHSBN0	Channel-dedicated sample-and-hold circuit unit 0 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN0 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
17	SHSBN1	Channel-dedicated sample-and-hold circuit unit 1 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN1 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
18	SHSBN2	Channel-dedicated sample-and-hold circuit unit 2 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR0.SHEN2 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
19	—	This bit is read as 0. The write value should be 0.	R
20	SHSBN4	Channel-dedicated sample-and-hold circuit unit 4 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADSHCR1.SHEN4 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W

Bit	Symbol	Function	R/W
21	SHSBN5	Channel-dedicated sample-and-hold circuit unit 5 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADShCR1.SHEN5 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
22	SHSBN6	Channel-dedicated sample-and-hold circuit unit 6 single-ended input bypass selection (odd analog channel) Can be written only when the target channel-dedicated sample-and-hold circuit setting is ADShCR1.SHEN6 = 0. 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

Note: Write access is permitted only when ADShCRx.SHENn (x = 0, 1, n = 0, 1, 2, 4, 5, 6) of the target sample-and-hold circuit is set to 0. If ADShCRx.SHENn is set to 1, writing 1 to this bit is ignored.

ADShSBPCR register controls the channel-dedicated sample-and-hold circuit single-ended bypass mode.

SHSBPn (Channel-dedicated sample-and-hold circuit unit n single-end input bypass control for even-numbered channel) (n = 0 to 2, 4 to 6)

SHSBNn (Channel-dedicated sample-and-hold circuit unit n single-end input bypass control for odd-numbered channel) (n = 0 to 2, 4 to 6)

The SHSBPn bit and the SHSBNn bit control the analog input for A/D conversion when the channel-dedicated sample-and-hold circuit is enabled.

When the bit is set to 0, the analog output held by the channel-dedicated sample-and-hold circuit is A/D-converted.

When the bit is set to 1, the analog input of the channel-dedicated sample-and-hold circuit is A/D-converted. In this case, the analog output held by the channel-dedicated sample-and-hold circuit is ignored on the A/D conversion.

When the channel-dedicated sample-and-hold circuit is used in single-ended input mode, the analog input of unused channel of the channel-dedicated sample-and-hold circuit can be A/D-converted without going through the channel-dedicated sample-and-hold circuit.

54.2.5.8 ADShDBPCR : Channel-dedicated Sample-and-hold Circuit Differential Input Mode Bypass Control Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x314

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SHDB P8	SHDB P7	SHDB P6	SHDB P5	SHDB P4	SHDB P3	SHDB P2	SHDB P1	SHDB P0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SHDBP0	Scan group 0 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
1	SHDBP1	Scan group 1 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W

Bit	Symbol	Function	R/W
2	SHDBP2	Scan group 2 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
3	SHDBP3	Scan group 3 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
4	SHDBP4	Scan group 4 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
5	SHDBP5	Scan group 5 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
6	SHDBP6	Scan group 6 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
7	SHDBP7	Scan group 7 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
8	SHDBP8	Scan group 8 channel-dedicated sample-and-hold circuit differential input bypass select 0: Use the sample-and-hold circuit 1: Bypass the sample-and-hold circuit	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

ADSHDBPCR register controls the channel-dedicated sample-and-hold circuit differential input bypass mode.

SHDBPn (Channel-dedicated sample-and-hold circuit differential input bypass control for scan group n) (n = 0 to 8)

When the bit is set to 0, the analog output held by the channel-dedicated sample-and-hold circuit specified in the scan group is A/D-converted.

When the bit is set to 1, the analog input of the channel-dedicated sample-and-hold circuit specified in the scan group is A/D-converted. In this case, the analog output held by the channel-dedicated sample-and-hold circuit is ignored on the A/D conversion.

In the case that the disconnection detection assist function is needed when the channel-dedicated sample-and-hold circuits are used in differential input mode, the input mode must be changed to single-ended mode. However, if the input mode is changed to use the disconnection detection assist function built in the channel-dedicated sample-and-hold circuit, calibration of the channel-dedicated sample-and-hold circuit is required.

This function allows bypassing the channel-dedicated sample-and-hold circuits to use the disconnection detection assist function without recalibration of the channel-dedicated sample-and-hold circuits.

54.2.5.9 ADHSDCR0 : Channel-dedicated Sample-and-hold Circuit Self-diagnosis Control Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x318

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	SHDIAG7[2:0]			—	SHDIAG6[2:0]			—	SHDIAG5[2:0]			—	SHDIAG4[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	SHDIAG3[2:0]			—	SHDIAG2[2:0]			—	SHDIAG1[2:0]			—	SHDIAG0[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SHDIAG0[2:0]	Scan group 0 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
6:4	SHDIAG1[2:0]	Scan group 1 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
7	—	This bit is read as 0. The write value should be 0.	R/W
10:8	SHDIAG2[2:0]	Scan group 2 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
14:12	SHDIAG3[2:0]	Scan group 3 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
15	—	This bit is read as 0. The write value should be 0.	R/W
18:16	SHDIAG4[2:0]	Scan group 4 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
22:20	SHDIAG5[2:0]	Scan group 5 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W

Bit	Symbol	Function	R/W
23	—	This bit is read as 0. The write value should be 0.	R/W
26:24	SHDIAG6[2:0]	Scan group 6 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
30:28	SHDIAG7[2:0]	Scan group 7 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: For more details about restrictions, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

The ADSSHSDCR0 register controls the diagnostic function of the channel-dedicated sample-and-hold circuit.

54.2.5.10 ADSSHSDCR1 : Channel-dedicated Sample-and-hold Circuit Self-diagnosis Control Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x31C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SHDIAG8[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SHDIAG8[2:0]	Scan group 8 channel-dedicated sample-and-hold circuit self-diagnostic voltage selection 000: Normal operation 001: Self-diagnostic voltage 0 V 010: Self-diagnostic voltage -VREFH 011: Self-diagnostic voltage +VREFH 100: Self-diagnostic voltage 0 V 101: Self-diagnostic voltage -VREFH/2 110: Self-diagnostic voltage +VREFH/2 111: Setting prohibited	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: For more details about restrictions, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

The ADSSHSDCR1 register controls the diagnostic function of the channel-dedicated sample-and-hold circuit.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC04

[illegible]

Bit	Symbol	Function	R/W
0	SHCSST0	Channel-dedicated sample-and-hold circuit constant sampling control bit for A/D converter unit 0 (ADC0) When SHCSST0 is enabled, channel-dedicated sample-and-hold circuit sampling starts. 0: Disable channel-dedicated sample-and-hold circuit constant sampling 1: Enable channel-dedicated sample-and-hold circuit constant sampling	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	SHCSST1	Channel-dedicated sample-and-hold circuit constant sampling control bit for A/D converter unit 1 (ADC1) When SHCSST1 is enabled, channel-dedicated sample-and-hold circuit sampling starts. 0: Disable channel-dedicated sample-and-hold circuit constant sampling 1: Enable channel-dedicated sample-and-hold circuit constant sampling	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

Note: Using this register is prohibited when channel-dedicated sample-and-hold circuit is not used.

Note: For more details about restrictions, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit.](#)

The ADSHCSCR register controls the constant sampling function of channel-dedicated sample-and-hold circuit.

54.2.5.12 ADREFCR : Internal Reference Voltage Monitor Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B NS = 0x5033_8000

Offset address: 0x320

Bit position: 31 0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	VDE
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----

Value after reset: 0

Bit	Symbol	Function	R/W
0	VDE	Internal Reference Voltage A/D Conversion Select 0: Disable A/D conversion of internal reference voltage 1: Enable A/D conversion of internal reference voltage	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

The ADREFCR register controls the function of the internal reference voltage circuit. When the A/D conversion of the internal reference voltage is performed, the VDE bit must be set to 1.

54.2.6 Digital Filter

54.2.6.1 ADDFSRn : A/D Converter Digital Filter Selection Register n (n = 0, 1)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: $0x340 + 0x04 \times n$

[illegible]

Bit	Symbol	Function	R/W
1:0	DFSEL0[1:0]	A/D Converter Unit n the 1st Digital Filter Characteristic Selection 0 1: Sinc filter Others: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	DFSEL1[1:0]	A/D Converter Unit n the 2nd Digital Filter Characteristic Selection 0 1: Sinc filter Others: Setting prohibited	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
17:16	DFSEL2[1:0]	A/D Converter Unit n the 3rd Digital Filter Characteristic Selection 0 1: Sinc filter Others: Setting prohibited	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
25:24	DFSEL3[1:0]	A/D Converter Unit n the 4th Digital Filter Characteristic Selection 0 1: Sinc filter Others: Setting prohibited	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADDFSRn register selects the characteristics of the digital filter for the A/D converter unit n. The selection of the digital filter to be used at A/D conversion is specified by the ADDOPCRAm register (m = 0 to 32). If the digital filter function is used with prohibited values set to the ADDFSRn.DFSELx[1:0] (x = 0 to 3) bits, the A/D conversion characteristics and A/D conversion results are not guaranteed.

54.2.7 Self-calibration

54.2.7.1 ADCALSTCR : A/D Converter Self-calibration State Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x264

[illegible]

Bit	Symbol	Function	R/W
9:0	CALADSST[9:0]	A/D Converter Self-calibration Sampling Time Configuration These bits set the sampling time in the range from 2 to 1023 states.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
21:16	CALADCST[5:0]	A/D Converter Self-calibration Successive Approximation Time Configuration. These bits set the successive approximation time in the range from 3 to 63 states.	R/W
31:22	—	These bits are read as 0. The write value should be 0.	R/W

The ADCALSTCR register specifies the sampling and the A/D successive approximation times at the self-calibration operation for all A/D converter (ADC0, ADC1). The sampling time and successive approximation time should be set by the number of clock cycles based on the A/D conversion clock (ADCLK) to meet the values specified in [section 70, Electrical Characteristics](#).

54.2.7.2 ADCALSHCR : Channel-dedicated Sample-and-hold Circuit Self-calibration State Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x2B0

Bit position:	31	18	16	7	0
Bit field:	—	—	—	—	—
Value after reset:	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CALSHSST[7:0]	Channel-dedicated Sample-and-hold Circuit Self-calibration Sampling Time Configuration These bits set the sampling time in the range from 4 to 255 states.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W
18:16	CALSHHST[2:0]	Channel-dedicated Sample-and-hold Circuit Self-calibration Hold Mode Switching Time Configuration These bits set the hold mode switching time in the range from 2 to 7 states.	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

The ADCALSHCR register specifies the sampling time and the hold mode switching time at the self-calibration operation for all channel-dedicated sample-and-hold circuits (SH0 to SH2, SH4 to SH6). The sampling time and the hold mode switching time should be set by the number of clock cycles based on the A/D conversion clock (ADCLK) to meet the values specified in [section 70, Electrical Characteristics](#).

54.2.7.3 ADUOFTRn : User Offset Table Register n (n = 0 to 7)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x360 + 0x04 × n

Bit position:	31	15	0
Bit field:	—	—	—
Value after reset:	0	0	0

Bit	Symbol	Function	R/W
15:0	UOFSET[15:0]	User Offset Table n 0x7FFF: + 32767 0x7FFE: + 32766 ⋮ 0x0002: + 2 0x0001: + 1 0x0000: 0 (without user offset) 0xFFFF: - 1 0xFFFE: - 2 ⋮ 0x8001: - 32767 0x8000: - 32768	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

The ADUOFTRn register specifies the offset value in the User Offset Table n.

54.2.7.4 ADUGTRn : User Gain Table Register n (n = 0 to 7)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: $0x380 + 0x04 \times n$

Bit position: 31 15 0

Bit field:		UGAIN[15:0]
------------	---	-------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	UGAIN[15:0]	User Gain Table n These bits set the gain value to be multiplied by the A/D conversion result. UGAIN[15:14]: Integer part of gain UGAIN[13:0]: Fractional part of gain	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

The ADUGTRn register specifies the gain value in the user gain table n.

54.2.8 Limiter Clip Function

54.2.8.1 ADLIMINTCR : Limiter Clip Interrupt Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x3A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
---------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	—	LIMIE8	LIMIE7	LIMIE6	LIMIE5	LIMIE4	LIMIE3	LIMIE2	LIMIE1	LIMIE0
------------	---	---	---	---	---	---	---	--------	--------	--------	--------	--------	--------	--------	--------	--------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
8:0	LIMIE0 to LIMIE8	Limiter Clip Interrupt n Enable bit The suffix number of each bit symbol corresponds to the limiter clip interrupt number n. 0: Disable the limiter clip interrupt n 1: Enable the limiter clip interrupt n	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

The ADLIMINTCR register enables/disables the limiter clip interrupt n.

54.2.8.2 ADLIMTRn : Limiter Clip Table Register n (n = 0 to 7)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: $0x3A4 + 0x04 \times n$

Bit position: 31 16 15 0

Bit field:	LIMU[15:0]	LIML[15:0]
------------	------------	------------

[illegible]

Bit	Symbol	Function	R/W
15:0	LIML[15:0]	Limiter clip table n : Lower-side limit value	R/W
31:16	LIMU[15:0]	Limiter clip table n : Upper-side limit value	R/W

The ADLIMTRn register specifies the lower- and upper-side limit value of the limiter clip table n.

54.2.8.3 ADLIMGRSR : Limiter Clip Scan Group Status Register

Base address: ADC_B = 0x4033_8000
ADC_B NS = 0x5033_8000

Offset address: 0xD28

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
---------------	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

[illegible][illegible]

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LIMGR F8	LIMGR F7	LIMGR F6	LIMGR F5	LIMGR F4	LIMGR F3	LIMGR F2	LIMGR F1	LIMGR F0

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
8:0	LIMGRF0 to LIMGRF8	Scan Group n Limiter Clip Flag The suffix number of each bit symbol corresponds to the scan group number n. 0: Limiter clip for scan group n is not detected 1: Limiter clip for scan group n is detected	R
31:9	—	These bits are read as 0.	R

The ADLIMGRSR register indicates whether the limiter clip occurred in the scanning operation for scan group n. Each flag can be cleared in the ADLIMGRSCR.

54.2.8.4 ADLIMCHSR0 : Limiter Clip Channel Status Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD2C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	LIMCH F22	LIMCH F21	LIMCH F20	LIMCH F19	LIMCH F18	LIMCH F17	LIMCH F16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LIMCH F15	LIMCH F14	LIMCH F13	LIMCH F12	LIMCH F11	LIMCH F10	LIMCH F9	LIMCH F8	LIMCH F7	LIMCH F6	LIMCH F5	LIMCH F4	LIMCH F3	LIMCH F2	LIMCH F1	LIMCH F0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	LIMCHF0 to LIMCHF22	Analog Channel n: Limiter Clip Flag The suffix number of each bit symbol corresponds to the analog channel number n. 0: Limiter clip is not detected 1: Limiter clip is detected	R
31:23	—	These bits are read as 0.	R

The ADLIMCHSR0 register indicates whether the limiter clip occurred when the A/D conversion for the analog channel n is performed. Each flag can be cleared in ADLIMCHSR0.

54.2.8.5 ADLIMEXSR : Extended Analog Limiter Clip Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD38

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	LIMEX F22	LIMEX F21	LIMEX F20	—	LIMEX F18	LIMEX F17	LIMEX F16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LIMEX F9	LIMEX F8	—	LIMEX F6	LIMEX F5	LIMEX F4	—	—	LIMEX F1	LIMEX F0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LIMEXF0	Self-diagnosis Channel for A/D unit 0: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
1	LIMEXF1	Temperature Sensor Channel for A/D unit 1: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
3:2	—	These bits are read as 0.	R
4	LIMEXF4	Temperature Sensor Channel: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
5	LIMEXF5	Internal Reference Voltage Channel: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R

Bit	Symbol	Function	R/W
6	LIMEXF6	VBATT 1/6 voltage monitor output Channel: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
7	—	This bit is read as 0.	R
8	LIMEXF8	D/A Converter 0 Channel: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
9	LIMEXF9	D/A Converter 1 Channel: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
15:10	—	These bits are read as 0.	R
16	LIMEXF16	Self-diagnosis Channel for Sample-and-hold circuit unit0: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
17	LIMEXF17	Self-diagnosis Channel for Sample-and-hold circuit unit1: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
18	LIMEXF18	Self-diagnosis Channel for Sample-and-hold circuit unit2: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
19	—	This bit is read as 0.	R
20	LIMEXF20	Self-diagnosis Channel for Sample-and-hold circuit unit4: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
21	LIMEXF21	Self-diagnosis Channel for Sample-and-hold circuit unit5: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
22	LIMEXF22	Self-diagnosis Channel for Sample-and-hold circuit unit6: Limiter Clip Flag 0: Limiter clip is not detected 1: Limiter clip is detected	R
31:23	—	These bits are read as 0.	R

The ADLIMEXSR register indicates whether the limiter clip occurred when the A/D conversion for the extended analog function is performed. Each flag can be cleared in the ADLIMEXSCR.

54.2.8.6 ADLIMGRSCR : Limiter Clip Scan Group Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD3C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	LIMGR C8	LIMGR C7	LIMGR C6	LIMGR C5	LIMGR C4	LIMGR C3	LIMGR C2	LIMGR C1	LIMGR C0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	LIMGRC0 to LIMGRC8	Scan Group n Limiter Clip Flag Clear The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: ADLIMGRSR.LIMGRFn is cleared	W
31:9	—	The write value should be 0.	W

The ADLIMGRSCR register clears the limiter clip flag for scan group n (ADLIMGRSR.LIMGRFn).

54.2.8.7 ADLIMCHSCR0 : Limiter Clip Channel Status Clear Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD40

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	LIMCH C22	LIMCH C21	LIMCH C20	LIMCH C19	LIMCH C18	LIMCH C17	LIMCH C16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LIMCH C15	LIMCH C14	LIMCH C13	LIMCH C12	LIMCH C11	LIMCH C10	LIMCH C9	LIMCH C8	LIMCH C7	LIMCH C6	LIMCH C5	LIMCH C4	LIMCH C3	LIMCH C2	LIMCH C1	LIMCH C0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	LIMCHC0 to LIMCHC22	Analog Channel n Limiter Clip Flag Clear bit The suffix number of each bit symbol corresponds to the analog channel number n. 0: No effect 1: ADLIMCHSR0.LIMCHFn is cleared	W
31:23	—	The write value should be 0.	W

The ADLIMCHSCR0 register clears the limiter clip flag for the analog channel n (ADLIMCHSR0.LIMCHFn).

54.2.8.8 ADLIMEXSCR : Extended Analog Limiter Clip Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD4C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	LIMEX C22	LIMEX C21	LIMEX C20	—	LIMEX C18	LIMEX C17	LIMEX C16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	LIMEX C9	LIMEX C8	—	LIMEX C6	LIMEX C5	LIMEX C4	—	—	LIMEX C1	LIMEX C0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LIMEXC0	Self-diagnosis channel for A/D unit 0: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC0 is cleared	W
1	LIMEXC1	Self-diagnosis channel for A/D unit 1: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC1 is cleared	W

Bit	Symbol	Function	R/W
3:2	—	The write value should be 0.	W
4	LIMEXC4	Temperature sensor channel: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC4 is cleared	W
5	LIMEXC5	Internal reference voltage channel: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC5 is cleared	W
6	LIMEXC6	VBATT 1/6 voltage monitor output channel: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC6 is cleared	W
7	—	The write value should be 0.	W
8	LIMEXC8	D/A converter 0 channel: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC8 is cleared	W
9	LIMEXC9	D/A converter 1 channel: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC9 is cleared	W
15:10	—	The write value should be 0.	W
16	LIMEXC16	Self-diagnosis channel for sample-and-hold circuit unit 0: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC16 is cleared	W
17	LIMEXC17	Self-diagnosis channel for sample-and-hold circuit unit 1: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC17 is cleared	W
18	LIMEXC18	Self-diagnosis channel for sample-and-hold circuit unit 2: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC18 is cleared	W
19	—	The write value should be 0.	W
20	LIMEXC20	Self-diagnosis channel for sample-and-hold circuit unit 4: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC20 is cleared	W
21	LIMEXC21	Self-diagnosis channel for sample-and-hold circuit unit 5: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC21 is cleared	W
22	LIMEXC22	Self-diagnosis channel for sample-and-hold circuit unit 6: Limiter Clip Flag Clear 0: No effect 1: ADLIMEXSR.LIMEXC22 is cleared	W
31:23	—	The write value should be 0.	W

The ADLIMEXSCR register clears the limiter clip flag for the extended analog function.

54.2.9 Compare Match Function

54.2.9.1 ADCMPENR : Compare Match Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CMPE N7	CMPE N6	CMPE N5	CMPE N4	CMPE N3	CMPE N2	CMPE N1	CMPE N0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CMPE0 to CMPE7	Compare Match n Enable The suffix number of each bit symbol corresponds to the compare match number n. 0: Disable the compare match n 1: Enable the compare match n	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

The ADCMPENR register enables/disables the compare match n.

54.2.9.2 ADCMPINTCR : Compare Match Interrupt Enable Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x404

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CMPIE 3	CMPIE 2	CMPIE 1	CMPIE 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CMPIE0 to CMPIE3	Compare Match Interrupt n Enable The suffix number of each bit symbol corresponds to the compare match interrupt number n. 0: Disable the compare match interrupt n 1: Enable the compare match interrupt n	R/W
31:4	—	These bits are read as 0. The write value should be 0.	R/W

The ADCMPINTCR register enables/disables the compare match interrupt n.

54.2.9.3 ADCCMPCRn : Composite Compare Match Configuration Register n (n = 0, 1)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x408 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	CCMP TBL7	CCMP TBL6	CCMP TBL5	CCMP TBL4	CCMP TBL3	CCMP TBL2	CCMP TBL1	CCMP TBL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CCMPCND[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CCMPCND[1:0]	Composite Compare Match Condition Selection 0 0: Logical disjunction (OR) conditions Others: Setting prohibited	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
23:16	CCMPTBL0 to CCMPTBL7	Composite Compare Match Condition Table Selection 0: Not use the compare match table m 1: Use the compare match table m	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

The ADCCMPCRn register configures the conditions for the composite compare match interrupt n.

CCMPCND[1:0] bits (Composite Compare Match Condition Selection)

The CCMPCND[1:0] bits select the generating condition for the composite compare match interrupt n.

CCMPTBLm bits (Composite Compare Match Condition Table Selection) (m = 0 to 7)

The CCMPTBLm bits select whether to use or not to use the compare match table m as the generating conditions of the composite compare match interrupt.

54.2.9.4 ADCMPMDR0 : Compare Match Mode Selection Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x448

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	CMPMD3[1:0]	—	—	—	—	—	—	—	CMPMD2[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CMPMD1[1:0]	—	—	—	—	—	—	—	CMPMD0[1:0]	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CMPMD0[1:0]	Compare Match 0 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W

Bit	Symbol	Function	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	CMPMD1[1:0]	Compare Match 1 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
17:16	CMPMD2[1:0]	Compare Match 2 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W
23:18	—	These bits are read as 0. The write value should be 0.	R/W
25:24	CMPMD3[1:0]	Compare Match 3 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

The ADCMPMDR0 register selects the generating condition of each compare match event.

54.2.9.5 ADCMPMDR1 : Compare Match Mode Selection Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x44C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	CMPMD7[1:0]	—	—	—	—	—	—	—	CMPMD6[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CMPMD5[1:0]	—	—	—	—	—	—	—	CMPMD4[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CMPMD4[1:0]	Compare Match 4 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
9:8	CMPMD5[1:0]	Compare Match 5 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
17:16	CMPMD6[1:0]	Compare Match 6 : Match Mode Selection 0 0: Generate the match event when high-side level or more 0 1: Generate the match event when low-side level or less 1 0: Generate the match event when high-side level or more, or low-side level or less 1 1: Generate the match event when low-side level or more and high-side level or less	R/W

The ADCMPMDR1 register selects the generating condition of each compare match event.

54.2.9.8 ADCMPTBSCR : Compare Match Table Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	CMPT BC7	CMPT BC6	CMPT BC5	CMPT BC4	CMPT BC3	CMPT BC2	CMPT BC1	CMPT BC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	CMPTBC0 to CMPTBC7	Compare Match Table n: Match Flag Clear The suffix number of each bit symbol corresponds to the compare match table number n. 0: No effect 1: ADCMPTBSR.CMPTBFn is cleared	W
31:8	—	The write value should be 0.	W

The ADCMPTBSCR register clears the match flag for the compare match n (ADCMPTBSR.CMPTBFn).

54.2.9.9 ADCMPCHSR0 : Compare Match Channel Status Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD08

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	CMPC HF22	CMPC HF21	CMPC HF20	CMPC HF19	CMPC HF18	CMPC HF17	CMPC HF16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMPC HF15	CMPC HF14	CMPC HF13	CMPC HF12	CMPC HF11	CMPC HF10	CMPC HF9	CMPC HF8	CMPC HF7	CMPC HF6	CMPC HF5	CMPC HF4	CMPC HF3	CMPC HF2	CMPC HF1	CMPC HF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	CMPCHF0 to CMPCHF22	Analog Channel n: Compare Match Flag The suffix number of each bit symbol corresponds to the analog channel number n. 0: Compare match is not detected 1: Compare match is detected	R
31:23	—	These bits are read as 0.	R

The ADCMPCHSR0 register indicates whether the compare match event for the Analog Channel n is detected. Each flag can be cleared in ADCMPCHSR0.

54.2.9.10 ADCMPEXSR : Extended Analog Compare Match Status Register

Base address: ADC_B = 0x4033_8000
 ADC_B_NS = 0x5033_8000

Offset address: 0xD14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	CMPE XF22	CMPE XF21	CMPE XF20	—	CMPE XF18	CMPE XF17	CMPE XF16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CMPE XF9	CMPE XF8	—	CMPE XF6	CMPE XF5	CMPE XF4	—	—	CMPE XF1	CMPE XF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CMPEXF0	Self-diagnosis channel for A/D unit 0: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
1	CMPEXF1	Self-diagnosis channel for A/D unit 1: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
3:2	—	These bits are read as 0.	R
4	CMPEXF4	Temperature sensor channel: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
5	CMPEXF5	Internal reference voltage channel: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
6	CMPEXF6	VBATT 1/6 voltage monitor output Channel: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
7	—	This bit is read as 0.	R
8	CMPEXF8	D/A converter 0 channel: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
9	CMPEXF9	D/A converter 1 channel: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
15:10	—	These bits are read as 0.	R
16	CMPEXF16	Self-diagnosis channel for sample-and-hold circuit unit 0: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
17	CMPEXF17	Self-diagnosis channel for sample-and-hold circuit unit 1: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
18	CMPEXF18	Self-diagnosis channel for sample-and-hold circuit unit 2: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
19	—	This bit is read as 0.	R
20	CMPEXF20	Self-diagnosis channel for sample-and-hold circuit unit 4: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R

Bit	Symbol	Function	R/W
21	CMPEXF21	Self-diagnosis channel for sample-and-hold circuit unit 5: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
22	CMPEXF22	Self-diagnosis channel for sample-and-hold circuit unit 6: Compare Match Flag 0: Compare match is not detected 1: Compare match is detected	R
31:23	—	These bits are read as 0.	R

The ADCMPEXSR register indicates whether the compare match event for the extended analog function channel is detected. Each flag can be cleared in ADCMPEXSCR.

54.2.9.11 ADCMPCHSCR0 : Compare Match Channel Status Clear Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD18

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	CMPC HC22	CMPC HC21	CMPC HC20	CMPC HC19	CMPC HC18	CMPC HC17	CMPC HC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMPC HC15	CMPC HC14	CMPC HC13	CMPC HC12	CMPC HC11	CMPC HC10	CMPC HC9	CMPC HC8	CMPC HC7	CMPC HC6	CMPC HC5	CMPC HC4	CMPC HC3	CMPC HC2	CMPC HC1	CMPC HC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	CMPCHC0 to CMPCHC22	Analog channel n: Compare Match Flag Clear bit The suffix number of each bit symbol corresponds to the analog channel number n. 0: No effect 1: ADCMPCHSR0.CMPCHFn is cleared	W
31:23	—	The write value should be 0.	W

The ADCMPCHSCR0 register clears the compare match flag for the Analog Channel n (ADCMPCHSR0.CMPCHFn).

54.2.9.12 ADCMPEXSCR : Extended Analog Compare Match Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD24

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	CMPE XC22	CMPE XC21	CMPE XC20	—	CMPE XC18	CMPE XC17	CMPE XC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	CMPE XC9	CMPE XC8	—	CMPE XC6	CMPE XC5	CMPE XC4	—	—	CMPE XC1	CMPE XC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CMPEXC0	Self-diagnosis channel for A/D unit 0: Compare Match Flag Clear 0: No effect 1: ADCMPEXSR.CMPEXF0 is cleared	W

Bit	Symbol	Function	R/W
1	CMPEXC1	Self-diagnosis channel for A/D unit 1: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF1 is cleared	W
3:2	—	The write value should be 0.	W
4	CMPEXC4	Temperature sensor channel: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF4 is cleared	W
5	CMPEXC5	Internal reference voltage channel: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF5 is cleared	W
6	CMPEXC6	VBATT 1/6 voltage monitor output channel: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF6 is cleared	W
7	—	The write value should be 0.	W
8	CMPEXC8	D/A converter 0 channel: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF8 is cleared	W
9	CMPEXC9	D/A converter 1 channel: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF9 is cleared	W
15:10	—	The write value should be 0.	W
16	CMPEXC16	Self-diagnosis channel for sample-and-hold circuit unit 0: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF16 is cleared	W
17	CMPEXC17	Self-diagnosis channel for sample-and-hold circuit unit 1: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF17 is cleared	W
18	CMPEXC18	Self-diagnosis channel for sample-and-hold circuit unit 2: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF18 is cleared	W
19	—	The write value should be 0.	W
20	CMPEXC20	Self-diagnosis channel for sample-and-hold circuit unit 4: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF20 is cleared	W
21	CMPEXC21	Self-diagnosis channel for sample-and-hold circuit unit 5: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF21 is cleared	W
22	CMPEXC22	Self-diagnosis channel for sample-and-hold circuit unit 6: Compare Match Flag Clear 0: No effect 1: ADCMPExSR.CMPEXF22 is cleared	W
31:23	—	The write value should be 0.	W

The ADCMPExSCR register clears the compare match flag for the Extended Analog Function Channel (ADCMPExSR.CMPEXF_n).

54.2.10 Start and Stop Control of A/D Conversion

54.2.10.1 ADCALSTR : A/D Converter Self-calibration Start Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	ADCALST1[2:0]			—	—	—	—	—	ADCALST0[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	ADCALST0[2:0] ^{*1}	A/D Converter Unit 0 (ADC0) Self-calibration Start Control [Function of each bit] b0: Start bit of the internal circuit calibration b1: Start bit of the gain and offset calibration b2: Start bit of the channel-dedicated sample-and-hold circuit calibration [Write value] 0: No effect (not start the calibration) 1: Start the calibration	W
7:3	—	The write value should be 0.	W
10:8	ADCALST1[2:0] ^{*1}	A/D Converter Unit 1 (ADC1) Self-calibration Start Control [Function of each bit] b0: Start bit of the internal circuit calibration b1: Start bit of the gain and offset calibration b2: Start bit of the channel-dedicated sample-and-hold circuit calibration [Write value] 0: No effect (not start the calibration) 1: Start the calibration	W
31:11	—	The write value should be 0.	W

Note 1. To perform a calibration operation, write 1 to each of the calibration start bits simultaneously.

The ADCALSTR register controls the start of the self-calibration for each A/D converter.

ADCALSTm[2:0] bits (A/D converter Unit m (ADCm) Self-calibration Start Control bits) (m = 0, 1)

The ADCALSTm[2:0] bits control the start of the self-calibration operation of the A/D converter Unit m. When any bit of ADCALSTm[2:0] is set to 1, the self-calibration operation corresponding to each bit is started. When multiple bits of ADCALSTm[2:0] are set to 1 simultaneously, each self-calibration operation is performed in the order as following:

1. Internal circuit calibration
2. A/D converter gain/offset calibration
3. Channel-specific sample and hold circuit gain/offset calibration

Note: The self-calibration operation corresponding to the bits set to 0 in ADCALSTm[2:0] is skipped.

The ADCALSTm[2:0] bits must be written when all A/D converters are stopped (ADSR.ADACTm = 0 and ADSR.CALACTm = 0);

Writing to the ADCALSTm[2:0] bits are prohibited when A/D converters are operating.

54.2.10.2 ADSYSTR : A/D Conversion Synchronous Software Start Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADSY ST8	ADSY ST7	ADSY ST6	ADSY ST5	ADSY ST4	ADSY ST3	ADSY ST2	ADSY ST1	ADSY ST0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	ADSYST0 to ADSYST8	Scan Group n: A/D Conversion start The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: Start the A/D conversion of scan group n	W
31:9	—	The write value should be 0.	W

The ADSYSTR register controls the start of the A/D conversion of scan group n. This register is used to start the A/D conversions of several scan groups simultaneously by software.

ADSYSTn bit (Scan Group n: A/D Conversion start) (n = 0 to 8)

The ADSYSTn bit controls the start of the A/D conversion of scan group n. When the ADSYSTn bit is set to 1, the A/D conversion of scan group n is started. Setting 0 to the ADSYSTn bit has no effect on the operation. Set 1 simultaneously to the ADSYSTn bit of the scan group to start A/D conversion simultaneously.

54.2.10.3 ADSTRn : A/D Conversion Software Start Register n (n = 0 to 8)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC20 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADST	Scan Group n A/D Conversion Start 0: No effect 1: Start the A/D conversion of scan group n	W
31:1	—	The write value should be 0.	W

The ADSTRn register controls the start of the A/D conversion of scan group n. This register is used to start the A/D conversion of a scan group by software.

If A/D conversions of several scan groups are to be started simultaneously by software, the ADSYSTR register should be used.

ADST bit (Scan Group n A/D Conversion Start)

The ADST bit controls the start of the A/D conversion of scan group n. When the ADST bit is set to 1, the A/D conversion of the scan group n is started. Setting 0 to the ADST bit has no effect on the operation.

54.2.10.4 ADSTOPR : A/D Conversion Stop Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC60

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ADST OP1	—	—	—	—	—	—	—	ADST OP0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADSTOP0	A/D Converter Unit 0 Force Stop 0: No effect 1: Force stop the operation of A/D converter unit 0	W
7:1	—	The write value should be 0.	W
8	ADSTOP1	A/D Converter Unit 1 Force Stop 0: No effect 1: Force stop the operation of A/D converter unit 1	W
31:9	—	The write value should be 0.	W

The ADSTOPR register forces each A/D converter to stop the operation. When the A/D converter is stopped with this register, the A/D conversion results with that A/D converter are not guaranteed.

ADSTOP0 bit (A/D Converter Unit 0 Force Stop)

The ADSTOP0 bit forces the A/D converter unit 0 to stop the A/D conversion operation. Setting 0 to the ADSTOP0 bit has no effect on the operation. When the ADSTOP0 bit is set to 1, the A/D converter unit 0 is forced to stop the operation. If the A/D converter unit 0 is forced to stop, the A/D conversion results with the A/D converter unit 0 are not guaranteed.

ADSTOP1 bit (A/D Converter Unit 1 Force Stop)

The ADSTOP1 bit forces the A/D converter unit 1 to stop the A/D conversion operation. Setting 0 to the ADSTOP1 bit has no effect on the operation. When the ADSTOP1 bit is set to 1, the A/D converter unit 1 is forced to stop the operation. If the A/D converter unit 1 is forced to stop, the A/D conversion results with the A/D converter unit 1 are not guaranteed.

54.2.11 Status Registers

54.2.11.1 ADNR : A/D Conversion Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC80

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CALACT1	CALACT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADACT1	ADACT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADACT0	A/D Converter Unit 0 (ADC0) A/D Conversion Status 0: ADC0 is not in A/D conversion 1: ADC0 is in A/D conversion	R
1	ADACT1	A/D Converter Unit 1 (ADC1) A/D Conversion Status 0: ADC1 is not in A/D conversion 1: ADC1 is in A/D conversion	R
15:2	—	These bits are read as 0.	R
16	CALACT0	A/D Converter Unit 0 (ADC0) : Calibration Status 0: ADC0 is not in the calibration operation 1: ADC0 is in the calibration operation	R
17	CALACT1	A/D Converter Unit 1 (ADC1) : Calibration Status 0: ADC1 is not in the calibration operation 1: ADC1 is in the calibration operation	R
31:18	—	These bits are read as 0.	R

The ADNR register indicates the status for each A/D converter operation.

54.2.11.2 ADGRSR : Scan Group Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC84

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ACTGR8	ACTGR7	ACTGR6	ACTGR5	ACTGR4	ACTGR3	ACTGR2	ACTGR1	ACTGR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	ACTGR0 to ACTGR8 ^{*1}	Scan Group n Status The suffix number of each bit symbol corresponds to the scan group number n. 0: Scan group n is idle 1: Scan group n is in the scanning operation	R

Bit	Symbol	Function	R/W
31:9	—	These bits are read as 0.	R

Note 1. In the group priority operation, if the scanning operation of a low priority group is interrupted, the ACTGRn bit of the corresponding scan group is set to 1.

The ADGRSR register indicates the operating status for each scan group.

54.2.11.3 ADSCANENDSR : Scan End Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD50

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SCEN DF8	SCEN DF7	SCEN DF6	SCEN DF5	SCEN DF4	SCEN DF3	SCEN DF2	SCEN DF1	SCEN DF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	SCENDF0 to SCENDF8*1	Scan Group n Scan End Flag The suffix number of each bit symbol corresponds to the scan group number n. 0: Scan group n has not been scanned 1: End of scan for scan group n is detected	R
31:9	—	These bits are read as 0.	R

Note 1. If the A/D conversion operation is stopped with the ADSTOPR register, the SCENDFn bit of the scan group in which the scan operation was stopped is not changed. (It is not set to 1)

The ADSCANENDSR register indicates whether the scanning operation for each scan group has been done. Each flag can be cleared in ADSCANENDSCR register.

54.2.11.4 ADSCANENDSCR : Scan End Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xD54

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SCEN DC8	SCEN DC7	SCEN DC6	SCEN DC5	SCEN DC4	SCEN DC3	SCEN DC2	SCEN DC1	SCEN DC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	SCENDC0 to SCENDC8	Scan Group n Scan End Flag Clear The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: ADSCANENDSR.SCENDFn is cleared	W
31:9	—	The write value should be 0.	W

The ADSCANENDSCR register clears the scan end flag for the scan group n.

54.2.11.5 ADERSR : A/D Conversion Error Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC88

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADER F1	ADER F0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADERF0	A/D Converter Unit 0 (ADC0) Error Flag 0: Error is not detected 1: Error is detected	R
1	ADERF1	A/D Converter Unit 1 (ADC1) Error Flag 0: Error is not detected 1: Error is detected	R
31:2	—	These bits are read as 0.	R

The ADERSR register indicates whether the operating error has occurred in each A/D converter. Each flag can be cleared in ADERSCR register.

The A/D conversion data is not guaranteed in the A/D converter operation with an error detected.

54.2.11.6 ADERSCR : A/D Conversion Error Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC8C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADER CLR1	ADER CLR0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADERCLR0	A/D Converter Unit 0 Error Flag Clear 0: No effect 1: ADERSR.ADERF0 is cleared	W
1	ADERCLR1	A/D Converter Unit 1 Error Flag Clear 0: No effect 1: ADERSR.ADERF1 is cleared	W
31:2	—	The write value should be 0.	W

The ADERSCR register clears the error flag for A/D converter unit 0 or unit 1.

54.2.11.7 ADCALENDSR : A/D Converter Self-calibration End Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC98

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CALE NDF1	CALE NDF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CALENDF0	A/D Converter Unit 0 Calibration End flag 0: End of the calibration is not detected 1: End of the calibration is detected	R
1	CALENDF1	A/D Converter Unit 1 Calibration End flag 0: End of the calibration is not detected 1: End of the calibration is detected	R
31:2	—	These bits are read as 0.	R

The ADCALENDSR register indicates the end of the calibration operation for the A/D converter unit 0 or unit 1. Each flag can be cleared in ADCALENDSR register.

54.2.11.8 ADCALENDSR : A/D Converter Self-calibration End Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xC9C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CALE NDC1	CALE NDC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CALENDC0	A/D Converter Unit 0 Calibration End Flag Clear 0: No effect 1: ADCALENDSR.CALENDF0 is cleared	W
1	CALENDC1	A/D Converter Unit 1 Calibration End Flag Clear 0: No effect 1: ADCALENDSR.CALENDF1 is cleared	W
31:2	—	The write value should be 0.	W

The ADCALENDSR register clears the Calibration End flag for the A/D converter unit 0 or unit 1.

54.2.11.9 ADOVFERSR : A/D Conversion Overflow Error Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCA0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADOV FEF1	ADOV FEF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADOVFEF0	A/D Converter Unit 0 (ADC0) Overflow Error Flag 0: ADC0 overflow error is not detected 1: ADC0 overflow error is detected	R
1	ADOVFEF1	A/D Converter Unit 1 (ADC1) Overflow Error Flag 0: ADC1 overflow error is not detected 1: ADC1 overflow error is detected	R
31:2	—	These bits are read as 0.	R

The ADOVFERSR register indicates whether the overflow error has occurred in an A/D conversion using A/D converter unit 0 or unit 1. Each flag can be cleared in ADOVFERSCR register.

54.2.11.10 ADOVFCHSR0 : A/D Conversion Overflow Channel Status Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCA4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OVFC HF22	OVFC HF21	OVFC HF20	OVFC HF19	OVFC HF18	OVFC HF17	OVFC HF16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVFC HF15	OVFC HF14	OVFC HF13	OVFC HF12	OVFC HF11	OVFC HF10	OVFC HF9	OVFC HF8	OVFC HF7	OVFC HF6	OVFC HF5	OVFC HF4	OVFC HF3	OVFC HF2	OVFC HF1	OVFC HF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	OVFCHF0 to OVFCHF22	Analog Channel n: Overflow Flag The suffix number of each bit symbol corresponds to the analog channel number n. 0: Overflow is not detected 1: Overflow is detected	R
31:23	—	These bits are read as 0.	R

The ADOVFCHSR0 register indicates whether the overflow has occurred in the A/D conversion of the analog channel n. Each flag can be cleared in ADOVFCHSCR0 register.

54.2.11.11 ADOVFEXSR : Extended Analog A/D Conversion Overflow Status Register

Base address: ADC_B = 0x4033_8000
 ADC_B_NS = 0x5033_8000

Offset address: 0xCB0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OVFE XF22	OVFE XF21	OVFE XF20	—	OVFE XF18	OVFE XF17	OVFE XF16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	OVFE XF9	OVFE XF8	—	OVFE XF6	OVFE XF5	OVFE XF4	—	—	OVFE XF1	OVFE XF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OVFEXF0	Self-diagnosis channel for A/D unit 0: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
1	OVFEXF1	Self-diagnosis channel for A/D unit 1: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
3:2	—	These bits are read as 0.	R
4	OVFEXF4	Temperature sensor channel: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
5	OVFEXF5	Internal reference voltage channel: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
6	OVFEXF6	VBATT 1/6 voltage monitor output channel: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
7	—	This bit is read as 0.	R
8	OVFEXF8	D/A converter 0 channel: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
9	OVFEXF9	D/A converter 1 channel: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
15:10	—	These bits are read as 0.	R
16	OVFEXF16	Self-diagnosis channel for sample-and-hold circuit unit 0: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
17	OVFEXF17	Self-diagnosis channel for sample-and-hold circuit unit 1: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
18	OVFEXF18	Self-diagnosis channel for sample-and-hold circuit unit 2: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
19	—	This bit is read as 0.	R
20	OVFEXF20	Self-diagnosis channel for sample-and-hold circuit unit 4: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R

Bit	Symbol	Function	R/W
21	OVFEXF21	Self-diagnosis channel for sample-and-hold circuit unit 5: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
22	OVFEXF22	Self-diagnosis channel for sample-and-hold circuit unit 6: Overflow Flag 0: Overflow is not detected 1: Overflow is detected	R
31:23	—	These bits are read as 0.	R

The ADOVFEXSR register indicates whether the overflow occurred when the A/D conversion for the extended analog function is performed. Each flag can be cleared in ADOVFEXSCR.

54.2.11.12 ADOVFERSCR : A/D Conversion Overflow Error Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCB4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ADOV FEC1	ADOV FEC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ADOVFEC0	A/D Converter Unit 0 (ADC0) Overflow Error Flag Clear 0: No effect 1: ADOVFERSR.ADOVFEC0 is cleared	W
1	ADOVFEC1	A/D Converter Unit 1 (ADC1) Overflow Error Flag Clear 0: No effect 1: ADOVFERSR.ADOVFEC1 is cleared	W
31:2	—	The write value should be 0.	W

The ADOVFERSCR register clears the Overflow Error flag for the A/D converter unit 0 or unit 1.

54.2.11.13 ADOVFCHSCR0 : A/D Conversion Overflow Channel Status Clear Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCB8

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OVFC HC22	OVFC HC21	OVFC HC20	OVFC HC19	OVFC HC18	OVFC HC17	OVFC HC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVFC HC15	OVFC HC14	OVFC HC13	OVFC HC12	OVFC HC11	OVFC HC10	OVFC HC9	OVFC HC8	OVFC HC7	OVFC HC6	OVFC HC5	OVFC HC4	OVFC HC3	OVFC HC2	OVFC HC1	OVFC HC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
22:0	OVFCHC0 to OVFCHC22	Analog Channel n: Overflow Flag Clear The suffix number of each bit symbol corresponds to the analog channel number n. 0: No effect 1: ADOVFCHSR0.OVFCHFn is cleared	W
31:23	—	The write value should be 0.	W

The ADOVFCHSR0 register clears the Overflow flag for the analog channel n.

54.2.11.14 ADOVFEXSCR : Extended Analog A/D Conversion Overflow Status Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCC4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OVFE XC22	OVFE XC21	OVFE XC20	—	OVFE XC18	OVFE XF17	OVFE XF16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	OVFE XF9	OVFE XF8	—	OVFE XC6	OVFE XC5	OVFE XC4	—	—	OVFE XC1	OVFE XC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OVFEXC0	Self-diagnosis Channel for A/D unit 0: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF0 is cleared	W
1	OVFEXC1	Self-diagnosis Channel for A/D unit 1: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF1 is cleared	W
3:2	—	The write value should be 0.	W
4	OVFEXC4	Temperature Sensor Channel: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF4 is cleared	W
5	OVFEXC5	Internal Reference Voltage Channel: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF5 is cleared	W
6	OVFEXC6	VBATT 1/6 voltage monitor output Channel: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF6 is cleared	W
7	—	The write value should be 0.	W
8	OVFEXF8	D/A Converter 0 Channel: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF8 is cleared	W
9	OVFEXF9	D/A Converter 1 Channel: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF9 is cleared	W
15:10	—	The write value should be 0.	W
16	OVFEXF16	Self-diagnosis Channel for Sample-and-hold circuit unit0: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF16 is cleared	W

Bit	Symbol	Function	R/W
17	OVFEXF17	Self-diagnosis Channel for Sample-and-hold circuit unit1: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF17 is cleared	W
18	OVFEXC18	Self-diagnosis Channel for Sample-and-hold circuit unit2: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF18 is cleared	W
19	—	The write value should be 0.	W
20	OVFEXC20	Self-diagnosis Channel for Sample-and-hold circuit unit4: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF20 is cleared	W
21	OVFEXC21	Self-diagnosis Channel for Sample-and-hold circuit unit5: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF21 is cleared	W
22	OVFEXC22	Self-diagnosis Channel for Sample-and-hold circuit unit6: Compare Match Flag Clear 0: No effect 1: ADOVFEXSR.OVFEXF22 is cleared	W
31:23	—	The write value should be 0.	W

The ADOVFEXSCR register clears the overflow flag for the extended analog function channels.

54.2.12 FIFO

54.2.12.1 ADFIFOOCR : FIFO Control Register

Base address: ADC_B = 0x4033_8000

Offset address: 0x4C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	FIFOC E8	FIFOC E7	FIFOC E6	FIFOC E5	FIFOC E4	FIFOC E3	FIFOC E2	FIFOC E1	FIFOC E0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FIFOE N8	FIFOE N7	FIFOE N6	FIFOE N5	FIFOE N4	FIFOE N3	FIFOE N2	FIFOE N1	FIFOE N0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	FIFOEN0 to FIFOEN8	Scan Group n FIFO Enable The suffix number of each bit symbol corresponds to the scan group number n. 0: Disable scan group n FIFO function 1: Enable scan group n FIFO function	R/W
15:9	—	These bits are read as 0. The write value should be 0.	R/W
24:16	FIFOCE0 to FIFOCE8	Scan Group n FIFO clear enable at scan start/resume 0: Scan group n FIFO clear disable 1: Scan group n FIFO clear enable	R/W
31:25	—	These bits are read as 0. The write value should be 0.	R/W

The ADFIFOOCR register enables/disables the FIFO function of scan group n.

FIFOENn bit (Scan Group n FIFO Enable) (n = 0 to 8)

The FIFOENn bit enables or disables the FIFO function of scan group n. When the FIFOENn bit is set to 1, the FIFO function of scan group n is enabled, and the result of A/D conversion is stored to the FIFO. When the FIFOENn bit is set to 0, the FIFO function of scan group n is disabled.

The A/D conversion result can also be read from the A/D data or extended data registers.

The FIFOCEn bit controls whether the scan group n FIFO clear function is enabled or disabled. When the FIFOCEn bit is set to 1, FIFO clear for scan group n is enabled and the FIFO is cleared when the scan group n scan starts/resumes. When the FIFOCEn bit is set to 0, FIFO clear for scan group n is disabled.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

[illegible]

The ADFIFOINTCR register enables/disables the FIFO data read request interrupt and FIFO overflow interrupt for scan group n.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

[illegible]

The ADFIFOINTLR0 register specifies the generation timing of the FIFO data read request interrupt for the scan group 0 and 1.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x4CC

[illegible]

The ADFIFOINTLR1 register specifies the generation timing of the FIFO data read request interrupt for the scan group 2 and 3.

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x4D0

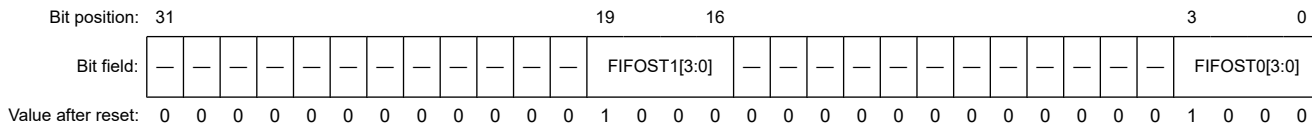
[illegible]

The ADFIFOINTLR2 register specifies the generation timing of the FIFO data read request interrupt for the scan group 4 and 5.

54.2.12.8 ADFIFOSR0 : FIFO Status Register 0

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCD0



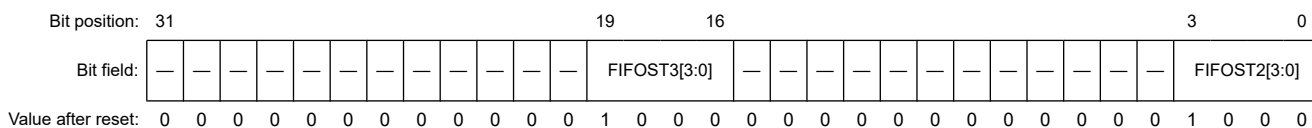
Bit	Symbol	Function	R/W
3:0	FIFOST0[3:0]	Number of vacant stages in FIFO for Scan Group 0	R
15:4	—	These bits are read as 0.	R
19:16	FIFOST1[3:0]	Number of vacant stages in FIFO for Scan Group 1	R
31:20	—	These bits are read as 0.	R

The ADFIFOSR0 register indicates the number of vacant stages in FIFO for scan group 0 and 1.

54.2.12.9 ADFIFOSR1 : FIFO Status Register 1

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCD4



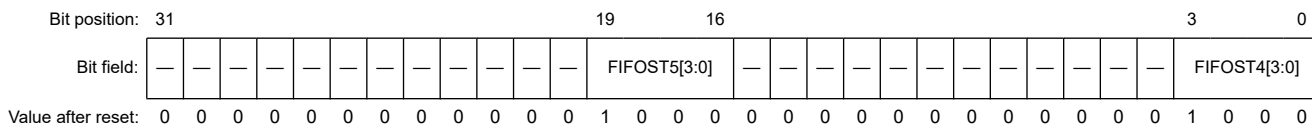
Bit	Symbol	Function	R/W
3:0	FIFOST2[3:0]	Number of vacant stages in FIFO for Scan Group 2	R
15:4	—	These bits are read as 0.	R
19:16	FIFOST3[3:0]	Number of vacant stages in FIFO for Scan Group 3	R
31:20	—	These bits are read as 0.	R

The ADFIFOSR1 register indicates the number of vacant stages in FIFO for scan group 2 and 3.

54.2.12.10 ADFIFOSR2 : FIFO Status Register 2

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCD8



Bit	Symbol	Function	R/W
3:0	FIFOST4[3:0]	Number of vacant stages in FIFO for Scan Group 4	R
15:4	—	These bits are read as 0.	R
19:16	FIFOST5[3:0]	Number of vacant stages in FIFO for Scan Group 5	R

Bit	Symbol	Function	R/W
31:20	—	These bits are read as 0.	R

The ADFIFOSR2 register indicates the number of vacant stages in FIFO for scan group 4 and 5.

54.2.12.11 ADFIFOSR3 : FIFO Status Register 3

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCDC

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
3:0	FIFOST6[3:0]	Number of vacant stages in FIFO for Scan Group 6	R
15:4	—	These bits are read as 0.	R
19:16	FIFOST7[3:0]	Number of vacant stages in FIFO for Scan Group 7	R
31:20	—	These bits are read as 0.	R

The ADFIFOSR3 register indicates the number of vacant stages in FIFO for scan group 6 and 7.

54.2.12.12 ADFIFOSR4 : FIFO Status Register 4

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCE0

[illegible]

Value after reset: 0 1 0 0 0

Bit	Symbol	Function	R/W
3:0	FIFOST8[3:0]	Number of vacant stages in FIFO for Scan Group 8	R
31:4	—	These bits are read as 0.	R

The ADFIFOSR4 register indicates the number of vacant stages in FIFO for scan group 8.

54.2.12.13 ADFIFODCR : FIFO Data Clear Register

Base address: ADC_B = 0x4033_8000
ADC_B NS = 0x5033_8000

Offset address: 0xCF0

[illegible]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FIFOD C8	FIFOD C7	FIFOD C6	FIFOD C5	FIFOD C4	FIFOD C3	FIFOD C2	FIFOD C1	FIFOD C0

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
8:0	FIFODC0 to FIFODC8	Scan Group n FIFO Data Clear The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: Clear the data of scan group n FIFO	W
31:9	—	The write value should be 0.	W

The ADFIFODCR register clears the data in FIFO for each scan group.

FIFODCn bit (Scan Group n FIFO Data Clear) (n = 0 to 8)

The FIFODCn bit clears the data in FIFO for scan group n. Writing 1 to the FIFODCn clears the data in the FIFO of scan group n. Writing 0 has no effect on the operation, and the data in FIFO is not cleared.

54.2.12.14 ADFIFOERSR : FIFO Error Status Register

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0xCF4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	FIFO LF8	FIFO LF7	FIFO LF6	FIFO LF5	FIFO LF4	FIFO LF3	FIFO LF2	FIFO LF1	FIFO LF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FIFO VF8	FIFO VF7	FIFO VF6	FIFO VF5	FIFO VF4	FIFO VF3	FIFO VF2	FIFO VF1	FIFO VF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	FIFOOVF0 to FIFOOVF8	Scan Group n FIFO Overflow Flag The suffix number of each bit symbol corresponds to the scan group number n. 0: No overflow 1: FIFO overflow is detected	R
15:9	—	These bits are read as 0.	R
24:16	FIFOFLF0 to FIFOFLF8	Scan Group n FIFO Data Read Request Flag The suffix number of each bit symbol corresponds to the scan group number n. 0: FIFO Data Read Request is not detected. 1: FIFO Data Read Request is detected.	R
31:25	—	These bits are read as 0.	R

The ADFIFOERSR register indicates the status of the FIFO. Each flag can be cleared in ADFIFOERSCR.

FIFOOVFn bit (Scan Group n FIFO Overflow Flag) (n = 0 to 8)

The FIFOOVFn bit indicates whether the overflow occurred in the FIFO of the scan group n.

When FIFOOVFn = 0, an overflow is not detected in the FIFO. When FIFOOVFn = 1, an overflow occurred in the FIFO and the A/D conversion result cannot be stored to the FIFO. FIFOOVFn can be cleared in ADFIFOERSCR.

FIFOFLFn bit (Scan Group n FIFO Data Read Request Flag) (n = 0 to 8)

The FIFOFLFn bit indicates whether a FIFO data read request in scan group n has been detected. The setting and clearing conditions of FIFOFLFn are shown as follows:

[Setting condition]

When the condition of $ADFIFOSRm.FIFOSTn[3:0] \leq ADFIFOINTLRm.FIFOILVn[3:0]$ is detected.

[Clearing condition]

When $\text{ADFIFOERSCR.FIFOFLC}_n$ is written to 1 under the condition of $\text{ADFIFOSRm.FIFOST}_n[3:0] > \text{ADFIFOINTLRm.FIFOILV}_n[3:0]$.

When the condition of $\text{ADFIFOSRm.FIFOST}_n[3:0] > \text{ADFIFOINTLRm.FIFOILV}_n[3:0]$ is occurred by the read access to ADFIFODR_n register with DMAC or DTC.

Note: $m = 0$ to 4, $n = 0$ to 8

54.2.12.15 ADFIFOERSCR : FIFO Error Status Clear Register

Base address: $\text{ADC_B} = 0x4033_8000$
 $\text{ADC_B_NS} = 0x5033_8000$

Offset address: $0xCF8$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	FIFOFLC8	FIFOFLC7	FIFOFLC6	FIFOFLC5	FIFOFLC4	FIFOFLC3	FIFOFLC2	FIFOFLC1	FIFOFLC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	FIFOVFC8	FIFOVFC7	FIFOVFC6	FIFOVFC5	FIFOVFC4	FIFOVFC3	FIFOVFC2	FIFOVFC1	FIFOVFC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
8:0	FIFOVFC0 to FIFOVFC8	Scan Group n FIFO Overflow Flag Clear The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: $\text{ADFIFOERSR.FIFOOVFC}_n$ is cleared	W
15:9	—	The write value should be 0.	W
24:16	FIFOFLC0 to FIFOFLC8	Scan Group n FIFO Data Read Request Flag Clear The suffix number of each bit symbol corresponds to the scan group number n. 0: No effect 1: $\text{ADFIFOERSR.FIFOFLFC}_n$ is cleared	W
31:25	—	The write value should be 0.	W

ADFIFOERSCR clears the FIFO overflow flags and the FIFO data read request flags for scan group n.

54.2.13 Data Register

54.2.13.1 ADDR_n : A/D Data Register n (n = 0 to 22)

Base address: $\text{ADC_B} = 0x4033_8000$
 $\text{ADC_B_NS} = 0x5033_8000$

Offset address: $0x2000 + 0x04 \times n$

Bit position:	31	15	0
Bit field:	ERR	DATA[15:0]	
Value after reset:	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA[15:0]	A/D conversion data	R
30:16	—	These bits are read as 0.	R
31	ERR	A/D conversion data error status 0: No error (the A/D conversion data is valid) 1: Error is detected (the A/D conversion data is not guaranteed)	R

The ADDR_n registers are read-only registers to read A/D conversion results.

DATA[15:0] bits (A/D conversion data)

The DATA[15:0] bits indicate the data of the A/D conversion result for the analog input channel *n*.

The data format of the A/D conversion results is determined by the setting of the virtual channel *m* (*m* = 0 to 32) to which the analog input channel *n* is assigned.

ERR bit (A/D conversion data error status)

The ERR bit indicates the error status in the A/D conversion for the analog input channel *n*.

When ERR = 0, the A/D conversion data is valid. When ERR = 1, the A/D conversion data is invalid, and the accuracy of the A/D conversion data is not guaranteed.

54.2.13.2 ADEXDR_n : A/D Extended Analog Data Register *n* (*n* = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x2180 + 0x04 × *n*

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ERR	—	—	—	—	DIAGSR[2:0]			—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA[15:0]	A/D conversion data	R
23:16	—	These bits are read as 0.	R
26:24	DIAGSR[2:0]*1	Self-diagnosis Status*1	R
30:27	—	These bits are read as 0.	R
31	ERR	A/D Conversion Error Status 0: No error (the A/D conversion data is valid) 1: Error is detected (the A/D conversion data is not guaranteed)	R

Note 1. The ADEXDR_n (*n* = 0, 1, 16 to 18, 20 to 22) registers have the DIAGSR[2:0] bits. The DIAGSR[2:0] bits in the ADEXDR_n (*n* = 4 to 6, 8, 9) registers are reserved bits.

The ADEXDR_n registers are read-only registers to read A/D conversion results of the extended analog function. The ADEXDR_n registers and the extended analog functions are as follows:

- ADEXDR0 : LIMEXF0 Self-diagnosis Channel for A/D unit0
- ADEXDR1 : LIMEXF1 Self-diagnosis Channel for A/D unit1
- ADEXDR4 : LIMEXF4 Temperature Sensor Channel
- ADEXDR5 : LIMEXF5 Internal Reference Voltage Channel
- ADEXDR6 : LIMEXF6 VBATT 1/6 voltage monitor output
- ADEXDR8 : LIMEXF8 D/A Converter 0 Channel
- ADEXDR9 : LIMEXF9 D/A Converter 1 Channel
- ADEXDR16 : LIMEXF16 Self-diagnosis Channel for Sample-and-hold circuit unit0
- ADEXDR17 : LIMEXF17 Self-diagnosis Channel for Sample-and-hold circuit unit1
- ADEXDR18 : LIMEXF18 Self-diagnosis Channel for Sample-and-hold circuit unit2

- ADEXDR20 : LIMEXF20 Self-diagnosis Channel for Sample-and-hold circuit unit4
- ADEXDR21 : LIMEXF21 Self-diagnosis Channel for Sample-and-hold circuit unit5
- ADEXDR22 : LIMEXF22 Self-diagnosis Channel for Sample-and-hold circuit unit6

DATA[15:0] bits (A/D conversion data)

The DATA[15:0] bits indicate the data of the A/D conversion result for the extended analog function channel. The data format of the A/D conversion results is determined by the setting of the virtual channel m (m = 0 to 36) to which the extended analog function channel is assigned.

DIAGSR[2:0] bits (Self-diagnosis Status)

The DIAGSR[2:0] bits indicate the voltage setting of the self-diagnosis channel.

The voltage setting value is selected among the setting values of the AD SGDCRm.DIAGVAL[2:0] bits (m = 0 to 8), AD SHSDCR0.SHDIAGm bits (m = 0 to 7) and AD SHSDCR1.SHDIAG8 bits according to the setting value of ADCHCRn.CNVCS[6:0].

The DIAGSR[2:0] bits are available in the ADEXDR0, ADEXDR1, ADEXDR16, ADEXDR17, ADEXDR18, ADEXDR20, ADEXDR21 and ADEXDR22 registers.

The DIAGSR[2:0] bits in ADEXDRn except above are reserved.

ERR bit (A/D Conversion Error Status)

The ERR bit indicates the error status in the A/D conversion for the extended analog functions.

When ERR = 0, the A/D conversion data is valid. When ERR = 1, the A/D conversion data is invalid, and the accuracy of the A/D conversion data is not guaranteed.

54.2.13.3 ADFIFODRn : FIFO Data Register n (n = 0 to 8)

Base address: ADC_B = 0x4033_8000
ADC_B_NS = 0x5033_8000

Offset address: 0x2200 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ERR	CH[6:0]							—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	DATA[15:0]	A/D Conversion Data	R
23:16	—	These bits are read as 0.	R
30:24	CH[6:0]	A/D Conversion Channel Number	R
31	ERR	A/D Conversion Data Error Status 0: No error (the A/D conversion data is valid) 1: Error is detected (the A/D conversion data is not guaranteed)	R

The ADFIFODRn register is used to read data stored in the FIFO of scan group n.

DATA[15:0] bits (A/D Conversion Data)

The DATA[15:0] bits are read-only bits that read the A/D conversion data stored in the scan group n FIFO.

CH[6:0] bits (A/D Conversion Channel Number)

The CH[6:0] bits indicate the channel number of the A/D conversion data that is read from the DATA[15:0] bits.

ERR bit (A/D Conversion Data Error Status)

The ERR bit indicates the error status in the A/D conversion data of the analog channel indicated by the CH[6:0] bits.

When ERR = 0, the A/D conversion data is valid. When ERR = 1, the A/D conversion data is invalid, and the accuracy of the A/D conversion data is not guaranteed.

54.3 Operation

54.3.1 A/D Conversion Clock

A/D conversion clock (ADCLK) is the operation clock of ADC. The A/D converters (ADC0 and ADC1) are operated and controlled by ADCLK as the basic clock. Figure 54.4 shows clock structure of ADC16H.

ADCLK is generated from the clock source and the division ratio selected in ADCLKCR register. The frequency of ADCLK should be set so that $PCLKA \geq ADCLK$. Also, the frequency of ADCLK should be set so that it is within the guaranteed operating range specified in the electrical characteristics. For details, see section 70, Electrical Characteristics.

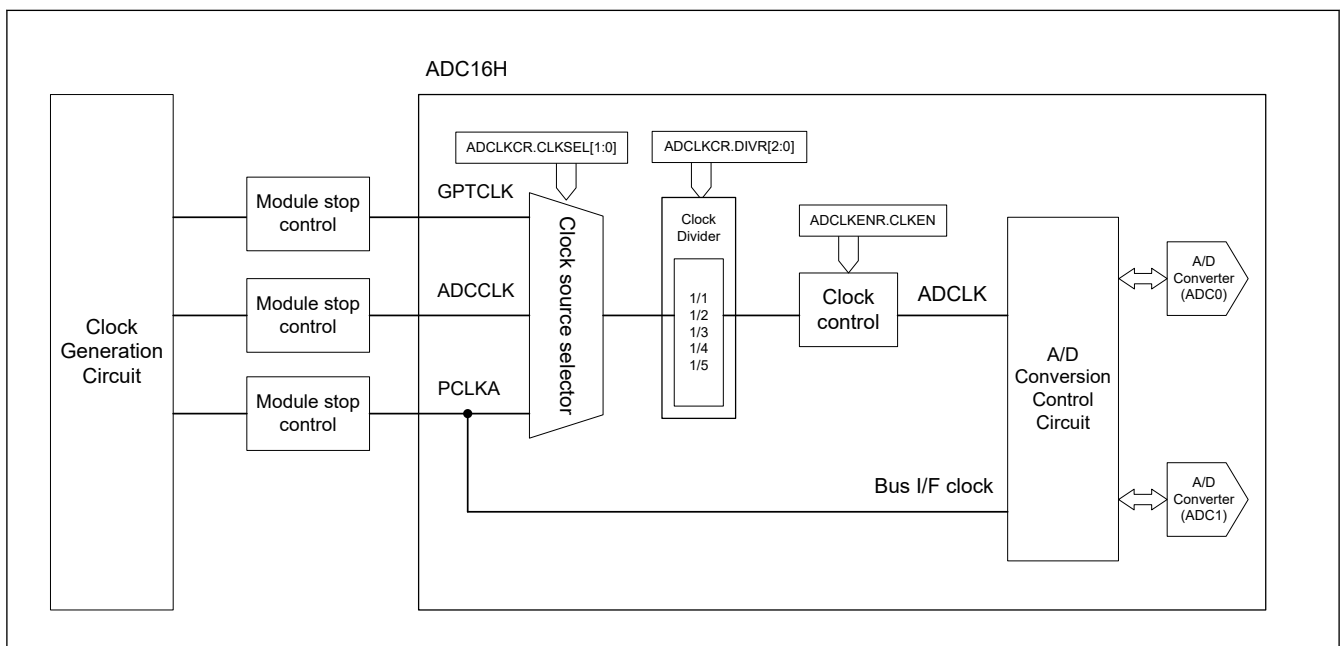


Figure 54.4 Clock structure

54.3.2 A/D Converter Operation Mode

54.3.2.1 SAR Mode

In SAR mode, A/D converter operates as a Nyquist type A/D converter. Analog channels are sampled one time and converted analog to digital by Successive Approximation Register (SAR) method. The features of SAR mode are shown in the following:

[Features of SAR mode]

- Fast conversion: A/D conversion data can be acquired with the shortest group delay.
- A/D conversion of up to 8 analog channels (virtual channels) is possible per scan group.
- Both single-ended input and differential input are supported (digital filter function is not supported)

For details about operation of A/D conversion in SAR mode, see section 54.3.7. Scanning Operation. For restrictions on SAR mode, see section 54.11.14. Restrictions on SAR Mode.

54.3.2.2 Oversampling Mode

In Oversampling mode, A/D converter operates as Oversampling-type A/D converter. Analog channels are oversampled and converted analog to digital by Noise Shaping Successive Approximation Register method (NS-SAR). The features of Oversampling mode are shown in the following:

[Features of Oversampling mode]

- High accuracy conversion: A/D conversion data can be acquired with Noise Shaping technology and Digital Filter function.*¹
- A/D conversion of up to 8 analog channels (virtual channels) is possible per scan group.
- Both of single-ended input and differential input are supported.

Note 1. Digital Filter function must be used in Oversampling mode. It is prohibited to operate without digital filter function in Oversampling mode.

For details about operation of A/D conversion in Oversampling mode, see [section 54.3.7. Scanning Operation](#). For restrictions on Oversampling mode, see [section 54.11.15. Restrictions on Oversampling Mode](#).

54.3.2.3 Hybrid Mode

Hybrid mode has both features of SAR mode and Oversampling mode. In Hybrid mode, A/D converter operates as Oversampling-type A/D converter. Analog channels are oversampled and converted analog to digital by Noise Shaping Successive Approximation Register (NS-SAR) method. In the hybrid mode, while switching channels for each sampling, multiple analog channels are oversampled. The features of Hybrid mode are shown in the following:

[Features of Hybrid mode]

- High accuracy conversion: A/D conversion data can be acquired with Noise Shaping technology and Digital Filter function.*¹
- Fast conversion: A/D conversion data can be acquired with the shortest group delay when in the Continuous scan mode and after initial delay time.
- A/D conversion of up to 4 analog channels (virtual channels) is possible per scan group.
- Both of single-ended input and differential input are supported.

Note 1. Digital Filter function must be used in Hybrid mode. It is prohibited to operate without digital filter function in Hybrid mode.

For details about operation of A/D conversion in Hybrid mode, see [section 54.3.7. Scanning Operation](#). For restrictions on Hybrid mode, see [section 54.11.16. Restrictions on Hybrid Mode](#).

54.3.3 Single-ended Input and Differential Input

ADC16H supports Single-ended input and Differential input. The selection of Single-ended input or Differential input is specified in ADCHCRn.AINMD (n = 0 to 32) bit. For the analog channels that support differential input, see [Table 54.4](#) and [Table 54.5](#).

Single-ended Input

In Single-ended input mode, the difference between the voltage of the analog channel (signal source) and the analog reference ground voltage (ADC0: VREFL0, ADC1: VREFL) is A/D-converted.

Differential Input

In Differential input mode, the even-numbered analog channel is used as the non-inverting input (+) (A_{INP}) and the odd-numbered analog channel is used as the inverting input (−) (A_{INN}), then the difference voltage between the non-inverting input (+) and the inverting input (−) ($A_{INP} - A_{INN}$) is A/D-converted.*¹

For A/D conversion with Differential input mode, select the non-inverting input (+) channel*² of the target of differential pair in ADCHCRn.CNVCS[6:0] (n = 0 to 32) bits, and select the Differential input mode in ADCHCRn.AINMD (n = 0 to 32) bit.

It is prohibited to perform A/D conversion for channels that do not support differential input with Differential input mode. In that case, A/D conversion result is not guaranteed.

Note 1. The differential input pair is a combination of channels with analog channel number $2i$ and $2i + 1$ ($i = 0, 1, 2, 3, \dots$), and they are named ANxxxP and ANxxxN (xxx = 000, 002, 004, ...). The combination of discontinuous analog channel numbers or the combination of analog channel numbers $2i - 1$ and $2i$ cannot be used as a differential input pair.

Note 2. It is the even-numbered analog channel which is named as ANxxxP (xxx = 000, 002, 004, ...).

54.3.4 Analog Channel

The analog channel is the source of the analog signal that is targeted for A/D conversion. The analog channels consist of the following elements:

- Analog input channels: A/D conversion channels for the analog input from MCU's I/O pin as the signal source.
- Extended analog channels: A/D conversion channels for the source of the analog signal inside the MCU.

To perform A/D conversion for the analog channels, assignments to the virtual channel and scan group are required. For each element, refer to each item.

54.3.5 Virtual Channel

The virtual channel is a group of registers that stores the A/D conversion configurations for an analog channel. The virtual channels can be specified the configurations for A/D conversion such as the selection of the analog channels, the optional settings for A/D conversion, the data processing method of the A/D conversion data, the assignment to scan groups, and so on.

To perform the A/D conversion of the analog channel, the analog channel should be assigned to any of the virtual channels. Then that virtual channel should be assigned to one of the scan groups. See Figure 54.5 for the relationship between the analog channels and the virtual channels and scan groups. Figure 54.6 shows the A/D conversion order in Figure 54.5 settings.

The virtual channel can be assigned to only one scan group. If the same analog channel is to be converted within the different scan group, or the same analog channel is to be converted several times within the same scan group, assign the analog channel to several virtual channels. When performing the A/D conversion more than once using more than one virtual channel within the same scan group for the same analog channel, the FIFO function should be used in combination.

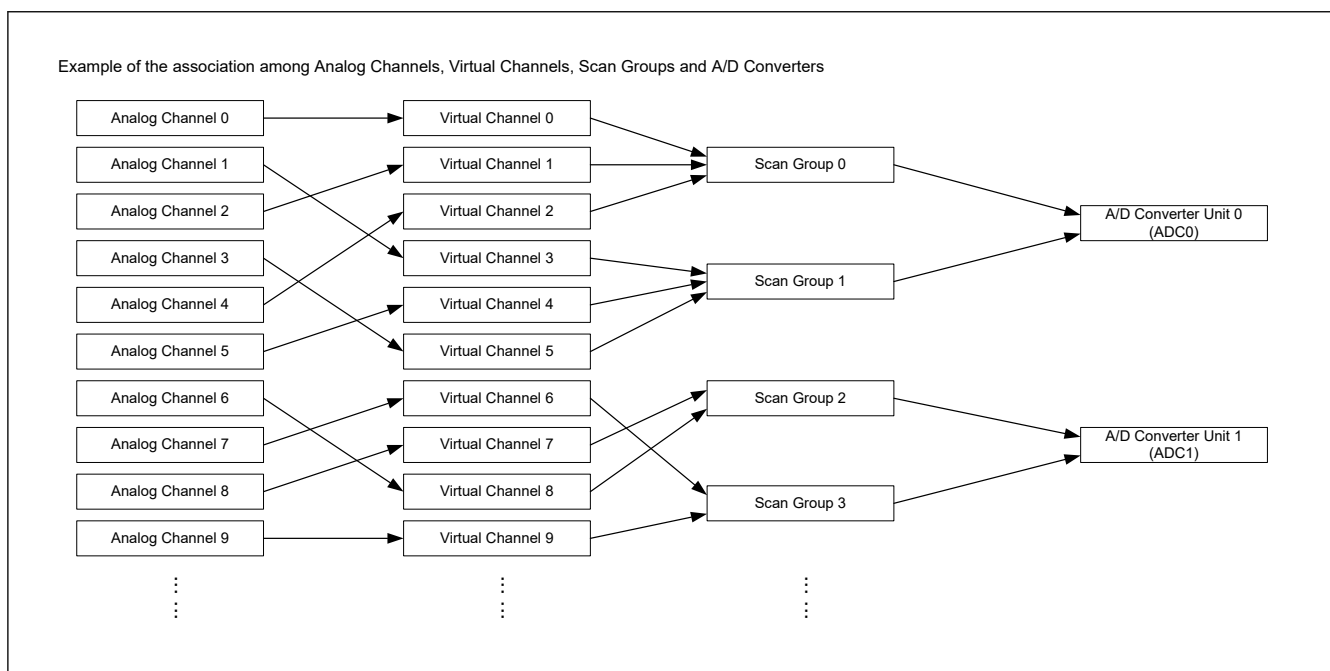


Figure 54.5 The concept of virtual channel

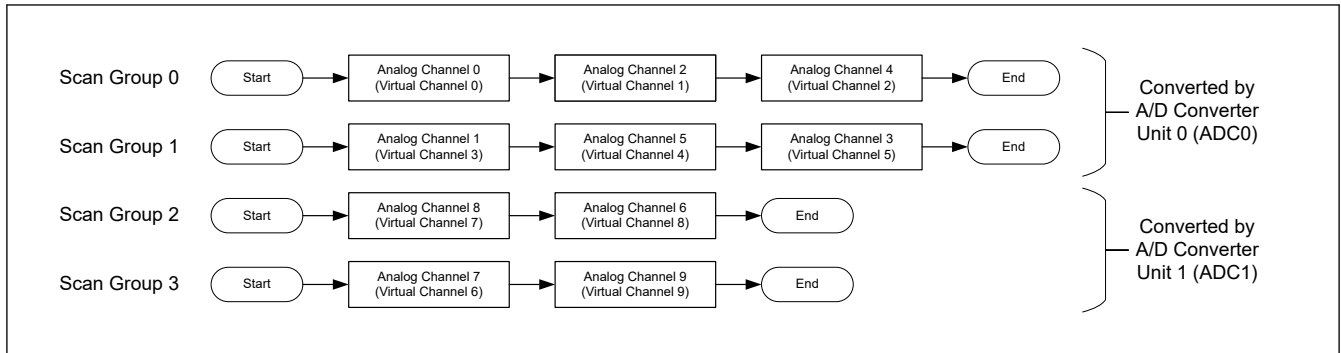


Figure 54.6 Example of A/D conversion order at the scanning operation

54.3.6 Scan Group

The scan group is a group of the analog channels that perform A/D conversion in a scanning operation. To perform the A/D conversion using the scan group, configure the following steps:

- Assign the analog channel for the A/D conversion to the virtual channel.
- Assign the virtual channel to the scan group.
- Assign the scan group to A/D converter.

See [Figure 54.5](#) and [Figure 54.6](#) for the relationship among the analog channels, the virtual channels, and the scan groups.

A single scan group can be assigned virtual channels in the following ranges, according to its operation mode.

- SAR mode: up to 8 virtual channels per scan group
- Oversampling mode: up to 8 virtual channels per scan group
- Hybrid mode: from 2 to 4 virtual channels per scan group

If more than the maximum number of virtual channels are allocated to one scan group, A/D conversion is performed on the 8th (SAR mode or Oversampling mode) or 4th (Hybrid mode) channels in ascending order of virtual channel numbers. Subsequent channels are not subject to A/D conversion.

Notes on the scan groups

The virtual channels assigned to one scan group should be assigned the analog channels that can be converted by the same A/D converter. If any analog channels cannot be converted by the specified A/D converter are included, the A/D conversion result of those channels are undefined.

In Hybrid mode, you must assign at least 2 virtual channels to a scan group. Operation is not guaranteed when any virtual channel is not assigned or only 1 virtual channel is assigned.

54.3.7 Scanning Operation

The scanning operation is the operation of sequential A/D conversion for the analog channels. The modes of the scanning operation are the followings:

- Single scan mode
 - In single scan mode, one scan group is scanned once for each A/D conversion start trigger input.
 - Each time an A/D conversion start trigger is input, A/D conversion is performed once for each analog channel assigned to that scan group.
- Continuous scan mode
 - The continuous scan mode repeats the scanning operation of one scan group.
 - When an A/D conversion start trigger is input, A/D conversion of each analog channel assigned to that scan group is repeated until A/D conversion stop processing is performed.
- Background continuous scan mode
 - The background continuous scan mode repeats the scanning operation of a single scanning group.

- Scanning operation starts when the first A/D conversion start trigger is input.
- A/D conversions of each analog channel assigned to that scan group are repeated until A/D conversion stop processing is performed. (Continuous scanning operation)
- When an A/D conversion start trigger is input during continuous scanning operation, the A/D conversion data at that point is output. (Unless an A/D conversion start trigger is input, the A/D conversion data are not output and the values of the A/D data registers and FIFO data registers are not updated.)
- One-channel continuous scan mode
 - One-channel continuous scan mode repeats the scanning operation of one analog channel and can change conversion channel by a start trigger.
 - The first start trigger activates A/D conversion, and the same channel is continuously converted until the next start trigger is input.
 - Whenever an A/D conversion channel is switched by a start trigger, A/D conversion is suspended and resumed.
- Fixed-channel continuous scan mode
 - Fixed-channel continuous scan mode operates with a combination of fixed channels operating as Hybrid and non-fixed channels operating as SAR.
 - Scan a fixed number of fixed channels first, then scan one non-fixed channel assigned to the scan group. One non-fixed channel is switched in sequence with each scan. And repeat these operations.
 - A/D conversion data will not be output until conversion is completed according to the number of TAPs in the digital filter.

The scan modes supported by each operation mode are shown in [Table 54.6](#).

Table 54.6 Scan mode and operation mode

Scan mode	Operation mode of A/D converter		
	SAR mode	Oversampling mode	Hybrid mode
Single scan mode	✓	✓	✓
Continuous scan mode	✓	✓	✓
Background continuous scan mode	—	—	✓
One-channel continuous scan mode	—	✓	—
Fixed-channel continuous scan mode	—	—	✓

Note: ✓ : available, —: not available

The scanning operations are performed on a scan group basis. When the scanning operation is started, A/D conversion is performed for each analog channel based on the virtual channel setting. If the scanning operation starts for 2 or more scan group assigned to the same A/D converter at the same time, the scanning operation of the group with the lowest scan group number is performed.

The A/D conversion order of the analog channel in the scanning operation is the ascending order of the virtual channel numbers assigned to the scan group. The channels with the lowest virtual channel numbers are performed the A/D conversion at the forward end of the scan group. The channels with larger virtual channel numbers are performed the A/D conversion at the backward of the scan group.

See [Figure 54.5](#) and [Figure 54.6](#) for the relationship of the conversion order of the respective analog channel in the scanning operation.

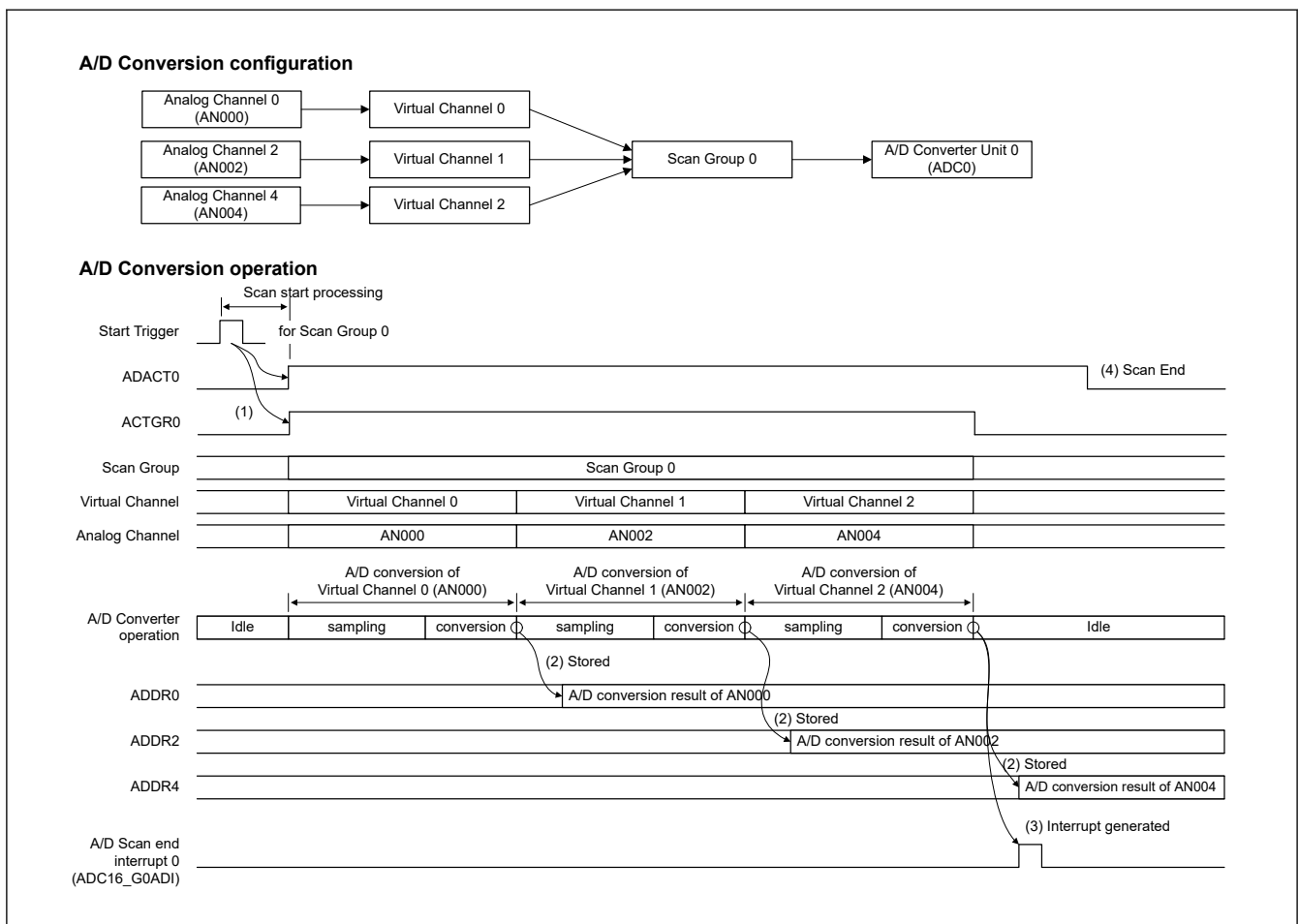
The scanning operations in each operation mode are shown in the following.

54.3.7.1 SAR Mode – Single Scan Mode

[Table 54.7](#) and [Figure 54.7](#) show the operation example in SAR mode – Single scan mode.

Table 54.7 Example of the scanning operation in SAR mode – Single scan mode

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started.
2	Each time an A/D conversion of one analog channel is completed, A/D conversion result is stored to the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, the A/D conversion result is also stored to the FIFO data register (ADFIFODRk (k = 0 to 8)).
3	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
4	During the scanning operation, ADGRSR.ACTGRn (n = 0 to 8) bit corresponding to that scan group is set to 1. ADSR.ADACTm (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1. When scanning operation is complete, each bit is cleared to 0 and A/D converter enters the idle state.

**Figure 54.7 Example of the scanning operation in SAR mode – Single scan mode**

54.3.7.2 SAR Mode – Continuous Scan Mode

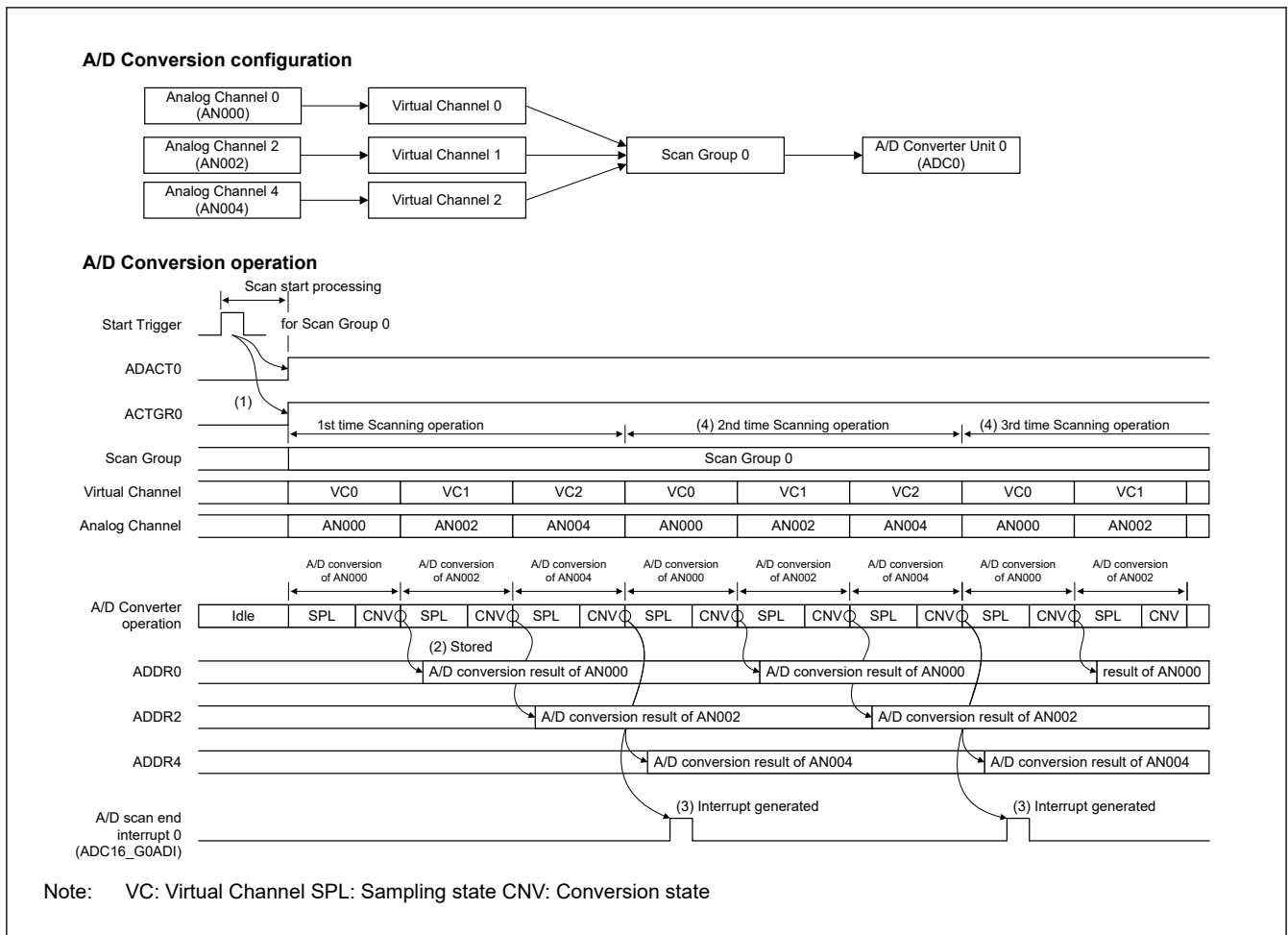
Table 54.8 and Figure 54.8 show the operation example in SAR mode – Continuous scan mode.

Table 54.8 Example of the scanning operation in SAR mode – Continuous scan mode (1 of 2)

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started. When scanning operation starts, ADGRSR.ACTGRn (n = 0 to 8) corresponding to that scan group is set to 1. ADSR.ADACTm (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1.

Table 54.8 Example of the scanning operation in SAR mode – Continuous scan mode (2 of 2)

Step	Operation
2	Each time an A/D conversion of one analog channel is completed, A/D conversion result is stored to the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, the A/D conversion result is also stored to the FIFO data register (ADFIFODRk (k = 0 to 8)).
3	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
4	Until the A/D conversion stop process is performed, Step 2 to 3 are repeated, and the scanning operation continues. To stop the A/D conversion, you must follow to section 54.6.4. Force Stops the A/D Conversion Operation .

**Figure 54.8 Example of scanning operation in SAR mode – Continuous scan mode**

54.3.7.3 Oversampling Mode – Single Scan Mode

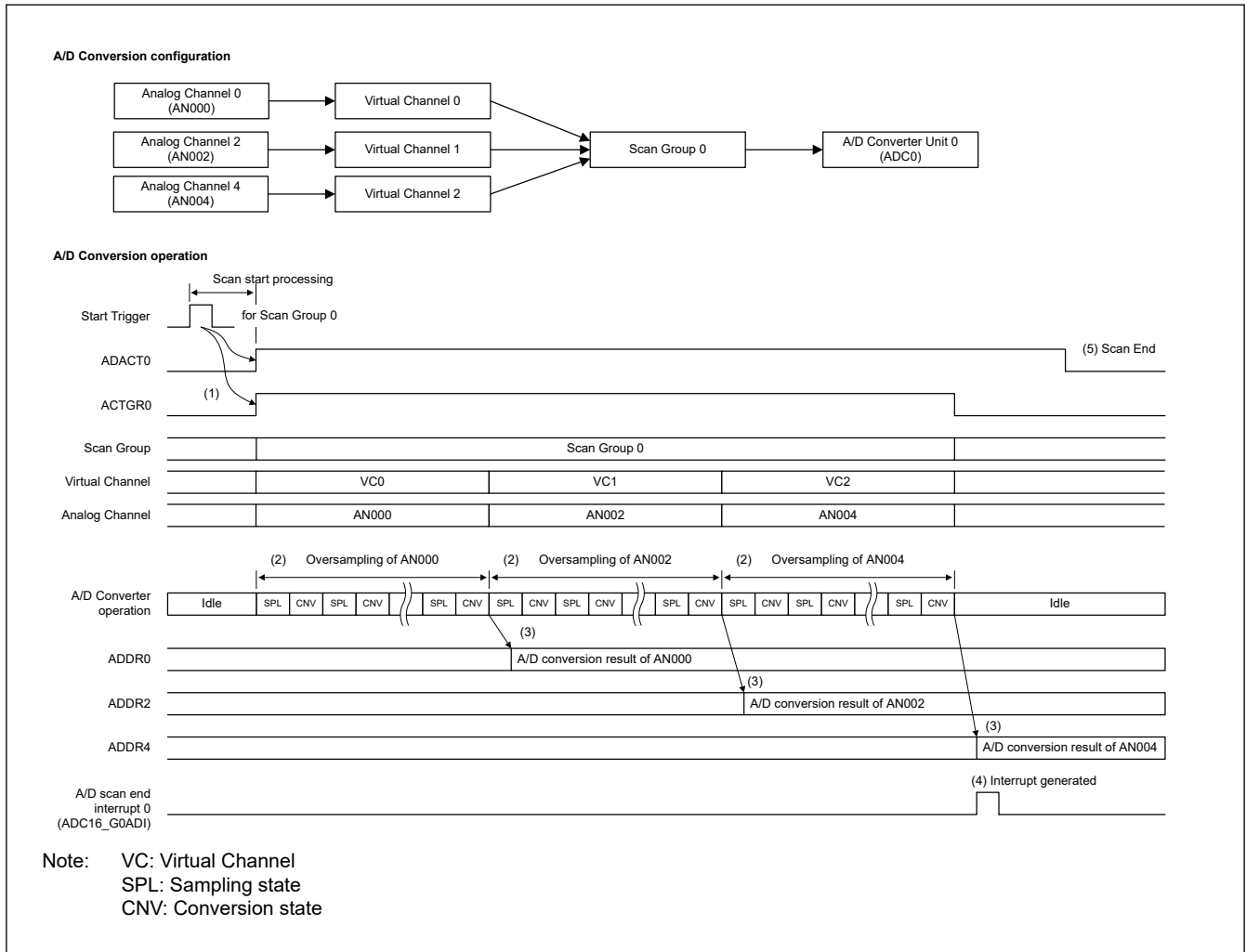
[Table 54.9](#) and [Figure 54.9](#) show the operation example in Oversampling mode – Single scan mode.

Table 54.9 Example of the scanning operation in Oversampling mode – Single scan mode (1 of 2)

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started.
2	Each analog channel is oversampled according to the number of TAPs in the digital filter and the number of A/D-converted value addition/averaging times. The oversampled data stored in the digital filter are discarded after the oversampling of each analog channel is completed.
3	A/D conversion data of each analog channel is output after oversampling. The A/D conversion data is stored in the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).

Table 54.9 Example of the scanning operation in Oversampling mode – Single scan mode (2 of 2)

Step	Operation
4	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
5	During the scanning operation, ADGRSR.ACTGRn (n = 0 to 8) bit corresponding to that scan group is set to 1. ADSR.ADACTm (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1. When scanning operation is complete, each bit is cleared to 0 and A/D converter enters the idle state.

**Figure 54.9 Example of scanning operation in Oversampling mode – Single scan mode**

54.3.7.4 Oversampling Mode – Continuous Scan Mode

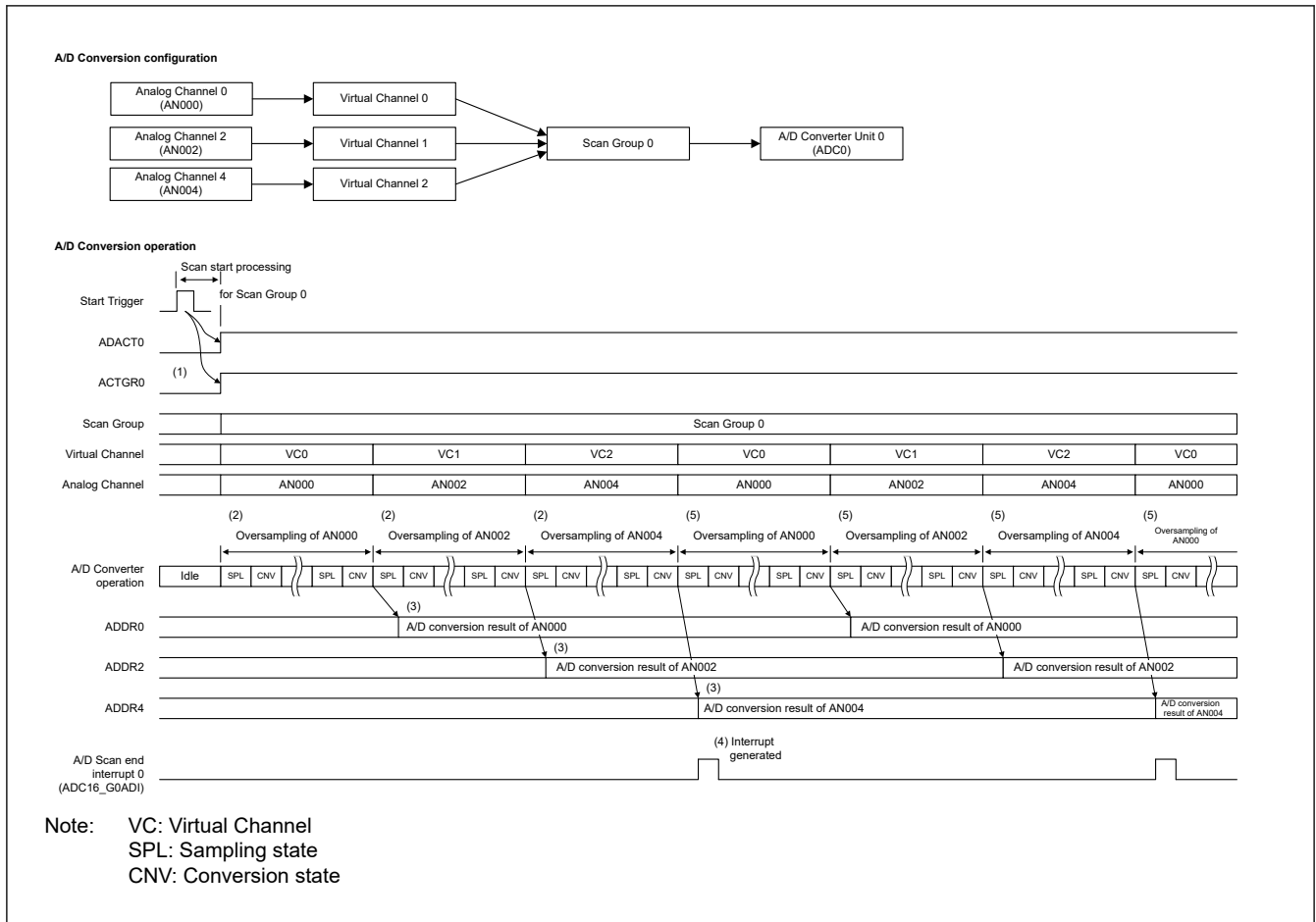
Table 54.10 and Figure 54.10 show the operation example in Oversampling mode – Continuous scan mode.

Table 54.10 Example of the scanning operation in Oversampling mode – Continuous scan mode (1 of 2)

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started. When scanning operation starts, ADGRSR.ACTGRn (n = 0 to 8) corresponding to that scan group is set to 1. ADSR.ADACTm (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1.
2	Each analog channel is oversampled according to the number of TAPs in the digital filter and the number of A/D-converted value addition/averaging times. The oversampled data stored in the digital filter are discarded after the oversampling of each analog channel is completed.

Table 54.10 Example of the scanning operation in Oversampling mode – Continuous scan mode (2 of 2)

Step	Operation
3	A/D conversion data of each analog channel is output after oversampling. The A/D conversion data is stored in the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).
4	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
5	Until the A/D conversion stop process is performed, Step 2 to 4 are repeated, and the scanning operation continues. To stop A/D conversion, you must follow to section 54.6.4. Force Stops the A/D Conversion Operation .

**Figure 54.10 Example of scanning operation in Oversampling mode – Continuous scan mode**

54.3.7.5 Oversampling Mode – One-channel Continuous Scan Mode

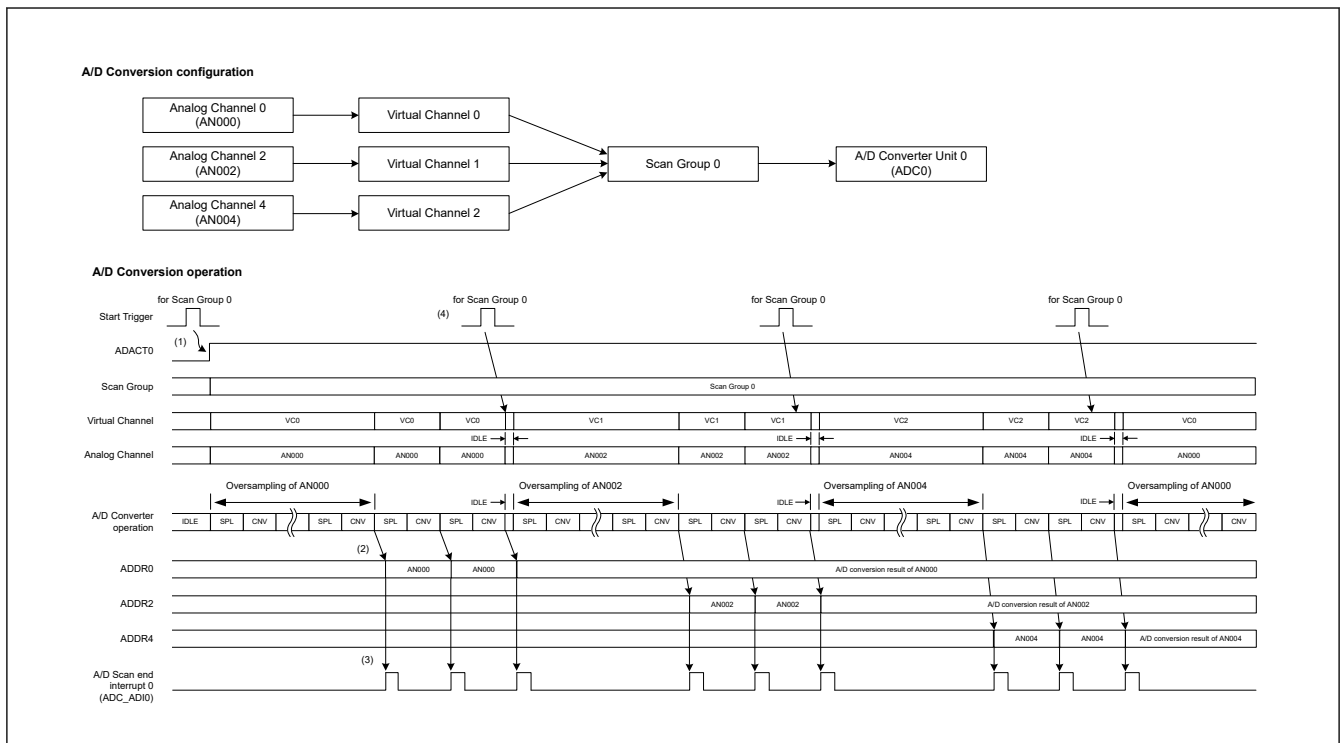
[Table 54.11](#) and [Figure 54.11](#) show the operation example in Oversampling mode – One-channel continuous scan mode.

Table 54.11 Example of scanning operation in Oversampling mode – One-channel continuous scan mode (1 of 2)

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started. When scanning operation starts, ADGRSR.ACTGRn (n = 0 to 8) corresponding to that scan group is set to 1. ADGRSR.ADACTm (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1.
2	Each time an A/D conversion of one analog channel is completed, A/D conversion result is stored to the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, the A/D conversion result is also stored to the FIFO data register (ADFIFODRk (k = 0 to 8)).

Table 54.11 Example of scanning operation in Oversampling mode – One-channel continuous scan mode (2 of 2)

Step	Operation
3	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of one analog channel is completed.
4	A/D conversion of the same virtual channel continues until trigger input is made. When the trigger input that is set as the start trigger of the startup scan group is performed, the A/D conversion of the next virtual channel starts.

**Figure 54.11 Example of scanning operation in Oversampling mode – One-channel continuous scan mode**

54.3.7.6 Hybrid Mode – Single Scan Mode

Table 54.12 and Figure 54.12 show the operation example in Hybrid mode – Single scan mode.

Table 54.12 Example of the scanning operation in Hybrid mode – Single scan mode

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started.
2	In Hybrid mode, the scanning operation is performed while switching analog channels every time oversampling is performed.
3	After the oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/averaging times is performed to each analog channel, A/D conversion data is output. (The time taken for the oversampling required to obtain the first A/D conversion data is called the initial delay time.) The A/D conversion data is stored in the data register (ADDR _i (i = 0 to 22), ADEXDR _j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODR _k (k = 0 to 8)).
4	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
5	During the scanning operation, ADGRSR.ACTGR _n (n = 0 to 8) bit corresponding to that scan group is set to 1. ADSCR.ADACT _m (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1. When scanning operation is complete, each bit is cleared to 0 and A/D converter enters the idle state. And the oversampled data in the digital filter are also discarded at the end of the scanning operation.

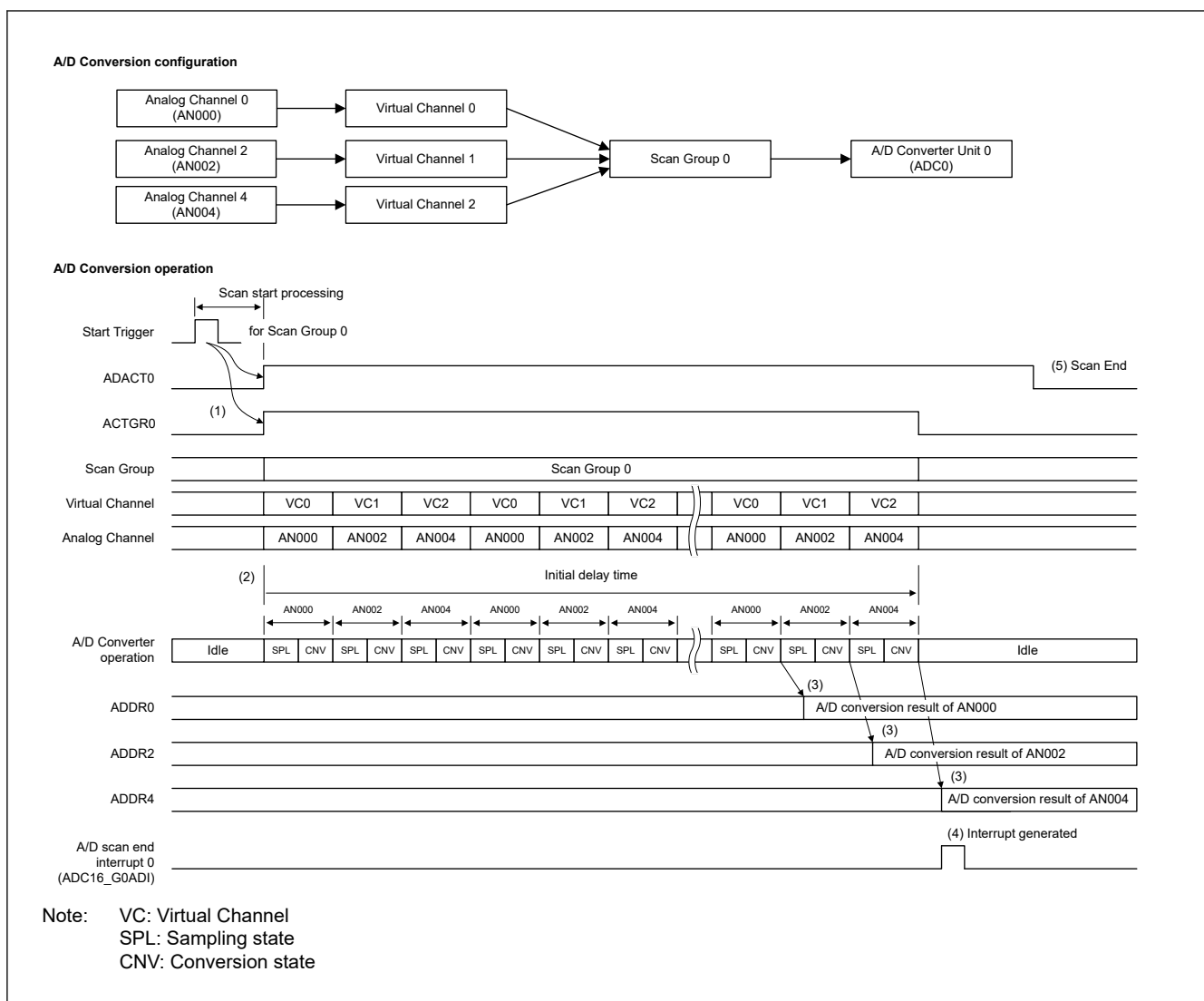


Figure 54.12 Example of scanning operation in Hybrid mode – Single scan mode

54.3.7.7 Hybrid Mode – Continuous Scan Mode

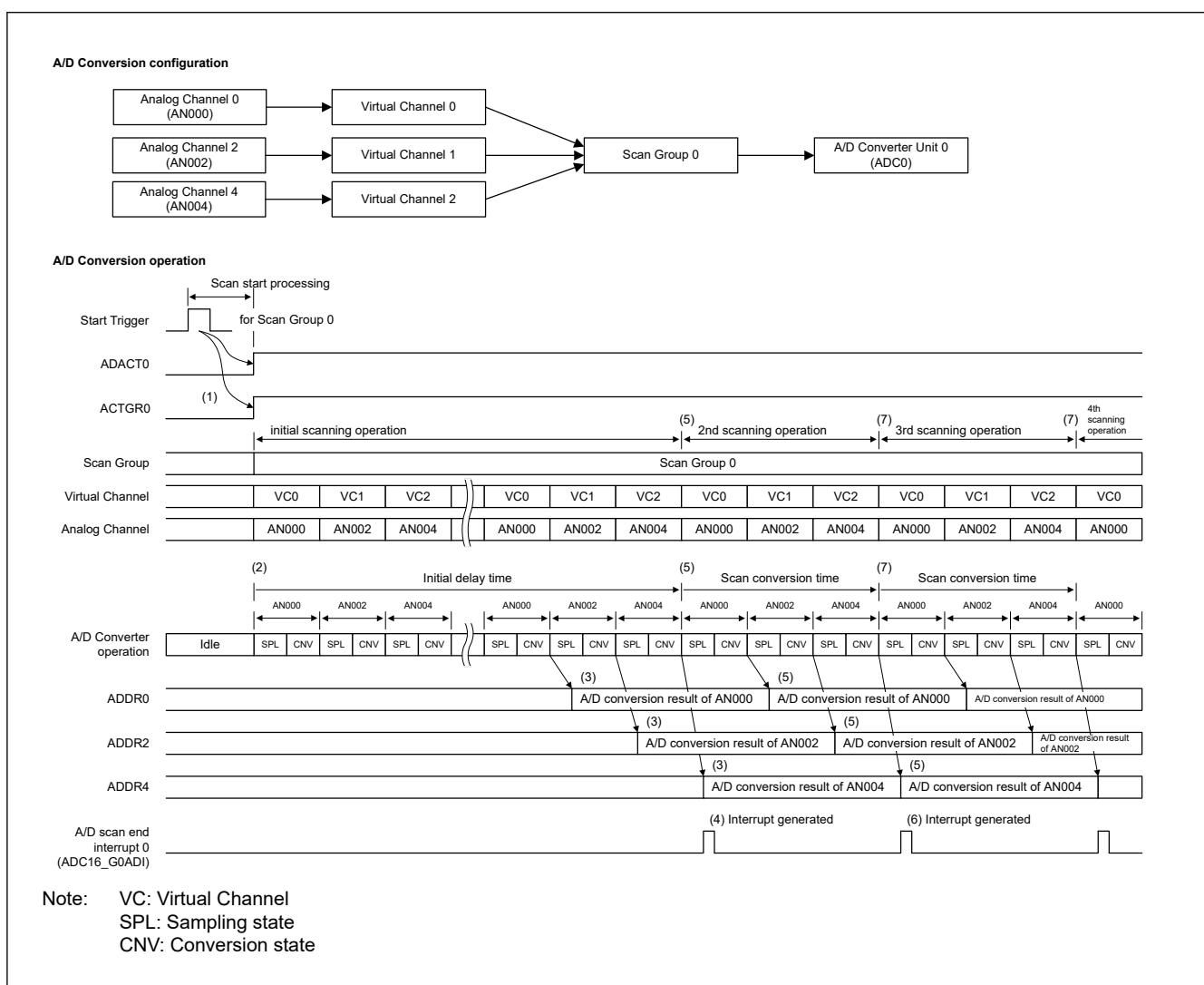
Table 54.13 and Figure 54.13 show the operation example in Hybrid mode – Continuous scan mode.

Table 54.13 Example of the scanning operation in Hybrid mode – Continuous scan mode (1 of 2)

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started. When scanning operation starts, ADGRSR.ACTGR _n (n = 0 to 8) corresponding to that scan group is set to 1. ADSCR.ADACT _m (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1.
2	In Hybrid mode, the scanning operation is performed while switching analog channels every time oversampling is performed.
3	After the oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/averaging times is performed to each analog channel, A/D conversion data is output. (The time taken for the oversampling required to obtain the first A/D conversion data is called the initial delay time.) The A/D conversion data is stored in the data register (ADDR _i (i = 0 to 22), ADEXDR _j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODR _k (k = 0 to 8)).
4	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.

Table 54.13 Example of the scanning operation in Hybrid mode – Continuous scan mode (2 of 2)

Step	Operation
5	In scanning operation after the initial delay time has elapsed, each time oversampling is performed for an analog channel, the data in the digital filter is updated. The updated A/D conversion data for each analog channel can be output each one time oversampling or each multiple times oversampling corresponding to the number of A/D-converted value addition/averaging times is performed. The A/D conversion data is stored in the data register (ADDRi (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).
6	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
7	Thereafter, until the A/D conversion stop process is performed, Step 5 to 6 are repeated, and the scanning operation continues. To stop the A/D conversion, you must follow to section 54.6.4. Force Stops the A/D Conversion Operation .

**Figure 54.13 Example of scanning operation in Hybrid mode – Continuous scan mode**

54.3.7.8 Hybrid Mode – Background Continuous Scan Mode

[Table 54.14](#) and [Figure 54.14](#) show the operation example in Hybrid mode – Background continuous scan mode.

Table 54.14 Example of the scanning operation in Hybrid mode – Background continuous scan mode

Step	Operation
1	When a software trigger or a trigger from a peripheral module is input, the scanning operation of the scan group corresponding to the trigger is started. When scanning operation starts, ADGRSR.ACTGR _n (n = 0 to 8) corresponding to that scan group is set to 1. ADSR.ADACT _m (m = 0, 1) bit corresponding to the A/D converter performing the A/D conversion is also set to 1.
2	In Hybrid mode, the scanning operation is performed while switching analog channels every time oversampling is performed.
3	After the oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/averaging times is performed to each analog channel, A/D conversion data becomes in ready to be output. (The time taken for the oversampling required to obtain the first A/D conversion data is called the initial delay time. In the background continuous scan operation, A/D conversion data can be obtained after the initial delay time has elapsed from the start of the scanning operation.)
4	In scanning operation after the initial delay time has elapsed, each time oversampling is performed for an analog channel, the data in the digital filter is updated. The updated A/D conversion data for each analog channel can be output each one time oversampling or each multiple times oversampling corresponding to the number of A/D-converted value addition/averaging times is performed.
5	When an A/D conversion start trigger is input during background continuous scanning operation, the most recent A/D conversion data at that time is stored in the data register (ADDR _i (i = 0 to 22), ADEXDR _j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)). If FIFO function is set to enabled, A/D conversion data is also stored in the FIFO data register (ADFIFODR _k (k = 0 to 8)).
6	If scan end interrupt is set to enable, scan end interrupt is generated.
7	Thereafter, until the A/D conversion stop process is performed, Background continuous scanning operation (Step 4) is repeated. And whenever the A/D conversion start trigger is input during Background continuous scanning operation, the A/D conversion data is output (Step 5 and Step 6). To stop A/D conversion, you must follow to section 54.6.4. Force Stops the A/D Conversion Operation .

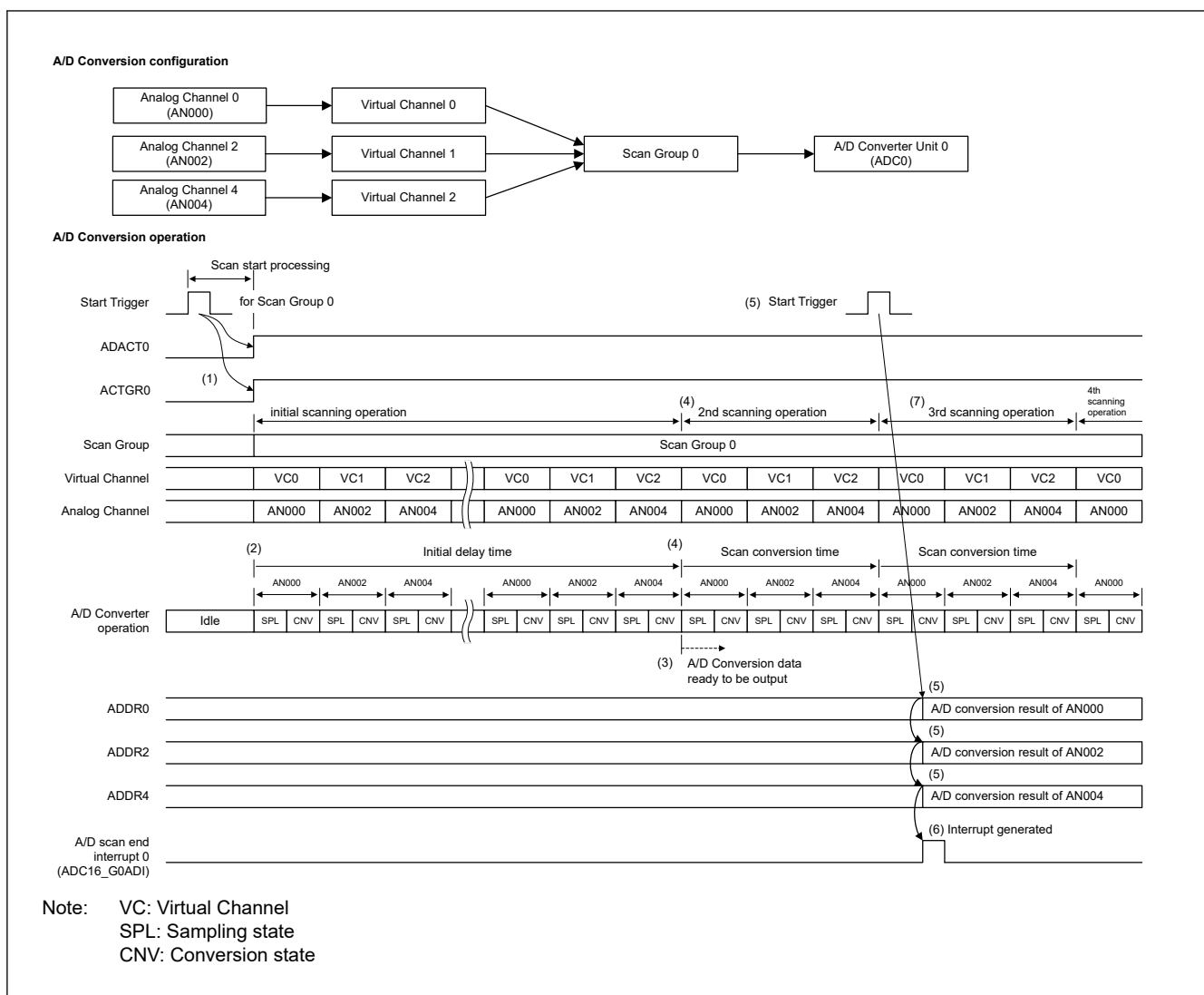


Figure 54.14 Example of scanning operation in Hybrid mode – Background continuous scan mode

54.3.7.9 Hybrid Mode – Fixed-channel Continuous Scan Mode

Table 54.15 and Figure 54.15 show the operation example in Hybrid mode – Fixed-channel continuous scan mode.

Table 54.15 Example of the scanning operation in Hybrid mode – Fixed-channel continuous scan mode (1 of 2)

Step	Operation
1	When a start trigger is input and the ADSCR.ADACTm (m = 0, 1) bit becomes 1 (A/D conversion in progress), A/D conversion of the scan group selected by the ADSCGRm (m = 0, 1) register is started.
2	First, the fixed channels are converted for the number of channels set in the ADSWNRm.SWFIXm (m = 0, 1) bit.*1
3	After that, non-fixed channels are converted in ascending order of virtual channel numbers for the number of channels set in the ADSWNRm.SWNOFIXm (m = 0, 1) bits. The waveform example in fig uses 3 virtual channels and the conversion order is virtual channel 20 → virtual channel 21 → virtual channel 22.
4	When A/D conversion for one channel is completed, the A/D conversion result is stored in the A/D data register (ADDRi.DATA (i = 0 to 22), ADEXDRj (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)) corresponding to the conversion target channel. If FIFO is used, the A/D conversion result is also stored to the FIFO data register (ADFIFODRk (k = 0 to 8)).

Table 54.15 Example of the scanning operation in Hybrid mode – Fixed-channel continuous scan mode (2 of 2)

Step	Operation
5	When the A/D data register is updated during the scan cycle, a scan group n interrupt request is generated if the ADINTCR.ADIEn (n = 0 to 8) bit of each scan group in Scan End Interrupt Enable Register is set to 1 (enable scan group n interrupt by end of scan) after conversion of the asynchronous channel ends.
6	After the conversion of fixed channels and non-fixed channels is completed, the conversion time of idle cycles set in the ADSWNRm.BCSSTSLm (m = 0, 1) registers is counted until the end of the scan cycle set in ADSWNRm.SWNUMm (m = 0, 1).
7	After that, the A/D conversion operation, the result stored in the A/D conversion result register, the scan group n interrupt caused by the end of conversion of the non-fixed channel, and the count until the end of the scan cycle are repeated. Fixed channels repeat A/D conversion of the same channel, and non-fixed channels repeat A/D conversion up to the last channel assigned to the scan group, then return to the first channel.*2

Note: It is necessary to set so that the same filter is not selected between channels.

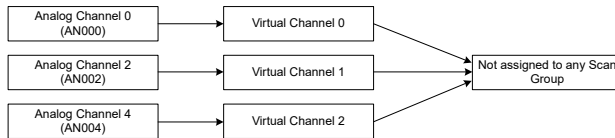
Note: When A/D conversion stop is set (ADSTOPR.ADSTOPm (m = 0, 1)), scan ends.

Note 1. Fixed channels in Fixed-channel continuous scan mode use the scan group and virtual channels determined for each unit. For details, see .

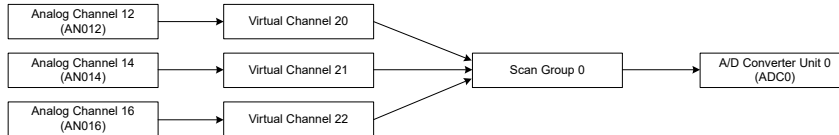
Note 2. If the scan period is set to less than the total number of conversions for fixed and non-fixed channels, conversion cycles beyond the scan period are not subject to A/D conversion.

A/D Conversion configuration

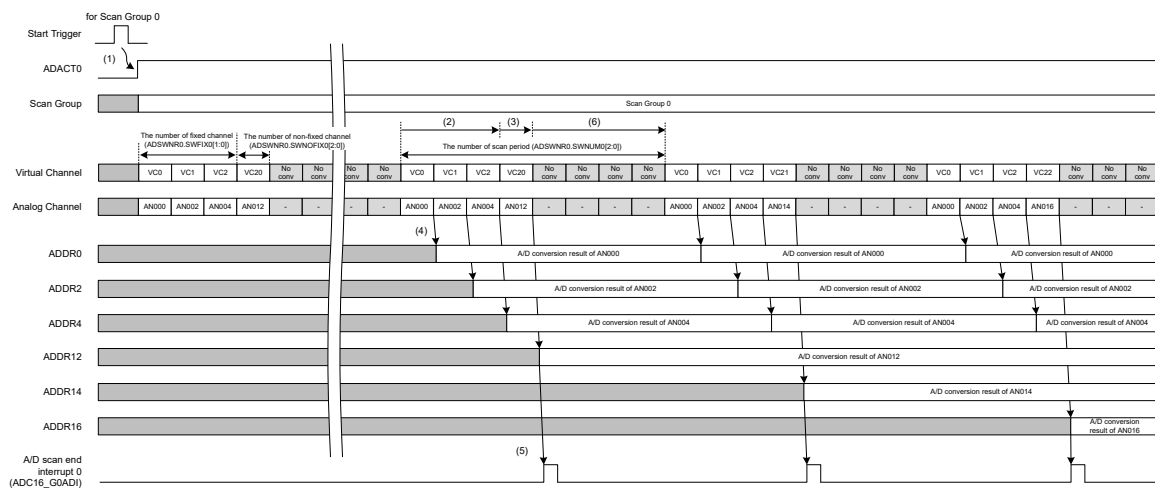
(For fixed channels *1)



(For non-fixed channels)



ADSWNR0.SWNUM0[2:0] = 0x7 : the number of scan period = 8
 ADSWNR0.SWFIX0[1:0] = 0x2 : the number of fixed channel = 3
 ADSWNR0.SWNOFIX0[2:0] = 0x1 : the number of non-fixed channel = 1

A/D Conversion operation

Note 1. Fixed channels in Fixed channel continuous scan mode use the scan group and virtual channels determined for each unit. For details, see .

Figure 54.15 Example of scanning operation in Hybrid mode – Fixed-channel continuous scan mode

54.3.8 Self-calibration

ADC16H has a built-in self-calibration function. Self-calibration function is used to calibrate the variations of characteristics caused by the chip-by-chip variations. Self-calibration function has the following operations.

1. Internal Circuit Calibration

- Adjusts the operation of A/D converter's internal circuitry.

2. Gain and Offset Calibration

- Measure the A/D converter's Gain Error and Offset Error.
- The calibration processing of the A/D conversion result based on the measured error data is performed after the A/D conversion.

For details, see [section 54.5.3.1. Gain Error and Offset Error Calibration](#).

- This self-calibration should be performed after Internal-circuit self-calibration is completed.

3. Channel-dedicated sample-and-hold circuit Gain and Offset Calibration

- Measure the Gain Error and the Offset Error when using the channel-dedicated sample-and-hold circuit.

- The calibration processing of the A/D conversion result based on the measured error data is performed after the A/D conversion.
For details, see [section 54.5.3.1. Gain Error and Offset Error Calibration](#).
- This self-calibration should be performed after A/D converter Gain/Offset self-calibration is completed.

54.3.8.1 Conditions Under Which Self-calibration is Required

The conditions under which self-calibration is required are shown in [Table 54.16](#). Under the conditions shown in [Table 54.16](#), self-calibration should be performed before using the A/D converter. If A/D converter is operating, stop all A/D converters and perform the self-calibration again. If the required self-calibration is not performed, the A/D conversion result is not guaranteed.

Table 54.16 List of conditions under which Self-calibration should be performed

Conditions under which self-calibration is required	Internal-circuit Calibration	Gain/Offset Calibration	Channel-dedicated sample-and-hold circuit Gain and Offset Calibration*1
After reset release	✓	✓	✓
After releasing the module stop	✓	✓	✓
When returning from Software Standby mode or Deep Software Standby mode	✓	✓	✓
When changed the ADCLK setting (When clock source or frequency is changed)	✓	✓	✓
When changed the operation mode or the scan mode of the A/D converter (When changed ADMDR.ADMdM bit (m = 0, 1))	✓	✓*2	✓
When changed A/D successive approximation time (When changed ADCNVSTR.CSTm bit (m = 0, 1))	✓	✓	✓
When changed the operation setting of the channel-dedicated sample-and-hold circuit *1	—	—	✓

Note: ✓ : Self-calibration should be performed.
— : Self-calibration is not required.

Note 1. When any of ADSHCRm.SHENn bits are set to 1, or any of ADSHCRm.SHENn bits or any of ADSHCRm.SHMDn bits are changed, perform the channel-dedicated sample-and-hold circuits self-calibration (m = 0, 1, n = 0 to 2, 4 to 6). This is not required if the channel-dedicated sample-and-hold circuits are not used.

Note 2. When using fixed-channel continuous scan mode, also perform gain/offset calibration for SAR mode (single scan mode). The interval between performing calibration in SAR mode (single scan mode) and Hybrid mode should be as short as possible.

54.3.8.2 Self-calibration Procedure

[Table 54.17](#) shows the self-calibration procedures.

Table 54.17 Procedure for self-calibration (1 of 2)

No.	Step	Description
1	Disable trigger input	Disables the trigger input from the peripheral module. (Write ADTRGENR.STTRGENn = 0)
2	Waiting for all A/D converters to stop	Check that all A/D converters are stopped. If A/D conversion is operating, wait until the A/D converter stops by either of the following: • Wait until all A/D conversions are complete • Force the A/D converter to stop the operation using the ADSTOPR register. For details on forced stopping of A/D conversion operation, see section 54.6.4. Force Stops the A/D Conversion Operation .
3	Setting the number of states for self-calibration	Set the number of states for the self-calibration of A/D converter to ADCALSTCR register. The values in ADCALSTCR register are commonly used for ADC0 and ADC1. Set the value to be set in ADCALSTCR register to satisfy the range defined in section 70, Electrical Characteristics .

Table 54.17 Procedure for self-calibration (2 of 2)

No.	Step	Description
4	Clearing the Error Status Flag	If error flags are detected, clear them before performing the self-calibration. (A/D converter error flag, A/D converter overflow flag, etc.)
5	Performing the self-calibration for ADC0	Performs the self-calibration for ADC0. - Internal-circuit Calibration - Gain/Offset Calibration After starting the self-calibration, wait until it is completed.
6	Performing the self-calibration for ADC1	Performs the self-calibration for ADC1 in the same as Step 5.
7	Setting the self-calibration of channel-dedicated sample-and-hold circuit ^{*1}	Make settings for self-calibration of channel-dedicated sample-and-hold circuit. Enable channel-dedicated sample-and-hold circuit. Sets the number of states for channel-dedicated sample-and-hold self-calibration (ADCALSTCR register ^{*2} and ADCALSHCR register).
8	Performing channel-dedicated sample-and-hold circuit self-calibration for SH0 to SH2. ^{*1}	Performs the self-calibration for channel-dedicated sample-and-hold circuit (SH0 to SH2) connected to ADC0. Channel-dedicated sample-and-hold circuit Gain/Offset Calibration After starting self-calibration, wait until it is completed.
9	Performing channel-dedicated sample-and-hold circuit self-calibration for SH4 to SH6 ^{*1}	Performs the self-calibration for channel-dedicated sample-and-hold circuit (SH4 to SH6) connected to ADC1 in the same as Step 8.
10	Error Status Check ^{*3}	Check for errors in self-calibration. If no operating error occurs, self-calibration is complete.

Note: In order to increase the effectiveness of self-calibration, it is recommended to perform self-calibration for the A/D converters one by one.

Note 1. Not required if the channel-dedicated sample-and-hold circuits are not used.

Note 2. When the self-calibration is performed for the channel-dedicated sample-and-hold circuitry, the number of sampling states at the self-calibration of the A/D converter (ADCALSTCR.CALADSST[9:0] bits) may also need to be changed. Set the value that satisfies the electrical characteristics.

Note 3. If the A/D converter error is detected by the self-calibration, it may be set outside the operation guaranteed range specified in the electric characteristics. Check the operation setting.

54.3.8.3 Restrictions on Self-calibration

(1) Prohibition of the scanning operation during self-calibration

Do not start the scanning operation during self-calibration. When performing the scanning operation, start the scanning operation after self-calibration is completed. Operation is not guaranteed when the scanning operation is started during self-calibration.

(2) Prohibition of additional writes to the ADCALSTR register during self-calibration

After the self-calibration is started, writing to ADCALSTR register is prohibited until self-calibration is completed. Write to ADCALSTR register after self-calibration is completed. Operation is not guaranteed if this restriction is violated.

(3) Restrictions on self-calibration for channel-dedicated sample-and-hold circuit

To perform self-calibration for the channel-dedicated sample-and-hold circuit, set ADSHCRm.SHENn = 1 (m = 0, 1, n = 0 to 2, 4 to 6) for the sample-and-hold circuit channel to be used before performing self-calibration for the channel-dedicated sample-and-hold circuit.

(4) Restrictions on self-calibration

Self-calibration should be performed for the A/D converters one by one. During the self-calibration operation, the other A/D converters should be in idle (not in scan operation or self-calibration operation).

If this restriction is violated, the A/D conversion accuracy will be degraded due to noise during the self-calibration operation. In this case, the A/D converter characteristics are not guaranteed.

In addition, in order to enhance the effect of self-calibration, self-calibration operation should be performed under conditions with as little noise as possible.

(5) Restrictions on setting of the number of status for self-calibration

The number of states to be set in the ADCALSTCR and ADCALSHCR registers should be set to satisfy the values specified in the Electrical Characteristics. In addition, the number of states involved in self-calibration should be set to satisfy the following restrictions.

If these restrictions are violated, A/D conversion results are not guaranteed.

[Restrictions on setting of ADCALSTCR register]

- ADCALSTCR.CALADSST[9:0] bits
 - According to the type of self-calibration, set the values to satisfy those specified in the electrical characteristics.
 - If the setting value differs according to the type of self-calibration, change the register setting value for each self-calibration.
- ADCALSTCR.CALADCST[5:0] bits
 - Set the same value as ADCNVSTR.CSTm[5:0] (m = 0, 1) bits.
(ADCALSTCR.CALADCST[5:0] = ADCNVSTR.CSTm[5:0])

[Restrictions on setting of ADCALSHCR register] (only when channel-dedicated sample-and-hold circuit is used)

- ADCALSHCR.CALSHSST[7:0] bits
 - Set the value equal to the value of the ADHSSTRm.SHSST[7:0] bits plus 1 (m = 0, 1).
(ADCALSHCR.CALSHSST[7:0] = ADHSSTRm.SHSST[7:0] + 1)
 - If the setting values of ADHSSTR0.SHSST[7:0] and ADHSSTR1.SHSST[7:0] are different, change the register setting values of CALSHSST[7:0] for each self-calibration of ADC0 (SH0 to SH2) and ADC1 (SH4 to SH6).
- ADCALSHCR.CALSHHST[2:0] bits
 - Set the same value as ADHSSTRm.SHHST[2:0] (m = 0, 1) bits.

54.3.9 Analog Input Channel

For the correspondence between the analog input pins and the analog channel numbers, the A/D converters, and channel-dedicated sample-and-hold circuit, see [Table 54.4](#).

54.3.10 Extended Analog Function

Extended analog function performs A/D conversion on the internal signal source of MCU. The available sources for extended analog function are as follows:

- Self-diagnosis voltage for A/D converters and channel-dedicated sample-and-hold circuits
- Internal reference voltage
- VBATT 1/6 voltage monitor output
- Temperature sensor
- D/A converter output (internal output path to ACMPHS)

54.3.11 Self-diagnosis Function for A/D Converters

The A/D converter has a built-in self-diagnosis function. The self-diagnosis function inputs the self-diagnosis voltage to the A/D converter and performs A/D conversion. The A/D conversion result of the self-diagnosis voltage can be used to confirm that the A/D converter is operating normally.

54.3.11.1 Self-diagnosis

The A/D converter has a built-in self-diagnosis function. The self-diagnosis function inputs the self-diagnosis voltage to the A/D converter and performs A/D conversion. The A/D conversion result of the self-diagnosis voltage can be used to confirm that the A/D converter is operating normally. [Table 54.18](#) shows the operation mode in which the self-diagnostic function is available.

Table 54.18 List of operation mode for which the self-diagnostic function is available

Operation mode – Scan mode	Self-diagnosis function
SAR mode – Single scan mode	✓
SAR mode – Continuous scan mode	✓
Oversampling mode – Single scan mode	✓
Oversampling mode – Continuous scan mode	—
Oversampling mode – One-channel continuous scan mode	—
Hybrid mode – Single scan mode	✓
Hybrid mode – Continuous scan mode	—
Hybrid mode – Background continuous scan mode	—
Hybrid mode – Fixed-channel continuous scan mode	—

Note: ✓: Available, —: Not Available

The procedure for performing self-diagnosis of A/D converter is shown in the following.

1. Assign the self-diagnosis channel to one of the virtual channels and configure the virtual channel settings.
2. Assign the virtual channel that the self-diagnosis channel is selected to one of the scan groups.
3. Set the self-diagnosis voltage in DIAGVAL[2:0] bits of AD SGDCR_n (n = 0 to 8) register corresponding to the scan group assigned in 2 above.
4. Start the scanning operation by inputting the A/D conversion start trigger. The self-diagnosis operation is executed at the A/D conversion of the self-diagnosis channel.

Table 54.19 lists the operation settings when the self-diagnosis function is used. When using the self-diagnosis function, follow the setting in Table 54.19. If self-diagnosis operation is performed with settings other than those shown in Table 54.19, the self-diagnosis result (A/D conversion result) is not guaranteed.

Table 54.19 List of operation setting and expected value for ADC0 self-diagnosis function

Self-diagnosis mode	Register setting value				Self-diagnosis operation
	ADSGDCR _x	ADCHCR _y		ADDOPCRC _y	Expected value of A/D conversion data *1 *2 *3
	DIAGVAL[2:0]	CNVCS[6:0]	AINMD	SIGNSEL	
Self-diagnosis mode 1	100b	0x60	1	0	0x0000
Self-diagnosis mode 2	101b	0x60	1	0	0x8000
Self-diagnosis mode 3	110b	0x60	1	0	0x7FFF

Note: x = 0 to 8, y = 0 to 32

Note 1. The expected value is the ideal A/D converter result stored in ADEXDR0.DATA[15:0] or ADFIFODRx.DATA[15:0].

Note 2. The A/D conversion data is the value when 16-bit data format (ADDOPCRC_y.ADPRC[1:0] = 00b) is selected.

Note 3. The A/D conversion result becomes (expected value + 1) or more when a positive accuracy error occurs in A/D conversion, and becomes (expected value - 1) or less when a negative accuracy error occurs.

Table 54.20 List of operation setting and expected value for ADC1 self-diagnosis function

Self-diagnosis mode	Register setting value				Self-diagnosis operation
	ADSGDCR _x	ADCHCR _y		ADDOPCRC _y	Expected value of A/D conversion data *1 *2 *3
	DIAGVAL[2:0]	CNVCS[6:0]	AINMD	SIGNSEL	
Self-diagnosis mode 1	100b	0x61	1	0	0x0000
Self-diagnosis mode 2	101b	0x61	1	0	0x8000
Self-diagnosis mode 3	110b	0x61	1	0	0x7FFF

Note: x = 0 to 8, y = 0 to 32

Note 1. The expected value is the ideal A/D converter result stored in ADEXDR1.DATA[15:0] or ADFIFODRx.DATA[15:0].

Note 2. The A/D conversion data is the value when 16-bit data format (ADDOPCRC_y.ADPRC[1:0] = 00b) is selected.

Note 3. The A/D conversion result becomes (expected value + 1) or more when a positive accuracy error occurs in A/D conversion, and becomes (expected value - 1) or less when a negative accuracy error occurs.

Table 54.21 List of operation setting and expected value for ADC0 channel-dedicated sample-and-hold circuit self-diagnosis function

Self-diagnosis mode	Register setting value					Self-diagnosis operation
	ADSHSDCRw	ADCHCRy		ADSHCRw	ADDOPCRCy	Expected value of A/D conversion data *1 *2 *3
	SHDIAGx[2:0]	CNVCS[6:0]	AINMD	SHMDz	SIGNSEL	
S&H self-diagnosis mode 1	001b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x0000
S&H self-diagnosis mode 2	010b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x8000
S&H self-diagnosis mode 3	011b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x7FFF
S&H self-diagnosis mode 4	100b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x0000
S&H self-diagnosis mode 5	101b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x8000
S&H self-diagnosis mode 6	110b	SH0: 0x70 SH1: 0x71 SH2: 0x72	1	1	0	0x7FFF

Note: x = 0 to 8, y = 0 to 32, z = 0 to 2

Note 1. The expected value is the ideal A/D converter result stored in ADEXDRn.DATA[15:0] (n = 16,17,18) or ADFIFODRx.DATA[15:0].

Note 2. The A/D conversion data is the value when 16-bit data format (ADDOPCRCy.ADP RC[1:0] = 00b) is selected.

Note 3. The A/D conversion result becomes (expected value + 1) or more when a positive accuracy error occurs in A/D conversion, and becomes (expected value - 1) or less when a negative accuracy error occurs.

Table 54.22 List of operation setting and expected value for ADC1 channel-dedicated sample-and-hold circuit self-diagnosis function (1 of 2)

Self-diagnosis mode	Register setting value					Self-diagnosis operation
	ADSHSDCRw	ADCHCRy		ADSHCRw	ADDOPCRCy	Expected value of A/D conversion data *1 *2 *3
	SHDIAGx[2:0]	CNVCS[6:0]	AINMD	SHMDz	SIGNSEL	
S&H self-diagnosis mode 1	001b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x0000
S&H self-diagnosis mode 2	010b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x8000
S&H self-diagnosis mode 3	011b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x7FFF
S&H self-diagnosis mode 4	100b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x0000
S&H self-diagnosis mode 5	101b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x8000

Table 54.22 List of operation setting and expected value for ADC1 channel-dedicated sample-and-hold circuit self-diagnosis function (2 of 2)

Self-diagnosis mode	Register setting value					Self-diagnosis operation
	ADSHSDCRw	ADCHCRy		ADSHCRw	ADDOPCRCy	Expected value of A/D conversion data *1 *2 *3
	SHDIAGx[2:0]	CNVCS[6:0]	AINMD	SHMDz	SIGNSEL	
S&H self-diagnosis mode 6	110b	SH4: 0x74 SH5: 0x75 SH6: 0x76	1	1	0	0x7FFF

Note: x = 0 to 8, y = 0 to 32, z = 4 to 6

Note 1. The expected value is the ideal A/D converter result stored in ADEXDRn.DATA[15:0] (n = 20,21, 22) or ADFIFODRx.DATA[15:0].

Note 2. The A/D conversion data is the value when 16-bit data format (ADDOPCRCy.ADPRC[1:0] = 00b) is selected.

Note 3. The A/D conversion result becomes (expected value + 1) or more when a positive accuracy error occurs in A/D conversion, and becomes (expected value - 1) or less when a negative accuracy error occurs.

54.3.11.2 Self-diagnosis Mode

(1) Self-diagnosis mode 1

In self-diagnosis mode 1, the non-inverting input (+) (A_{INP}) and inverting input (−) (A_{INN}) of A/D converter are input VREFL0 level (ADC0) or VREFL level (ADC1). Since the differential voltage is 0, the ideal value of the self-diagnosis result is 0x0000 (± 0). The A/D conversion result becomes 0x0001 (+1) or more when a positive accuracy error occurs in A/D conversion, and becomes 0xFFFF (−1) or less when a negative accuracy error occurs.

(2) Self-diagnosis mode 2

In self-diagnosis mode 2, the non-inverting input (+) (A_{INP}) of A/D converter is input VREFL0 level (ADC0) or VREFL level (ADC1), and the inverting input (−) (A_{INN}) of A/D converter is input VREFH0 level (ADC0) or VREFH level (ADC1). Since the differential voltage is $-VREFH0$ (if $VREFL0 = 0V$) or $-VREFH$ (if $VREFL = 0V$), the ideal value of the self-diagnosis result is 0x8000 (−32768). When an A/D conversion error occurs, the A/D conversion result of self-diagnosis is greater than 0x8000.

(3) Self-diagnosis mode 3

In self-diagnosis mode 3, the non-inverting input (+) (A_{INP}) of A/D converter is input VREFH0 level (ADC0) or VREFH level (ADC1), and the inverting input (−) (A_{INN}) of A/D converter is input VREFL0 level (ADC0) or VREFL level (ADC1). Since the differential voltage is $+VREFH0$ (if $VREFL0 = 0V$) or $+VREFH$ (if $VREFL = 0V$), the ideal value of the self-diagnosis result is 0x7FFF (+32767). When an A/D conversion error occurs, the A/D conversion result of self-diagnosis is less than 0x7FFF.

54.3.11.3 Notes on Self-diagnosis Function

(1) Notes on the results of the self-diagnosis

The self-diagnostic result of A/D converter unit 0 can be read from the ADEXDR0 register. The self-diagnostic result of A/D converter unit 1 can be read from the ADEXDR1 register.

(2) Notes on data format during self-diagnosis

When self-diagnosis is performed, if a data format other than 16-bit length is set, an overflow occurs due to A/D conversion data rounding, and the A/D conversion data error is detected. To perform self-diagnosis, select the 16-bit data format.

54.3.12 Internal Reference Voltage

ADC16H can perform A/D conversion of Internal Reference Voltage. Internal Reference Voltage outputs a constant voltage that is independent of the power supply voltage (V_{CC} , V_{SS} , AV_{CC} , AV_{SS} , $VREFH0$, $VREFL0$, $VREFH$ and $VREFL$). The monitoring of Internal Reference Voltage can be used to detect variations in the analog reference supply voltage ($VREFH0$, $VREFL0$, $VREFH$ and $VREFL$), or abnormalities or faults in the system.

When performing A/D conversion of Internal Reference Voltage, it is required to set the Internal Reference Voltage Monitor Enable Register (ADREFCR) and assign the analog channel of Internal Reference Voltage to any of the virtual channels. See [Table 54.5](#) for the correspondence between Internal Reference Voltage function and the analog channels. For the sampling time when A/D conversion of Internal Reference Voltage is performed, see [section 70, Electrical Characteristics](#).

54.3.13 Temperature Sensor

ADC16H can perform A/D conversion of Temperature Sensor. The chip temperature can be estimated from the A/D conversion result of Temperature Sensor. For more details on Temperature Sensor, see [section 56, Temperature Sensor \(TSN\)](#).

When A/D conversion of Temperature Sensor is performed, Temperature Sensor should be configured and the analog channel of Temperature Sensor should be assigned to any of the virtual channels. See [Table 54.5](#) for the correspondence between Temperature Sensor and analog channels. For the sampling time when A/D conversion of Temperature Sensor is performed, see [section 70, Electrical Characteristics](#).

54.3.14 D/A Converter

ADC16H can perform A/D conversion of the voltage that the D/A converter is outputting to the internal module (ACMPHS). When the output of the D/A converter is used for the reference input voltage of High-Speed Analog Comparator (ACMPHS), this function can be used to monitor the output voltage of the D/A converter. For details of the D/A converter function, see [section 55, 12-Bit D/A Converter \(DAC12\)](#).

When A/D conversion is performed on the output of D/A converter (DA0 and DA1) (output to ACMPHS), the D/A converter should be configured and the analog channel of the D/A converter should be assigned to one of the virtual channels. For the sampling time when A/D conversion is performed on the D/A converter output voltage, see [section 70, Electrical Characteristics](#).

54.3.15 VBATT 1/6 Voltage Monitor Output

ADC16H can perform A/D conversion of VBATT 1/6 voltage monitor output. VBATT 1/6 voltage monitor output is a function that can monitor the input voltage level to the VBATT pin at 1/6 level. For more details on VBATT voltage monitor function, see [section 12, Battery Backup Function](#).

When A/D conversion of VBATT 1/6 voltage monitor output is performed, VBATT should be configured and the analog channel of VBATT 1/6 voltage monitor output should be assigned to any of the virtual channels. See [Table 54.5](#) for the correspondence between VBATT 1/6 voltage monitor output and analog channels. For the sampling time when A/D conversion of VBATT 1/6 voltage monitor output is performed, see [section 70, Electrical Characteristics](#).

54.3.16 Channel-dedicated Sample-and-hold Circuit

ADC16H has a built-in channel-dedicated sample-and-hold circuit. By using multiple channel-dedicated sample-and-hold circuits, it is possible to sample multiple analog inputs simultaneously.

The conditions under which channel-dedicated sample-and-hold circuit can be used are shown in [Table 54.23](#).

Table 54.23 Operation mode in which channel-dedicated sample-and-hold circuits can be used

Operation mode	Channel-dedicated sample-and hold circuits
SAR mode – Single scan mode	✓*1
SAR mode – Continuous scan mode	—
Oversampling mode – Single scan mode	—
Oversampling mode – Continuous scan mode	—
Oversampling mode – One-channel continuous scan mode	—
Hybrid mode – Single scan mode	✓
Hybrid mode – Continuous scan mode	✓
Hybrid mode – Background continuous scan mode	✓
Hybrid mode – Fixed-channel continuous scan mode	✓*2

Note: ✓: available, —: not available

Note 1. Constant Sampling Function is available.

Note 2. Only fixed channels are supported when fixed channels are set to 3ch and all fixed channels enable channel-dedicated sample-and-hold circuit. Non-fixed channels cannot use channel-dedicated sample-and-hold circuit.

54.3.16.1 Configuration of Channel-dedicated Sample-and-hold Circuit

(1) Configuration in Single-ended input mode

This subsection describes the internal configuration when using channel-dedicated sample-and-hold circuit with single-ended inputs.

Figure 54.16 shows the configuration of channel-dedicated sample-and-hold circuit in Single-ended input mode.

Channel-dedicated sample-and-hold circuits are connected to two analog input channels.

In the case that one of the channels is used as an input for channel-dedicated sample-and-hold circuit and the other is assigned as an input for normal A/D conversion when the channel-dedicated sample-and-hold circuit is used in Single-ended input mode, one of the following methods can be used.

- Assign each input channel to a different scan group
- Use the function of ADHSBPCR register for AD conversion in one scan group

For example, in the case of using the function of ADHSBPCR register for AD conversion in one scan group, set 0 to ADHSBPCR.SHSP0 bit and set 1 to ADHSBPCR.SHSDN0 bit when AN000 is used as an input channel of SH0 in Single-ended input mode and AN001 is used as an input channel of normal AD conversion.

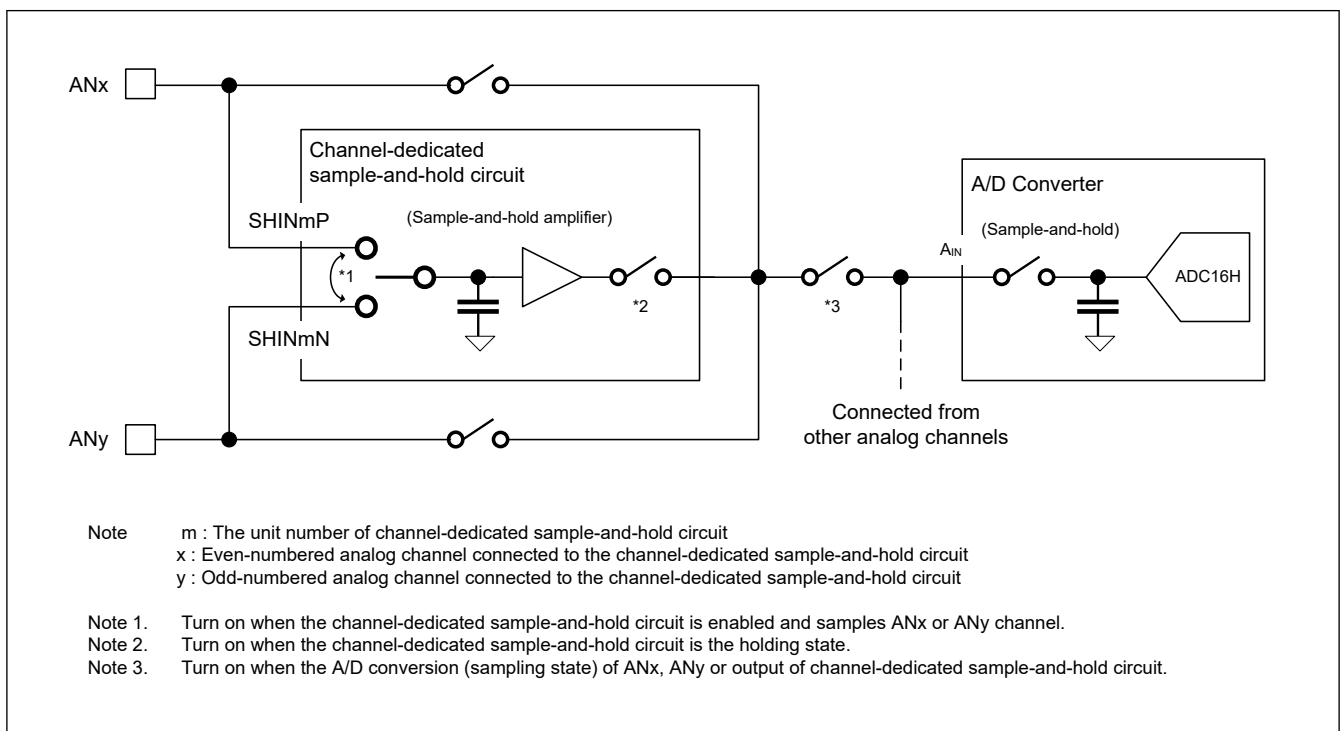


Figure 54.16 Internal configuration of channel-dedicated sample-and-hold circuit in Single-ended input mode

(2) Configuration in Differential input mode

This subsection describes the internal configuration when channel-dedicated sample-and-hold circuit is used with differential inputs.

Figure 54.17 shows the configuration of channel-dedicated sample-and-hold circuit in Differential input mode.

When channel-dedicated sample-and-hold circuit is operating in Differential input mode, the even analog channel is connected to the non-inverting input (+) (SHINmP) and the odd-numbered analog channel is connected to the inverting input (-) (SHINmN). Two analog inputs (differential input) are sampled and held in one scanning operation, and

output to the A/D converter. The A/D conversion setting should be set to convert a differential input signal source. (ADCHCRn.AINMD = 1 (n = 0 to 32))

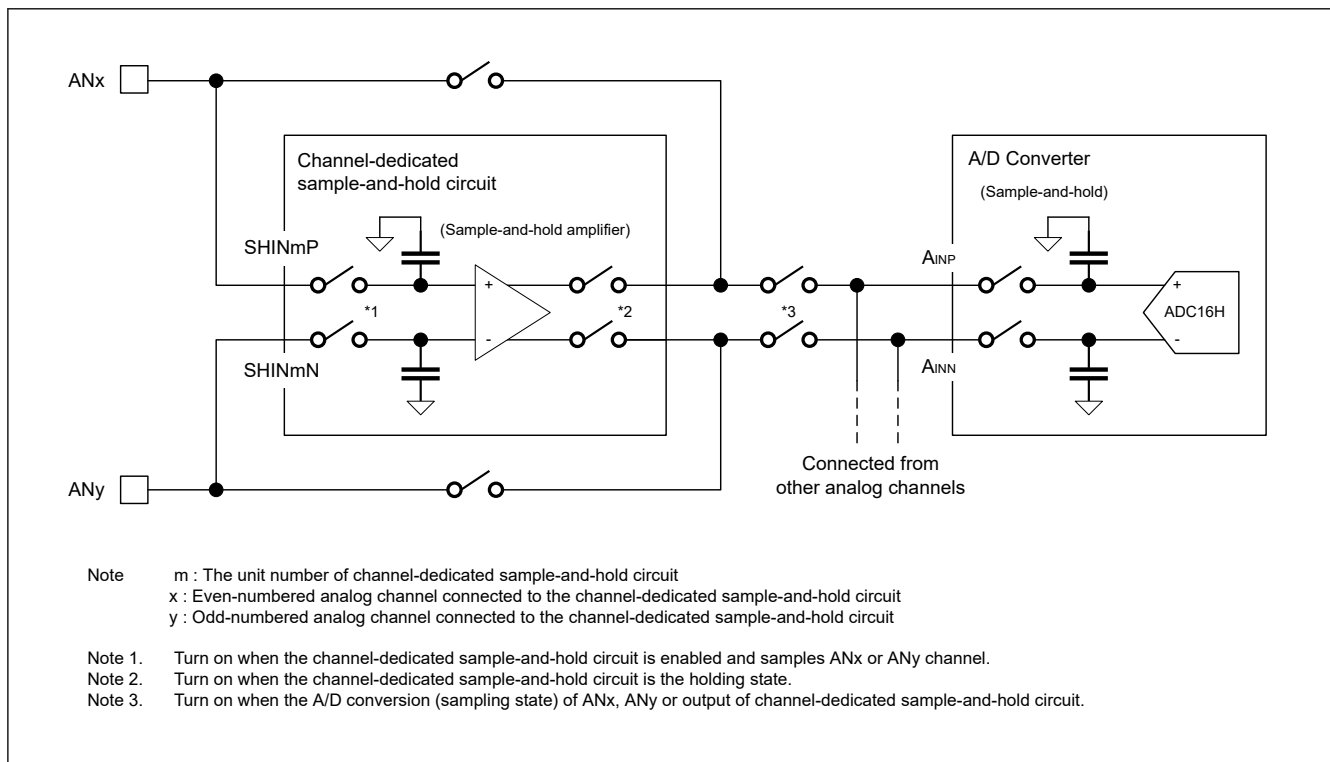


Figure 54.17 Internal configuration of channel-dedicated sample-and-hold circuit in Differential input mode

54.3.16.2 Operation in SAR Mode with Channel-dedicated Sample-and-hold Circuit

(1) Basic operation in SAR mode – Single scan mode (Single-ended input mode)

This section describes the basic operation of channel-dedicated sample-and-hold circuit in SAR mode. The configuration and behavior of example operations are shown in [Table 54.24](#), [Table 54.25](#) and [Figure 54.18](#).

Table 54.24 Configuration of example operation for channel-dedicated sample-and-hold circuit in SAR mode – Single scan mode

Item	Configuration
A/D converter	Set ADC0 to SAR mode – Single scan mode
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Single-ended input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)
Virtual channel*1	Assign AN000, AN002, AN004 and AN020 to virtual channel 0 to 3, and set to Single-ended input mode.
Scan group	Assign virtual channel 0 to 3 to scan group 0. Set scan group 0 to convert with ADC0.

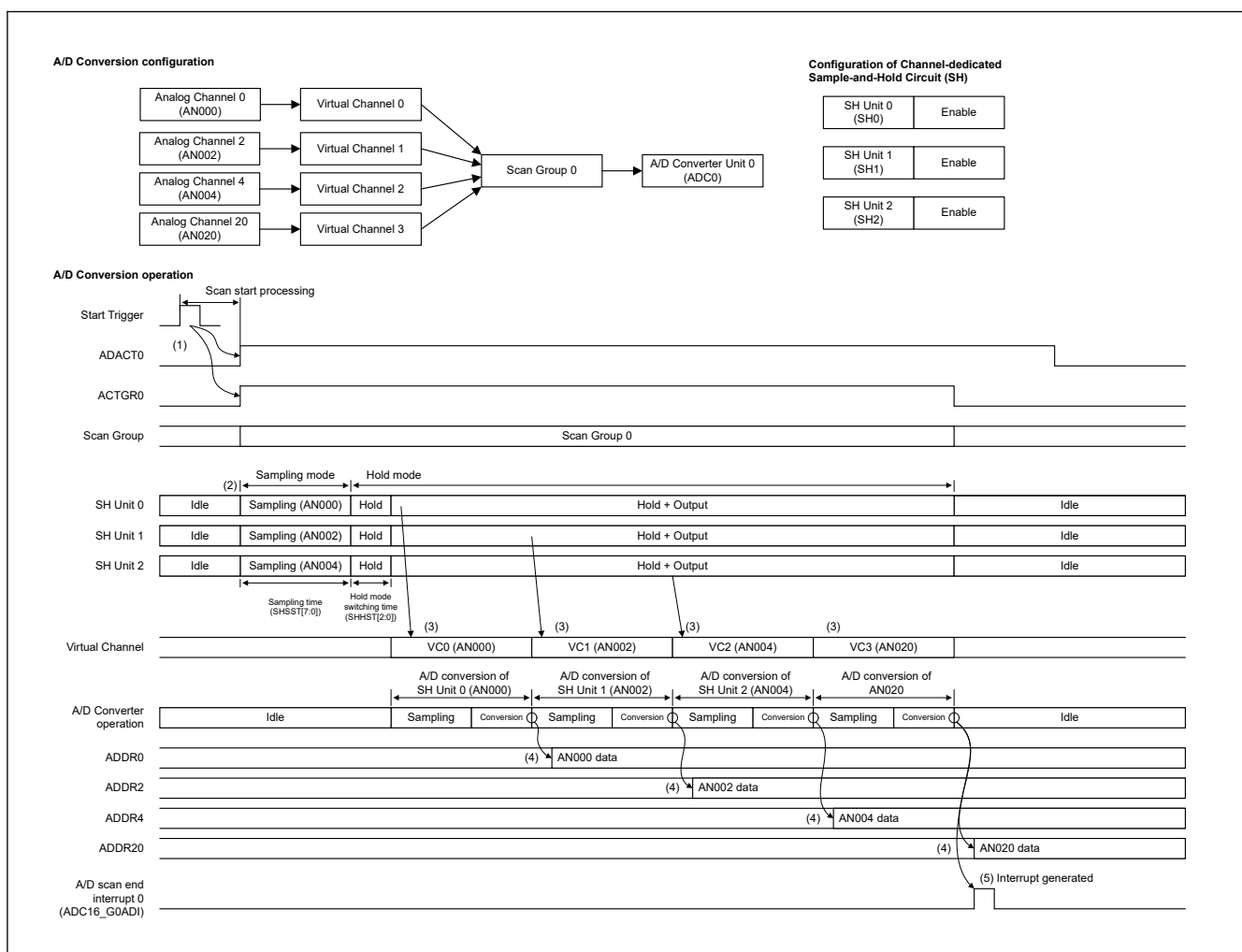
Note 1. When channel-dedicated sample-and-hold circuit is used, there are restrictions on the setting of virtual channels. For more details, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

Table 54.25 Example operation of channel-dedicated sample-and-hold circuit in SAR mode – Single scan mode (1 of 2)

Step	Detail behavior of example operation
1	The scanning operation of scan group 0 starts when a trigger for scan group 0 is input.
2	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.

Table 54.25 Example operation of channel-dedicated sample-and-hold circuit in SAR mode – Single scan mode (2 of 2)

Step	Detail behavior of example operation
3	A/D conversion is performed on the voltages held by each channel-dedicated sample-and-hold circuit and on other analog input channels (AN020).
4	When A/D conversion of each channel completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
5	If scan end interrupt in scan group 0 is set to enabled, a scan end interrupt occurs.

**Figure 54.18 Basic operation of channel-dedicated sample-and-hold circuit in SAR mode****(2) Advanced operation in SAR mode - Single scan mode**

This section describes the operation of A/D conversion that shares one channel-dedicated sample-and-hold circuit between two analog channels. The operation shown in this section is possible only when all of the following conditions are met.

[Operating Conditions]

- Operation mode: SAR mode – Single scan mode
- Input method: Single-ended input mode

The following shows an example of A/D conversions of even-numbered analog channels and odd-numbered analog channels using channel-dedicated sample-and-hold circuit in SAR mode – Single scan mode. The configuration example is shown in Table 54.26, and the operation detail is shown in Table 54.27 and Figure 54.19.

Table 54.26 Configuration example of advanced operation for channel-dedicated sample-and-hold circuit

Item	Configuration
A/D converter	Set ADC0 to SAR mode – Single scan mode
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Single-ended input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)
Virtual channel*1	- Assign AN000, AN002 and AN004 to virtual channel 0 to 2, and set to single-ended input mode. - Assign AN001, AN003 and AN005 to virtual channel 3 to 5, and set to single-ended input mode.
Scan group	- Scan group 0: Assign AN000, AN002, AN004 (virtual channels 0 to 2). - Scan group 1: Assign AN001, AN003, AN005 (virtual channels 3 to 5). - Set scan group 0 and scan group 1 to convert with ADC0.

Note 1. When channel-dedicated sample-and-hold circuit is used, there are restrictions on the setting of virtual channels. For more details, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

Table 54.27 Example advanced operation of channel-dedicated sample-and-hold circuit

Step	Detail behavior of example operation
1	The scanning operation of scan group 0 starts when a trigger for scan group 0 is input.
2	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.
3	A/D conversion is performed on the voltages held by channel-dedicated sample-and-hold circuit.
4	When A/D conversion of each channel in scan group 0 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
5	If scan end interrupt in scan group 0 is set to enabled, a scan end interrupt occurs.
6	After scanning of scan group 0 is completed, the scanning operation of scan group 1 starts when a trigger for scan group 1 is input.
7	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN001, AN003, AN005) and holds them.
8	A/D conversion is performed on the voltages held by channel-dedicated sample-and-hold circuit.
9	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
10	If scan end interrupt for scan group 1 is set to enabled, a scan end interrupt occurs.

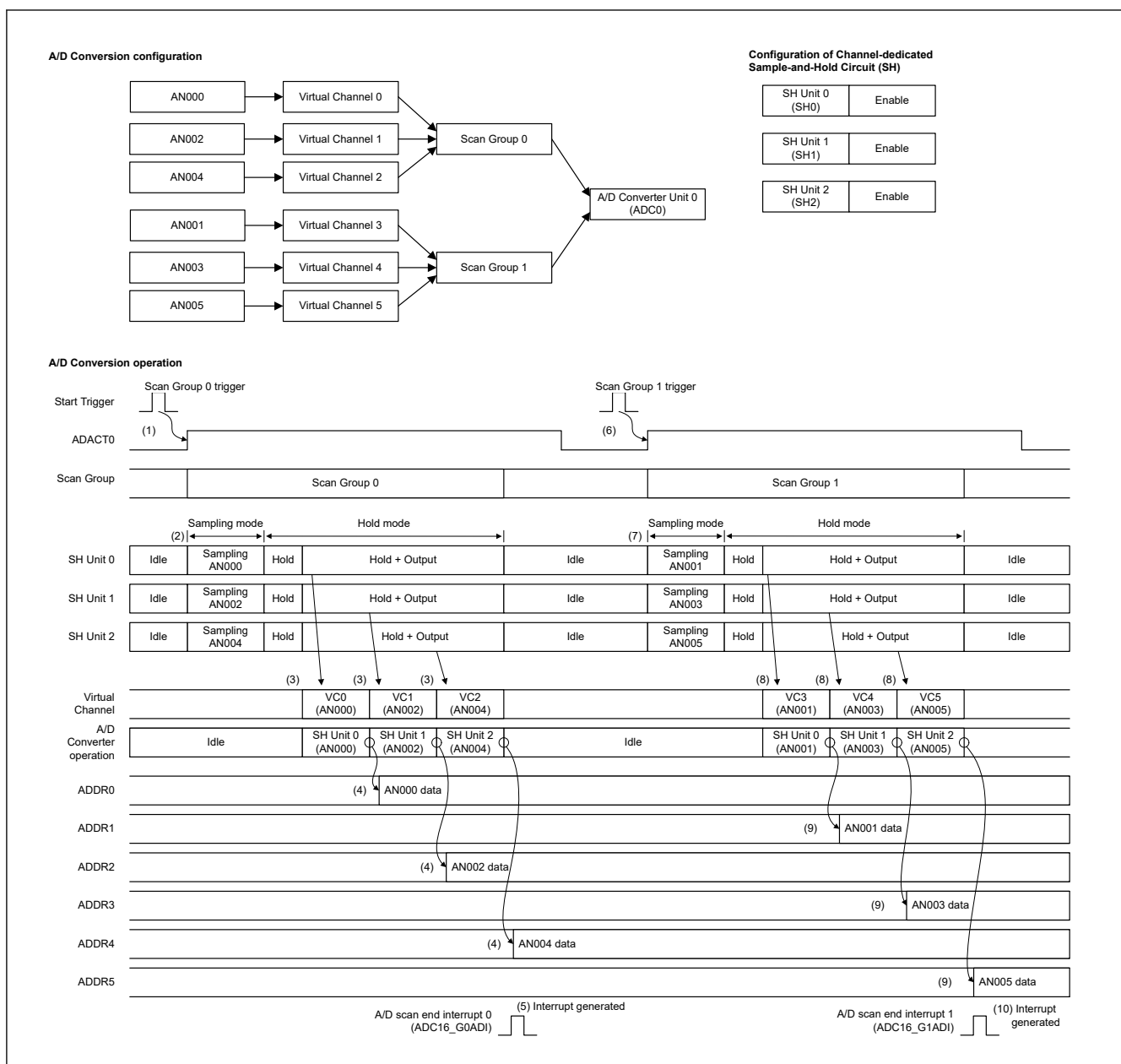


Figure 54.19 Example operation when channel-dedicated sample-and-hold circuit is shared with two analog channels in SAR mode

54.3.16.3 Operation in Hybrid Mode with Channel-dedicated Sample-and-hold Circuit

(1) Basic Operation in Hybrid Mode - Single Scan Mode

This section describes the basic operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Single scan mode. Since Differential input mode is supported in Hybrid mode, this example describes a case that channel-dedicated sample-and-hold circuits are set to Differential input mode. The configuration and behavior of example operation are shown in [Table 54.28](#), [Table 54.29](#) and [Figure 54.20](#).

Table 54.28 Configuration of example operation for channel-dedicated sample-and-hold circuit in Hybrid mode – Single scan mode (1 of 2)

Item	Configuration
A/D converter	Set ADC0 to Hybrid mode – Single scan mode
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Differential input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)

Table 54.28 Configuration of example operation for channel-dedicated sample-and-hold circuit in Hybrid mode – Single scan mode (2 of 2)

Item	Configuration
Virtual channel*1	Assign AN000, AN002 and AN004 to virtual channel 0 to 2, and set to Differential input mode. (This will function AN000, AN002 and AN004 as the non-inverting inputs (+) of the differential input, and AN001, AN003, and AN005 as the inverting inputs (–) of the differential input.)
Scan group*1	- Assign virtual channel 0 to 2 to scan group 0. - Set scan group 0 to convert with ADC0.

Note 1. When channel-dedicated sample-and-hold circuit is used, there are restrictions on the setting of virtual channels. For more details, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

Table 54.29 Example operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Single scan mode

Step	Behavior
1	The scanning operation of scan group 0 starts when a trigger for scan group 0 is input.
2	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.
3	The A/D converter performs oversampling once for each output of channel-dedicated sample-and-hold circuit (SH0 to SH2).
4	After one oversampling for each analog channel, channel-dedicated sample-and-hold circuit releases the held voltage and enters the idle state.
5	Step 2 to 4 is repeated until oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/average times is performed for each analog channel.
6	A/D conversion data corresponding to each analog channel is output. The A/D conversion data is stored in the data register (ADDRi (i = 0, 2, 4)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).
7	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
8	A/D converter and channel-dedicated sample-and-hold circuit enter the idle state when scanning operation is completed. The oversampling data in digital filter are discarded at the end of the scanning operation.

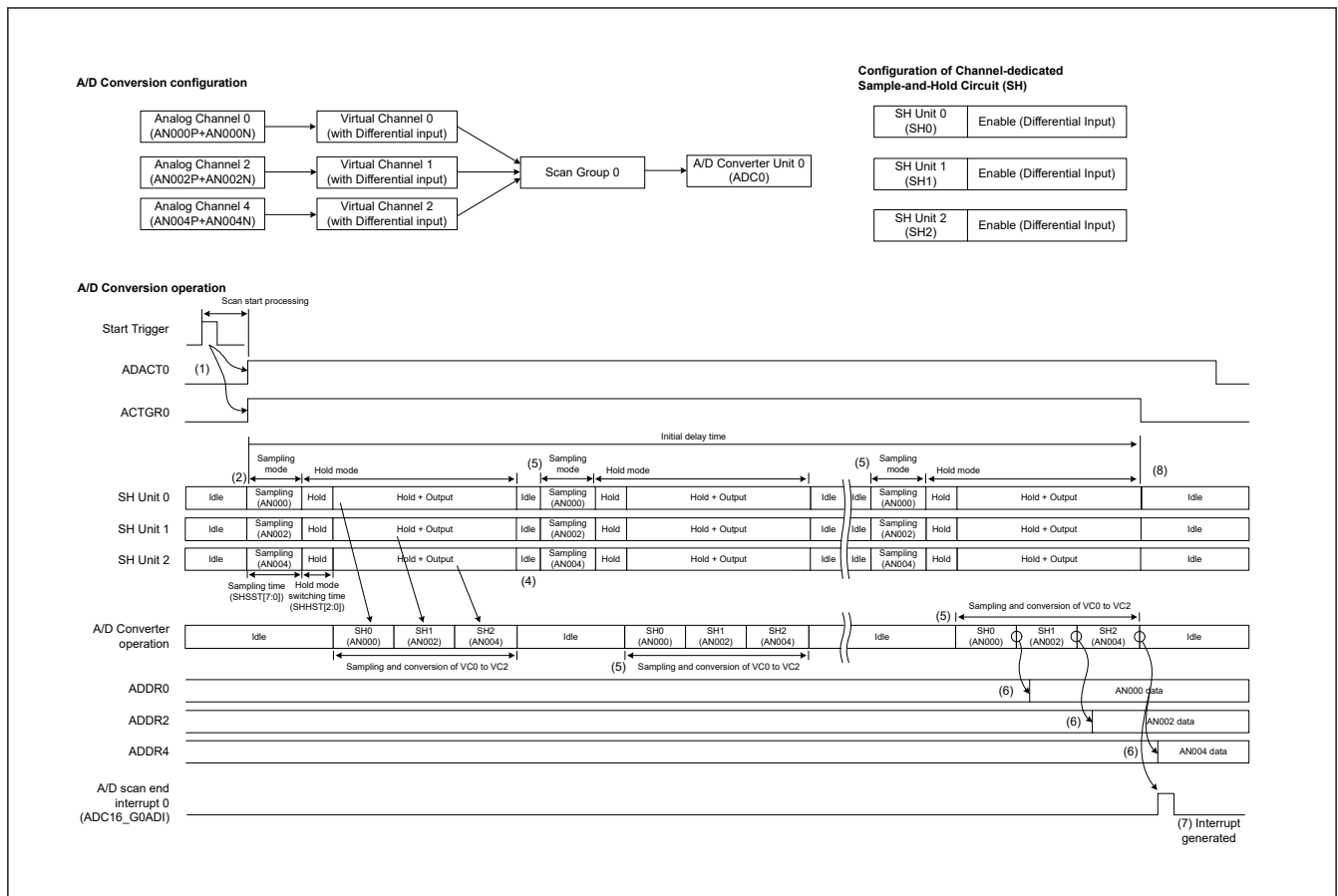


Figure 54.20 Basic operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Single scan mode

(2) Basic Operation in Hybrid Mode - Continuous Scan Mode

This section shows the basic operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Continuous scan mode. Since Differential input mode is supported in Hybrid mode, this example describes a case that channel-dedicated sample-and-hold circuits are set to Differential input mode. The configuration and behavior of example operation are shown in [Table 54.30](#), [Table 54.31](#) and [Figure 54.21](#).

Table 54.30 Configuration of example operation for channel-dedicated sample-and-hold circuit in Hybrid mode – Continuous scan mode

Item	Configuration
A/D converter	Set ADC0 to Hybrid mode – Continuous scan mode
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Differential input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)
Virtual channel* ¹	Assign AN000, AN002 and AN004 to virtual channel 0 to 2, and set to Differential input mode. This treats AN000, AN002 and AN004 as the non-inverting inputs (+) of the differential input, and AN001, AN003, and AN005 as the inverting inputs (–) of the differential input.
Scan group* ¹	- Assign virtual channel 0 to 2 to scan group 0. - Set scan group 0 to convert with ADC0.

Note 1. When channel-dedicated sample-and-hold circuit is used, there are restrictions on the setting of virtual channels. For more details, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

Table 54.31 **Example operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Continuous scan mode (1 of 2)**

Step	Behavior
1	The scanning operation of scan group 0 starts when a trigger for scan group 0 is input.

Step	Behavior
2	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.
3	The A/D converter performs oversampling once for each output of channel-dedicated sample-and-hold circuit (SH0 to SH2).
4	After one oversampling for each analog channel, channel-dedicated sample-and-hold circuit releases the held voltage and enters the idle state.
5	Step 2 to 4 is repeated until oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/average times is performed for each analog channel.
6	A/D conversion data corresponding to each analog channel is output. The A/D conversion data is stored in the data register (ADDRi (i = 0, 2, 4)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).
7	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
8	The second and subsequent rounds of scanning operations repeat the operation of Step 2 to 4, retaining the oversampling data stored in the digital filter. Each one time oversampling or each multiple times oversampling corresponding to the number of A/D-converted value addition/averaging times is performed for each analog channel, the next A/D conversion data is output. The A/D conversion data is stored in the data register (ADDRi (i = 0, 2, 4)). If FIFO is used, A/D conversion data is also stored in FIFO data register (ADFIFODRk (k = 0 to 8)).
9	If the scan end interrupt is enabled, the scan end interrupt corresponding to that scan group is generated when the A/D conversion of all virtual channels assigned to that scan group is completed.
10	Thereafter, until the forcibly stop process is performed, Step 8 to 9 are repeated, and the scanning operation continues. To perform the forcibly stop, follow to section 54.6.4. Force Stops the A/D Conversion Operation . If scanning operation is aborted by the forcibly stop process, A/D converter and channel-dedicated sample-and-hold circuit enter the idle state. The oversampling data in digital filter are discarded at the end of the scanning operation.



(3) Basic Operation in Hybrid Mode - Background Continuous Scan Mode

This section shows the basic operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Background continuous scan mode. Since Differential input mode is supported in Hybrid mode, this example describes a case that channel-dedicated sample-and-hold circuits are set to Differential input mode. The configuration and behavior of example operation are shown in [Table 54.32](#), [Table 54.33](#) and [Figure 54.22](#).

Table 54.32 Configuration of example operation for channel-dedicated sample-and-hold circuit in Hybrid mode – Background continuous scan mode

Item	Configuration
A/D converter	Set ADC0 to Hybrid mode – Continuous scan mode
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Differential input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)
Virtual channel*1	Assign AN000, AN002 and AN004 to virtual channel 0 to 2, and set to Differential input mode. This treats AN000, AN002 and AN004 as the non-inverting inputs (+) of the differential input, and AN001, AN003, and AN005 as the inverting inputs (–) of the differential input.
Scan group*1	- Assign virtual channel 0 to 2 to scan group 0. - Set scan group 0 to convert with ADC0.

Note 1. When channel-dedicated sample-and-hold circuit is used, there are restrictions on the setting of virtual channels. For more details, see [section 54.3.16.6. Restrictions on Channel-dedicated Sample-and-hold Circuit](#).

Table 54.33 Example operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Background continuous scan mode

Step	Behavior
1	The scanning operation of scan group 0 starts when a trigger for scan group 0 is input.
2	Channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.
3	The A/D converter performs oversampling once for each output of channel-dedicated sample-and-hold circuit (SH0 to SH2).
4	After one oversampling for each analog channel, channel-dedicated sample-and-hold circuit releases the held voltage and enters the idle state.
5	Step 2 to 4 is repeated. After the oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/averaging times is performed to each analog channel, first A/D conversion data becomes in ready to be output. Step 2 to 4 is repeated until oversampling corresponding to the number of TAPs of the digital filter and the number of A/D-converted value addition/average times is performed for each analog channel.
6	The second and subsequent rounds of scanning operations repeat the operation of Step 2 to 4, retaining the oversampling data stored in the digital filter. Each time oversampling is performed for each analog channel, the data in the digital filter is updated. Each one time oversampling or each multiple times oversampling corresponding to the number of A/D-converted value addition/averaging times is performed for each analog channel, the next A/D conversion data becomes in ready to be output.
7	When an A/D conversion start trigger is input during background continuous scanning operation, the most recent A/D conversion data at that time is stored in the data register (ADDRi (i = 0, 2, 4)). When FIFO function is set to enabled, A/D conversion data is also stored in the FIFO data register (ADFIFODRk (k = 0 to 8)). Also, if scan end interrupt is set to enable, scan end interrupt is generated. If FIFO interrupt is set to enable, FIFO data read request is generated when the condition for generating it is met.
8	Thereafter, until the forcibly stop process is performed, background continuous scanning operation (Step 6) is repeated. And whenever the A/D conversion start trigger is input during background continuous scanning operation, the A/D conversion data is output (Step 7). To perform the forcibly stop, follow to section 54.6.4. Force Stops the A/D Conversion Operation . If scanning operation is aborted by the forcibly stop process, A/D converter and channel-dedicated sample-and-hold circuit enter the idle state. The oversampling data in digital filter are discarded at the end of the scanning operation.



This section shows the basic operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Fixed-channel continuous scan mode. The configuration and behavior of example operation are shown in [Table 54.34](#), [Table 54.35](#). This mode is available only when the number of fixed channels is set to 3 and the fixed channels are used only for channel-dedicated sample-and-hold circuit.

Item	Configuration
A/D converter	Set ADC0 to Hybrid mode – Fixed-channel continuous scan mode.
Channel-dedicated sample-and-hold circuit	Set SH0 to SH2 to enabled and Differential input mode. (AN000 to AN005 are subject to A/D conversion using channel-dedicated sample-and-hold circuit.)
Virtual channel*1	Assign AN000, AN002 and AN004 to virtual channel 0 to 2, and set to Differential input mode. This treats AN000, AN002 and AN004 as the non-inverting inputs (+) of the differential input, and AN001, AN003, and AN005 as the inverting inputs (–) of the differential input.
Scan group*1	- Assign virtual channel 0 to 2 to scan group 0. - Set scan group 0 to convert with ADC0.

Page 3435 of 4304

Table 54.35 Example operation of channel-dedicated sample-and-hold circuit in Hybrid mode – Fixed-channel continuous scan mode

Step	Behavior
1	When a start trigger is input and the AD _{SR} .ADACT _m (m = 0, 1) bit becomes 1 (A/D conversion in progress), A/D conversion of the scan group selected by the AD _{SGCR} _m (m = 0, 1) register is started.
2	Three fixed-channel channel-dedicated sample-and-hold circuit (SH0 to SH2) starts sampling the analog inputs (AN000, AN002, AN004) and holds them.
3	The A/D converter performs oversampling once for each output of channel-dedicated sample-and-hold circuit (SH0 to SH2).
4	One non-fixed channel is A/D converted in SAR mode. Non-fixed channels are converted from the virtual channels assigned to the relevant scan group in ascending order of channel number.
5	After performing A/D conversions for the three fixed channels and one non-fixed channel, channel-dedicated sample-and-hold circuit releases the held voltage and enters the idle state.
6	When A/D conversion for one channel is completed, the A/D conversion result is stored in the A/D data register (AD _{DRi} .DATA (i = 0 to 22), AD _{EXDRj} (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)) corresponding to the conversion target channel. If FIFO is used, the A/D conversion result is also stored to the FIFO data register (AD _{FIFODRk} (k = 0 to 8)).
7	If scan end interrupt is set to enable, scan end interrupt is generated.
8	The conversion time of idle cycles is counted until the end of the scan cycle.
9	Thereafter, until the A/D conversion stop process is performed, Background continuous scanning operation (Step 2 to 6) is repeated. For fixed channel scan, conversion of the same channel is repeated, and for non-fixed channel scan, conversion is repeated one channel at a time up to the last channel assigned to the scan group, and then back to the first channel. To stop A/D conversion, you must follow to section 54.6.4. Force Stops the A/D Conversion Operation .

54.3.16.4 Constant Sampling Function

Constant sampling is a function in which the channel-dedicated sample-and-hold circuit continues sampling before and after A/D conversion starts. This makes it possible to simultaneously sample multiple channels and acquire A/D conversion data at the same time.

[Figure 54.23](#) shows the difference in A/D conversion operation when constant sampling is enabled (AD_{SHCSCR}.SHCSST_m = 1 (x = 0, 1)) and when it is disabled (AD_{SHCSCR}.SHCSST_m = 0 (x = 0, 1)).

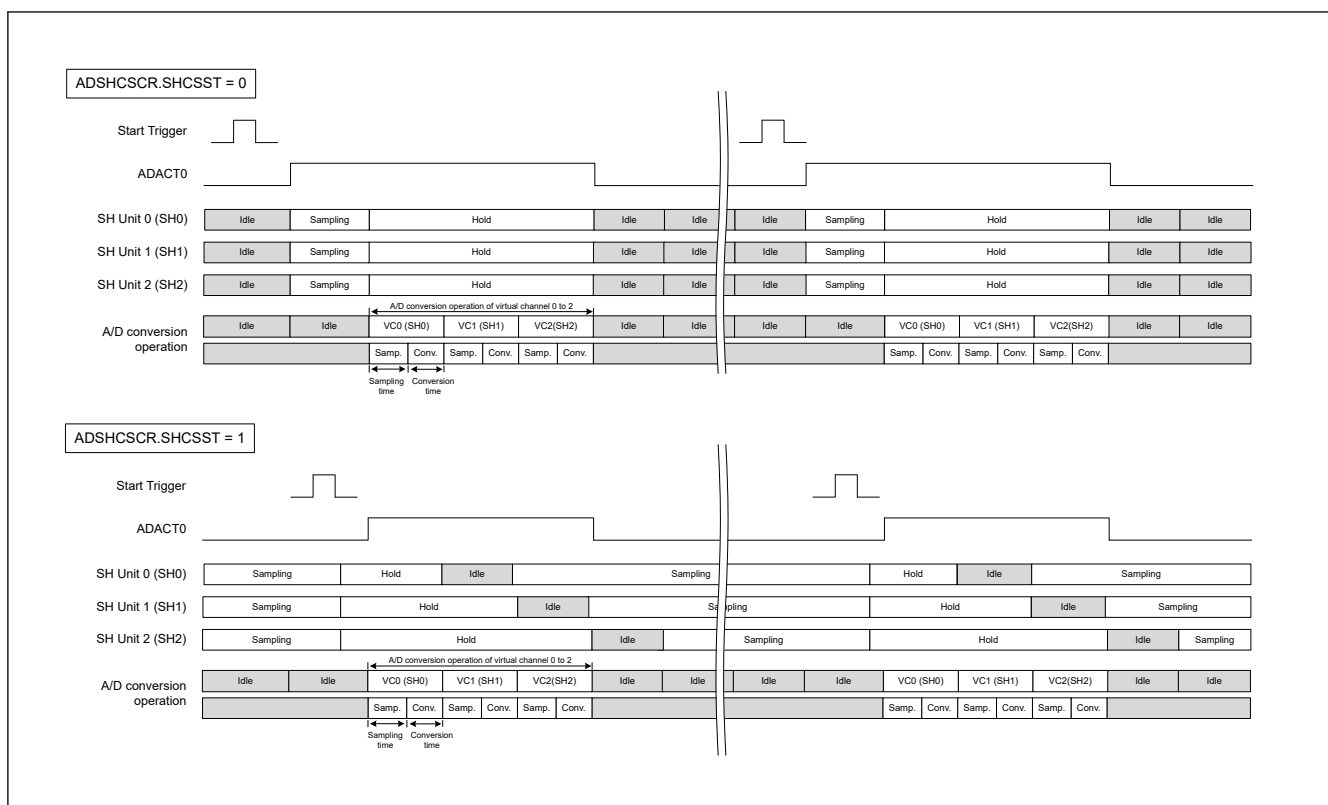


Figure 54.23 Difference between the function of channel-dedicated sample-and-hold circuit when using or not using constant sampling

54.3.16.5 Channel-dedicated Sample-and-hold Circuit Bypass Function

When channel-dedicated sample-and-hold (S&H) circuit is in differential input mode, the differential input bypass function allows disconnection detection with the disconnection detection assist function of A/D converter (single-ended input mode).

This function is intended to avoid the complex calibration procedure that occurs when the S&H switches from differential input mode to single-ended input mode, since the differential input bypass function is supported only in single-ended input mode.

When the ADSDHDBPCR.SHDBP_n ($n = 0$ to 8) bit is set to 1, the hold output of the S&H circuit is not selected for the AD conversion during the scan group. Instead, the input channel of the S&H circuit is selected for the AD conversion. The disconnection detection assist function can be executed in the scan group. For example, the following settings are possible in the order:

1. SH0 is enabled.
2. AN000 is assigned to scan group 0. The input mode is set to differential mode.
3. AN000 and AN001 are assigned to scan group 1. The input mode is set to single-ended mode and the disconnection detection assist function is enabled.
4. The ADSDHDBPCR.SHDBP0 bit is set to 0 and the ADSDHDBPCR.SHDBP1 bit is set to 1.

In the AD conversion of scan group 0, the sample-and-hold operation is performed with SH0, and the hold output is AD converted.

In the AD conversion of scan group 1, the disconnection detection assist function for the input channels of SH0, AN000 and AN001, is performed.

In this case, the SH0 setting is not changed and recalibration for SH0 is not required.

54.3.16.6 Restrictions on Channel-dedicated Sample-and-hold Circuit

When using channel-dedicated sample-and-hold circuit, observe the restrictions described in the sections that follow.

(1) Limitations on usage of channel-dedicated sample-and-hold circuit

Channel-dedicated sample-and-hold circuit is only available in the following:

- SAR mode – Single scan mode
- Hybrid mode – Single scan mode
- Hybrid mode – Continuous scan mode
- Hybrid mode – Background continuous scan mode
- Hybrid mode – Fixed channel continuous scan mode *1

In the operation mode other than above, channel-dedicated sample-and-hold circuit is prohibited.

In addition, the constant sampling function can only be used in SAR mode – Single scan mode.

The sampling time, hold mode switching time, and A/D sampling time of the channel-dedicated sample-and-hold circuit should be set as follows.

$$- 3.5 \times (\text{ADSHSTR0.SHHST}[2:0] + \text{ADSSTRx.SSTy}[9:0]) \geq \text{ADSHSTR0.SHSST}[7:0]$$

Note 1. Only fixed channels are supported when fixed channels are set to 3ch and all fixed channels enable channel-dedicated sample-and-hold circuit. Non-fixed channels cannot use channel-dedicated sample-and-hold circuit.

(2) Restrictions on self-calibration operation of channel-dedicated sample-and-hold circuit

Self-calibration operation is required when channel-dedicated sample-and-hold circuit is used. For more details on self-calibration procedures for channel-dedicated sample-and-hold circuit and the restrictions on it, see [section 54.3.8. Self-calibration](#).

(3) Restrictions on assigning virtual channels and scan groups

For analog channels using channel-dedicated sample-and-hold circuit, assign them to virtual channels so that A/D conversion is performed at the front of the scan group. Specifically, assign an analog channel that uses channel-dedicated sample-and-hold circuit from the virtual channel that has the smallest virtual channel number among the virtual channels assigned to that scan group. (Assign the analog channels of channel-dedicated sample-and-hold circuit to the front virtual channels in ascending order of the virtual channel number.)

If this restriction is violated, the operation of channel-dedicated sample-and-hold circuit and A/D converter is not guaranteed.

(4) Prohibition of multiple assignments of channels using the same channel-dedicated sample-and-hold circuit in the same scan group

It is prohibited to assign more than one virtual channel that uses the same channel-dedicated sample-and-hold circuit to a single scan group. Specific restrictions are the followings:

[Restrictions]

- The virtual channel whose signal source is selected as the even-numbered analog channel connected to channel-dedicated sample-and-hold circuit and the virtual channel whose signal source is selected as the odd-numbered analog channel must not be assigned to the same scan group without using the bypass function.
- It is prohibited to assign the even-numbered analog channels or odd-numbered analog channels connected to channel-dedicated sample-and-hold circuit to multiple virtual channels, and assign them to the same scan group.

Operation is not guaranteed if this restriction is violated.

To avoid this restriction, assign virtual channels that use the same channel-dedicated sample-and-hold circuit to separate scan groups. This restriction can be avoided by preventing A/D conversion from occurring more than once for the same channel-dedicated sample-and-hold circuit during one scanning operation. For the example workaround for this restriction, see [\(2\) Advanced operation in SAR mode - Single scan mode](#).

(5) Restriction on differential input bypass function

When a scan group is set as differential input bypass, all channel-dedicated sample-and-hold circuits including another scan group cannot be used for SH disconnection detection function and constant sampling function.

(6) Restrictions on group priority operation

Channel-dedicated sample-and-hold circuit is available only for the highest priority scan group in group priority operation.

(7) Restrictions on channel-dedicated sample-and-hold circuit (S&H) constant sampling function

- S&H constant sampling function is only available in SAR mode – Single scan mode. In the operation mode other than specified, S&H constant sampling function is prohibited.
- When using the S&H constant sampling function, the virtual channel settings of the channel that converts S&H, use the virtual channel number register corresponding to the S&H unit as shown in [Table 54.36](#).

Table 54.36 Virtual channel number associated with sample-and-hold circuit unit

A/D converter unit	Channel-dedicated sample-and-hold circuit unit	Assigned virtual channel number
ADC0	S&H unit 0	Virtual Channel 0
	S&H unit 1	Virtual Channel 1
	S&H unit 2	Virtual Channel 2
ADC1	S&H unit 4	Virtual Channel 4
	S&H unit 5	Virtual Channel 5
	S&H unit 6	Virtual Channel 6

- When using the S&H constant sampling function, set the S&H sampling time (ADSHSTRm.SHSST[7:0](m = 0, 1)) according to the following rules:
 - When synchronization between units is disabled — $\text{ADSHSTRx.SHSSTx} (x = 0 \text{ to } 3) = 8'h4$ (reset value)
 - When synchronization between units is enabled — The value for which the total time of S&H sampling time and S&H hold time ($\text{ADSHSTRm.SHSST}[7:0] + \text{ADSHSTRm.SHHST}[2:0] (m = 0, 1)$) is equal to the synchronization period.
- When S&H constant sampling function is enabled, it is prohibited to use any of the scan groups with S&H bypass input by setting the ADShDBPCR register.
- When unit synchronization is enabled and S&H is used with constant sampling, set the synchronous period to 6 ADCLK or longer.
- Set S&H constant sampling (ADSHCSCR.SHCSTm (m = 0, 1)) before starting A/D conversion after calibration.
- When S&H is used with constant sampling, the S&H sampling period is the period from the end of the S&H conversion to the start of the next S&H conversion. When you enter a conversion start trigger continuously, ensure that the trigger interval is sufficient to ensure the sampling period of S&H.

(8) Restrictions on channel-dedicated sample-and-hold circuit (S&H) disconnection detection assist function

- S&H disconnection detection assist function is only available in SAR mode – Single scan mode. In the operation mode other than specified, S&H disconnection detection assist function is prohibited.
- When S&H disconnection detection assist function is enabled, it is prohibited to use any of the scan groups with S&H bypass input by setting the ADShDBPCR register.

(9) Restrictions on channel-dedicated sample-and-hold circuit (S&H) self-diagnosis function

- S&H self-diagnosis function is only available in SAR mode – Single scan mode. In the operation mode other than specified, S&H self-diagnosis function is prohibited.
- It is prohibited to set up normal conversion and S&H self-diagnostic channels of the same S&H unit in one scan group.

54.3.17 Disconnection Detection Assist Function

Disconnection detection assist function is the function that fixes the charges of the sampling capacitance of the A/D converter to a specified condition prior to starting A/D conversion. This function can be used to detect the wiring disconnection connected to analog input.

Table 54.37 shows the operation mode in which disconnection detection assist function is available. And when using disconnection detection assist function, it is recommended to enable the high accuracy mode by setting the ADACMDR.ADHACMDn (n = 0,1) bits to 1.

Table 54.37 List of operation mode for which disconnection detection assist function is available

Operation mode – Scan mode	When channel-dedicated sample-and-hold circuit is not used	When channel-dedicated sample-and-hold circuit is used
SAR mode – Single scan mode	✓	✓
SAR mode – Continuous scan mode	✓	—
Oversampling mode – Single scan mode	—	—
Oversampling mode – Continuous scan mode	—	—
Oversampling mode – One-channel continuous scan mode	—	—
Hybrid mode – Single scan mode	✓	—
Hybrid mode – Continuous scan mode	✓	—
Hybrid mode – Background continuous scan mode	—	—
Hybrid mode – Fixed-channel continuous scan mode	—	—

Note: ✓: available, —: not available

Figure 54.24 shows an example operation of A/D conversion when disconnection detection assist function is used, and Figure 54.25 shows an example operation of A/D conversion when disconnection detection assist function and channel-dedicated sample-and-hold circuit are used. Figure 54.26 shows an example of the disconnection detection when precharge is selected, and Figure 54.27 shows an example of the disconnection detection when discharge is selected.

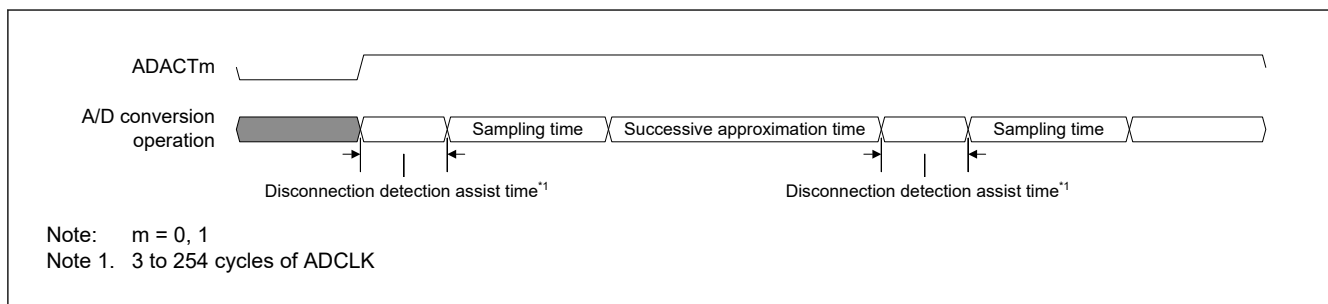


Figure 54.24 Operation of A/D conversion when disconnection detection assist function is used

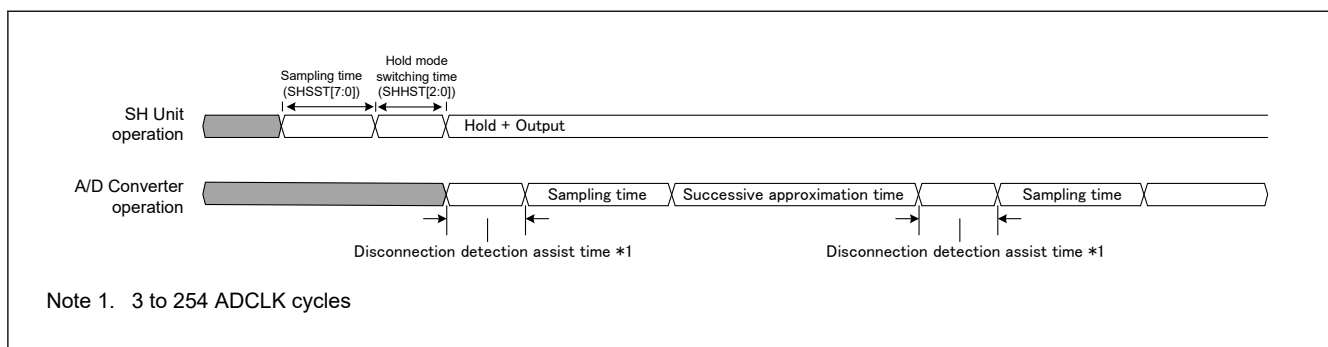


Figure 54.25 Operation of A/D conversion when the disconnection detection assist function and channel-dedicated sample-and-hold circuit are used

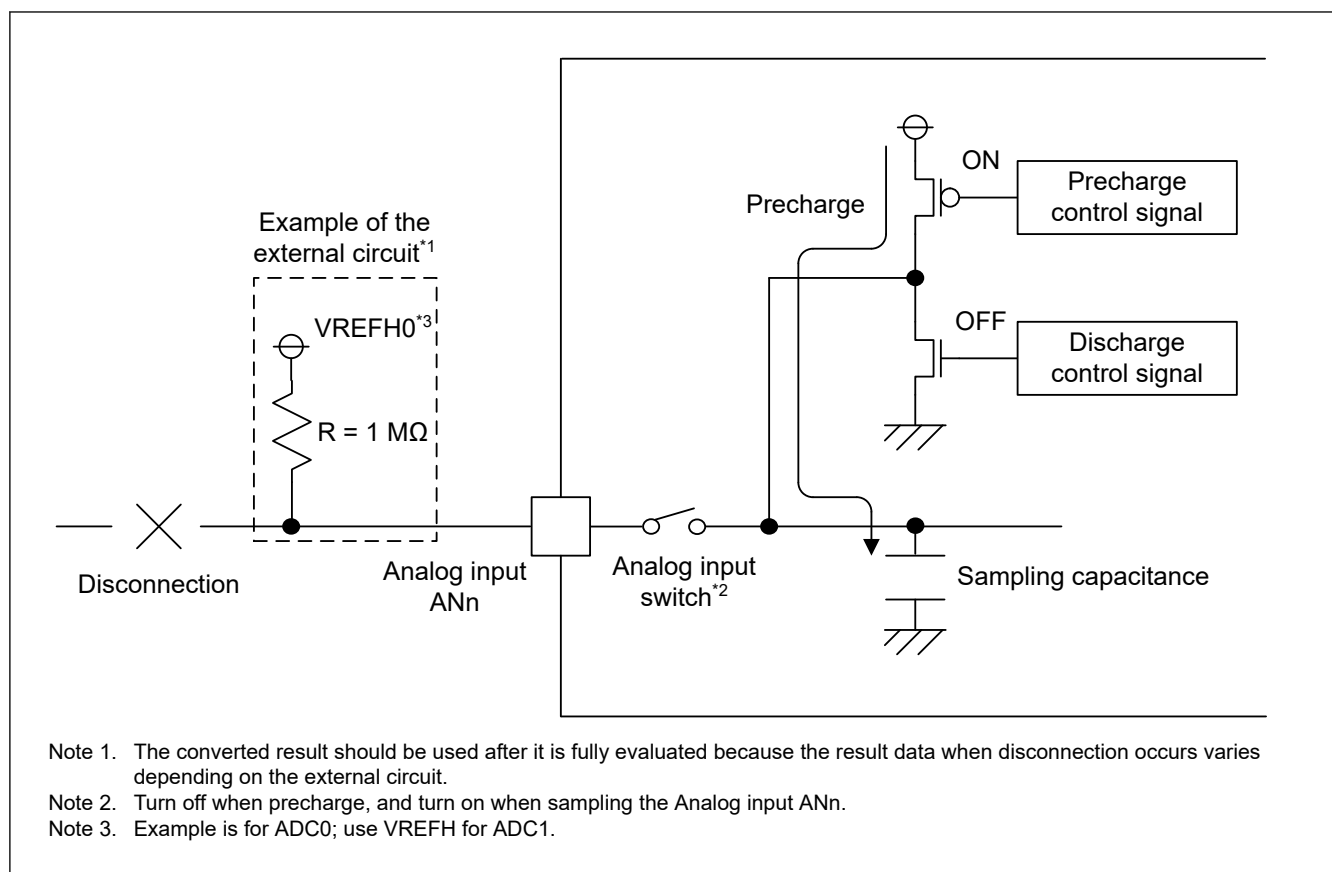


Figure 54.26 Example of disconnection detection when precharge is selected

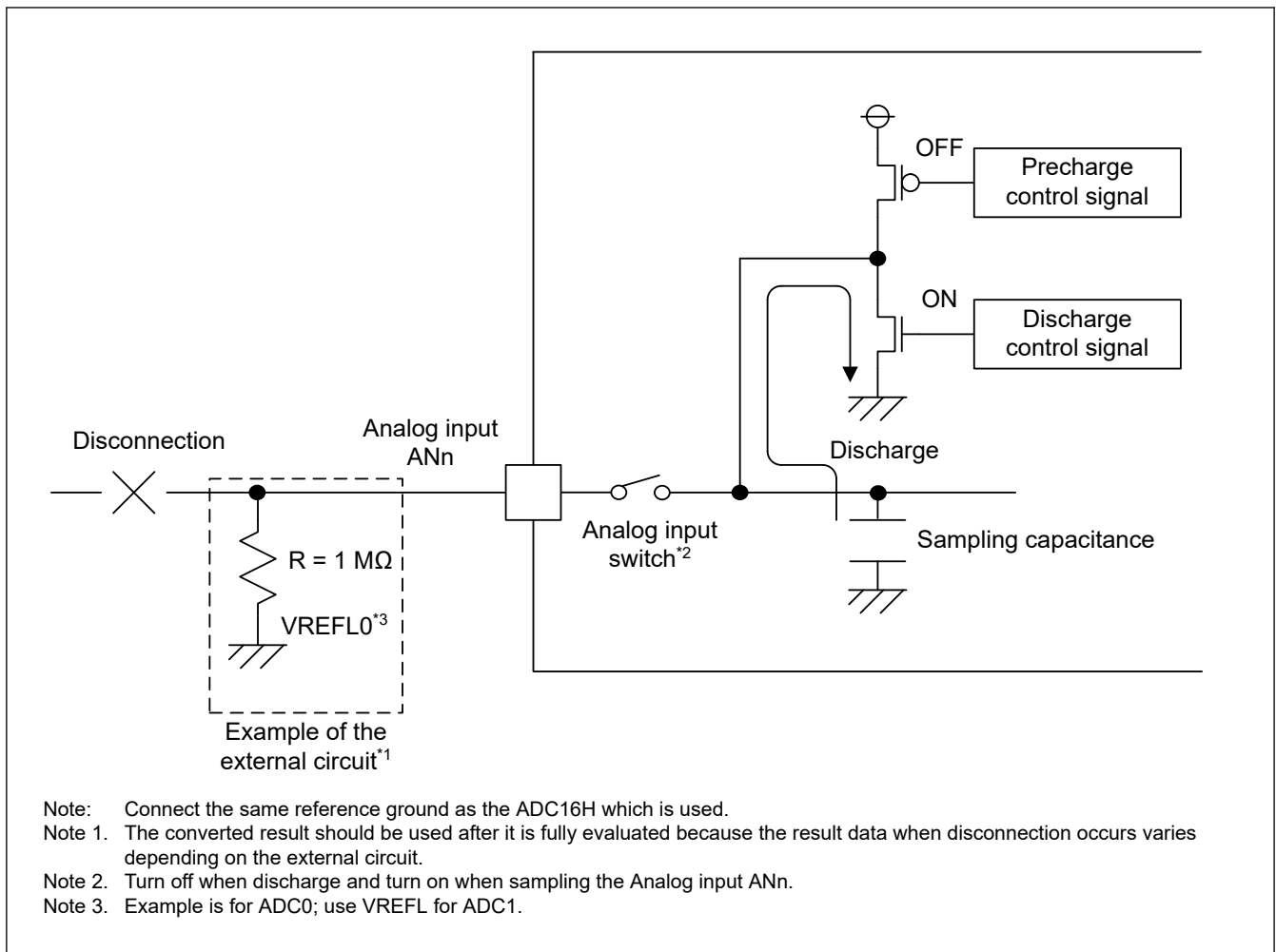


Figure 54.27 Example of disconnection detection when discharge is selected

54.3.17.1 Restrictions on Disconnection Detection Assist Function

When using disconnection detection assist function, observe the following restrictions:

- Disconnection detection assist function is only available in the following modes:
 - SAR mode – Single scan mode
 - SAR mode – Continuous scan mode
 - Hybrid mode – Single scan mode
 - Hybrid mode – Continuous scan mode

In the operation mode other than above, disconnection detection assist function is prohibited.

- Disconnection detection assist function with channel-dedicated sample-and-hold circuit is only available in the following modes:
 - SAR mode – Single scan mode

In the operation mode other than above, disconnection detection assist function with channel-dedicated sample-and-hold circuit is prohibited.

54.3.18 Group Priority Operation

Group priority operation function is the function that performs the scanning operation (A/D conversion) for a scan group with a higher priority based on the priority of scan group.

The priority of scan groups is group 0 > group 1 > ... > group n (n: the number of maximum scan group - 1).

Table 54.38 shows the conditions when group priority operation can be used and the corresponding register settings. Set the operation according to Table 54.38.

Table 54.38 Group priority operation usable conditions and corresponding register settings

Operating conditions		Group priority operation				
Operation mode of A/D converter		Operation	Setting for ADGSPCR register *1			
			PGSm	RSCNm	LGRRSm	GRPm
SAR mode	Single scan mode	✓	1	1	1	0
	Continuous scan mode	✓	1	1	1	1
Oversampling mode		—	0	0	0	0
Hybrid mode		—	0	0	0	0

Note: ✓: Available, —: Not Available (Prohibited)
m = 0, 1

Note 1. Settings other than the specified value are prohibited.

54.3.18.1 Group Priority Operation in SAR Mode – Single Scan Mode

In group priority operation in SAR mode – Single scan mode, the single scanning operation is performed according to the priority of scan group.

If an A/D conversion start trigger for the high priority group is input during a scanning operation of the low priority group, the scanning operation of the low priority group is interrupted, and the scanning operation of the high priority group is started. After the scanning operation of the high priority group is completed, the scanning operation of the low priority group is resumed.

When an A/D conversion start trigger of the low priority group during a scanning operation of the high priority group is input, the scanning operation of the low priority group is performed after the scanning operation of the high priority group is completed.

Example of the Group Priority Operation in SAR Mode – Single Scan Mode

Table 54.39 and Figure 54.28 show the example of group priority operation in SAR mode – Single scan mode when analog channel 0 (AN000) is assigned to scan group 0 (high priority) and analog channel 1 to 3 (AN001 to AN003) are assigned to scan group 1 (low priority).

Table 54.39 Example of the group priority operation in SAR mode – Single scan mode (2 groups)

Step	Operation
1	The scanning operation of scan group 1 starts when a trigger for scan group 1 is input.
2	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
3	If a trigger for scan group 0 is input during A/D conversion for scan group 1, the scanning operation of scan group 1 is suspended and the scanning operation of scan group 0 is started. *1
4	When A/D conversion of each channel in scan group 0 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
5	If a scan end interrupt for scan group 0 is enabled, a scan end interrupt occurs.
6	The scanning operation of scan group 1 is resumed after scanning of scan group 0 is completed. The scanning operation is resumed from the A/D conversion incomplete channel. *1
7	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
8	If a scan end interrupt for scan group 1 is enabled, a scan end interrupt occurs.
9	When all scanning operations are completed, ADSR.ADACTm (m = 0, 1) bits are cleared and the A/D converter enters the idle status.

Note 1. The timing of interruption and restart of scanning operation depends on the operating conditions. For details, see section 54.3.18.3. [Restrictions on Group Priority Operation.](#)

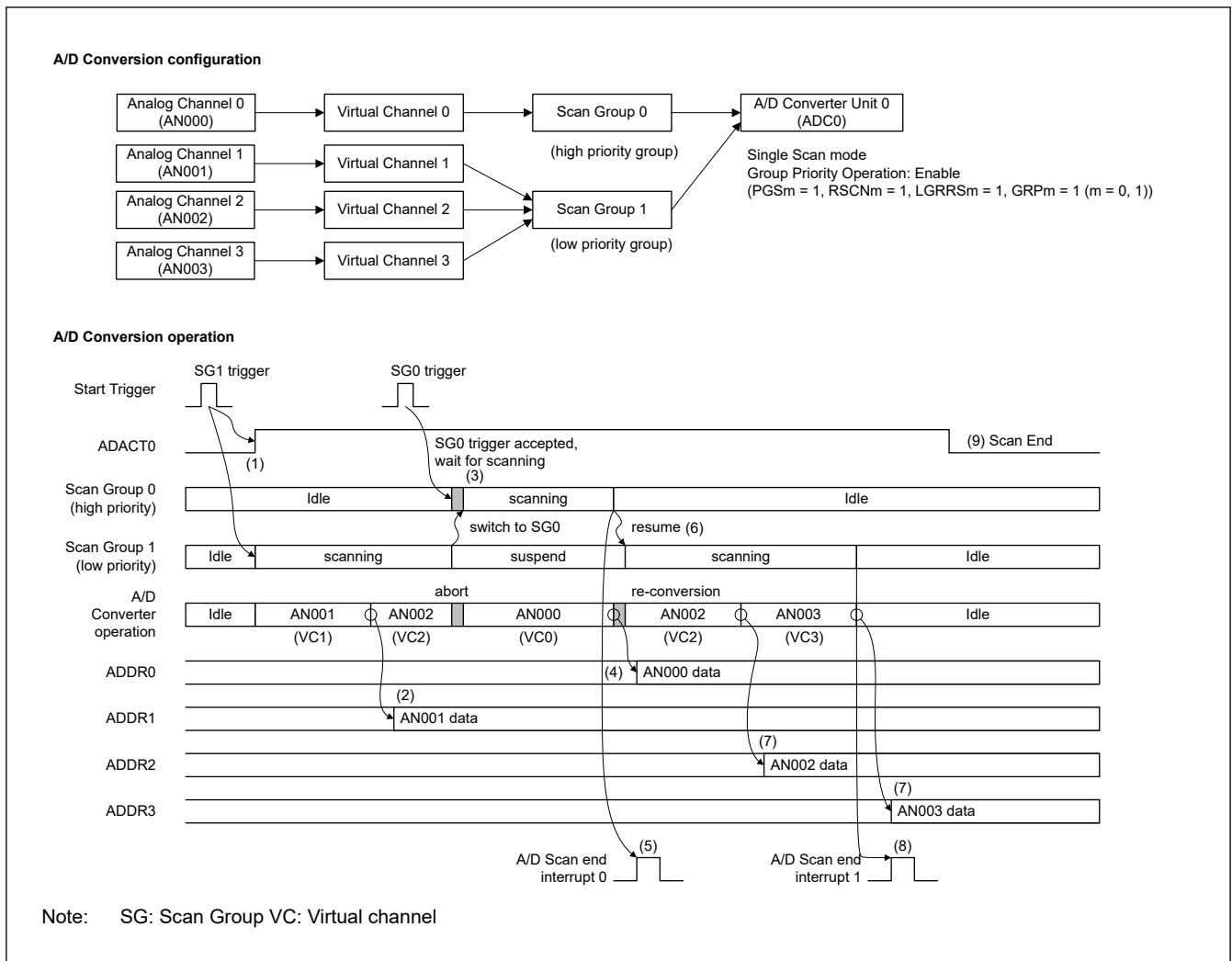


Figure 54.28 Example of group priority operation in SAR mode – Single scan mode (2 groups)

54.3.18.2 Group Priority Operation in SAR Mode – Continuous Scan Mode

In group priority operation in SAR mode – Continuous scan mode, the continuous scanning operation is performed for the scan group (continuous-scan group) in which scanning operation was first started. If an A/D conversion start trigger of another scan group (interrupt-scan group) is input during the continuous scanning operation of the continuous-scan group, the following operations are performed according to the priority of the interrupt-scan group.

1. Priority: When Interrupt-scan group (high) > Continuous-scan group (low)
 - The scanning operation of the continuous-scan group is interrupted, and the single scanning operation of the interrupt-scan group is started.
 - The continuous scanning operation of the continuous-scan group is restarted after the single scanning operation of the interrupt-scan group is completed.
2. Priority: When Continuous-scan group (high) > Interrupt-scan group (low)
 - This operation is prohibited. Operation is not guaranteed.

In case 1. above, if an A/D conversion start trigger for another scan group (interrupt-scan group B) is input during the single scanning operation of the interrupt-scan group (interrupt-scan group A), the following operation is performed according to the priority of interrupt-scan group B.

3. Priority: When Interrupt-scan group B (high) > Interrupt-scan group A (middle) > Continuous-scan group (low)
 - The single scanning operation of interrupt-scan group A is suspended, and the single scanning operation of interrupt-scan group B is started.

- After the single scanning operation of interrupt-scan group B is completed, the single scanning operation of interrupt-scan group A is resumed.
 - After the scanning operation of interrupt-scan group A is completed, the continuous scanning operation of the continuous-scan group is resumed.
4. Priority: When interrupt-scan group A (high) > interrupt-scan group B (medium) > continuous-scan group (low)
- After the single scanning operation of interrupt-scan group A is completed, the single scanning operation of interrupt-scan group B is started.
 - After the single scanning operation of interrupt-scan group B is completed, the continuous scanning operation of the continuous-scan group is resumed.
5. Priority: When interrupt-scan group A (high) > continuous-scan group (medium) > interrupt-scan group B (low)
- This operation is prohibited. Operation is not guaranteed.

Example operations are shown in the followings:

(1) Example of two-group priority operation in SAR mode – Continuous scan mode

Table 54.40 and Figure 54.29 show the example of group priority operation in SAR mode – Continuous scan mode when analog channel 0 (AN000) is assigned to scan group 0 (high-priority group) and analog channel 1 to 3 (AN001 to AN003) is assigned to scan group 1 (low-priority group).

Table 54.40 Example of the group priority operation in SAR mode – Continuous scan mode (2 groups)

Step	Operation
1	The continuous scanning operation of scan group 1 (low priority group) is started by the trigger input of scan group 1.
2	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
3	If a trigger for scan group 0 is input during A/D conversion for scan group 1, the scanning operation of scan group 1 is suspended and scanning operation of scan group 0 is started. *1
4	When A/D conversion of each channel in scan group 0 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
5	If a scan end interrupt for scan group 0 is enabled, a scan end interrupt occurs.
6	The scanning operation of scan group 1 is resumed after scanning of scan group 0 is completed. The scanning operation is resumed from the A/D conversion incomplete channel. *1
7	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
8	If a scan end interrupt for scan group 1 is enabled, a scan end interrupt occurs.
9	Scan group 1 continues the continuous scanning operation. *2

Note 1. The timing of interruption and restart of scanning operation depends on the operating conditions. For details, see [section 54.3.18.3. Restrictions on Group Priority Operation](#).

Note 2. To stop the continuous scanning operation, follow the procedures described in [section 54.6.4. Force Stops the A/D Conversion Operation](#).

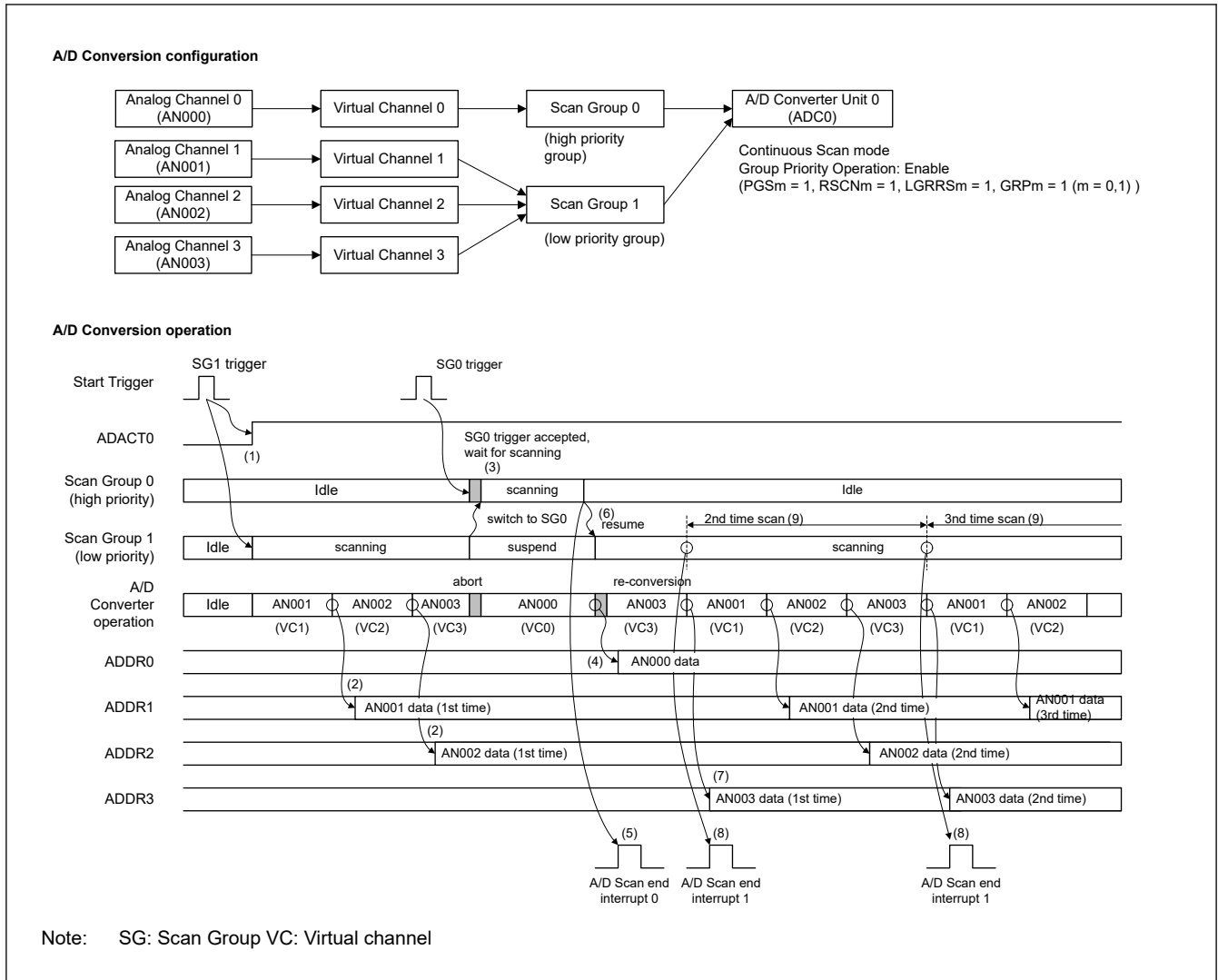


Figure 54.29 Example of group priority operation in SAR mode – Continuous scan mode (2 groups)

(2) Example of three-groups priority operation in SAR mode – Continuous scan mode

Table 54.41 and Figure 54.30 show the example of group priority operation when analog channel 0 (AN000) is assigned to scan group 0 (high priority), analog channels 1 and 2 (AN001 and AN002) are assigned to scan group 1 (middle priority), and analog channel 3 to 5 (AN003 to AN005) are assigned to scan group 2 (low priority).

Table 54.41 Example of the group priority operation in SAR mode – Continuous scan mode (3 groups) (1 of 2)

Step	Operation
1	The continuous scanning operation of scan group 2 is started by the trigger input of scan group 2.
2	When A/D conversion of each channel in scan group 2 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
3	If a trigger for scan group 1 is input during A/D conversion on scan group 2, the scanning operation of scan group 2 is suspended and the scanning operation of scan group 1 is started. ^{*1}
4	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
5	If a trigger for scan group 0 is input during A/D conversion for scan group 1, the scanning operation of scan group 1 is suspended and the scanning operation of scan group 0 is started. ^{*1}
6	When A/D conversion of each channel in scan group 0 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
7	If a scan end interrupt for scan group 0 is enabled, a scan end interrupt occurs.

Table 54.41 Example of the group priority operation in SAR mode – Continuous scan mode (3 groups) (2 of 2)

Step	Operation
8	The scanning operation of scan group 1 is resumed after scanning of scan group 0 is completed. The scanning operation is restarted from the A/D conversion incomplete channel. ^{*1}
9	When A/D conversion of each channel in scan group 1 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
10	If a scan end interrupt for scan group 1 is enabled, a scan end interrupt occurs.
11	Scan group 2 is resumed after scanning of scan group 1 is completed. The scanning operation is resumed from the A/D conversion incomplete channel. ^{*1}
12	When A/D conversion of each channel in scan group 2 completes, the A/D conversion result is stored in the corresponding A/D data register n (ADDRn).
13	If a scan end interrupt for scan group 2 is enabled, a scan end interrupt occurs.
14	Scan group 1 continues the continuous scanning operation. ^{*2}

Note 1. The timing of interruption and restart of scanning operation depends on the operating conditions. For details, see [section 54.3.18.3. Restrictions on Group Priority Operation](#).

Note 2. To stop the continuous scanning operation, follow the procedures described in [section 54.6.4. Force Stops the A/D Conversion Operation](#).

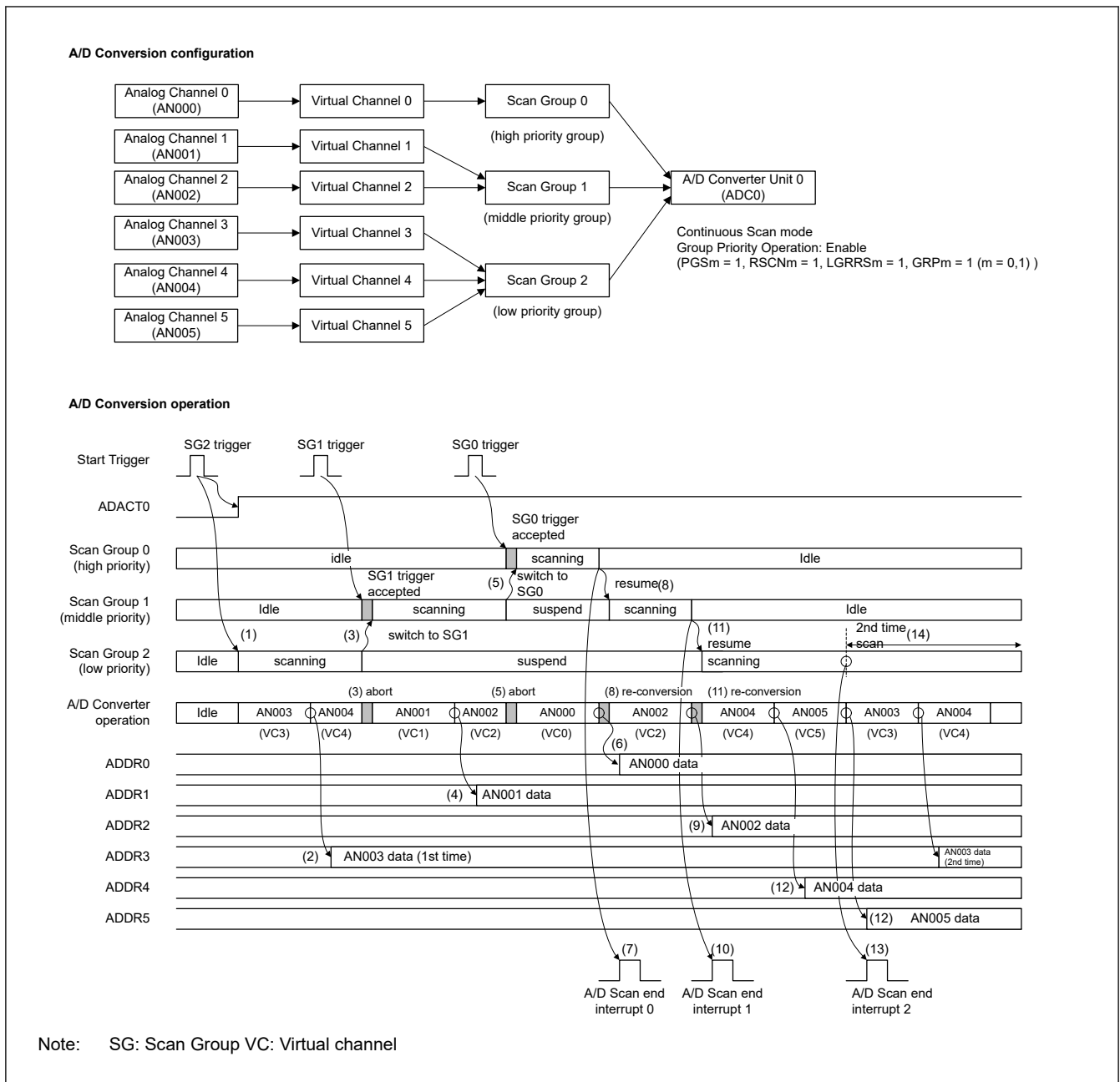


Figure 54.30 Example of group priority operation in SAR mode – Continuous scan mode (3 groups)

54.3.18.3 Restrictions on Group Priority Operation

(1) Channel-dedicated sample-and-hold circuit restrictions

Channel-dedicated sample-and-hold circuit is available only for the highest priority scan group in group priority operation.

(2) Restrictions on the operation mode and operation setting of the A/D converter

Group priority operation is only available in SAR mode; Group priority operation is prohibited in Oversampling mode or Hybrid mode.

If these restrictions are violated, operation is not guaranteed.

(3) Restrictions on group priority operation in SAR mode – Continuous scan mode

In group priority operation in SAR mode – Continuous scan mode, do not input A/D conversion start triggers for scan groups that have a lower priority than the scan group that started continuous scanning operation. If this restriction is violated, the operation is not guaranteed.

(4) Restrictions on trigger input for the same scan group

Input the A/D conversion start triggers for the same scan group in group priority operation for the following periods.

[When synchronous operation is disabled (ADSYCR.ADSYDISm = 1)]

- 2 ADCLK + 2 PCLK cycles after the scan end flag is set ^{*1}

[When synchronous operation is enabled (ADSYCR.ADSYDISm = 0)]

- Synchronous operation period cycle × 2 after the scan end flag is set ^{*1}

Note: m = 0, 1

Note 1. ADSCANENDSR.SCENDFn = 1 (n = 0 to 8)

If A/D conversion start triggers are input without waiting for the above period, the A/D conversion start triggers may be lost. To accept A/D conversion start triggers reliably, input the trigger after the above period has elapsed.

(5) Restrictions on suspend and resume for scanning operation of low-priority groups

When an A/D conversion start trigger for the high priority group is input during scanning of the low priority group in group priority operation, the scanning operation of the low priority group is interrupted at the timing shown in the following:

[Scan interruption timing of low priority group]

- When synchronous operation is enabled
 - After the A/D conversion start trigger of the high-priority group is accepted, the scanning operation of the low-priority group is suspended after waiting until the next synchronous operation period timing.
- When synchronous operation is disabled
 - The scanning operation of the low-priority group is immediately suspended after the A/D conversion start trigger of the high-priority group is accepted.

If A/D conversion is being performed on channels for which A/D-converted value addition/averaging function is to be used at the timing of scan interruption of a low-priority group, the midway result of A/D-converted value addition/averaging value at that time is discarded. When the scanning operation of the low-priority group is resumed, A/D conversion starts over from the beginning of the number of times specified for A/D-converted value addition/averaging function.

If the scanning operation is suspended by group priority operation, ADGRSR.ACTGRn (n = 0 to 8) bit of the suspended scan group remains set to 1.

54.3.19 Synchronous Operation

Synchronous operation is the function that controls the operations of multiple A/D converters so that they are synchronized.

54.3.19.1 Synchronous Operation Examples

(1) Basic synchronous operation

The basic operation of synchronous operation is shown in the following. The configuration example is shown in [Table 54.42](#), and the detail of operation is shown in [Table 54.43](#) and [Figure 54.31](#).

Table 54.42 Configuration example of basic synchronous operation (1 of 2)

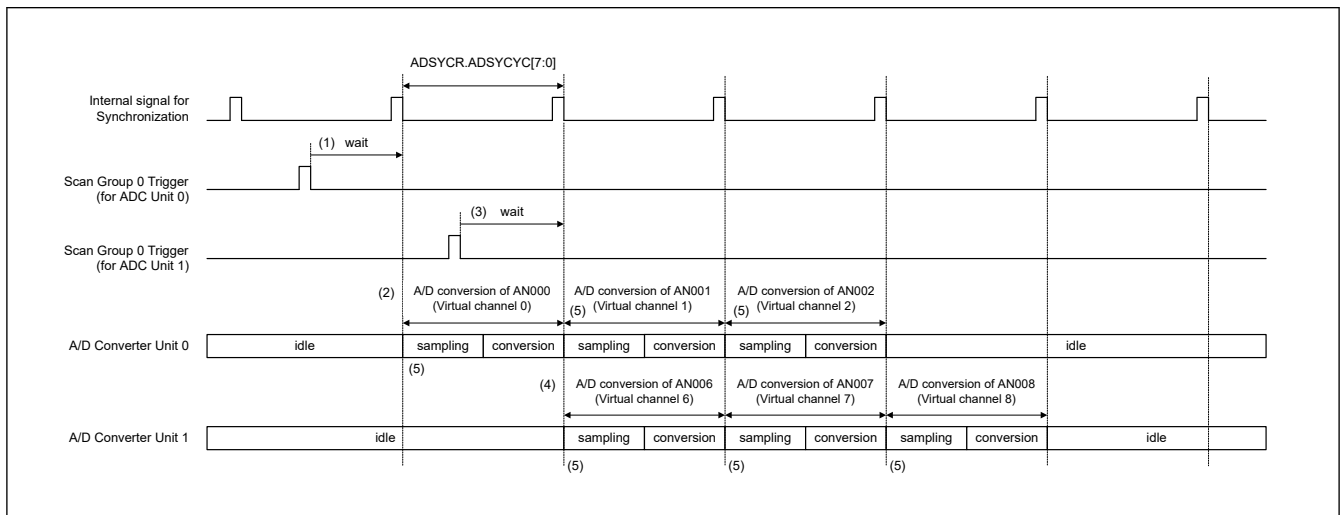
Item	Configuration
A/D converter	• ADC0: Set to SAR mode – Single scan mode • ADC1: Set to SAR mode – Single scan mode
Virtual channel	• Virtual channel 0 to 2: Assign AN000 to AN002 and set to Single-ended input mode. • Virtual channel 6 to 8: Assign AN006 to AN008 and set to Single-ended input mode.

Table 54.42 Configuration example of basic synchronous operation (2 of 2)

Item	Configuration
Scan group	[Scan Group 0] - Assign AN000 to AN002 (Virtual channel 0 to 2) - Set to convert with ADC0 [Scan Group 1] - Assign AN006 to AN008 (Virtual channel 6 to 8) - Set to convert with ADC1

Table 54.43 Example of basic synchronous operation

Step	Detail behavior of example operation
1	When a trigger of scan group 0 is input, a wait is inserted until the A/D conversion synchronization timing.
2	At the timing of synchronous operation period, the trigger input of scan group 0 is accepted, and the scanning operation of scan group 0 (for ADC0) starts.
3	When a trigger of scan group 1 is input, a wait is inserted until the A/D conversion synchronization timing.
4	At the timing of synchronous operation period, the trigger input of scan group 1 is accepted, and the scanning operation of scan group 1 (for ADC1) starts.
5	During synchronous operation, the A/D converter starts the sampling operation in accordance with synchronous operation period.

**Figure 54.31 Example of synchronous operation in SAR mode – Single scan mode****(2) Synchronous operation using with channel-dedicated sample-and-hold circuit**

An example of synchronous operation using with channel-dedicated sample-and-hold circuit is shown in the following. The configuration example is shown in [Table 54.44](#), and the detail of operation is shown in [Table 54.45](#) and [Figure 54.32](#).

Table 54.44 Configuration example of basic synchronous operation (1 of 2)

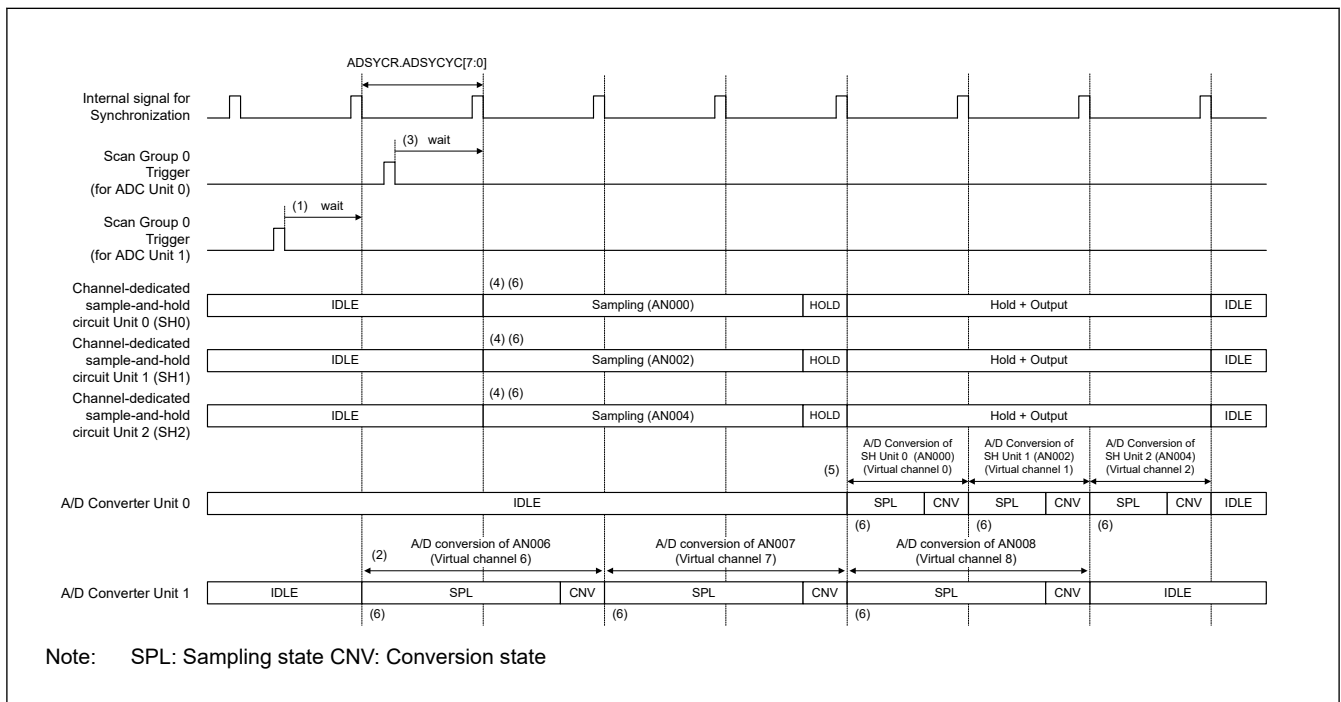
Item	Configuration
A/D converter	- ADC0: Set to SAR mode – Single scan mode - ADC1: Set to SAR mode – Single scan mode
Channel-dedicated sample-and-hold circuit	- Set SH0 to SH2 to enabled and Single-ended input mode. - Set SH4 to SH6 to enabled and Single-ended input mode.
Virtual channel	- Virtual channel 0 to 2: Assign AN000, AN002 and AN004, and set to Single-ended input mode. - Virtual channel 6 to 8: Assign AN006, AN008 and AN010, and set to Single-ended input mode.

Table 54.44 Configuration example of basic synchronous operation (2 of 2)

Item	Configuration
Scan group	[Scan Group 0] - Assign AN000, AN002 and AN004 (Virtual channel 0 to 2) - Set to convert with ADC0 [Scan Group 1] - Assign AN006, AN008 and AN010 (Virtual channel 6 to 8) - Set to convert with ADC1

Table 54.45 Example of synchronous operation with channel-dedicated sample-and-hold circuit

Step	Detail behavior of example operation
1	When a trigger of scan group 1 is input, a wait is inserted until the A/D conversion synchronization timing.
2	At the timing of synchronous operation period, the trigger input of scan group 1 is accepted, and the scanning operation of scan group 1 (for ADC1) starts.
3	When a trigger of scan group 0 is input, a wait is inserted until the A/D conversion synchronization timing.
4	At the timing of synchronous operation period, the trigger input of scan group 0 is accepted, and the scanning operation of scan group 0 (for ADC0) starts. Channel-dedicated sample-and-hold circuit (SH0 to SH2) samples and holds the analog-channels of AN000/AN002/AN004 at the beginning of the scanning operation in scan group 0.
5	After the holding by channel-dedicated sample-and-hold circuit, A/D converter starts the samplings and the conversions.
6	During synchronous operation, channel-dedicated sample-and-hold circuit and A/D converter starts the sampling operation in accordance with synchronous operation period.

**Figure 54.32 Example of synchronous operation using with channel-dedicated sample-and-hold circuit**

54.3.19.2 Restrictions on Synchronous Operation

When synchronous operation is used, observe the following restrictions. If the following restrictions are violated, operations are not guaranteed.

1. Basic restrictions

- Set the synchronous operation period to a value larger than successive approximation time of ADC_m ($m = 0, 1$).
 - $\text{ADSYCR.ADSYCYC}[7:0] \geq \text{ADCNVSTR.CSTM}[5:0] + 1$ ($m = 0, 1$)

- Set so that the sum of the sampling times of the analog channels and successive approximation time of ADC_m (m = 0, 1) is an integral multiple of the synchronous operation period.
 - $\text{ADSSTRx.SSTy}[9:0] + \text{ADCNVSTR.CSTm}[5:0] = \text{ADSYCR.ADSYCYC}[7:0] \times i$
(x = 0 to 7, y = 0 to 15, m = 0, 1, i: Any integer greater than or equal to 1 (i = 1, 2, 3...))
- Set the synchronous operation period to a value larger than successive approximation time at self-calibration.^{*1}
 - $\text{ADSYCR.ADSYCYC}[7:0] \geq \text{ADCALSTCR.CALADCST}[5:0] + 1$
- Set the sampling time and successive approximation time at self-calibration to be integral multiples of the synchronous operation period.^{*1}
 - $\text{ADCALSTCR.CALADSST}[9:0] + \text{ADCALSTCR.CALADCST}[5:0] = \text{ADSYCR.ADSYCYC}[7:0] \times i$
(i: Any integer greater than or equal to 1 (i = 1, 2, 3...))

2. Restrictions when using channel-dedicated sample-and-hold circuit

- Set the synchronous operation period to a value greater than the hold mode switching time of channel-dedicated sample-and-hold circuit.
 - $\text{ADSYCR.ADSYCYC}[7:0] \geq \text{ADSHSTRm.SHHST}[2:0] + 1$
(m = 0, 1)
- Set the sum of channel-dedicated sample-and-hold circuit sampling time and the hold mode switching time to be an integral multiple of the synchronous operation period.
 - $\text{ADSHSTRm.SHSST}[7:0] + \text{ADSHSTRm.SHHST}[2:0] = \text{ADSYCR.ADSYCYC}[7:0] \times i$
(m = 0, 1, i: Any integer greater than or equal to 1 (i = 1, 2, 3...))
- Set the hold mode switching time of channel-dedicated sample-and-hold circuit so that it is the same as successive approximation time of ADC_m (m = 0, 1).
 - $\text{ADSHSTRm.SHHST}[2:0] = \text{ADCNVSTR.CSTm}[5:0]$
(m = 0, 1)
- Set the synchronous operation period to a value greater than the hold mode switching time in channel-dedicated sample-and-hold circuit's self-calibration operation.^{*1}
 - $\text{ADSYCR.ADSYCYC}[7:0] \geq \text{ADCALSHCR.CALSHHST}[2:0] + 1$
- Set so that the sum of channel-dedicated sample-and-hold circuit sampling time and the hold mode switching time in self-calibration operation is an integral multiple of the synchronous operation period.^{*1}
 - $\text{ADCALSHCR.CALSHSST}[7:0] + \text{ADCALSHCR.CALSHHST}[2:0] = \text{ADSYCR.ADSYCYC}[7:0] \times i$
(i: Any integer greater than or equal to 1 (i = 1, 2, 3...))

3. Restrictions when using disconnection detection assist function

- Set Disconnection Detection Assist period so that it is the same as the synchronous operation period.
 - $\text{ADSGDCRn.ADNDIS}[7:0] = \text{ADSYCR.ADSYCYC}[7:0]$
(n = 0 to 8)

4. Restriction when using high accuracy mode

- Set so that the sum of the sampling times of the analog channels and successive approximation time of ADC_m (m = 0, 1) is an even or equal multiple of the synchronous operation period.
 - $\text{ADSSTRx.SSTy}[9:0] + \text{ADCNVSTR.CSTm}[5:0] = \text{ADSYCR.ADSYCYC}[7:0] \times i$
(x = 0 to 7, y = 0 to 15, m = 0, 1, i: Any even number or equal to 1 (i = 1, 2, 4, 6...))

Note 1. In relation to this restriction, the restrictions on the settings of self-calibration should be satisfied at the same time. For restrictions on the setting of self-calibration, see [section 54.3.8.3. Restrictions on Self-calibration](#).

54.4 A/D Conversion Monitor Signals

This section describes the internal operation of A/D conversion.

The A/D conversion monitor signals are output from pins that are configurable by the function of I/O ports.

The operating timing of A/D conversions can be checked by observing these signals with an oscilloscope.

The following A/D conversion monitor signals are available.

ADSYNC

ADSYNC is a reference signal of the synchronous operation of A/D conversion.

This signal indicates the synchronous timing of A/D conversion. When the synchronous operation function is enabled, the A/D converter operation is executed in synchronization with this signal. When the synchronous operation function is disabled, this signal is ignored. See [section 54.3.19. Synchronous Operation](#) for more details about synchronous operation of A/D conversion.

ADnFLAG1 (n = 0,1)

ADnFLAG1 indicates the period of the conversion of A/D converter unit n. While this signal is asserted, the A/D converter is performing the conversion operation.

ADSTn (n = 0,1)

ADSTn indicates the period of the scanning operation of A/D converter unit n. This signal is asserted during the execution of the A/D scanning operation.

[Figure 54.33](#) shows the operation timing of each signal. See [section 21, I/O Ports](#) for information on how to select output signals.

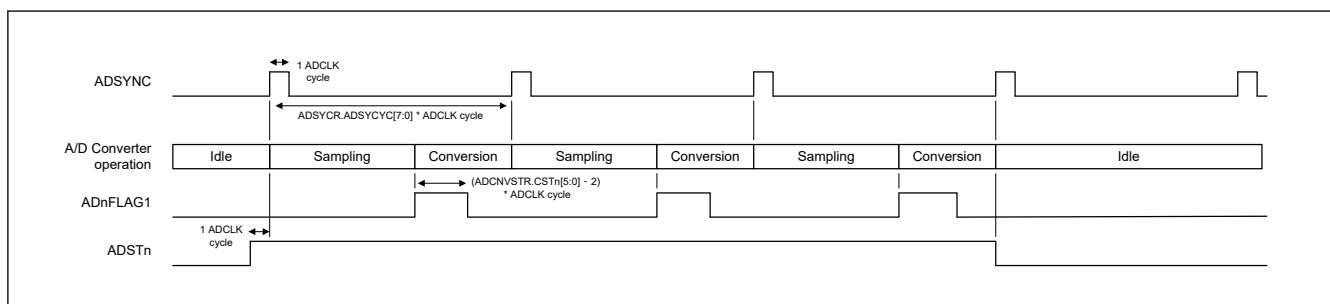


Figure 54.33 A/D conversion monitor signals

54.5 A/D Conversion Data

This section describes the internal processing for A/D conversion data. For details about the operation of A/D conversion, see [section 54.3. Operation](#).

54.5.1 Internal Data Processing Flow

[Figure 54.34](#) shows the internal processing-flow of A/D conversion data.

The A/D conversion data output from the A/D converter is temporarily stored in the data buffer. And then, various data processing is performed based on the register setting. If the function that corresponds to each data processing is not used, its data processing is skipped.

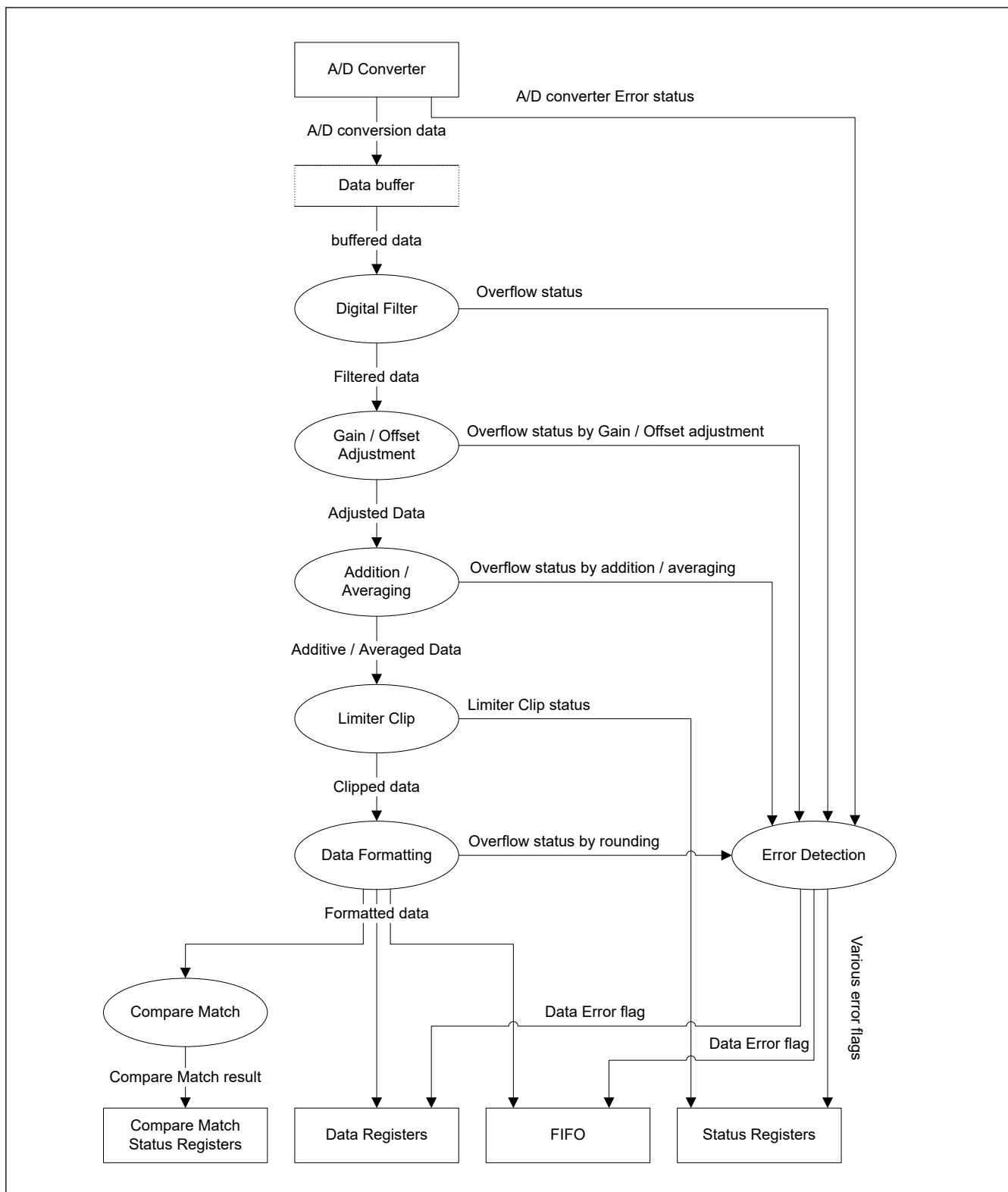


Figure 54.34 Internal data flow diagram of A/D conversion data

54.5.2 Digital Filter Function

ADC16H has 4 digital filters per 1 unit of A/D converter. High-precision A/D conversion results can be obtained by using a digital filter. The digital filter function must be used in Oversampling mode or Hybrid mode. The digital filter function in SAR mode is prohibited.

54.5.2.1 Configuration and Characteristics

The digital filter consists of a FIR type filter with 22 taps. The digital filter is set by ADDOPCRAn.DFSEL[2:0] ($n = 0$ to 32) and ADDFSRm ($m = 0, 1$) registers. The configuration of the digital filter is shown in Figure 54.35.

The type of the digital filter (FIR filter) can be selected from the following. For the characteristics of the digital filters, see section 70, Electrical Characteristics.

[List of digital filter characteristics]

- Sinc filter

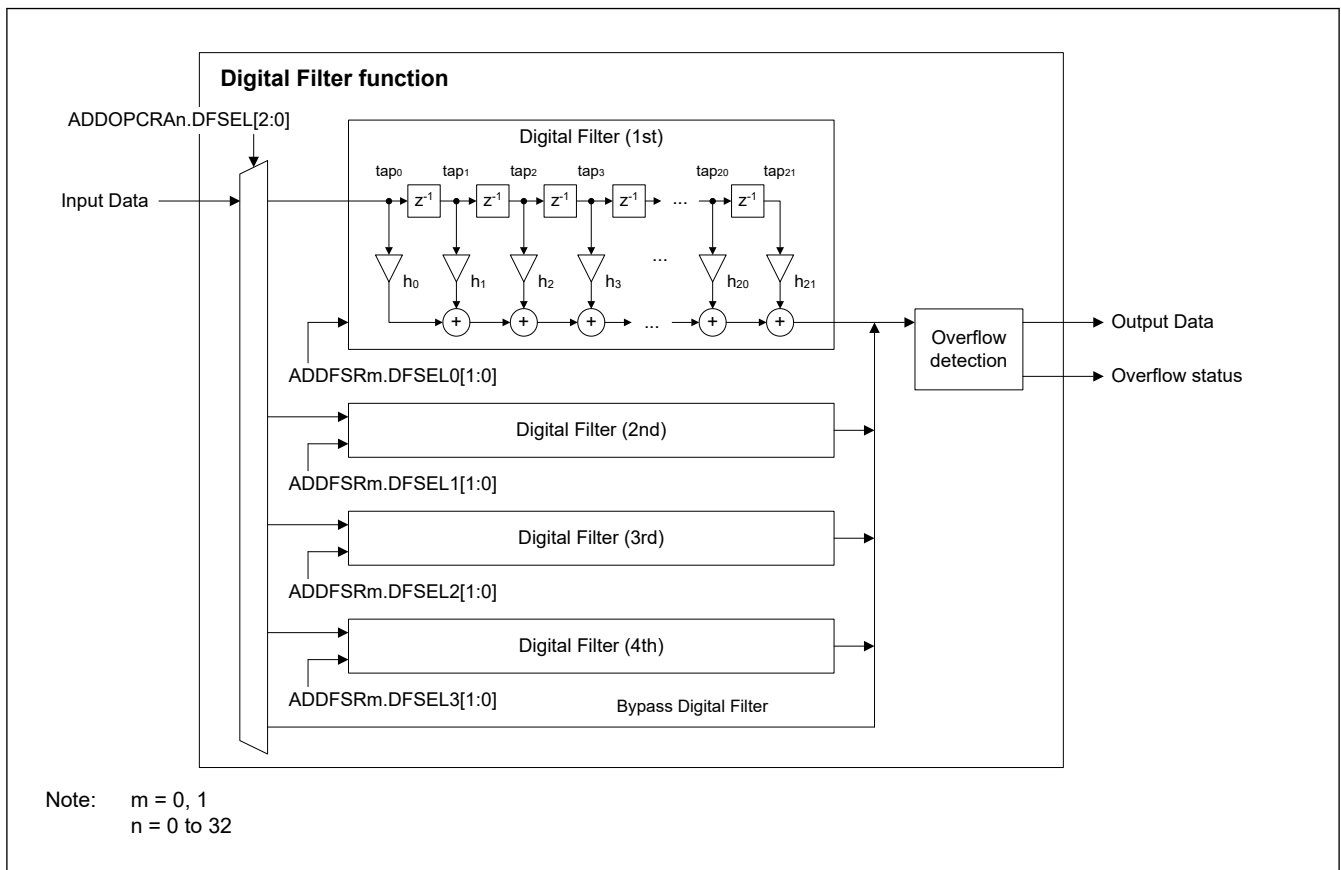


Figure 54.35 Block diagram of digital filter

54.5.2.2 Operation of Digital Filter Function

When the digital filter function is enabled, A/D conversion data is sequentially input to the digital filter. When all the taps in the digital filter are filled with data, the calculation result is output and sent to the next data processing.

The operation of the digital filters in Oversampling mode and Hybrid mode is shown in the followings.

(1) Digital filter operation in Oversampling mode

In Oversampling mode, oversampling (A/D conversion) is performed continuously for one analog channel. Each time oversampling is performed, A/D conversion data is sequentially input to the digital filter. When all the taps in the digital filter are filled with data, the calculation result is output and sent to the next data processing.

In Oversampling mode, the data of the taps in the digital filter is discarded when the A/D conversion data is output to the next data processing. However, when A/D-converted value addition/averaging function is used, the data of taps in the digital filter is kept until the data required to calculate the addition value or average value of the A/D converted value is collected. When the A/D-converted value addition/averaging value is calculated, the tap data in the digital filter is discarded.

(2) Digital filter operation in Hybrid mode

Hybrid mode can process oversampled data (A/D converted data) of up to four analog channels in parallel by using several digital filter circuits at the same time. Each time oversampling is performed, A/D conversion data is sequentially input to the

digital filter. When all the taps in the digital filter are filled with data, the calculation result is output and sent to the next data processing.

In Hybrid mode – Single scan mode, taps in the digital filter are discarded at the end of scanning operation.

In Hybrid mode – Continuous scan mode, the data of the taps in the digital filter continues to be updated as long as the continuous scanning operation is continued. Therefore, after the data is filled in all taps in the digital filter, a new calculation result is output for each oversampling. When scanning operation is aborted due to a forced stop of A/D conversion, the data of taps in the filter is discarded.

The operation of the digital filter in Hybrid mode – Background continuous scan mode is the same as in Hybrid mode – Continuous scan mode. During the background continuous scanning operation, the digital filters and other data processing continue to take place in the background. When scanning operation is aborted due to a forced stop of A/D conversion, the data of taps in the filter is discarded.

54.5.3 Calibration and Adjustment

In this process, the following processing is performed for the results output from the A/D converter or for the data output from Digital Filter.

1. Calibrate Gain Error and Offset Error
2. Adjust User's Gain
3. Adjust User's Offset

For details on each process, refer to each item.

54.5.3.1 Gain Error and Offset Error Calibration

This process calibrates the A/D converter's Gain Error and Offset Error due to chip-by-chip characteristic variations.

Calibration for the A/D converter's Gain Error and Offset Error is performed by calculation based on error data measured inside the ADC16H by self-calibration operation.

Using gain error and offset error calibration when the input to the A/D converter is near the VREFH0 (unit 0) and VREFH (unit 1) upper limit or VREFL0 (unit 0) and VREFL (unit 1) lower limit, an overflow of A/D conversion data may occur.

54.5.3.2 User's Gain Adjustment

User's Gain adjusting function multiplies the A/D conversion data by an arbitrary coefficient value.

The coefficient value of User's Gain is set in ADUGTRn.UGAIN[15:0] (n = 0 to 7). User's Gain can be set for each virtual channel. The User's Gain Table to be used is selected in ADDOPCRAm.GAINSEL[3:0] (m = 0 to 32). The coefficient value of User's Gain is the sum of the gain corresponding to the bits that are set to 1 in ADUGTRn.UGAIN[15:0]. [Table 54.46](#) shows the gain factors corresponding to the respective bit in ADUGTRn.UGAIN[15:0]. An example of setting User's Gain is shown in [Table 54.47](#). [Figure 54.36](#) and [Figure 54.37](#) show the relationship between before adjustment (input) and after adjustment (output) to the A/D conversion result when User's Gain adjustment function is used.

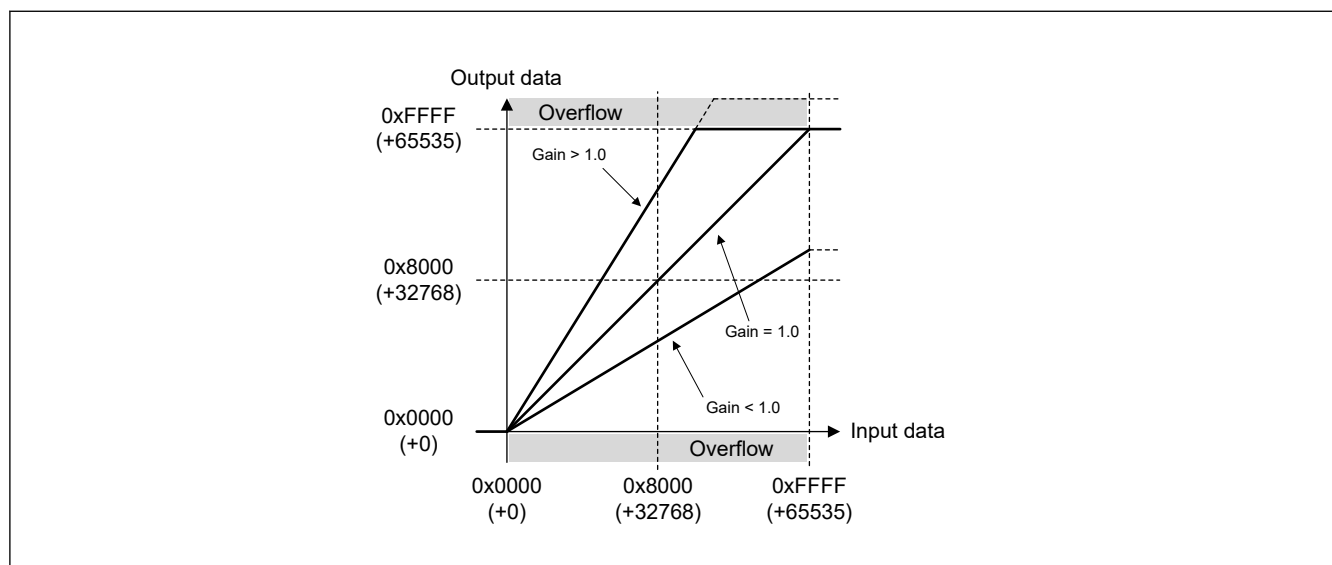
The overflow of A/D conversion data may occur when User's Gain adjusting function is used.

Table 54.46 List of gains corresponding to each gain setting bit in User's Gain setting table register

UGAIN[15:8]	Gain value	UGAIN[7:0]	Gain value
b15	$2^1 = 2.0$	b7	$2^{-7} = 7.813\text{E-}03$
b14	$2^0 = 1.0$	b6	$2^{-8} = 3.906\text{E-}03$
b13	$2^{-1} = 0.5$	b5	$2^{-9} = 1.953\text{E-}03$
b12	$2^{-2} = 0.25$	b4	$2^{-10} = 9.766\text{E-}04$
b11	$2^{-3} = 0.125$	b3	$2^{-11} = 4.883\text{E-}04$
b10	$2^{-4} = 0.0625$	b2	$2^{-12} = 2.441\text{E-}04$
b9	$2^{-5} = 0.03125$	b1	$2^{-13} = 1.221\text{E-}04$
b8	$2^{-6} = 1.563\text{E-}02$	b0	$2^{-14} = 6.104\text{E-}04$

Table 54.47 Example of User's Gain Settings

ADUGTRn.UGAIN[15:0] (n = 0 to 7)	Gain value
0x0000	x0.0000
⋮	⋮
0x0400	x0.0625
⋮	⋮
0x0800	x0.1250
⋮	⋮
0x1000	x0.2500
⋮	⋮
0x2000	x0.5000
⋮	⋮
0x4000 (initial value)	x1.0000
⋮	⋮
0x8000	x2.0000
⋮	⋮
0xFFFF	x3.9999

**Figure 54.36** User's Gain Adjustment (16-bit data length format and Unsigned data format)

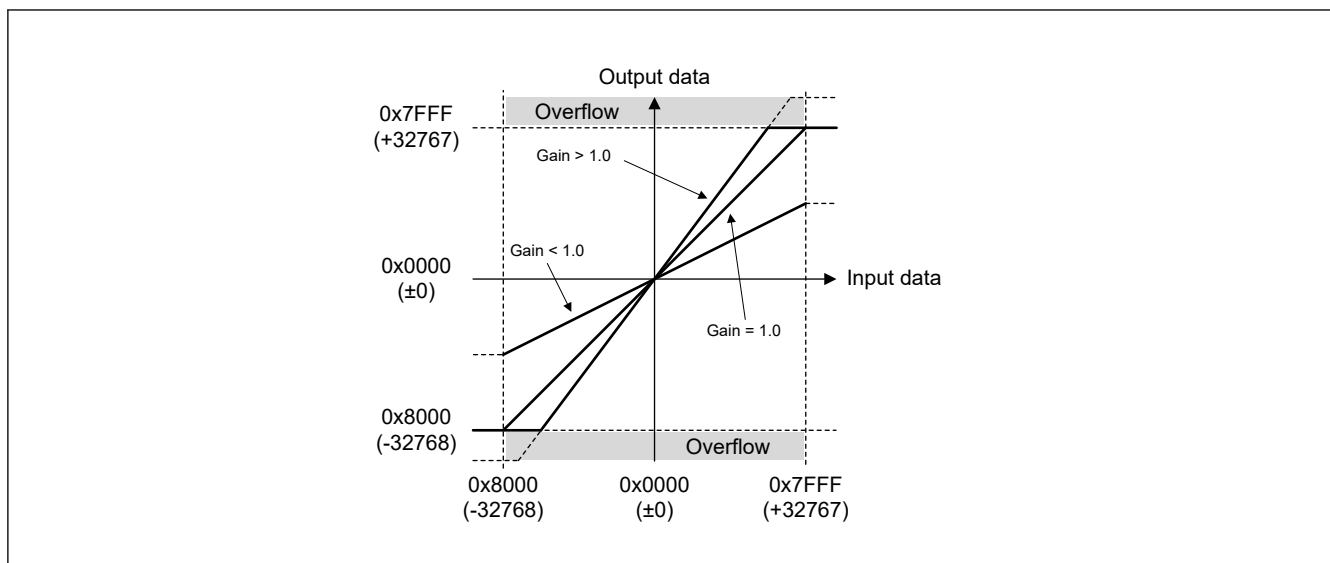


Figure 54.37 User's Gain Adjustment (16-bit data length format and Signed data format)

54.5.3.3 User's Offset Adjustment

User's Offset adjustment function adds or subtracts any constant value to or from the A/D conversion data.

The constant value of User's Offset is set in ADUOFTRn.UOFSET[15:0] (n = 0 to 7). User's Offset can be set for each virtual channel. User's Offset Table to be used is selected in ADDOPCRAM.OFSETSEL[3:0] (m = 0 to 32). [Table 54.48](#) shows the relationship between the register setting value of User's Offset and the offset value. [Figure 54.38](#) and [Figure 54.39](#) show the relationship between before adjustment (input) and after adjustment (output) to the A/D conversion result when User's Offset adjustment function is used.

When User's Offset adjusting function is used, overflow of A/D conversion data may occur.

Table 54.48 Relationship between User's Offset register value and offset value

ADUOFTRn.UOFSET[15:0] (n = 0 to 7)	Offset value (16bit data length format)
0x7FFF	+32767
0x7FFE	+32766
0x7FFD	+32765
⋮	⋮
0x0003	+3
0x0002	+2
0x0001	+1
0x0000 (initial value)	0
0xFFFF	-1
0xFFFE	-2
0xFFFD	-3
⋮	⋮
0x8002	-32766
0x8001	-32767
0x8000	-32768

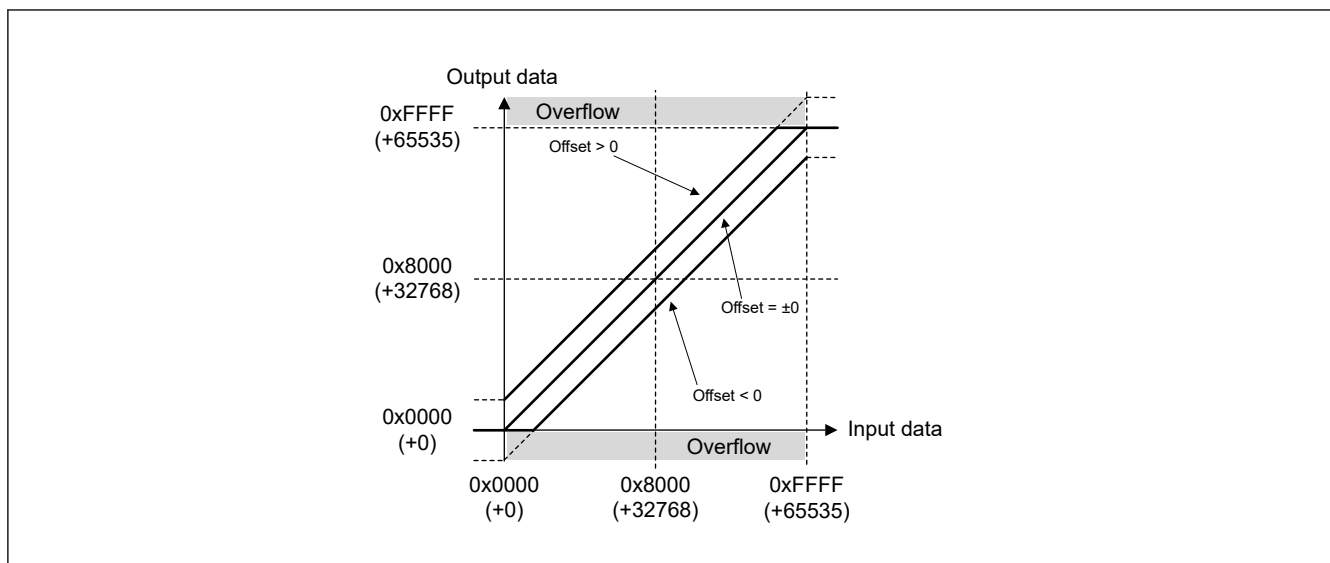


Figure 54.38 User's Offset Adjustment (16-bit data length format and Unsigned data format)

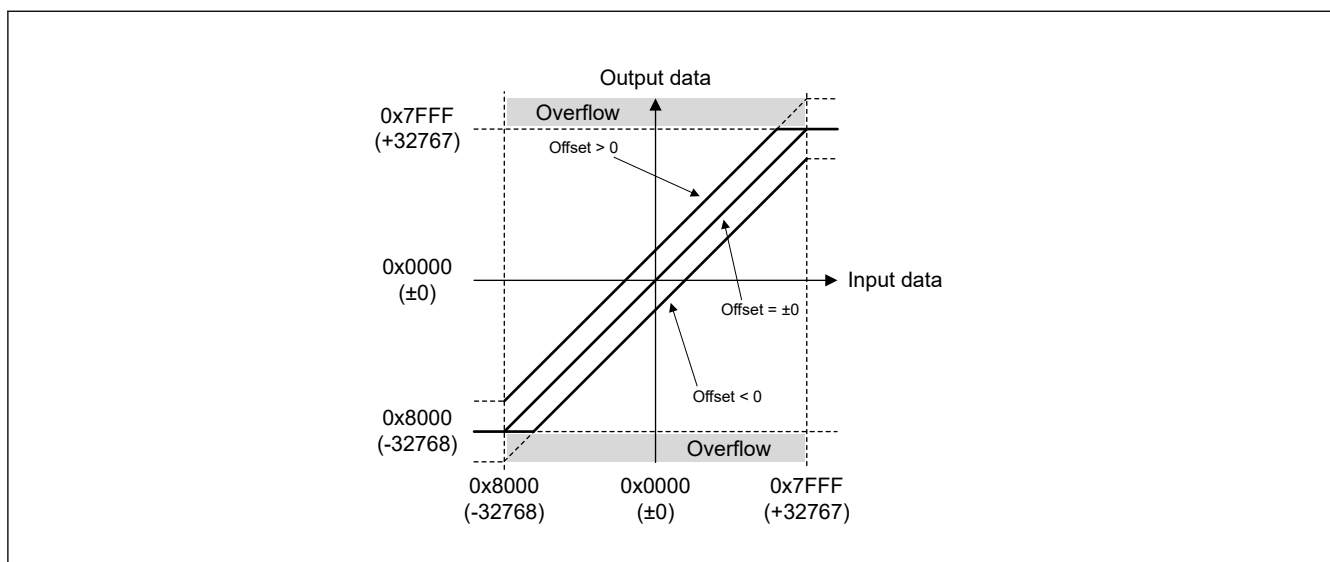


Figure 54.39 User's Offset Adjustment (16-bit data length format and Signed data format)

54.5.4 A/D-converted Value Addition/Averaging Function

A/D-converted value addition/averaging function is a function to calculate the sum value or average value of the result of A/D conversion of analog channels continuously for the specified number of times. A/D-converted value addition/averaging function can be configured for each virtual channel in ADDOPCRBn.AVEMD[1:0] bits and ADDOPCRBn.ADC[3:0] bits ($n = 0$ to 32).

A/D-converted value addition/averaging function inputs the output of the previous data processing. When the specified number of times of data is input, the added value or average value is calculated, and the result is output to the next data processing. When A/D-converted value addition/averaging function is used, the A/D converter repeats the A/D conversion until the data is collected for the number of times specified by the ADDOPCRBn.ADC[3:0] ($n = 0$ to 32) bits. One calculation result (total value or average value) is output every time the specified number of times of data is input to A/D-converted value addition/averaging. Figure 54.40 shows the operation of A/D-converted value addition/averaging function in SAR mode or Oversampling mode. Figure 54.41 shows the operation of A/D-converted value addition/averaging function in Hybrid mode.

When the A/D-converted value addition/averaging function is used, overflow of the A/D conversion data may occur. However, A/D conversion overflow may not be detected under certain conditions. For more information on A/D conversion overflow, see [section 54.7.2. A/D Conversion Overflow](#).

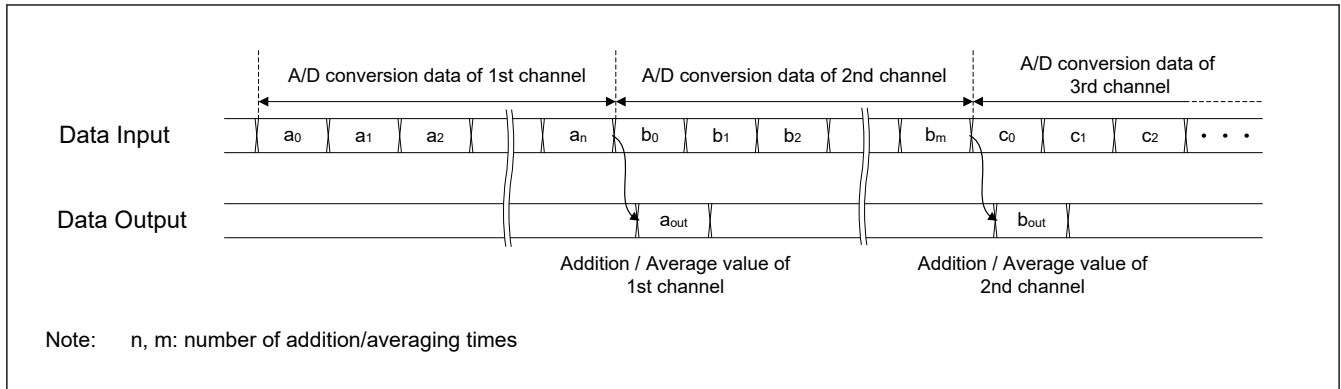


Figure 54.40 Example operation of A/D-converted value addition/averaging function (SAR mode or Oversampling mode)

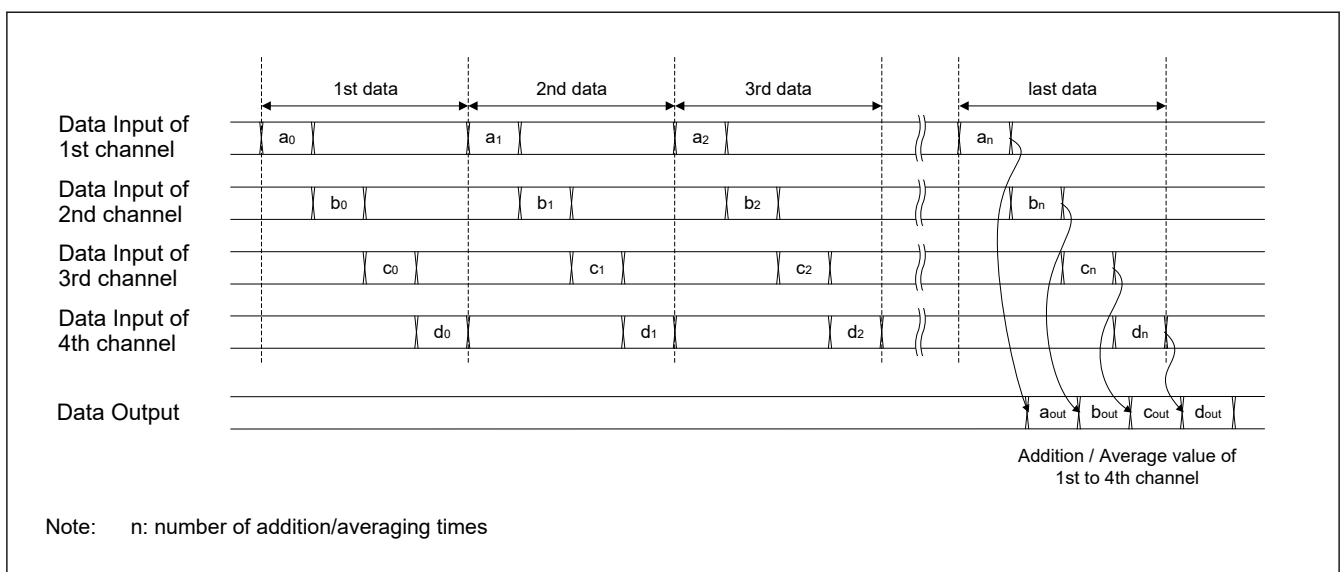


Figure 54.41 Example operation of A/D-converted value addition/averaging function (Hybrid mode)

54.5.5 Limiter Clip Function

Limiter Clip Function is the function to set the upper and lower limits of A/D conversion data. When the A/D conversion data exceeds the specified upper limit value, it is clipped to the upper limit value. If the A/D conversion data falls below the specified lower limit, it is clipped to the lower limit value. Figure 54.42 shows an example of Limiter Clip Function operation.

The upper and lower limits of Limiter Clip Function can be specified in ADLIMTRn.LIMU[15:0] (n = 0 to 7) and ADLIMTRn.LIML[15:0] (n = 0 to 7). The Limiter Clip Function can be selected use or not (enable or disable) for each virtual channel in ADDOPCRCm.LIMTBLS[3:0] (m = 0 to 32).

Limiter Clip Function is processed by 16-bit length. When 14-/12-/10-bit is selected as the data length of the A/D conversion data (when ADDOPCRCm.ADPRC[1:0] = 01b, 10b, or 11b (m = 0 to 32)), the rounding to the specified data length is performed after the clipping by Limiter Clip Function. For details, see section 54.5.6. Data Formatting Process.

The upper and lower limits of the limiter clip are treated as signed or unsigned depending on the setting of the sign selection bit (ADDOPCRCm.SIGNSEL (m = 0 to 32)) in the A/D conversion data.

For Limiter Clip Function, the upper limit value should be greater than the lower limit value (ADLIMTRn.LIMU[15:0] > ADLIMTRn.LIML[15:0] (n = 0 to 7)). If the upper limit value is less than or equal to the lower limit value, the A/D conversion data always becomes 0x0000.

If a limiter clip occurs, the flags are set in the following status registers:

- ADLIMGRSR: The flag is set to the bit corresponding to the scan group where the limiter clip occurred.

- ADLIMCHSR0: If a limiter clip occurs during A/D conversion of an analog input channel, the flag is set to the corresponding bit.
- ADLIMEXSR: If a limiter clip occurs during A/D conversion of Extended Analog Function, the flag is set to the corresponding bit.

To clear the flags in the status registers, write 1 to the corresponding bit in ADLIMGRSCR, ADLIMCHSCR0, ADLIMEXSCR.

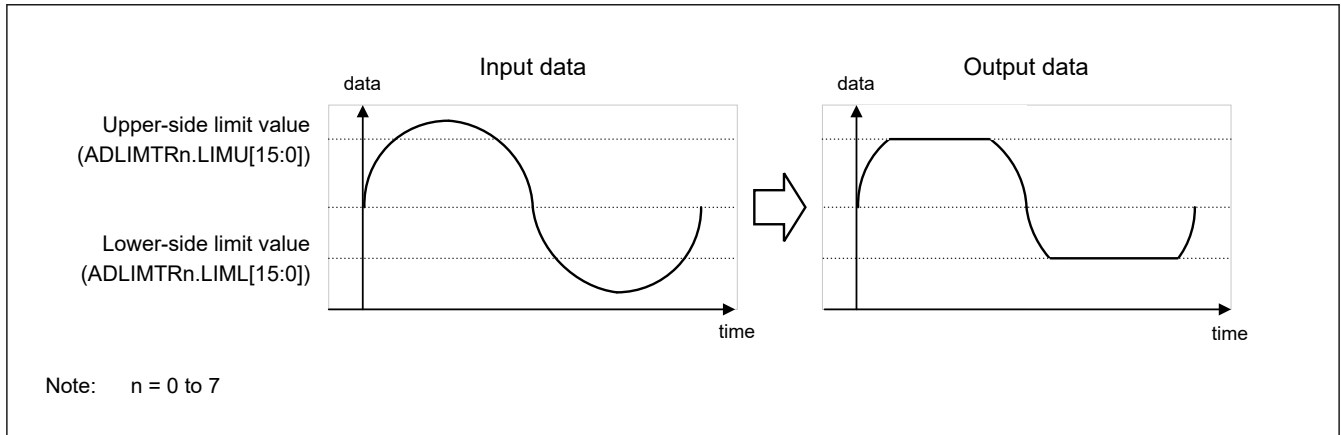


Figure 54.42 Example of limiter clip function operation

54.5.6 Data Formatting Process

In this process, the following formatting is performed on the A/D conversion data.

- Signed/Unsigned data processing
- Data Rounding process

The data formatted in this process is stored to DATA[15:0] bits of the A/D data register (ADDR_i (i = 0 to 22)), the extended A/D data register (ADEXDR_j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)), and the FIFO data register (ADFIFODR_k (k = 0 to 8)) as the result of A/D conversion.

For details on the data format, see [section 54.5.7. Data Format](#).

54.5.6.1 Signed/Unsigned Data Process

The A/D conversion data is formatted into Signed or Unsigned data format based on the setting of the ADDOPCRN.SIGNSEL (n = 0 to 32) bits. Select the signed data format (SIGNSEL = 0) for Differential input mode and the unsigned format (SIGNSEL = 1) for Single-ended input mode.

54.5.6.2 Data Rounding Process

The A/D conversion data is processed as a fraction of the data length based on the setting of the ADDOPCRN.ADPRC[1:0] (n = 0 to 32) bits. If 14-/12-/10-bit data format is selected (ADPRC[1:0] = 01b, 10b, 11b), the lower bits of the A/D conversion data will be rounding. (If the most significant bit of the digit to be cut off is 0, the digit is rounded down; if it is 1, the digit is rounded up.) If the 16-bit data format is selected (ADPRC[1:0] = 00b), the A/D conversion data is not rounded.

54.5.7 Data Format

This subsection describes the data format of the A/D conversion result stored to DATA[15:0] bits in A/D data register (ADDR_i (i = 0 to 22)), Extended A/D data register (ADEXDR_j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)), and FIFO data register (ADFIFODR_k (k = 0 to 8)).

Note: The data format does not guarantee the resolution and accuracy of the A/D converter itself. See [section 70, Electrical Characteristics](#) for the characteristics of the A/D converter.

(1) 16-bits data length format

Table 54.49 shows the alignment of A/D conversion data on 16-bit data format. Figure 54.43 and Figure 54.44 show the data range for 16-bit data format.

For 16-bit data length signed data format, the differential input voltage is A/D-converted in the range of 0x8000 to 0x7FFF. This range corresponds to the voltage range of -VREFH0 to +VREFH0 for ADC0 or the voltage range of -VREFH to +VREFH for ADC1. In 16-bit data length unsigned data format, the single-ended input voltage is A/D-converted in the range of 0x0000 to 0xFFFF. This range corresponds to the voltage range of VREFL0 to VREFH0 for ADC0 or the voltage range of VREFL to VREFH for ADC1.

Table 54.49 Alignment of A/D conversion result data (16-bit data length)

Bit position:	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Bit field:	DATA [15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W:	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

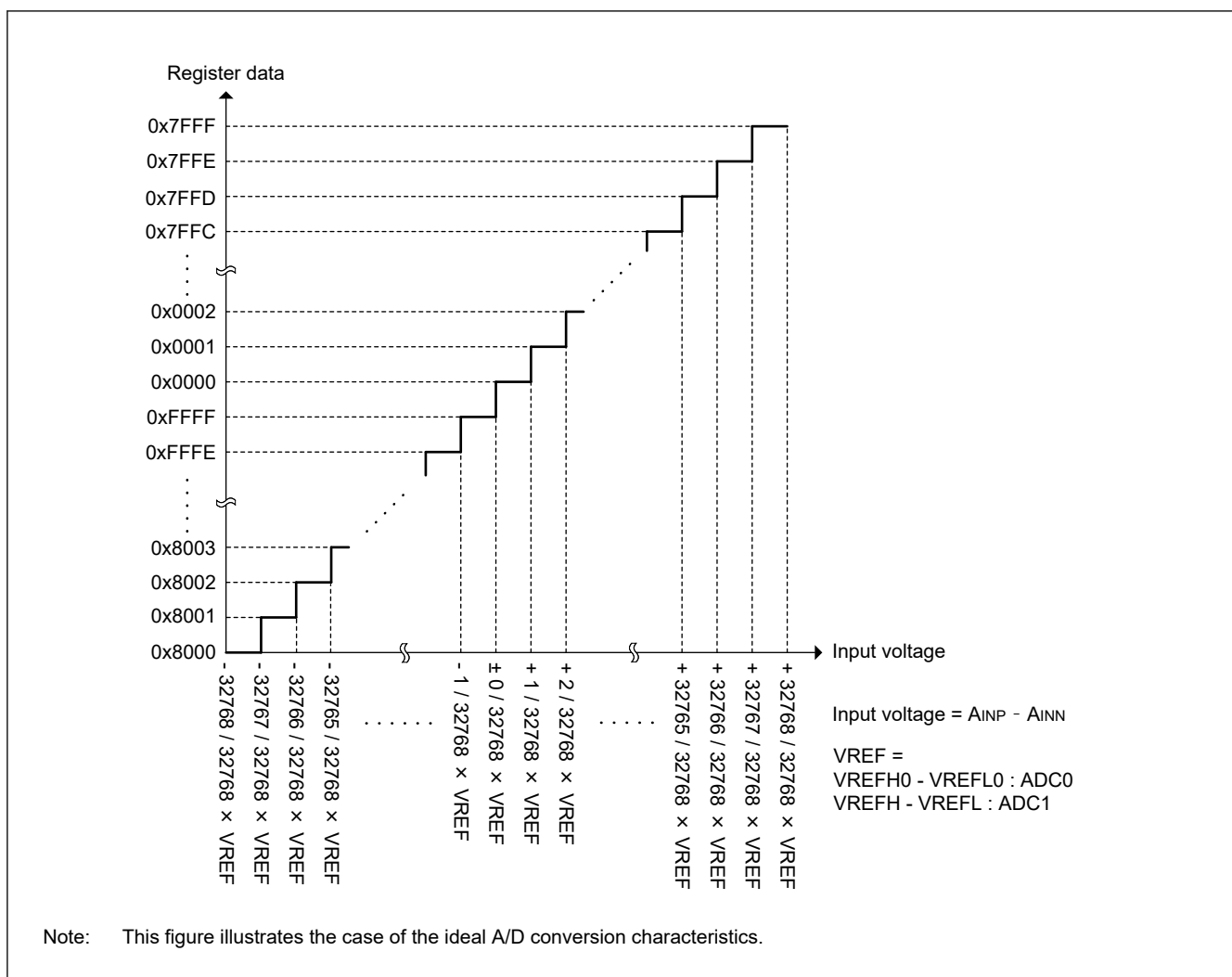


Figure 54.43 Data range of A/D conversion result (16-bit data length, Signed data, Differential input)

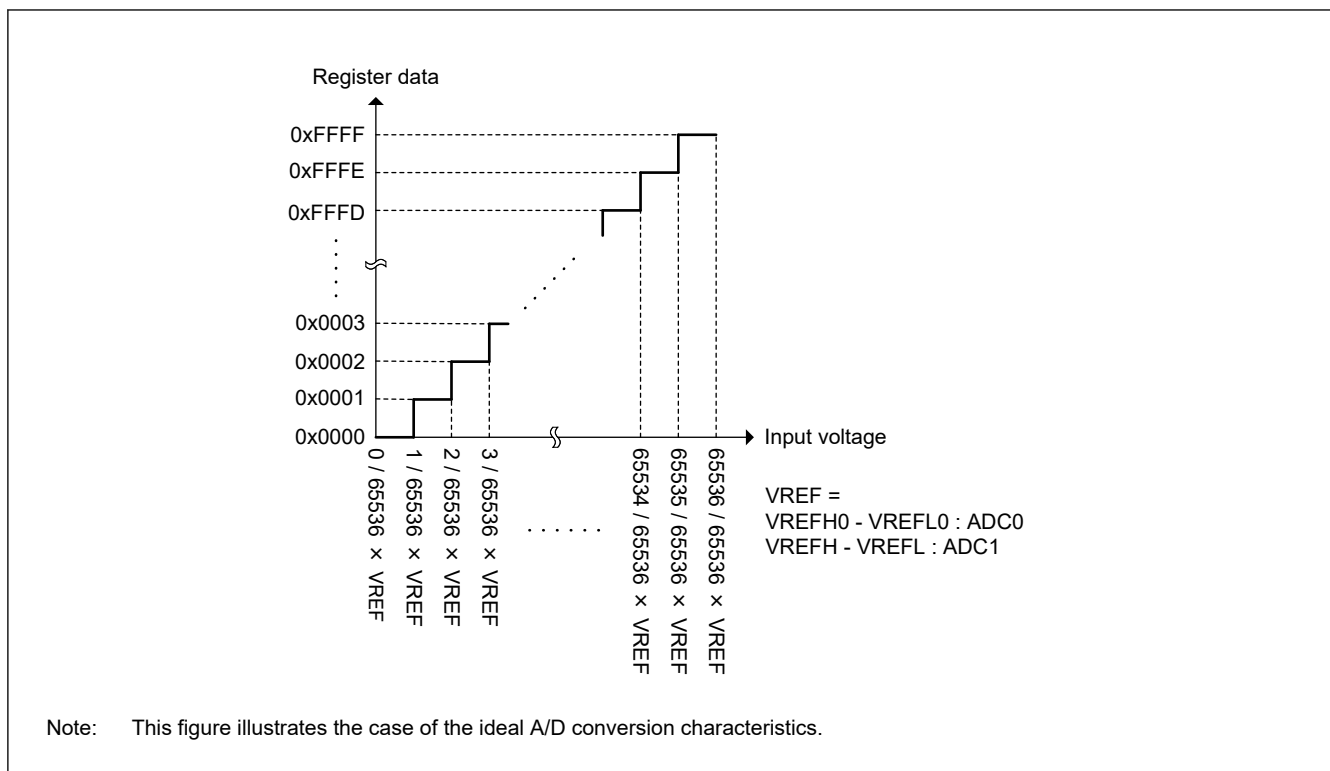


Figure 54.44 Data range of A/D conversion result (16-bit data length, Unsigned data, Single-ended input)

(2) 14-bits data length format

Table 54.50 shows the alignment of A/D conversion data on 14-bit data format. Figure 54.45 and Figure 54.46 show the data range for 14-bit data format.

For 14-bit data length signed data format, the differential input voltage is A/D-converted in the range of 0x2000 to 0x1FFF as the range of -VREFH0 to +VREFH0 for ADC0 or the range of -VREFH to +VREFH for ADC1. In 14-bit data length unsigned data format, the single-ended input voltage is A/D-converted in the range of 0x0000 to 0x3FFF as the range of VREFL0 to VREFH0 for ADC0 or the range of VREFL to VREFH for ADC1.

For 14-bit data length, the upper 2 bits (bit15 to bit14) are always 0.

Table 54.50 Alignment of A/D conversion result data (14-bit data length)

Bit position:	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Bit field:	0	0	DATA[13:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W:	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

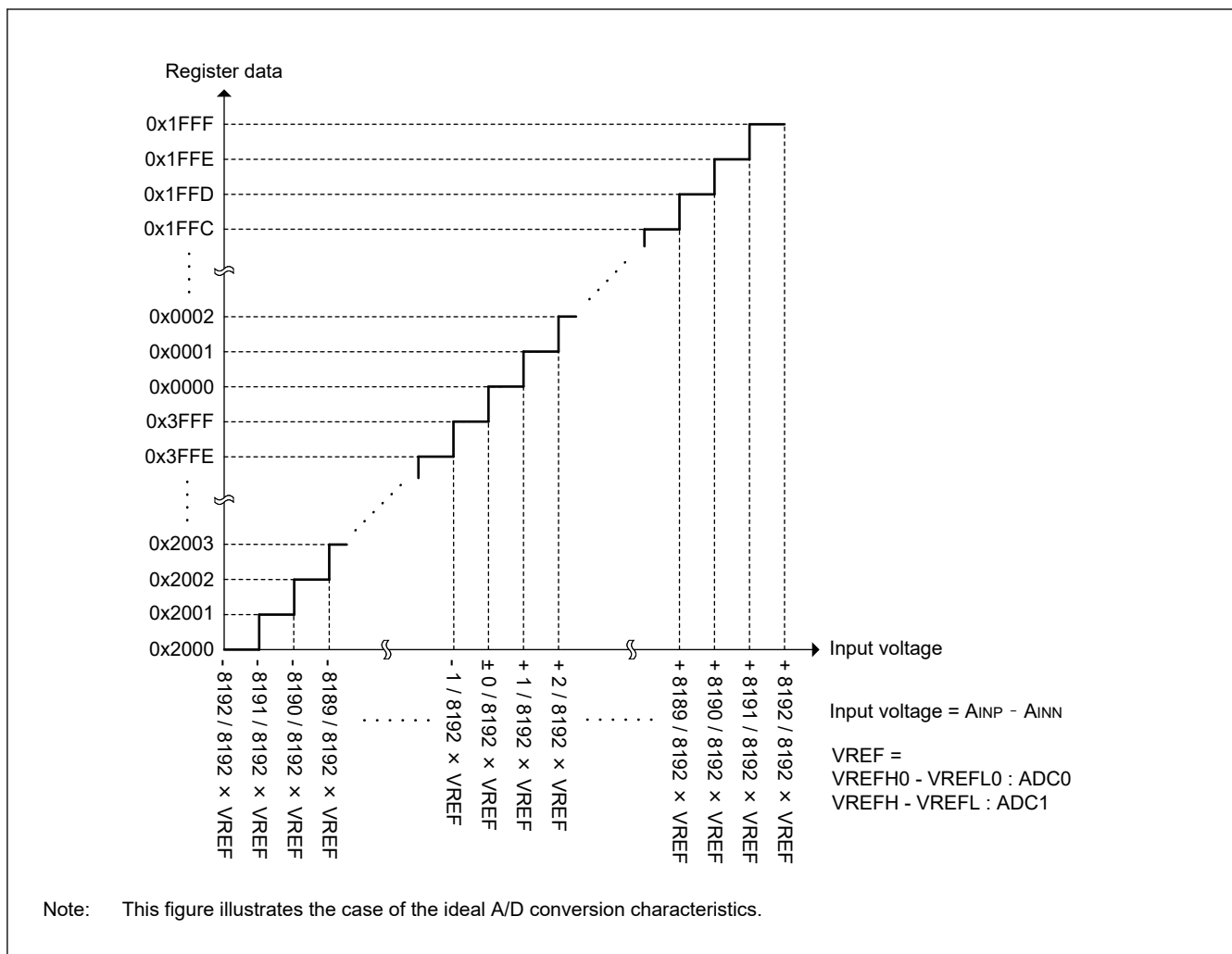


Figure 54.45 Data range of A/D conversion result (14-bit data length, Signed data, Differential input)

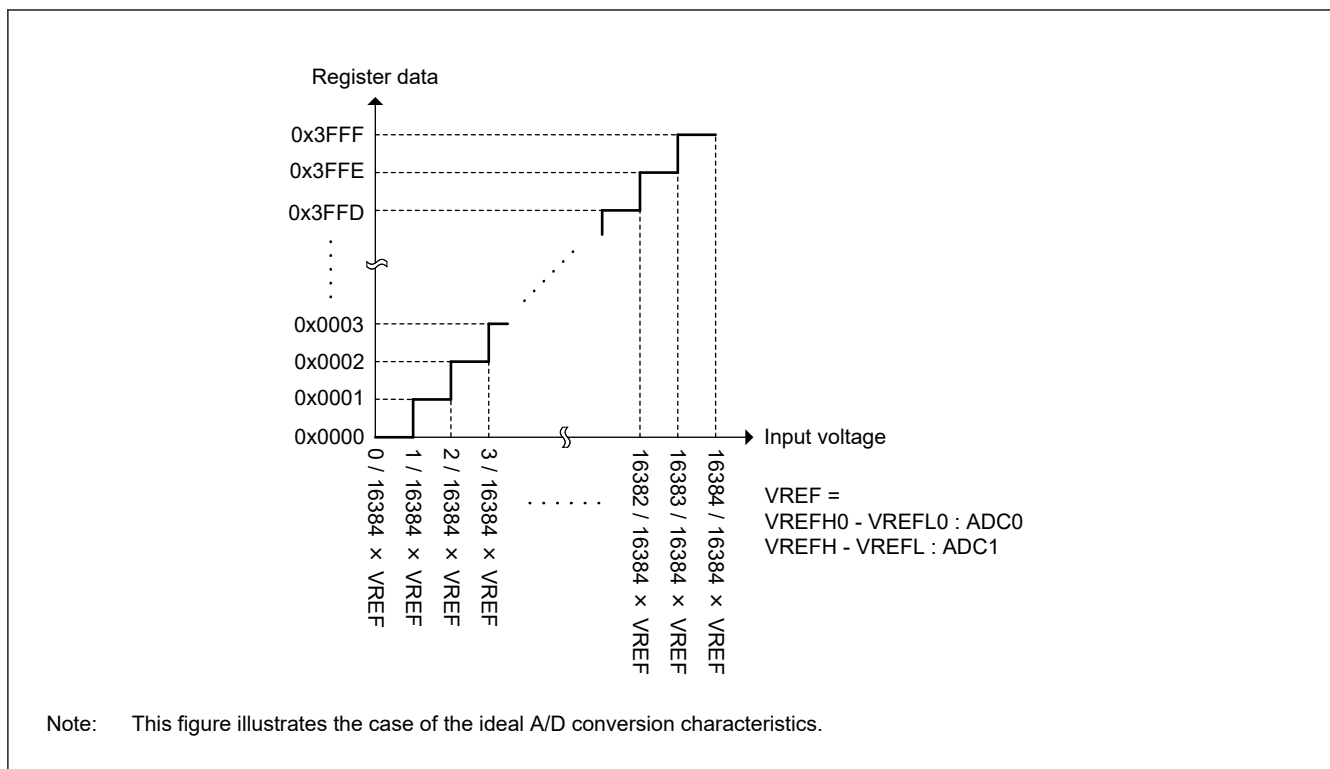


Figure 54.46 Data range of A/D conversion result (14-bit data length, Unsigned data, Single-ended input)

(3) 12-bits data length format

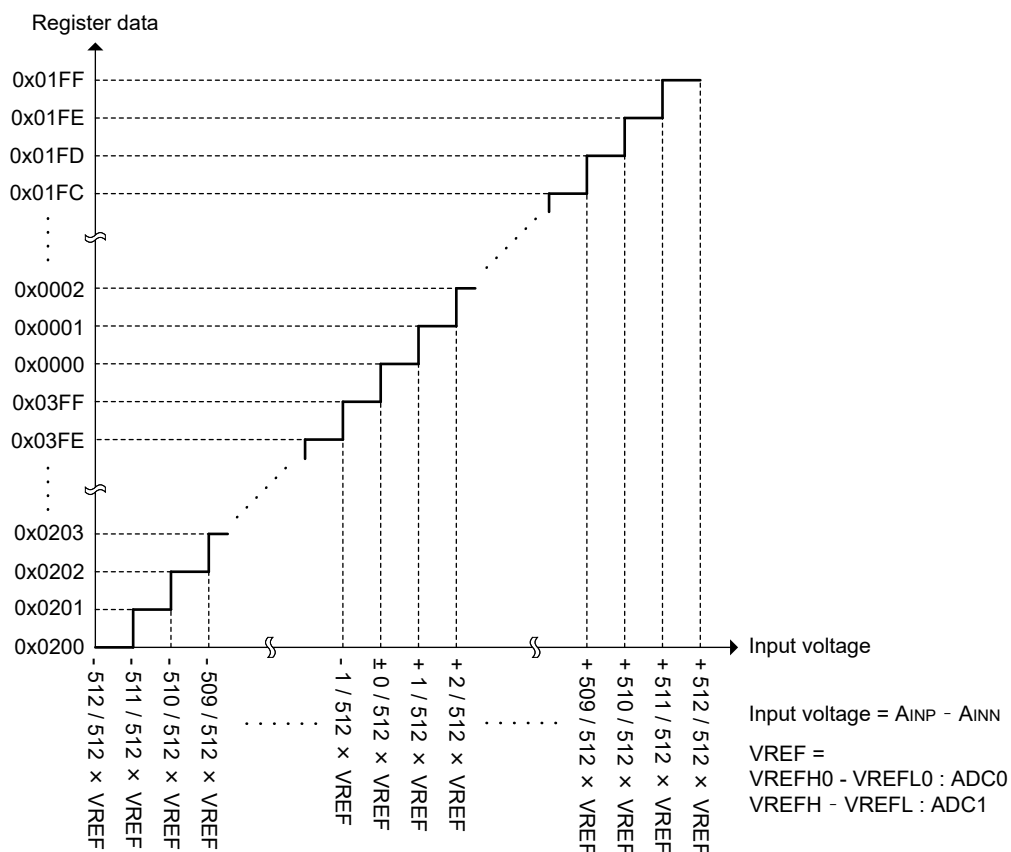
Table 54.51 shows the alignment of A/D conversion data on 12-bit data format. Figure 54.47 and Figure 54.48 show the data range for 12-bit data format.

For 12-bit data length signed data format, the differential input voltage is A/D-converted in the range of 0x0800 to 0x07FF as the range of -VREFH0 to +VREFH0 for ADC0 or the range of -VREFH to +VREFH for ADC1. In 12-bit data length unsigned data format, the single-ended input voltage is A/D-converted in the range of 0x0000 to 0x0FFF as the range of VREFL0 to VREFH0 for ADC0 or the range of VREFL to VREFH for ADC1.

For 12-bit data length, the upper 4 bits (bit15 to bit12) are always 0.

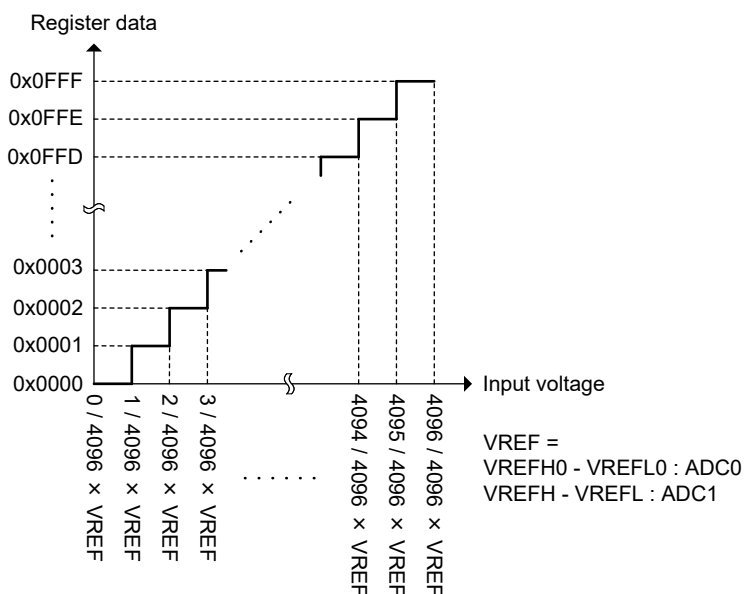
Table 54.51 Alignment of A/D conversion result data (12-bit data length)

Bit position:	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Bit field:	0	0	0	0	DATA[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W:	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R



Note: This figure illustrates the case of the ideal A/D conversion characteristics.

Figure 54.47 Data range of A/D conversion result (12-bit data length, Signed data, Differential input)



Note: This figure illustrates the case of the ideal A/D conversion characteristics.

Figure 54.48 Data range of A/D conversion result (12-bit data length, Unsigned data, Single-ended input)

(4) 10-bits data length format

Table 54.52 shows the alignment of A/D conversion data on 10-bit data format. Figure 54.49 and Figure 54.50 show the data range for 10-bit data format.

For 10-bit data length signed data format, the differential input voltage is A/D-converted in the range of 0x0200 to 0x01FF as the range of -VREFH0 to +VREFH0 for ADC0 or the range of -VREFH to +VREFH for ADC1. In 10-bit data length unsigned data format, the single-ended input voltage is A/D-converted in the range of 0x0000 to 0x03FF as the range of VREFL0 to VREFH0 for ADC0 or the range of VREFL to VREFH for ADC1.

For 10-bit data length, the upper 6 bits (bit15 to bit10) are always 0.

Table 54.52 Alignment of A/D conversion result data (10-bit data length)

Bit position:	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Bit field:	0	0	0	0	0	0	DATA [9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W:	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

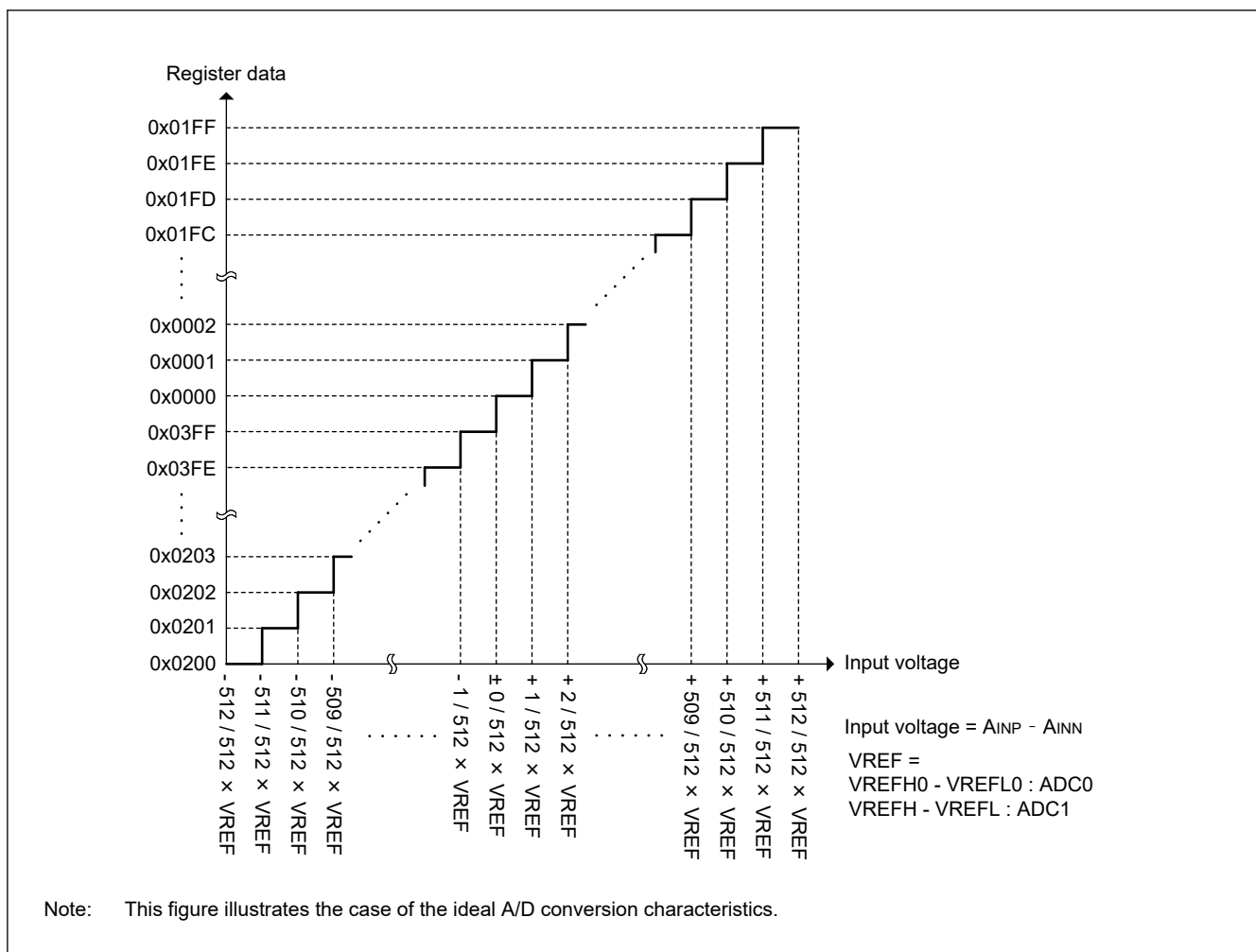


Figure 54.49 Data range of A/D conversion result (10-bit data length, Signed data, Differential input)

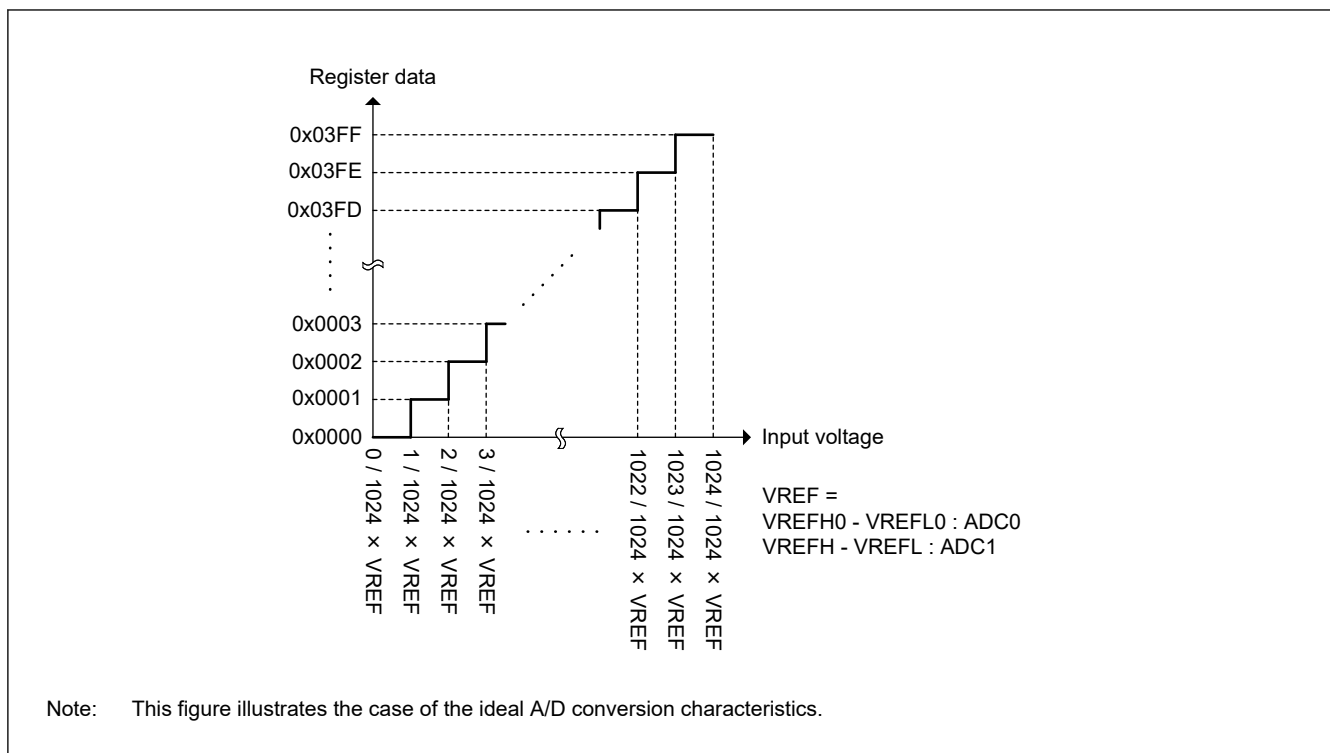


Figure 54.50 Data range of A/D conversion result (10-bit data length, Unsigned data, Single-ended input)

54.5.8 Compare Match Function

54.5.8.1 Compare Match

The compare match function compares the A/D conversion result with the reference value set in the compare match table register (ADCMPTBRn (n = 0 to 7)). The compare match function compares the A/D conversion data after the data formatting process.

(1) Compare match mode

The comparison mode for detecting the compare match is selected in ADCMPMDRm.CMPMDn[1:0] (m = 0, 1. n = 0 to 7). The comparison mode can be selected from the following four modes.

1. Compare match is detected when the A/D conversion value is greater than or equal to the specified upper limit value.
2. Compare match is detected when the A/D conversion value is less than or equal to the specified lower limit.
3. Compare match is detected when the A/D conversion value is greater than or equal to the specified upper limit value, or less than or equal to the lower limit value.
4. Compare match is detected when the A/D conversion value is within the specified upper and lower limit values (when the value is lower than or equal to the upper limit value and greater than or equal to the lower limit value).

Figure 54.51 shows an example of compare match detection.

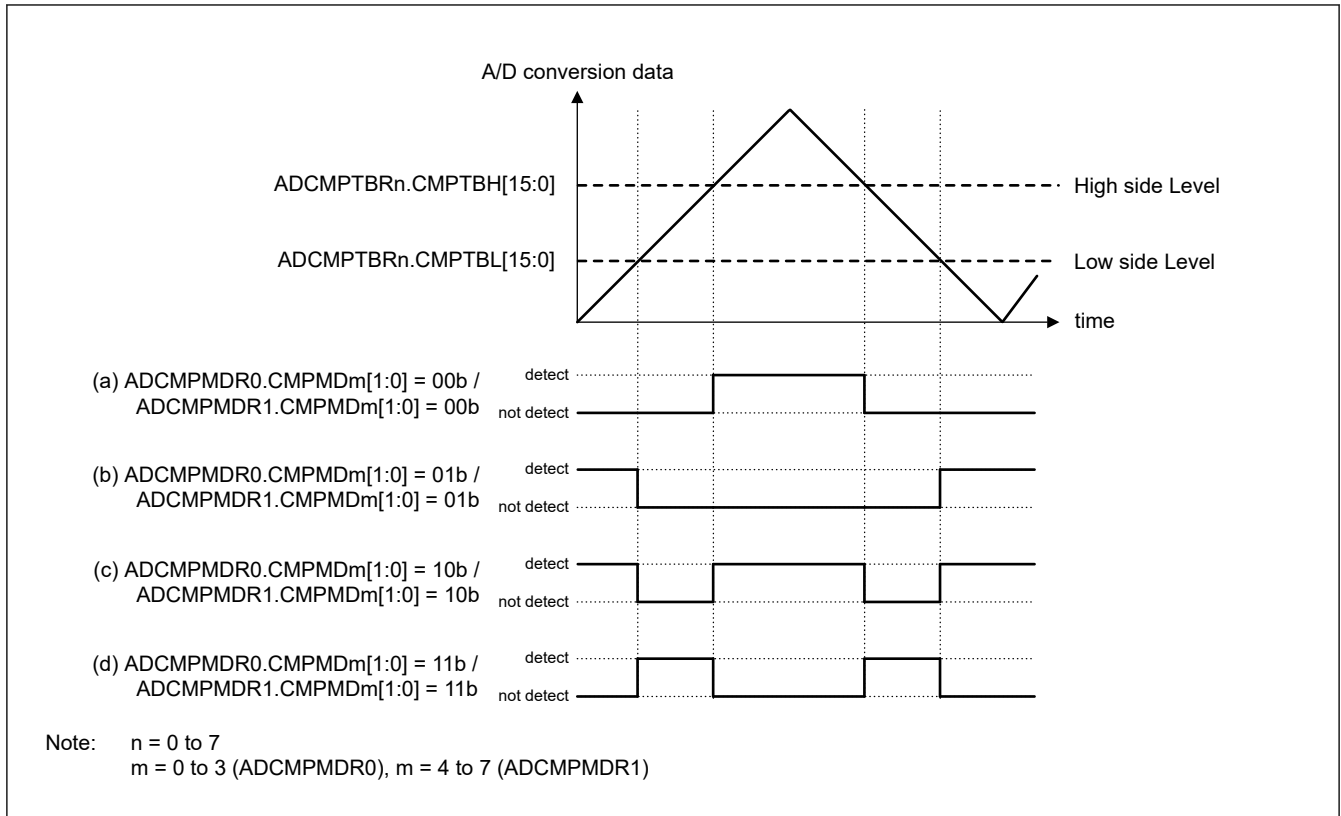


Figure 54.51 Compare match detection

(2) Compare match detection

To use the compare match function, enable the compare match table used in ADCMPENR register, and select the compare match table in ADDOPCRBn.CMPTBLEm (n = 0 to 32, m = 0 to 7) for each virtual channel.

When A/D conversion is performed on a virtual channel for which the compare match function is enabled, a compare match is judged based on the A/D conversion result of its channel.

When a compare match is detected, the flags are set in the following status registers:

- ADCMPTBSR: The flag is set to the bit corresponding to the compare match table used when a compare match is detected.
- ADCMPCHSR0: The flag is set to the bit corresponding to the analog input channel that detected the compare match.
- ADCMPEXSR: The flag is set to the bit corresponding to Extended Analog Function where a compare match is detected.

To clear the flag of each status register, write 1 to the corresponding bit of ADCMPTBSCR, ADCMPCHSCR0, ADCMPEXSCR.

If compare match interrupt is enabled in ADCMPINTCR register and a compare match using compare match table 0 to 3 is detected, the corresponding interrupt is generated.

54.5.8.2 Composite Compare Match

The composite compare match function generates interrupts and ELC events by combining the comparison results of multiple compare match tables. The combination and condition of the compare match table for the composite compare match function is set in ADCCMPCR0 or ADCCMPCR1.

Composite compare match interrupt is generated when a compare match is detected that matches the specified condition.

Figure 54.52 shows the relationship between composite compare match and compare match function. Table 54.53 shows the relationship between the composite compare match function and the control register.

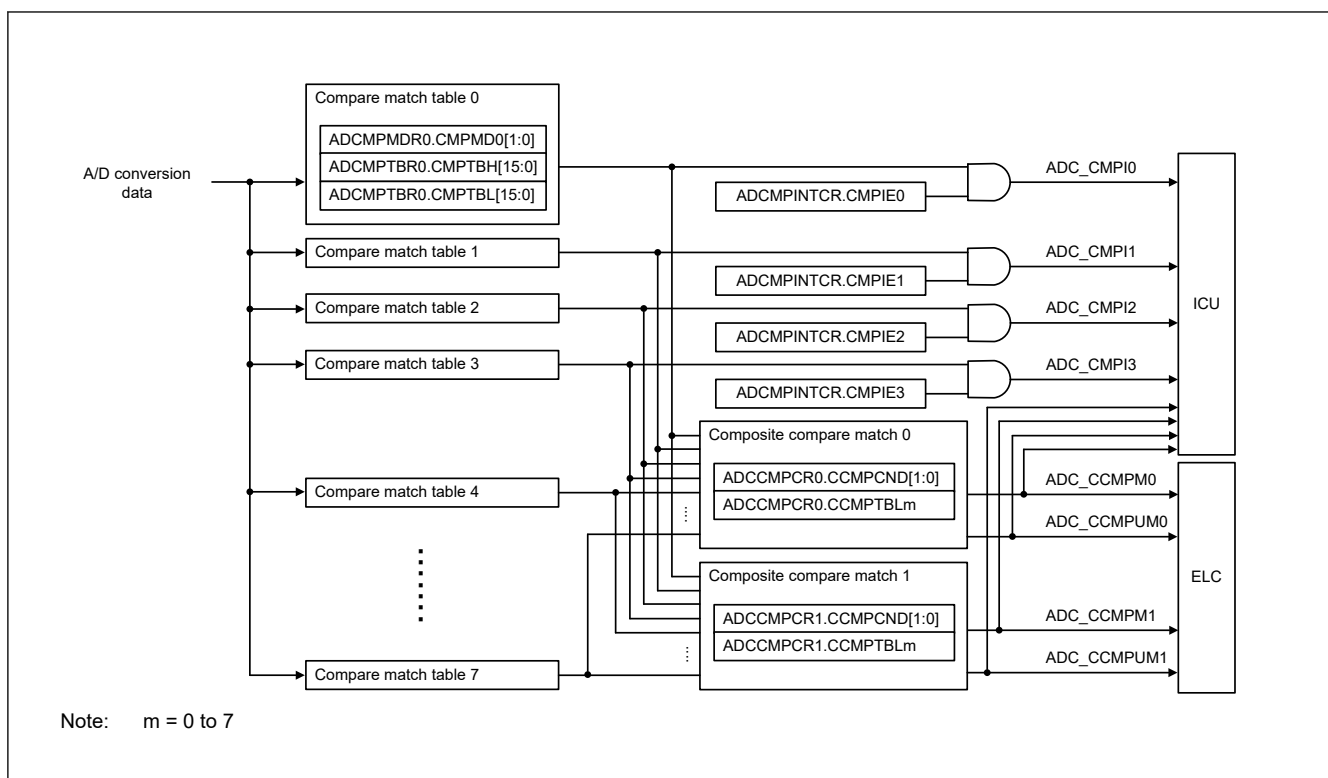


Figure 54.52 Relationship between composite compare match function and compare match function

Table 54.53 Composite compare match and control register

Function name	Enable composite compare function	Composite compare mode selection	Composite compare match interrupt
Composite compare match 0	ADCCMPCR0.CCMPTBLm (m = 0 to 7)	ADCCMPCR0.CCMPCND[1:0]	ADC_CCMPM0 ADC_CCMPUM0
Composite compare match 1	ADCCMPCR1.CCMPTBLm (m = 0 to 7)	ADCCMPCR1.CCMPCND[1:0]	ADC_CCMPM1 ADC_CCMPUM1

54.5.9 Data Registers

A/D data register (ADDR_i (i = 0 to 22)) and Extended A/D data register (ADEXDR_j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)) store the A/D conversion data of the analog channels corresponding to each register. These registers are updated (overwritten) when A/D conversion for the analog channel corresponding to each register is completed.

54.5.10 FIFO Function

FIFO consists of 8-stage registers and can hold up to 8 A/D conversion data. One FIFO is implemented for each scan group. The A/D conversion data stored in FIFO can be read from ADFIFODR_n register (n = 0 to 8).

FIFO acts as a ring buffer. Writing and reading A/D conversion data to and from FIFO is controlled by the write pointer and read pointer inside FIFO. The block diagram of FIFO is shown in [Figure 54.53](#).

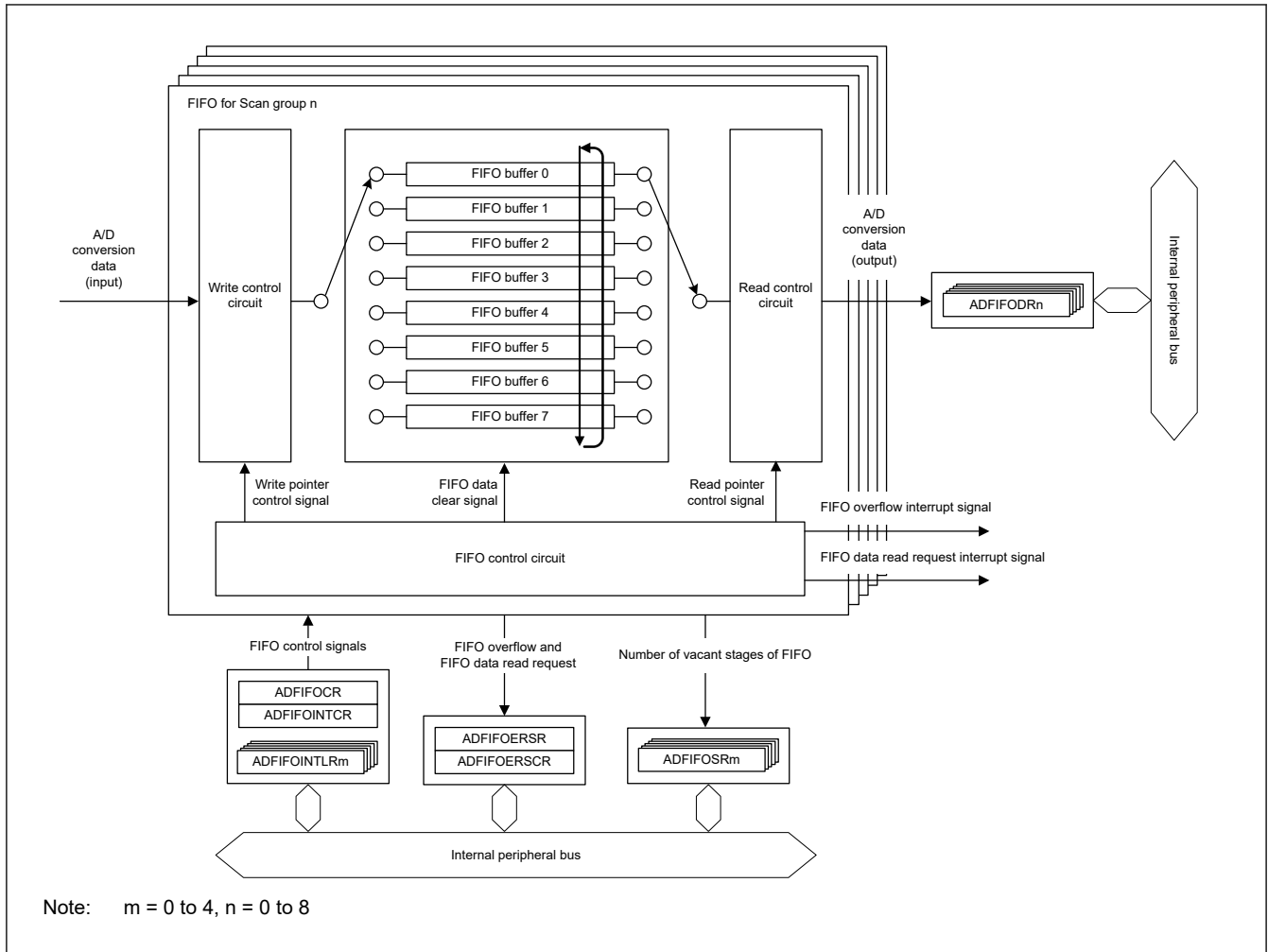


Figure 54.53 The block diagram of FIFO

54.5.10.1 Operation When Writing A/D Conversion Data to FIFO

The A/D conversion data is stored sequentially to the register pointed by the write pointer. When A/D conversion data is written to a register in FIFO, the write pointer is switched to the next register. At this time, the number of vacant stages (ADFIFOSRm.FIFOSTn[3:0] ($m = 0$ to 4, $n = 0$ to 8) bits) is decreased 1.

A FIFO overflow occurs when FIFO is full (ADFIFOSRm.FIFOSTn[3:0] bits are 0000b) and an additional A/D conversion data is written to FIFO. When a FIFO overflow occurs, the A/D conversion data is not written and the write pointer does not change.

54.5.10.2 Operation When Reading A/D Conversion Data from FIFO

When A/D conversion data is read from FIFO data register (ADFIFODRn ($n = 0$ to 8)), the data in the register pointed to by FIFO read pointer is read. When A/D conversion data is read from FIFO, the read pointer is switched to the next register. At this time, the number of vacant stages (ADFIFOSRm.FIFOSTn[3:0] ($m = 0$ to 4, $n = 0$ to 8) bits) is increased 1.

If FIFO is empty (ADFIFOSRm.FIFOSTn[3:0] bits are 1111b) and the reading from the FIFO is occurred, an invalid data is read. (If FIFO is read after a reset or after FIFO data is cleared, 0x00000000 is read. If data had been written to register in FIFO, the previous data is read.) The read pointer does not change in the read operation when FIFO is empty.

Read A/D conversion data from FIFO data register (ADFIFODRn) by 32-bit access. For details, see [section 54.5.10.5. Usage Notes on FIFO.](#)

54.5.10.3 FIFO Data Read Request and Overflow

When A/D conversion data is written to or read from FIFO, the write pointer or read pointer changes. Depending on the state of the write pointer and the read pointer, the number of vacant stages in FIFO is reflected in ADFIFOSRm.FIFOSTn[3:0] (m = 0 to 4, n = 0 to 8).

FIFO data read request flag is set (ADFIFOERSR.FIFOFLFn = 1 (n = 0 to 8)) when the number of vacant stages become less than or equal to the value specified in ADFIFOINTLRm.FIFOILVn[3:0] (m = 0 to 4, n = 0 to 8) bit. If a FIFO data read request occurs, FIFO data read request interrupt can be generated.

When FIFO is full (ADFIFOSRm.FIFOSTn[3:0] bits are 0000b) and additional A/D conversion data are written, FIFO overflow flag is set (ADFIFOERSR.FIFOOVFn = 1 (n = 0 to 8)). If a FIFO overflow occurs, FIFO data overflow interrupt can be generated.

54.5.10.4 Clearing FIFO Data

The data stored in the registers in FIFO can be cleared by writing to ADFIFODCR.FIFODCn (n = 0 to 8) bit. When writing 1 to ADFIFODCR.FIFODCn bit, the registers, read pointer, and write pointer in FIFO of the scan group corresponding to FIFODCn bit are initialized. Clearing FIFO data should be performed while A/D conversion is not being performed.

54.5.10.5 Usage Notes on FIFO

- **Reading**
Reading the A/D conversion data from the FIFO data register (ADFIFODRn (n = 0 to 8)) should be performed with a single 32-bit access. Access to each bit of ADFIFODRn, 16-bit read access, and 8-bit read access are prohibited. If these restrictions are violated, the operation of FIFO is not guaranteed as the read pointer may not change normally, and correct data may not be able to read. Alternatively, the read pointer may advance unintentionally, and A/D conversion data may be lost.
- **Rescanning**
When rescanning from the first channel by resuming the scan in group priority operation (ADGSPCR.PGS_m = 1 (m = 0, 1), ADGSPCR.LGRPS_m = 0 (m = 0, 1)), enable FIFO clear (ADFIFOOCR.FIFOCEn = 1 (n = 0 to 8)) for scan groups other than the highest priority.
- **Clearing**
When FIFO clear is enabled (ADFIFOOCR.FIFOCEn = 1 (n = 0 to 8)), all data remaining in the FIFO at the start of scan or restart of a scan is cleared. Control the scan start timing so that the FIFO is not cleared before the necessary FIFO data is read.
Note that the clearing FIFO data function by starting/resuming scanning is prohibited for other than SAR mode and group priority operation.

54.5.11 Data Output Decimation Function

In Oversampling mode — One-channel continuous scan mode, the output of conversion data can be decimated. The conversion data output interval can be set for each unit using ADDECCR.DCIM_m (m = 0, 1). When conversion data output is decimated using this function, the interrupt output is decimated as well.

When this function is used in addition/average mode, the conversion data output interval is the period when the number of times the conversion data has been added or averaged and then subtracted by this function. In addition/average mode, the data to be subtracted by this function is the data before addition/averaging.

Figure 54.54 and Figure 54.55 show examples of operation when data is output every 4 times.

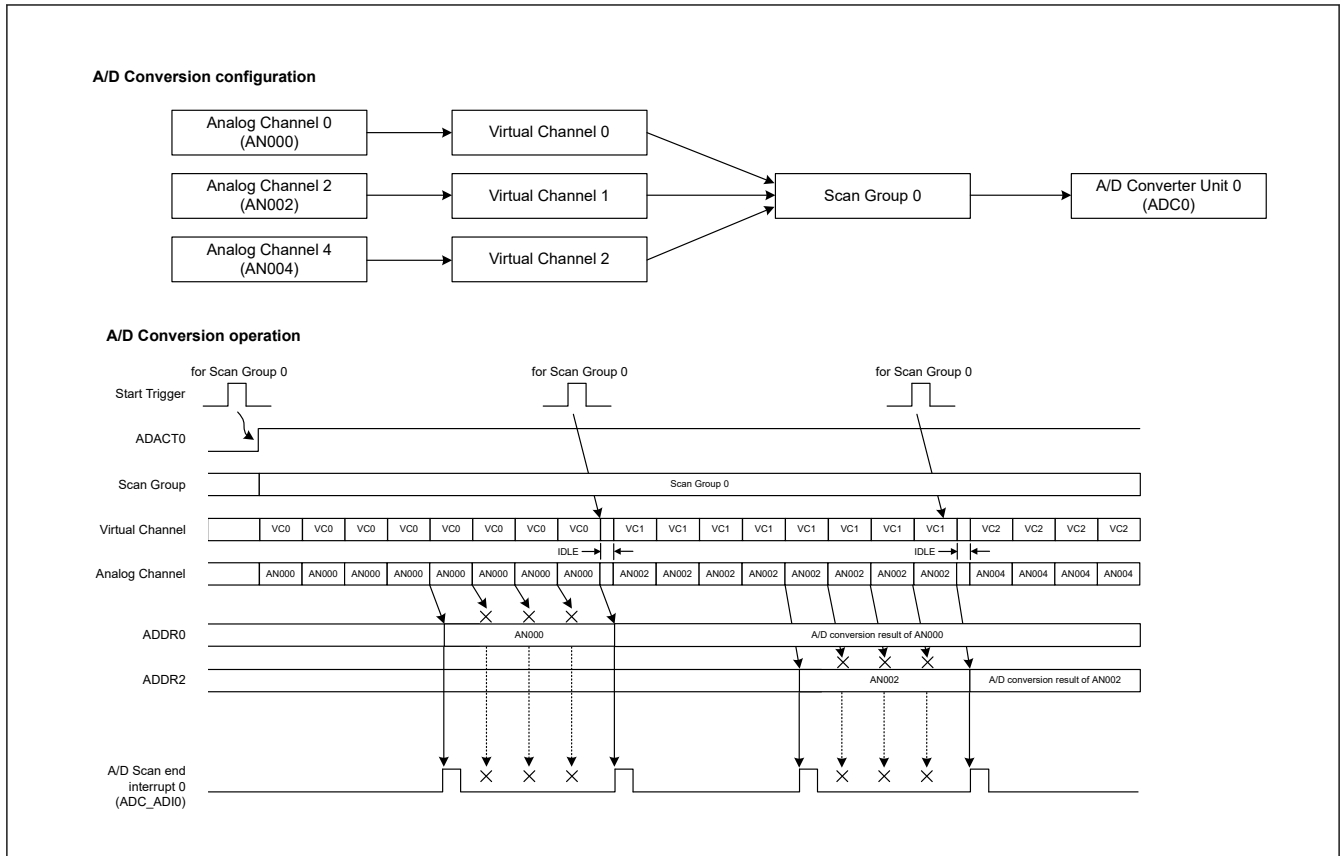


Figure 54.54 Example of data output decimation function (output every 4 times)

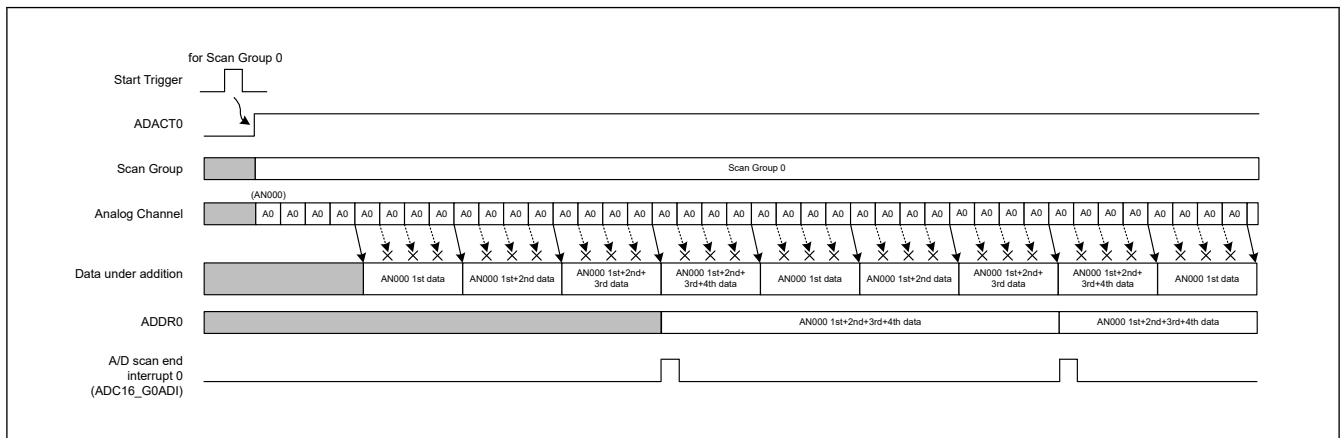


Figure 54.55 Example of data output decimation function (output every 4 times) when addition function is enabled (addition count is 4 times)

54.5.12 A/D Conversion Data Error Detection

The ERR bit in A/D data register (ADDR_i (i = 0 to 22)), Extended A/D data register (ADEXDR_j (j = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22)), and FIFO data register (ADFIFODR_k (k = 0 to 8)) is a flag that indicates an A/D conversion data error. An A/D conversion data error indicates that an invalid A/D conversion data has been detected. If an A/D conversion data error occurs (ERR = 1), the A/D conversion data cannot be guaranteed.

A/D conversion data error is generated by the following factors.

- A/D converter error
- A/D conversion overflow

For more details about these errors, see [section 54.7. Error Detection](#).

54.6 Start and Stop Control of A/D Conversion

54.6.1 Software Trigger

A/D conversion of scan group n can be started by writing 1 to ADSTR n .ADST ($n = 0$ to 8) bit or ADSYSTR.ADSYST n ($n = 0$ to 8) bit.

ADSTR n .ADST bit is used to start A/D conversion individually for each scan group. ADSYSTR.ADSYST n bit is used to start A/D conversion on the scan group assigned to ADC0 or ADC1 at the same time.

Except when group priority operation, writing to ADSTR n .ADST bit or ADSYSTR.ADSYST n bit for the scan groups, which use A/D converter that is in A/D conversion, is ignored.

54.6.2 Peripheral Module Triggers

A/D conversion can be started by triggers from the peripheral modules listed in the followings.

- ELC Trigger
- GPT Trigger
- External Trigger (I/O port)

To perform A/D conversion by a trigger from a peripheral module, configure the trigger for each scan group and enable the trigger input from the peripheral module in ADTRGENR register.

54.6.2.1 ELC Trigger

A/D conversion can be started by an event (ELC event) from Event Link Controller. To start A/D conversion using an ELC event, configure the scan group using the ELC event in ADTRGELC n ($n = 0$ to 8) register and enable the A/D conversion start trigger in ADTRGENR register. [Table 54.54](#) shows the correspondence between ADTRGELC n register and ELC event.

Table 54.54 Correspondence between ELC Trigger Enable Register and ELC Event

Resister bit	Event Name
ADTRGELC n .TRGELC0	ELC_AD00
ADTRGELC n .TRGELC1	ELC_AD01
ADTRGELC n .TRGELC2	ELC_AD02
ADTRGELC n .TRGELC3	ELC_AD10
ADTRGELC n .TRGELC4	ELC_AD11
ADTRGELC n .TRGELC5	ELC_AD12

Note: $n = 0$ to 8

54.6.2.2 GPT Trigger

A/D conversion can be started by an interrupt source from the general-purpose PWM Timer (GPT). To start A/D conversion using an interrupt source from the GPT, configure the scan group using the GPT interrupt source in ADTRGGPT n ($n = 0$ to 8) register and enable the A/D conversion start trigger in ADTRGENR register. [Table 54.55](#) shows the correspondence between ADTRGGPT n register and GPT interrupt sources.

Table 54.55 Correspondence between GPT Trigger Enable Registers and GPT Interrupt Sources (1 of 2)

Resister bit	Event Name
ADTRGGPT n .TRGGPTA0	GPT0_ADTRGA
ADTRGGPT n .TRGGPTA1	GPT1_ADTRGA
ADTRGGPT n .TRGGPTA2	GPT2_ADTRGA
ADTRGGPT n .TRGGPTA3	GPT3_ADTRGA
ADTRGGPT n .TRGGPTA4	GPT4_ADTRGA
ADTRGGPT n .TRGGPTA5	GPT5_ADTRGA

Table 54.55 Correspondence between GPT Trigger Enable Registers and GPT Interrupt Sources (2 of 2)

Resister bit	Event Name
ADTRGGPTn.TRGGPTA6	GPT6_ADTRGA
ADTRGGPTn.TRGGPTA7	GPT7_ADTRGA
ADTRGGPTn.TRGGPTA8	GPT8_ADTRGA
ADTRGGPTn.TRGGPTA9	GPT9_ADTRGA
ADTRGGPTn.TRGGPTA10	GPT10_ADTRGA
ADTRGGPTn.TRGGPTA11	GPT11_ADTRGA
ADTRGGPTn.TRGGPTA12	GPT12_ADTRGA
ADTRGGPTn.TRGGPTA13	GPT13_ADTRGA
ADTRGGPTn.TRGGPTB0	GPT0_ADTRGB
ADTRGGPTn.TRGGPTB1	GPT1_ADTRGB
ADTRGGPTn.TRGGPTB2	GPT2_ADTRGB
ADTRGGPTn.TRGGPTB3	GPT3_ADTRGB
ADTRGGPTn.TRGGPTB4	GPT4_ADTRGB
ADTRGGPTn.TRGGPTB5	GPT5_ADTRGB
ADTRGGPTn.TRGGPTB6	GPT6_ADTRGB
ADTRGGPTn.TRGGPTB7	GPT7_ADTRGB
ADTRGGPTn.TRGGPTB8	GPT8_ADTRGB
ADTRGGPTn.TRGGPTB9	GPT9_ADTRGB
ADTRGGPTn.TRGGPTB10	GPT10_ADTRGB
ADTRGGPTn.TRGGPTB11	GPT11_ADTRGB
ADTRGGPTn.TRGGPTB12	GPT12_ADTRGB
ADTRGGPTn.TRGGPTB13	GPT13_ADTRGB

Note: n = 0 to 8

54.6.2.3 External Trigger

A/D conversion can be started by the input from the external trigger pin (ADTRG0 and ADTRG1). To start A/D conversion using an external trigger, configure the scan group using the external trigger in ADTRGEXTn (n = 0 to 8) register and enable the A/D conversion start trigger in ADTRGENR register.

External trigger is active low. Before setting to enable the external trigger (ADTRG0 and ADTRG1), you should input the High level to the external trigger pin.

54.6.3 Trigger Delay

The trigger delay function adds a delay to the A/D conversion start trigger in order to adjust the A/D conversion start timing for each scan group. This function adds a delay to the internal trigger for each scan group generated by accepting an external trigger, ELC trigger or GPT trigger. This function cannot be used to add a delay to a software trigger.

The delay value to be added to the internal trigger is set for each scan group in ADTRGDLRi (i = 0 to 4) register. The delay value to be added is the register setting value × A/D conversion clock (ADCLK) cycle.

54.6.4 Force Stops the A/D Conversion Operation

The scanning operation of the A/D converter can be forcibly stopped by writing 1 to ADSTOPR.ADSTOPm (m = 0, 1) bit during scanning operation. If the scanning operation is forcibly stopped, the A/D conversion data is not guaranteed.

54.6.4.1 Forced Stop Procedure

Table 54.56 shows the procedure for forced stop of A/D conversion operation. Observe Table 54.56 when performing the forced stop. If this procedure is violated, the A/D converter may not be able to stop and then it may not operate normally. In that case, a reset is required for recovery.

Table 54.56 Procedure for forced stop of A/D conversion operation

No.	Step Name	Description
1	Disable trigger input	Disables the trigger input from the peripheral module. Write ADTRGENR.STTRGENn = 0.
2	Waiting time	After setting the above Step 1, a wait time is required to safely stop the A/D converter. Proceed to the next processing after the specified waiting time has elapsed. For details on the wait time, see section 54.6.4.2. Waiting Time after Disabling Trigger Input for Forced Stop Processing .
3	Checking the A/D converter status	Check if the A/D converter is operating after the waiting time of the above Step 2 has elapsed. If the A/D converter is running (ADSR.ADACTm = 1), proceed to Step 4. If the A/D converter is stopped (ADSR.ADACTm = 0), no further processing is required (proceed to Step 6).
4	Forced stop of A/D converter	Forcibly stop the A/D converter by ADSTOPR register. Write ADSTOPR.ADSTOPm = 1.
5	Waiting for A/D converter to stop	Wait until the A/D converter stops. ADSR.ADACTm = 0.
6	End	Processing of forced stop is completed.

Note: m = 0, 1, n = 0 to 8

Do not write to ADSYSTR register or ADSTRn register while this forced stop procedure is in progress.

54.6.4.2 Waiting Time after Disabling Trigger Input for Forced Stop Processing

In the forced stop procedure shown in Table 54.56, after disabling the trigger input from the peripheral module, a wait time is required to perform the forced stop (Step 2 of Table 54.56). This wait time is the time that must be secured to perform a forced stop safely.

The wait time after disabling the trigger input for forced stop processing is calculated as follows.

[Waiting time after disabling trigger input in forced stop processing]

- When synchronous operation is enabled (ADSYCR.ADSYDISm = 0)
 - $(\text{ADTRGDLRi.TRGDLYn}[7:0] \text{ setting value} + \text{ADSYCR.ADCSYCYC}[7:0] \text{ setting value} \times 2) \times t_{\text{ADCLK}}$
- When synchronous operation is disabled (ADSYCR.ADSYDISm = 1)
 - $(4 + \text{ADTRGDLRi.TRGDLYn}[7:0] \text{ setting value}) \times t_{\text{ADCLK}}$

Note: i = 0 to 4, n = 0 to 8, m = 0, 1 t_{ADCLK} : ADCLK period

After the waiting time calculated by the above expressions has elapsed, proceed to the Step 3 in Table 54.56. There are two ways to generate the waiting time: (a) Use the timer function built into MCU. (b) Read the status register multiple times. If you choose (b), the number of register reads to generate the waiting time required is calculated by the following equation:

[Number of register reads to generate the waiting time]

$$N_{\text{RD}} = (\text{Wait time after disabling trigger input} \div (t_{\text{PCLK}} \times N_{\text{RDCYC}})) + 1$$

N_{RD} : Number of register reads required to generate the trigger input disable wait time (truncate after the decimal point)

t_{PCLK} : The period of the Bus I/F clock (PCLKA)

N_{RDCYC} : Minimum number of register read access cycles (see section x, I/O Registers [DEADBEEF].)

54.7 Error Detection

54.7.1 A/D Converter Error

An A/D converter error is detected when an abnormal operation of the A/D converter occurs. If an A/D converter error is detected, the A/D conversion result is not guaranteed.

An A/D converter error occurs in the following cases:

[A/D Converter Error Generation Conditions]

- When A/D conversion is performed without self-calibration.
- When the operating frequency of ADCLK exceeds the guaranteed operating frequency range specified in the electric characteristics.
- When the successive approximation time of the A/D converter exceeds the guaranteed operating range specified in the electric characteristics.
- When an accidental abnormal operation occurs in the A/D converter due to external factors.

If an A/D converter error is detected, the flag is set in ADERSR.ADERFn (n = 0, 1) bit. To clear the A/D converter error flag, write 1 to the corresponding bit in ADERSCR register.

Notes on A/D Converter Error

The A/D converter error is an auxiliary function used to detect abnormal operation of the A/D converter. It does not guarantee the reliable detection of abnormal operation. Even under the A/D converter error occurrence conditions described above, an A/D converter error may not be detected depending on the individual chip differences and chip operating conditions.

54.7.2 A/D Conversion Overflow

A/D conversion overflow is detected when A/D conversion data exceeds the range that can be handled in the specified data format. If A/D conversion overflow occurs, the A/D conversion data becomes the upper or lower limit value of the specified data format. For the range that can be handled as A/D conversion data, see [section 54.5.7. Data Format](#).

A/D conversion overflow is detected in the following cases:

- When the input to the A/D converter exceeds the range between VREFH0 to VREFL0 or VREFH to VREFL
 - Single-ended input: When the input to the A/D converter unit 0 exceeds VREFH0 or falls below VREFL0. When the input to the A/D converter unit 1 exceeds VREFH or falls below VREFL.
 - Differential input: When the differential input to the A/D converter unit 0 exceeds +VREFH0 or falls below -VREFH0. When the differential input to the A/D converter unit 1 exceeds +VREFH or falls below -VREFH.
- When overflow occurs by the internal processing (calculation) for the A/D conversion data shown in the following:
 - Gain error and Offset error calibration
 - User's Gain/User's Offset adjustment function
 - When using A/D-converted value addition function
 - Data formatting process

When an A/D conversion overflow is detected, the flag is set in the following status register:

- ADOVFERSR: The flag is set to the bit corresponding to the A/D converter that detected the A/D conversion overflow.
- ADOVFCHSR0: The flag is set to the bit corresponding to the analog input channel on which A/D conversion overflow is detected.
- ADOVFEXSR: The flag is set to the bit corresponding to the extended analog channel on which A/D conversion overflow is detected.

To clear the flags in the status registers, write 1 to the corresponding bit in ADOVFERSCR, ADOVFCHSR0, ADOVFEXSCR.

54.7.3 FIFO Overflow

If A/D conversion data is written (added) to FIFO while FIFO is full, a FIFO overflow is detected. For more details about FIFO overflow, see [section 54.5.10. FIFO Function](#).

54.8 Procedure for Setting up and Changing A/D Conversion

54.8.1 Initial Setup Procedure for A/D Conversion

[Table 54.57](#) shows the initial setup procedure for A/D conversion.

Table 54.57 Initial setup procedure for A/D conversion

No.	Step	Description
1	Release module-stop	Release the module-stop bit for ADC16H in MSTPCR register.
2	I/O port configuration	Set ASEL bit of the pin used as analog input to 1.
3	Synchronous operation configuration	Set the synchronous operation function. The synchronous operation function is disabled at the initial value of the register after reset release. If the synchronous operation function is used, be sure to enable the synchronous operation function (ADSYCR.ADSYDISm = 0 (m = 0, 1)).
4	ADCLK configuration	Set the clock source and division ratio for ADCLK.*1 Then, set ADCLKENR.CKEN bit to 1, and wait for ADCLK to supply (ADCLKSR.CLKSR = 1).
5	A/D conversion configuration	Configure the A/D conversion settings.
6	Wait for operation stabilization	Wait until the operating stabilization times specified in the Electrical Characteristics have elapsed.
7	Self-calibration	Self-calibration must be executed prior to starting A/D conversion. Set up for self-calibration and execute it. For details, see section 54.3.8. Self-calibration .
8	Trigger configuration	To start A/D conversion by a trigger from peripheral modules, configure the triggers for each scan group.
9	Start of A/D conversion	When a software trigger or a trigger from peripheral modules is input, A/D conversion (scanning operation) starts.

Note 1. When GPTCLK is selected, release the GPT module-stop state using the Module Stop Register E (MSTPCRE). For details, see [section 11, Low Power Mode](#).

54.8.2 Procedure for Changing ADCLK Settings

[Table 54.58](#) shows the procedure for changing ADCLK settings.

Table 54.58 Procedure for changing ADCLK settings (1 of 2)

No.	Step	Description
1	Disable trigger input	Disables the trigger input from the peripheral module. (Write ADTRGENR.STTRGENn = 0 (n = 0 to 8))
2	Stop A/D conversion	Check that all A/D converter is stopped. When A/D conversion is in progress, wait until all A/D conversion is completed or forcibly stop A/D conversion operation. For details about forcibly stop A/D conversion operation, see section 54.6.4. Force Stops the A/D Conversion Operation .
3	Stop ADCLK supplies	Set ADCLKENR.CKEN bit to 0. Then, wait for ADCLK to stop (ADCLKSR.CLKSR = 0).
4	Change ADCLK setting	Change the clock-source and the division ratio for ADCLK.*1.
5	Started supplying ADCLK	Set ADCLKENR.CKEN bit to 1, and wait for ADCLK to supply (ADCLKSR.CLKSR = 1).

Table 54.58 Procedure for changing ADCLK settings (2 of 2)

No.	Step	Description
6	Change the A/D conversion configuration	Change the following settings according to the changed ADCLK frequency. • Successive approximation time for A/D converter • Number of sampling states for A/D conversion • Number of sampling states and hold mode switching states for channel-dedicated sample-and-hold circuit *2 • Number of states for self-calibration operation (A/D converter and channel-dedicated sample-and-hold circuit *2) • Synchronous operation period *3 • Disconnection detection assist period *4 If other settings related to A/D conversion are also to be changed, change them in this step.
7	Wait for operation stabilization	Wait until the operating stabilization times specified in the section 70, Electrical Characteristics have elapsed.
8	Self-calibration	Execute self-calibration operation prior to starting A/D conversion. For details, see section 54.3.8. Self-calibration .
9	Trigger configuration	To start A/D conversion by a trigger from peripheral modules, configure the triggers for each scan group.
10	Start of A/D conversion	When a software trigger or a trigger from peripheral modules is input, A/D conversion (scanning operation) starts.

Note 1. When GPTCLK is selected, release the GPT module stop state using the module stop register E (MSTPCRE). For details, see [section 11, Low Power Mode](#).

Note 2. Setting is not required when channel-dedicated sample-and-hold circuit is not used.

Note 3. No change is required when synchronous operation setting is disabled (ADSYCR, ADSYDISm = 1 (m = 0, 1)).

Note 4. Setting is not required when disconnection detection assist function is not used.

54.8.3 Procedure for Changing the Settings of the A/D Converter

[Table 54.59](#) shows the procedure for changing the A/D conversion configuration. If there is a change in ADCLK setting, follow the procedures in [section 54.8.2. Procedure for Changing ADCLK Settings](#).

Table 54.59 Procedure for changing the settings of the A/D converter

No.	Step	Description
1	Disable trigger input	Disables the trigger input from the peripheral module. (Write ADTRGENR.STTRGENn = 0 (n = 0 to 8))
2	Stop A/D conversion	Check that all A/D converter is stopped. When A/D conversion is in progress, wait until all A/D conversion is completed or forcibly stop A/D conversion operation. For details about forcibly stop A/D conversion operation, see section 54.6.4. Force Stops the A/D Conversion Operation .
3	Change the A/D conversion configuration	Change the A/D conversion settings. (except the settings for ADCLK)
4	Wait for operation stabilization	Wait until the various operating stabilization times specified in the Electrical Characteristics have elapsed.
5	Self-calibration	Execute self-calibration operation prior to starting A/D conversion. For details, see section 54.3.8. Self-calibration .
6	Trigger configuration	To start A/D conversion by a trigger from peripheral modules, configure the triggers for each scan group.
7	Start of A/D conversion	When a software trigger or a trigger from peripheral modules is input, A/D conversion (scanning operation) starts.

54.9 Interrupt Sources and ELC Events

[Table 54.60](#) lists ADC16H interrupt sources or ELC event requests.

For more details about interrupts, see [section 14, Interrupt Controller Unit \(ICU\)](#).

For more details about ELC events, see [section 20, Event Link Controller \(ELC\)](#).

Table 54.60 Interrupt sources (1 of 2)

Interrupt request or ELC event	Symbol	Description	Status flag
A/D converter error interrupt	ADC_ERR0	Generated when abnormal operation of ADC0 is detected.	ADERSR.ADERF0
	ADC_ERR1	Generated when abnormal operation of ADC1 is detected.	ADERSR.ADERF1
A/D converter self-calibration end interrupt	ADC_CALEND0	Generated when self-calibration operation of ADC0 is finished.	ADCALENSR.CALENDF0
	ADC_CALEND1	Generated when self-calibration operation of ADC1 is finished.	ADCALENSR.CALENDF1
A/D scan end interrupt	ADC_ADI0	Generated at the scan end of scan group 0	ADSCANENDSR.SCENDF0
	ADC_ADI1	Generated at the scan end of scan group 1	ADSCANENDSR.SCENDF1
	ADC_ADI2	Generated at the scan end of scan group 2	ADSCANENDSR.SCENDF2
	ADC_ADI3	Generated at the scan end of scan group 3	ADSCANENDSR.SCENDF3
	ADC_ADI4	Generated at the scan end of scan group 4	ADSCANENDSR.SCENDF4
	ADC_ADI5	Generated at the scan end of scan group 5	ADSCANENDSR.SCENDF5
	ADC_ADI6	Generated at the scan end of scan group 6	ADSCANENDSR.SCENDF6
	ADC_ADI7	Generated at the scan end of scan group 7	ADSCANENDSR.SCENDF7
	ADC_ADI8	Generated at the scan end of scan group 8	ADSCANENDSR.SCENDF8
Limiter clip interrupt	ADC_LIMCLPI	Generated when a limiter clip using limiter table 0 to 7 is detected for A/D conversion.	ADLIMGRSR.LIMGRFn ADLIMCHSR0.OVFCHFi ADLIMEXSR.LIMEXFj
A/D conversion overflow interrupt	ADC_RESOVF0	Generated when the overflow is detected in A/D conversion result with ADC0.	ADOVFERSR.ADOVFEF0 ADOVFCHSR0.OVFCHFi ADOVFEXSR.OVFEXFj
	ADC_RESOVF1	Generated when the overflow is detected in A/D conversion result with ADC1.	ADOVFERSR.ADOVFEF1 ADOVFCHSR0.OVFCHFi ADOVFEXSR.OVFEXFj
Compare match interrupt	ADC_CMPi0	Generated when a compare match with compare match table 0 is detected.	ADCMPBTSR.CMPTBF0
	ADC_CMPi1	Generated when a compare match with compare match table 1 is detected.	ADCMPBTSR.CMPTBF1
	ADC_CMPi2	Generated when a compare match with compare match table 2 is detected.	ADCMPBTSR.CMPTBF2
	ADC_CMPi3	Generated when a compare match with compare match table 3 is detected.	ADCMPBTSR.CMPTBF3
Composite compare match interrupt	ADC_CCMPM0 ADC_CCMPM1	Generated when a composite compare match with combined conditions using compare match table 0 to 7 is detected.	ADCMPBTSR.CMPTBF0 to ADCMPBTSR.CMPTBF8
	ADC_CCMPUM0 ADC_CCMPUM1	Generated when a composite compare match with combined conditions using compare match table 0 to 7 is not detected.	ADCMPBTSR.CMPTBF0 to ADCMPBTSR.CMPTBF8

Table 54.60 Interrupt sources (2 of 2)

Interrupt request or ELC event	Symbol	Description	Status flag
FIFO data read request interrupt	ADC_FIFOREQ0	Generated when the number of vacant stages in FIFO for scan group 0 become less than or equal to the specified value.	ADFIFOSR0.FIFOST0[3:0]
	ADC_FIFOREQ1	Generated when the number of vacant stages in FIFO for scan group 1 become less than or equal to the specified value.	ADFIFOSR0.FIFOST1[3:0]
	ADC_FIFOREQ2	Generated when the number of vacant stages in FIFO for scan group 2 become less than or equal to the specified value.	ADFIFOSR1.FIFOST2[3:0]
	ADC_FIFOREQ3	Generated when the number of vacant stages in FIFO for scan group 3 become less than or equal to the specified value.	ADFIFOSR1.FIFOST3[3:0]
	ADC_FIFOREQ4	Generated when the number of vacant stages in FIFO for scan group 4 become less than or equal to the specified value.	ADFIFOSR2.FIFOST4[3:0]
	ADC_FIFOREQ5	Generated when the number of vacant stages in FIFO for scan group 5 become less than or equal to the specified value.	ADFIFOSR2.FIFOST5[3:0]
	ADC_FIFOREQ6	Generated when the number of vacant stages in FIFO for scan group 6 become less than or equal to the specified value.	ADFIFOSR3.FIFOST6[3:0]
	ADC_FIFOREQ7	Generated when the number of vacant stages in FIFO for scan group 7 become less than or equal to the specified value.	ADFIFOSR3.FIFOST7[3:0]
	ADC_FIFOREQ8	Generated when the number of vacant stages in FIFO for scan group 8 become less than or equal to the specified value.	ADFIFOSR4.FIFOST8[3:0]
FIFO data overflow interrupt	ADC_FIFOOVF	Generated when FIFO overflow is detected in any of FIFO for scan group 0 to 8.	ADFIFOERSR.FIFOOVF0 to ADFIFOERSR.FIFOOVF8

Note: n = 0 to 8.
i = 0 to 28.
j = 0 to 2, 5 to 8.

(1) A/D converter error interrupt

A/D converter error interrupt can be generated when an A/D converter error is detected.

A/D converter error interrupt (ADC_ERRm (m = 0, 1)) is generated when ADERINTCR.ADEIEm (m = 0, 1) bit is 1 and ADERSR.ADERFm (m = 0, 1) bit is 1.

(2) A/D converter self-calibration end interrupt

A/D converter self-calibration end interrupt can be generated at the end of A/D converter self-calibration operation.

A/D converter self-calibration end interrupt (ADC_CALENDm (m = 0, 1)) is generated when ADCALINTCR.CALENDIEm (m = 0, 1) bit is 1 and ADCALENSR.CALENDFm (m = 0, 1) bit is 1.

(3) A/D scan end interrupt

A/D scan end interrupt can be generated at the end of scanning operation of scan group n (n = 0 to 8).

A/D scan end interrupt for scan group 0 to 8 (ADC_ADi0 to ADC_ADi8) are generated when ADINTCR.ADIEn ($n = 0$ to 8) bit is set to 1 and ADSCANENDSR.SCENDFn ($n = 0$ to 8) bit is set to 1.

However, A/D scan end interrupt is not generated when A/D conversion operation (scanning operation) is forcibly stopped by ADSTOPR register.

(4) Limiter clip interrupt

Limiter clip interrupt (ADC_LIMCLPI) can be generated when a limiter clip with limiter table i ($i = 0$ to 7) is detected.

Limiter clip interrupt is generated when either ADLIMINTCR.LIMIEi ($i = 0$ to 8) bit is 1 and ADLIMGRSR.LIMGRFi ($i = 0$ to 8) bit is 1 is satisfied.

(5) A/D conversion overflow interrupt

A/D conversion overflow interrupt can be generated when A/D conversion overflow in the A/D conversion result using ADCm ($m = 0, 1$) is detected. For details about A/D conversion overflow, see [section 54.7.2. A/D Conversion Overflow](#).

A/D conversion overflow interrupt (ADC_RESOVFm ($m = 0, 1$)) is generated when ADOVFINTCR.ADOVFIEm ($m = 0, 1$) bit is 1 and ADOVFERSR.ADOVFEFm ($m = 0, 1$) bit is 1.

(6) Compare match interrupt

Compare match interrupt can be generated when a compare match is detected.

Compare match interrupt (ADC_CMPIj ($j = 0$ to 3)) is generated when ADCMPINTCR.CMPIEj ($j = 0$ to 3) bit is 1 and ADCMPTBSR.CMPTBFj ($j = 0$ to 3) bit is 1.

There are no interrupts corresponding to the compare match tables 4 to 7.

(7) Composite compare match interrupt

Composite compare match interrupt (ADC_CCMPMk ($k = 0, 1$)) can be generated by combining the comparison results of two or more compare match tables. For details about composite compare match, see [section 54.5.8.2. Composite Compare Match](#).

(8) FIFO data read request interrupt

FIFO data read request interrupt can be generated when the number of vacant stages in FIFO become less than or equal to the specified value.

FIFO data read request interrupt for scan group 0 to 8 (ADC_FIFOREQ0 to ADC_FIFOREQ8) are generated when ADFIFOINTCR.FIFOIE n ($n = 0$ to 8) is set to 1 and ADFIFOSRm.FIFOST n [3:0] $\leq$ ADFIFOINTLRm.FIFOILV n [3:0] ($m = 0$ to 4, $n = 0$ to 8).

However, FIFO read request interrupt is not generated while the ADFIFOERSR.FIFOFLF n ($n = 0$ to 8) bit corresponding to its interrupt source is set to 1.

(9) FIFO data overflow interrupt

FIFO data overflow interrupt (ADC_FIFOOVF) can be generated when a FIFO overflow is detected in any of FIFO in scan groups 0 to 8.

FIFO data overflow interrupt is generated when ADFIFOINTCR.FIFOIE n ($n = 0$ to 8) bit is 1 and ADFIFOERSR.FIFOOVF n ($n = 0$ to 8) bit is 1.

54.10 Scan Conversion Time

54.10.1 Scan Start Processing Time

The processing time from the input of the A/D conversion start trigger to the start of the scanning operation is shown in [Table 54.61](#) and [Figure 54.56](#).

Table 54.61 Scan start processing time

Item	Symbol	Processing time
Peripheral module trigger input processing time ^{*1}	t_{D_TRG}	[When GPIO trigger is selected] 2 to 4 ADRCCLK + (ADTRGDLRi.TRGDLYn[7:0] + (0 to 1)) × ADCLK [When other than above] 1 ADRCCLK + (ADTRGDLRi.TRGDLYn[7:0] + (0 to 1)) × ADCLK
Software trigger input processing time	t_{D_SW}	[When ADCLK = PCLKA/1] ^{*2} - Number of access cycles to I/O registers^{*3} + (0 to 1 ADCLK) [Other than above] (Number of access cycles to I/O registers^{*3}) + 1 PCLKA + (1 to 3 ADCLK)
Internal Trigger processing time	t_{D_ITRG}	[When synchronous operation is disabled] (ADSYCR.ADSYDISm = 1) 6 ADCLK [Idle start] 7 to 13 ADCLK [Suspend start]^{*4} [When synchronous operation is enabled] (ADSYCR.ADSYDISm = 0) 7 ADCLK [Idle start] (3+ADSYCR.ADSYCYC[7:0]) × ADCLK to (2+2×ADSYCR.ADSYCYC[7:0]) × ADCLK [Suspend start]^{*5}
Wait Time for Synchronous Operation	t_{D_SYOP}	[When synchronous operation is disabled] (ADSYCR.ADSYDISm = 1) 0 [When synchronous operation is enabled] (ADSYCR.ADSYDISm = 0) 0 to ADSYCR.ADSYCYC[7:0] × ADCLK
Total scan start processing time	Started by Peripheral Module Trigger	t_{D_ADST}
	Started by Software Trigger	$t_{D_SW} + t_{D_ITRG} + t_{D_SYOP}$

Note: ADRCCLK: Clock source of ADCLK
n = 0 to 8, m = 0, 1, i = 0 to 4

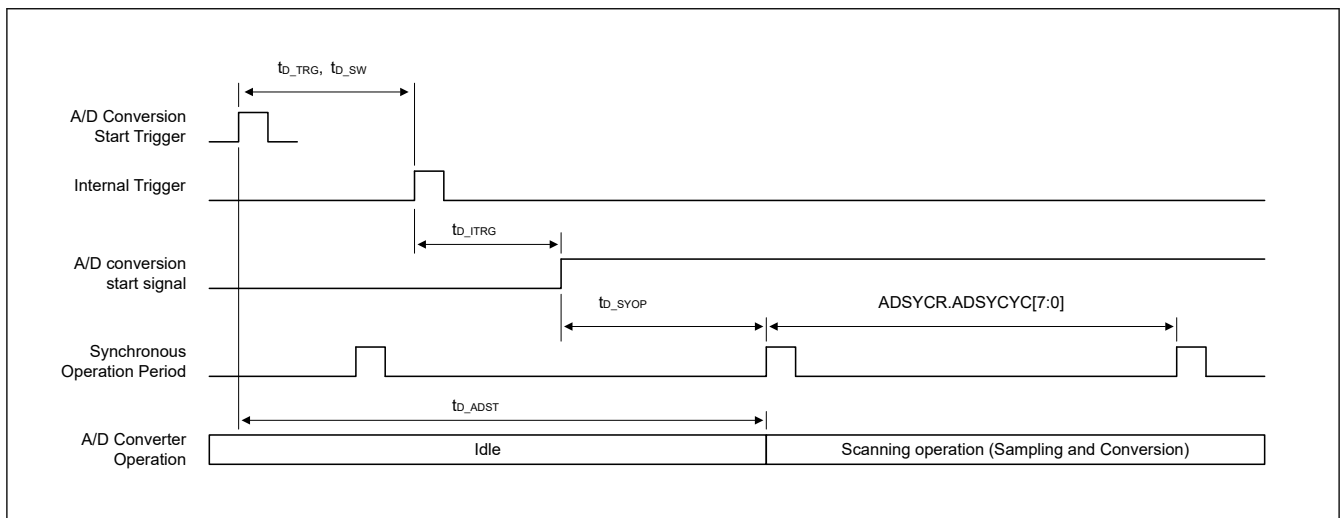
Note 1. It does not include the input delay time from the source of the peripheral module trigger to ADC16H.

Note 2. When ADCLKCR.CLKSEL[1:0] = 10b and ADCLKCR.DIVR[2:0] = 000b are set.

Note 3. It is the number of access cycles to ADSTRn register (n = 0 to 8) or ADSYSTR register. For more details about the number of access cycles for I/O registers, see section x, I/O Registers [DEADBEEF].

Note 4. When constant sampling is enabled, it is 7 to (6 + ADHSSTRm.SHSST[7:0]) × ADCLK. When constant sampling is enabled, by setting the sampling time setting of the channel-dedicated sample-and-hold circuit to ADHSSTRm.SHSST[7:0] = 0x4 (reset value), the effect of the ADHSSTRm.SHSST[7:0] setting on the conversion start time can be ignored.

Note 5. When constant sampling is enabled, (3+ADSYCR.ADSYCYC[7:0]) × ADCLK to (2+ADHSSTRm.SHSST[7:0] + ADSYCR.ADSYCYC[7:0]) × ADCLK. When constant sampling is enabled, by setting the sampling time setting of the channel-dedicated sample-and-hold circuit to ADHSSTRm.SHSST[7:0] = 0x4 (reset value), the effect of the ADHSSTRm.SHSST[7:0] setting on the conversion start time can be ignored.

**Figure 54.56 Scan start processing time**

54.10.2 Conversion Processing Time

The various processing times in the conversion operation are shown in [Table 54.62](#) and [Figure 54.57](#) to [Figure 54.61](#).

Table 54.62 A/D conversion processing time (1 of 2)

Item			Symbol	Processing Time		
Channel-dedicated sample-and-hold processing time	Sampling time		t _{SH_SPL}	ADSHSTRm.SHSST[7:0] × ADCLK		
	Hold mode switching time		t _{SH_HLD}	ADSHSTRm.SHHST[2:0] × ADCLK		
	Sampling mode switching time (only when Hybrid mode)		t _{SH_D}	(ADSYCR.ADSYCYC[7:0] - 1) × ADCLK		
Disconnection detection assist processing time			t _{DDA}	[When disconnection detection assist function is disabled] 0 [When disconnection detection assist function is enabled] - ADSGDCRn.ADNDIS[7:0] × ADCLK		
A/D conversion time	Sampling time		t _{AD_SPL}	ADSSTRp.SSTq[9:0] × ADCLK		
	Successive approximation time		t _{AD_CNV}	ADCNVSTR.CSTm[5:0] × ADCLK		
A/D conversion data processing time	SAR mode (Not using digital filter function)	When ADCLK = PCLKA/1 is set*1	t _{ADDP}	[When A/D-converted value addition/averaging function is not used] 6 ADCLK + 2 PCLKA [When A/D-converted value addition/averaging function is used] 7 ADCLK + 2 PCLKA		
		Other than above		[When A/D-converted value addition/averaging function is not used] 7 ADCLK + (5 to 6 PCLKA) [When A/D-converted value addition/averaging function is used] 8 ADCLK + (5 to 6 PCLKA)		
	Oversampling mode or Hybrid mode (Using with digital filter function)	When ADCLK = PCLKA/1 is set*1		[When A/D-converted value addition/averaging function is not used] 8 ADCLK + 2 PCLKA [When A/D-converted value addition/averaging function is used] 9 ADCLK + 2 PCLKA		
		Other than above		[When A/D-converted value addition/averaging function is not used] 9 ADCLK + 5 to 6 PCLKA [When A/D-converted value addition/averaging function is used] 10 ADCLK + 5 to 6 PCLKA		
	Total A/D conversion time (SAR mode)*3	Channel conversion time*2		t _{ADCH_S}	(t _{DDA} + t _{AD_SPL} + t _{AD_CNV}) × N _{ADC} × ADCLK	
		Scan conversion time*4		t _{SCAN_S}	[When channel-dedicated sample-and-hold circuit is not used] - Σt_{ADCH_S} [When channel-dedicated sample-and-hold circuit is used] - t_{SH_SPL} + t_{SH_HLD} + Σt_{ADCH_S}	
Total A/D conversion time (Oversampling mode)*3	Oversampling period		t _{OV_OS}	(t _{DDA} + t _{AD_SPL} + t _{AD_CNV}) × ADCLK		
	Channel conversion time		t _{ADCH_O}	(t _{DDA} + t _{AD_SPL} + t _{AD_CNV}) × (N _{TAP} + N _{ADC}) × ADCLK		
	Scan conversion time*5		t _{SCAN_O}	Σt _{ADCH_O}		

Table 54.62 A/D conversion processing time (2 of 2)

Item		Symbol	Processing Time
Total A/D conversion time (Hybrid mode)* ³	Channel-dedicated sample-and-hold processing time in Hybrid mode	t_{HY_SH}	$t_{SH_SPL} + t_{SH_HLD} + t_{SH_D}$
	Oversampling period	t_{HY_OS}	$(t_{DDA} + t_{AD_SPL} + t_{AD_CNV}) \times ADCLK$
	Scan conversion time	t_{HY_ID}	[When channel-dedicated sample-and-hold circuit is not used] $(N_{TAP} + N_{ADC}) \times \Sigma t_{HY_OS}^*{}^6$ [When channel-dedicated sample-and-hold circuit is used] $(t_{HY_SH} \times (N_{ADC} + N_{TAP}) - t_{SH_D}) + (N_{TAP} + N_{ADC}) \times \Sigma t_{HY_OS}^*{}^6$
	After initial delay time	t_{SCAN_HY}	[When channel-dedicated sample-and-hold circuit is not used] $\Sigma t_{HY_OS}^*{}^6$ [When channel-dedicated sample-and-hold circuit is used] $t_{HY_SH} + \Sigma t_{HY_OS}^*{}^6$

Note: $n = 0$ to 8 , $m = 0$, 1 , $p = 0$ to 7 , $q = 0$ to 15

N_{ADC} : This value is the number of the times of the addition/averaging according to the setting value of $ADDOPCRBx.ADC[3:0]$ ($x = 0$ to 32). If A/D-converted value addition/averaging function is not used, this value is 1 .

N_{TAP} : This value is the number of TAP of the digital filter that selected in $ADDOPCRAX.DFSEL[2:0]$ ($x = 0$ to 32) and

$ADDFSRm.DFSELy[1:0]$ ($m = 0, 1$, $y = 0$ to 4).

N_{SGCH} : This value is the number of the channels in the scan group.

Note 1. When $ADCLKCR.CLKSEL[1:0] = 10b$ and $ADCLKCR.DIVR[2:0] = 000b$ are set.

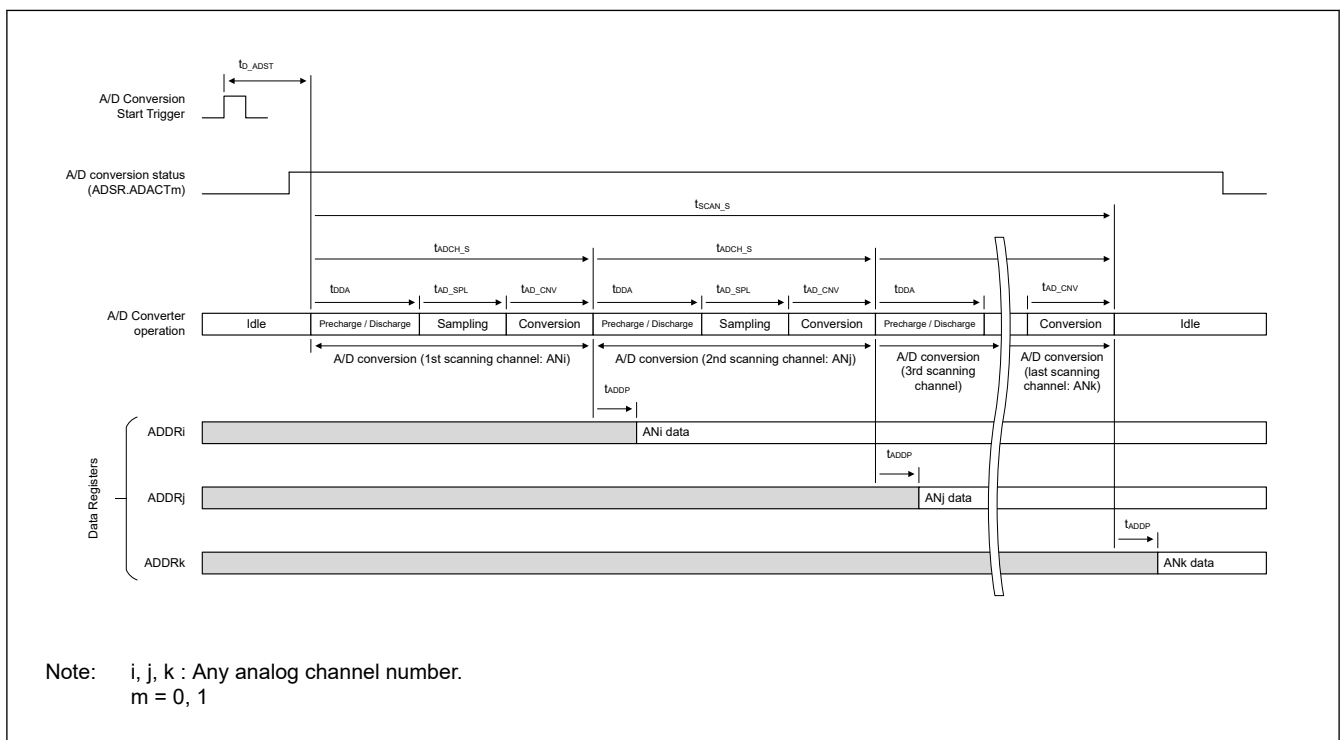
Note 2. It does not include the channel-dedicated sample-and-hold processing time.

Note 3. It does not include the A/D conversion data processing time.

Note 4. This is the sum of the channel conversion times (t_{ADCH_N}) that calculated from the conversion settings for each analog channel assigned to the scan group. If channel-dedicated sample-and-hold circuits are used, channel-dedicated sample-and-hold processing times (t_{SH_SPL} and t_{SH_HLD}) are also added.

Note 5. This is the sum of the channel conversion times (t_{ADCH_O}) that calculated from the conversion settings for each analog channel assigned to the scan group.

Note 6. This is the sum of the oversampling period (t_{HY_OS}) that calculated from the conversion settings for each analog channel assigned to the scan group.

**Figure 54.57 A/D conversion processing time (SAR mode)**

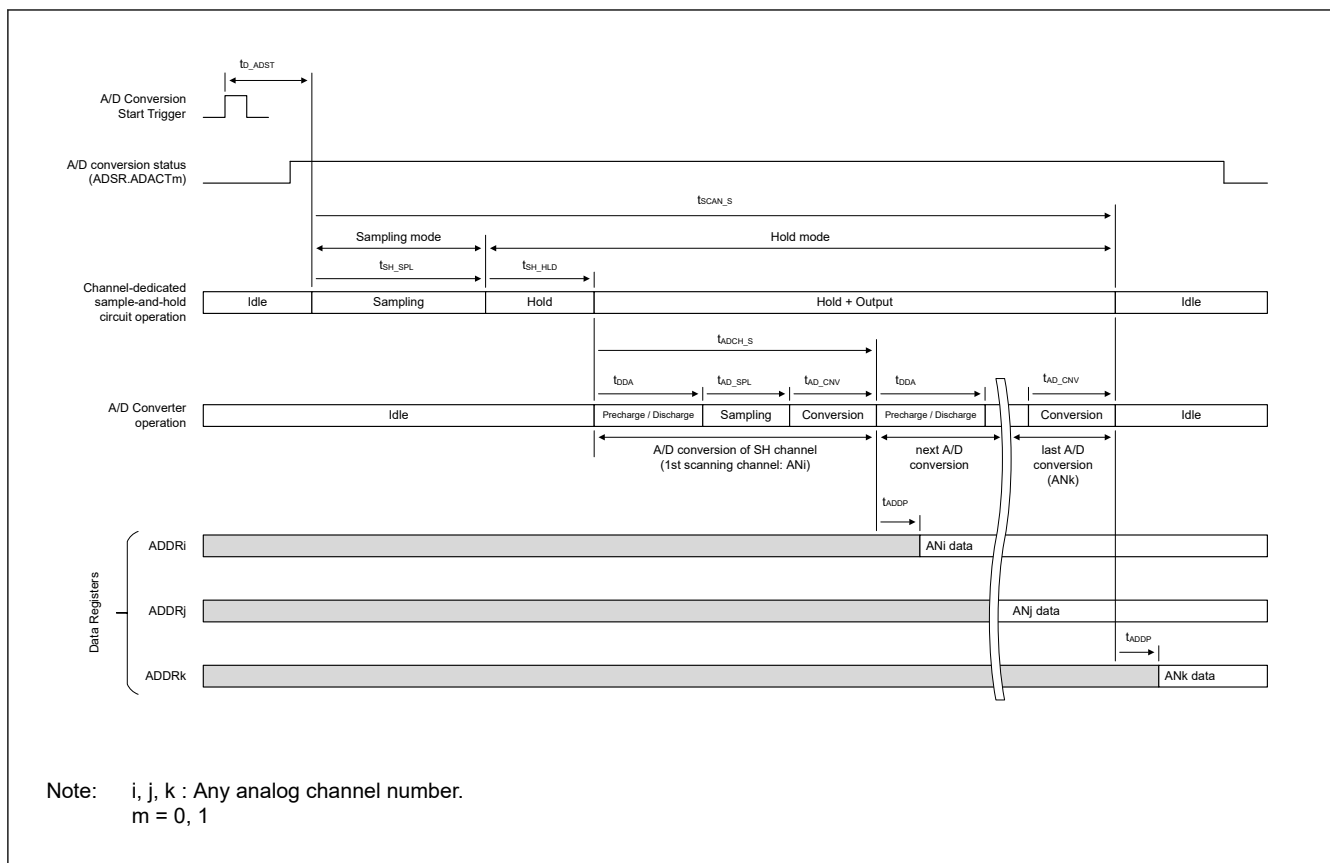


Figure 54.58 A/D conversion processing time (SAR mode with channel-dedicated sample-and-hold circuit)

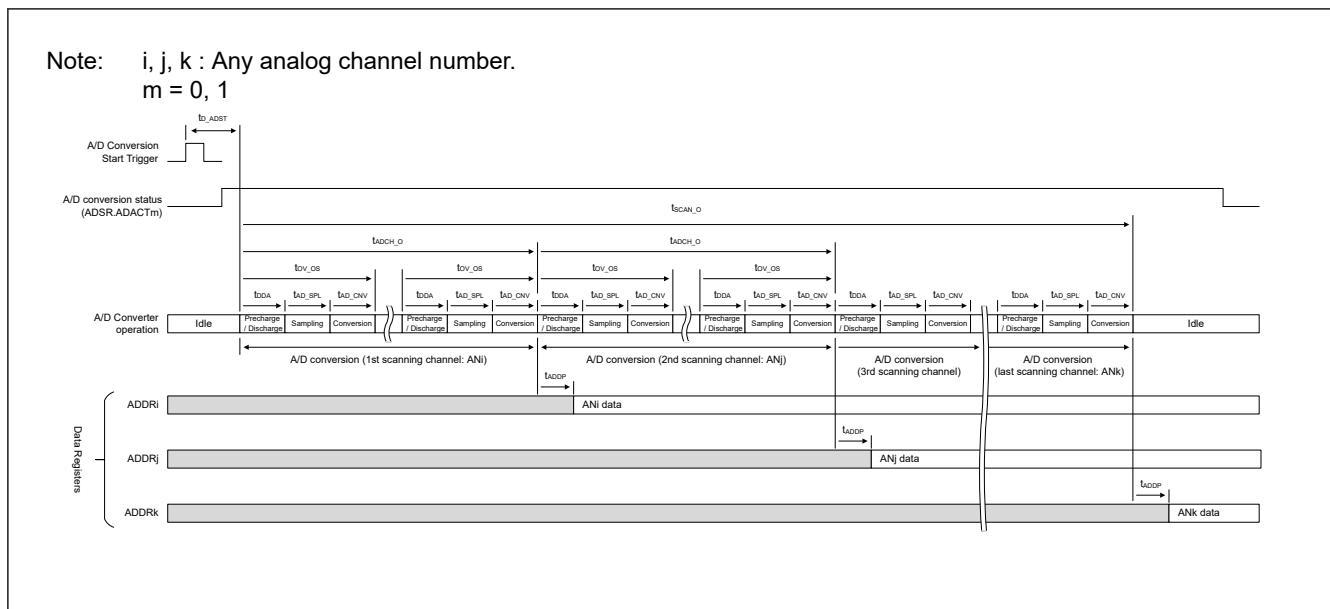


Figure 54.59 A/D conversion processing time (Oversampling mode)

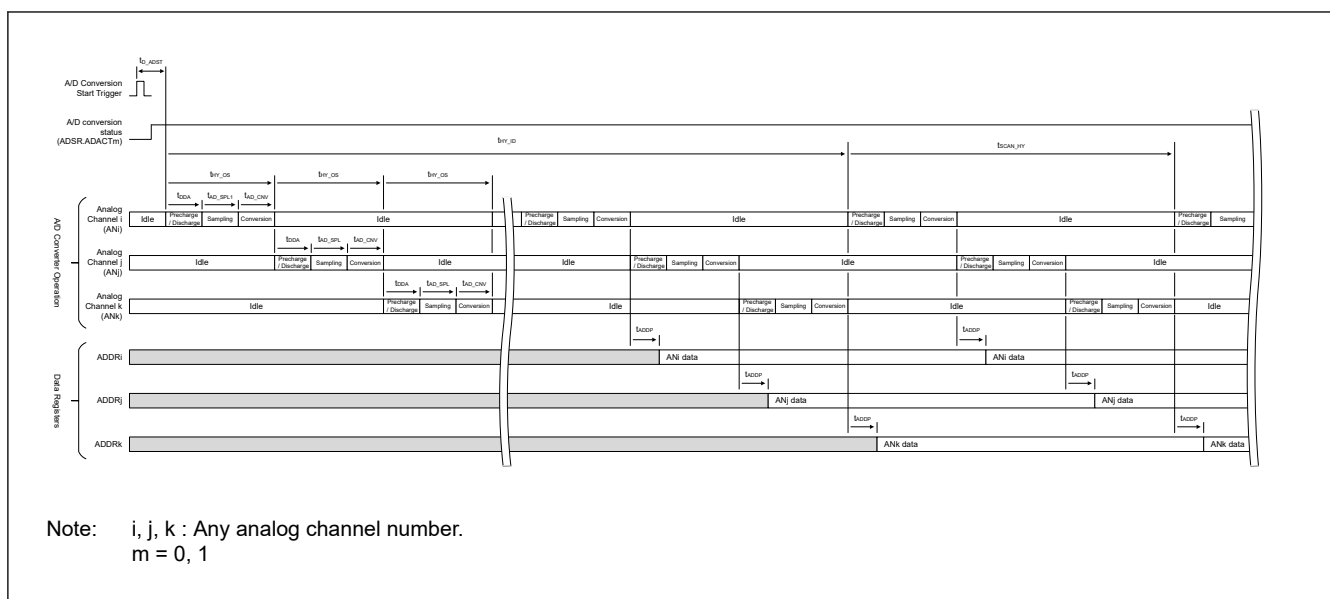


Figure 54.60 A/D conversion processing time (Hybrid mode)

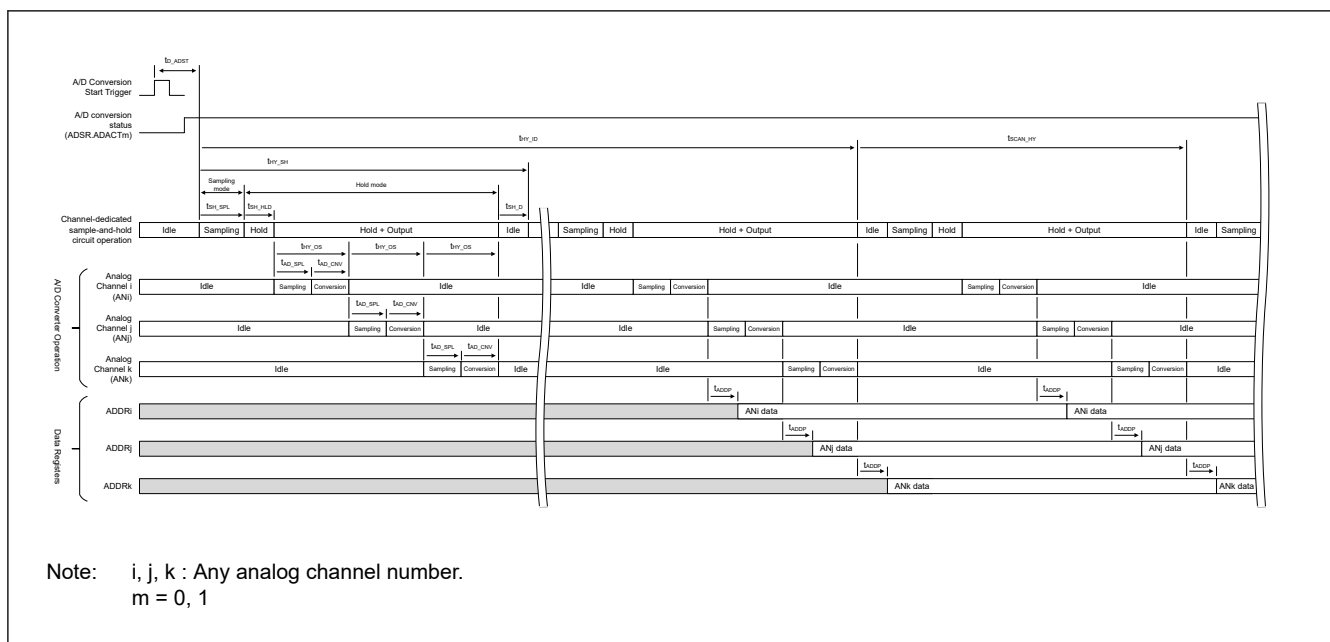


Figure 54.61 A/D conversion processing time (Hybrid mode with channel-dedicated sample-and-hold circuit)

54.10.3 Scan End Processing Time

The scan end processing time and the forcibly stop processing time are shown in Table 54.63, Figure 54.62, Figure 54.63.

Table 54.63 Scan end processing time and forcibly stop processing time (1 of 2)

Item	Symbol	Processing Time
Scan end processing time	Until reflected in the status register*1	t_{ED1}
	Until output the FIFO data read request interrupt	t_{ED2}
		$t_{ED1} + 2 \text{ PCLKA}$

Table 54.63 Scan end processing time and forcibly stop processing time (2 of 2)

Item		Symbol	Processing Time
Forcibly Stop processing time	Forcibly stop trigger input processing time	$t_{\text{STOP_TRG}}$	[When ADCLK = PCLKA/1 is set] ^{*2} - Number of access cycles to I/O registers^{*3} + 1 PCLKA [Other than above] (Number of access cycles to I/O registers^{*3}) + 1 PCLKA + (3 to 4 ADCLK)
	Wait time for synchronous operation	$t_{\text{STOP_SYNC}}$	[When synchronous operation is disabled] (ADSYCR.ADSYDISm = 1) 0 [When synchronous operation is enabled] (ADSYCR.ADSYDISm = 0) 0 to ADSYCR.ADSYCYC[7:0] × ADCLK
	Forcibly stop processing time	t_{STOP}	[When ADCLK = PCLKA/1 is set] ^{*2} (ADSYCR.ADSYDISm = 1) 4 PCLKA (ADSYCR.ADSYDISm = 0) 3 PCLKA [Other than above] (ADSYCR.ADSYDISm = 1) 4 PCLKA + (2 to 3 PCLKA) (ADSYCR.ADSYDISm = 0) 3 PCLKA + (2 to 3 PCLKA)

Note: $m = 0, 1$

Note 1. This is the time it takes for ADSCANENDSR.SCENDFn = 1 (n = 0 to 8) or until an end-of-scan interrupt is generated.

Note 2. When ADCLKCR.CLKSEL[1:0] = 10b and ADCLKCR.DIVR[2:0] = 000b are set.

Note 3. It is the number of access cycles to ADSTOPR register. For more details about the number of access cycles for I/O registers, see section x, I/O Registers [DEADBEEF].

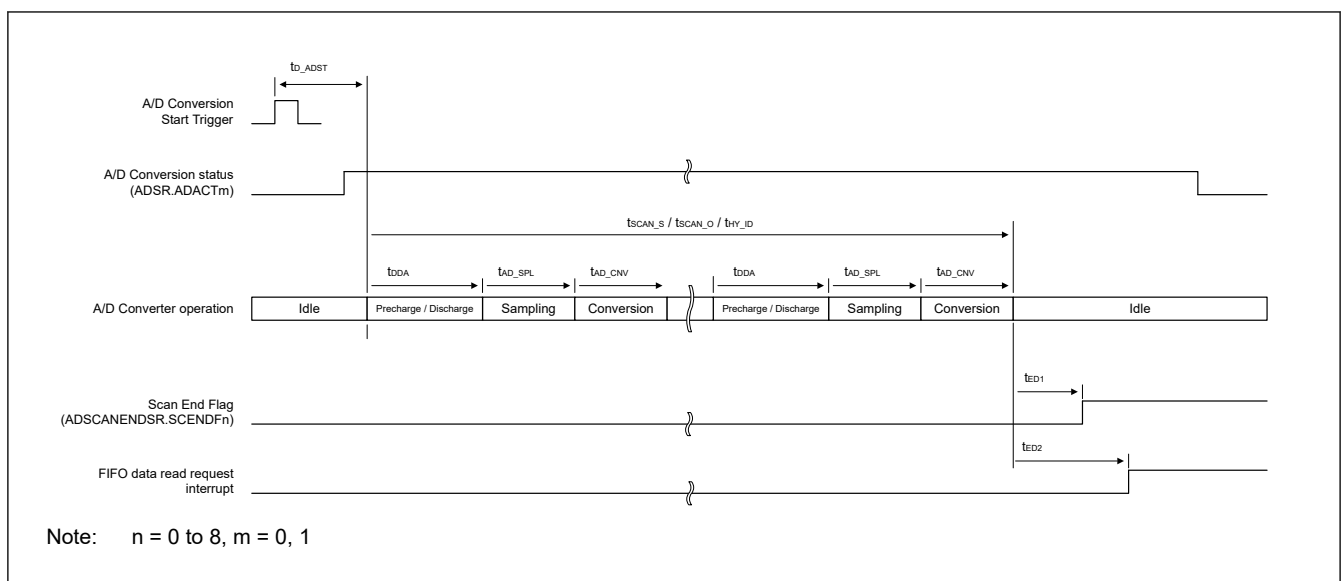


Figure 54.62 Scan end processing time (SAR/Oversampling/Hybrid mode – Single scan mode)

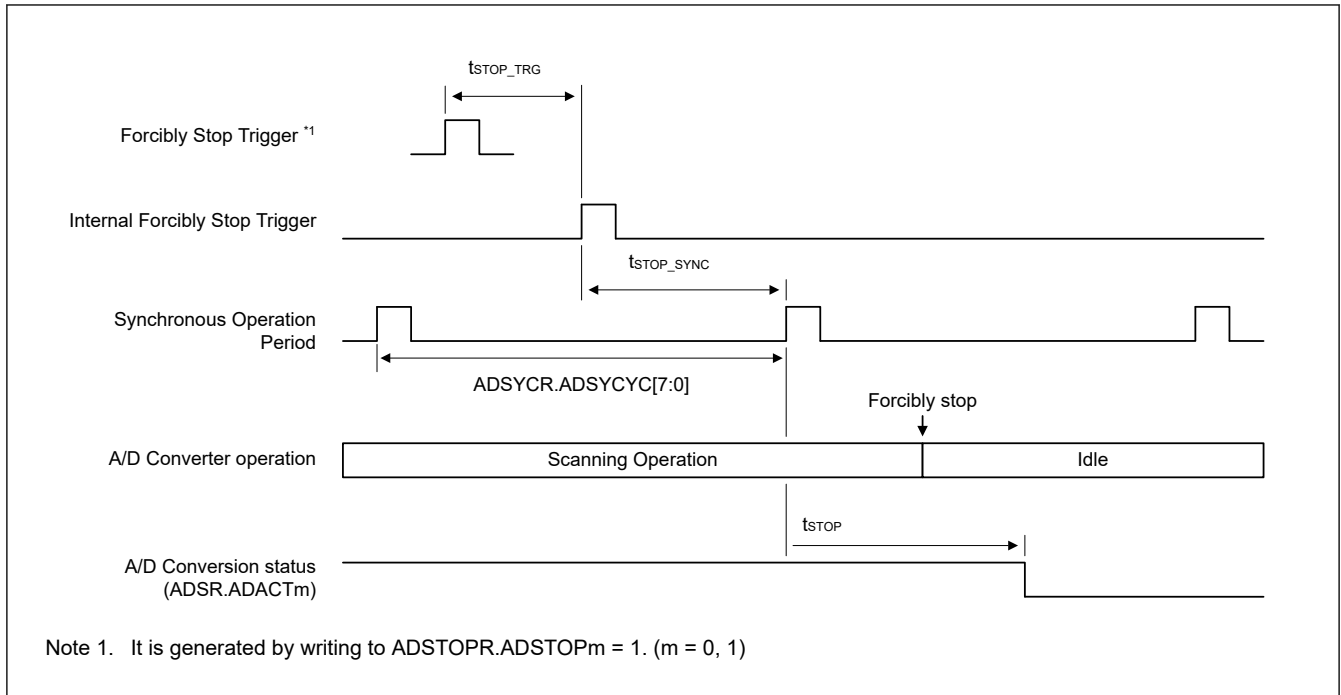


Figure 54.63 Forcibly stop processing time (SAR/Oversampling/Hybrid mode – Single scan mode)

54.11 Usage Notes

54.11.1 Prohibition of Changing the Operation Settings During A/D Conversion Operation

The registers related to the operation setting of the A/D converter should be set while all A/D converters are stopped (ADSR.ADACTm = 0 and ADSR.CALACTm = 0 (m = 0, 1)). Changing (writing) of registers except for those listed below is prohibited during A/D conversion. If the operation setting is changed during A/D conversion, operation is not guaranteed.

[Registers that can be written while the A/D converter is operating]

- Status clear registers
 - Status clear registers related to A/D converter operation (ADERSCR, ADCALENDSR, ADSCANENDSCR)
 - A/D conversion overflow status clear registers (ADOVFERSCR, ADOVFCHSCR0, ADOVFEXSCR)
 - Limiter clip status clear register (ADLIMGRSCR, ADLIMCHSCR0, ADLIMEXSCR)
 - Compare match status clear register (ADCMPBTBSCR, ADCMPCHSCR0, ADCMPPEXSCR)
 - FIFO error status clear register (ADFIFOERSCR)
- Software trigger register (ADSYSTR, ADSTRn (n = 0 to 8))
- A/D converter stop register (ADSTOPR)
- A/D converter start trigger enable register (ADTRGENR) *1

Note 1. Writing during operation is only permitted for disabling the trigger input (ADTRGENR.STTRGENn = 0 (n = 0 to 8)) in order to stop the A/D conversion. To avoid unintended operation, do not change the trigger input to Enable (ADTRGENR.STTRGENn = 1) during operation.

54.11.2 Usage Notes on Forced Stop of A/D Conversion

If you want to forcibly stop the A/D conversion operation, follow the procedure in [section 54.6.4. Force Stops the A/D Conversion Operation](#).

54.11.3 Usage Notes on A/D Data Registers

When A/D conversion is performed multiple times on the same analog channel, A/D data register ADDR_n (n = 0 to 22) or A/D extended analog data register ADEXDR_m (m = 0, 1, 4, 5, 6, 8, 9, 16, 17, 18, 20, 21, 22) corresponding to that analog channel is overwritten at the later A/D conversion.

If you intend to keep the data for each A/D conversion for the same analog channel, perform one of the following methods.

- Read out data from registers at each scanning operation
 - At the end of scanning each scan group, read out the A/D conversion data of the analog channel that was subject to A/D conversion from ADDR_n or ADEXDR_m.
 - This method is effective if A/D conversion of the same analog channel is not performed more than once in the same scan group, and A/D conversion data can be read from the registers before the next scanning operation.
- Use FIFO function to hold data for each A/D conversion
 - By using FIFO, multiple A/D conversion data for the same analog channel can be kept.
 - Read out the A/D conversion data before FIFO overflow.

54.11.4 Settings for the Module-stop Function

The Module Stop Control Register can enable or disable ADC16H operation. The ADC16H is initially stopped after a reset. The registers become accessible on release from the module-stop state.

See [section 54.11.6. Restrictions on Entering and Releasing the Low-Power States](#) in connection with this usage note.

54.11.5 Settings for the Software Standby Mode

Before entering Software Standby mode, stop ADC16H operation according to the procedures described in [Figure 54.64](#) and [Figure 54.65](#).

Note: For details on the wait time, see [section 54.6.4.2. Waiting Time after Disabling Trigger Input for Forced Stop Processing](#).

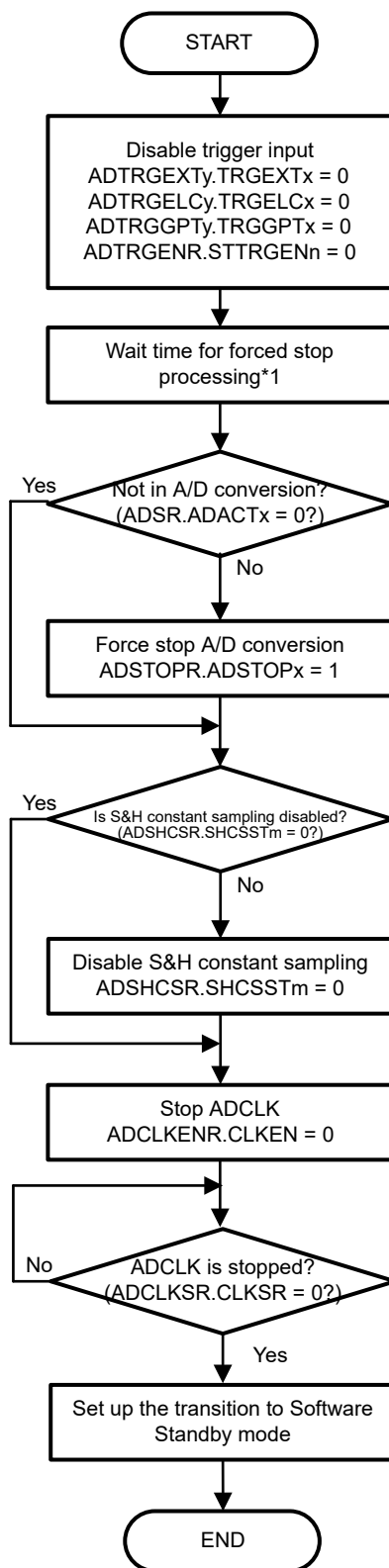


Figure 54.64 Example procedure for entering Software Standby mode

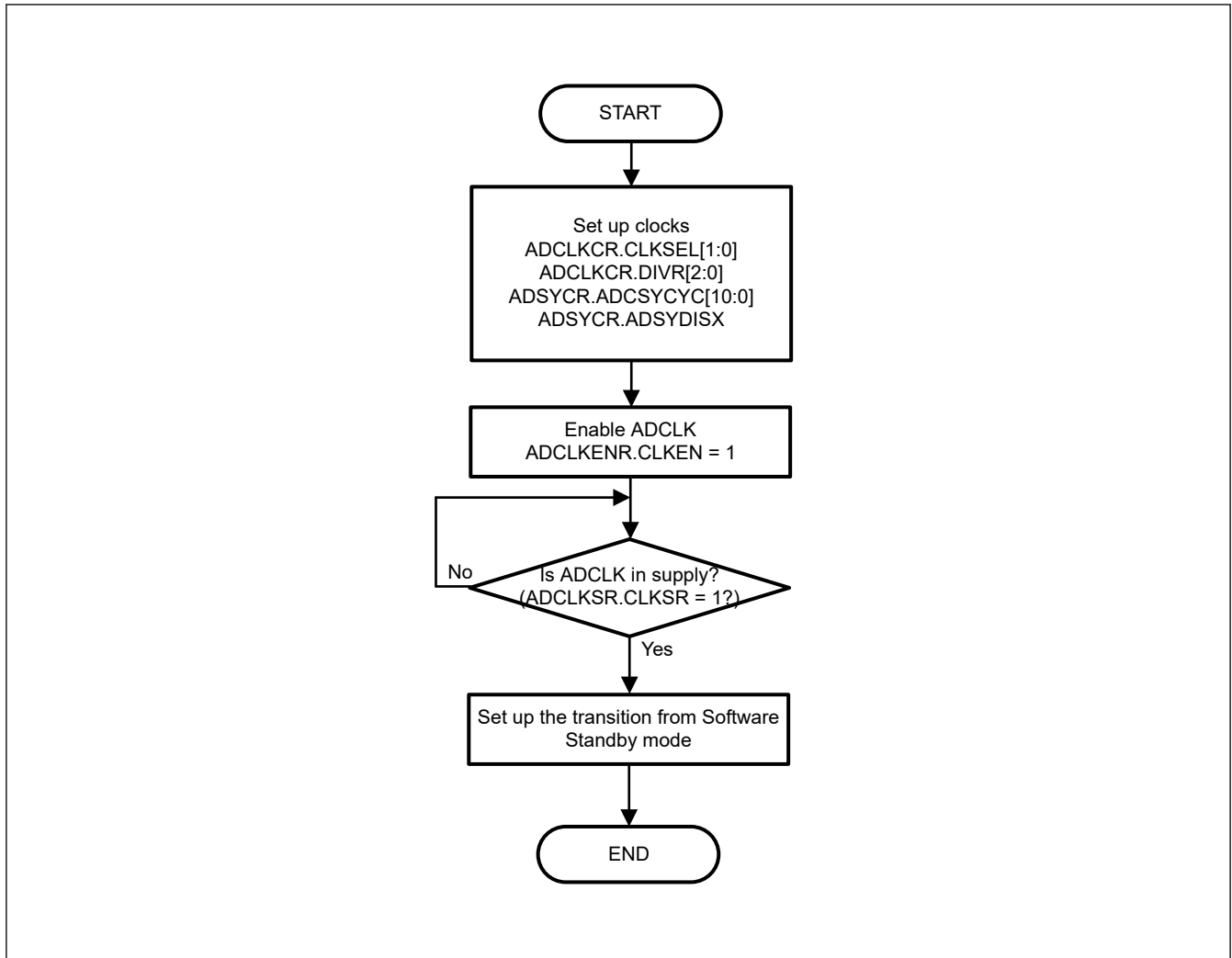


Figure 54.65 Example procedure for resuming Software Standby mode

54.11.6 Restrictions on Entering and Releasing the Low-Power States

Before entering the module-stop state or Software Standby mode, be sure to stop A/D conversion. In Addition, configure the A/D conversion not to start during the entering to the low power state.

To operate the A/D converter after the module-stop state or software standby mode is released, wait for the operation stabilization time specified in the Electrical Characteristics, execute self-calibration operation, and then start A/D conversion.

Also, when releasing the module-stop state again after entering to the module-stop state, make sure the shutdown time specified in the Electrical Characteristics has elapsed before releasing the module stop state.

Operations are not guaranteed if these restriction are violated.

54.11.7 Notes on Board Design

(1) Protection Circuit

To prevent the analog input pins from being destroyed by abnormal voltage such as excessive surge, take the following measures.

- Insert a capacitor between the power pin and the ground pin. The insertion points are below.
 - between AVCC0 and AVSS0
 - between VREFH0 and VREFL0
 - between VREFH and VREFL

- Connect a protection circuit to protect the analog input pins.

An example of a protection circuit is shown in [Figure 54.66](#).

(2) Board design to ensure A/D conversion accuracy

In order to ensure the accuracy of A/D conversion, design the board considering the following:

- Analog circuits and digital circuits should be separated from each other as far as possible.
- Analog circuit signal lines and digital circuit signal lines should not intersect or be placed near each other.
- The analog input, analog reference power supply (VREFH0, VREFH), analog reference ground (VREFL0, VREFL), and analog power supply (AVCC0) should be separated from digital circuits using the analog ground (AVSS0).
- The analog ground (AVSS0) should be connected to a stable digital ground (VSS) on the board (single-point ground plane connection).
- Place a capacitor for noise filters between analog power supply pin and the ground pin and between analog reference voltage supply pin and the ground pin close to the pin and connect them. Also, connect AVSS0, VREFL0 and VREFL pins to the analog ground on the board as close to the pins as possible. An example of connection is shown in [Figure 54.66](#).

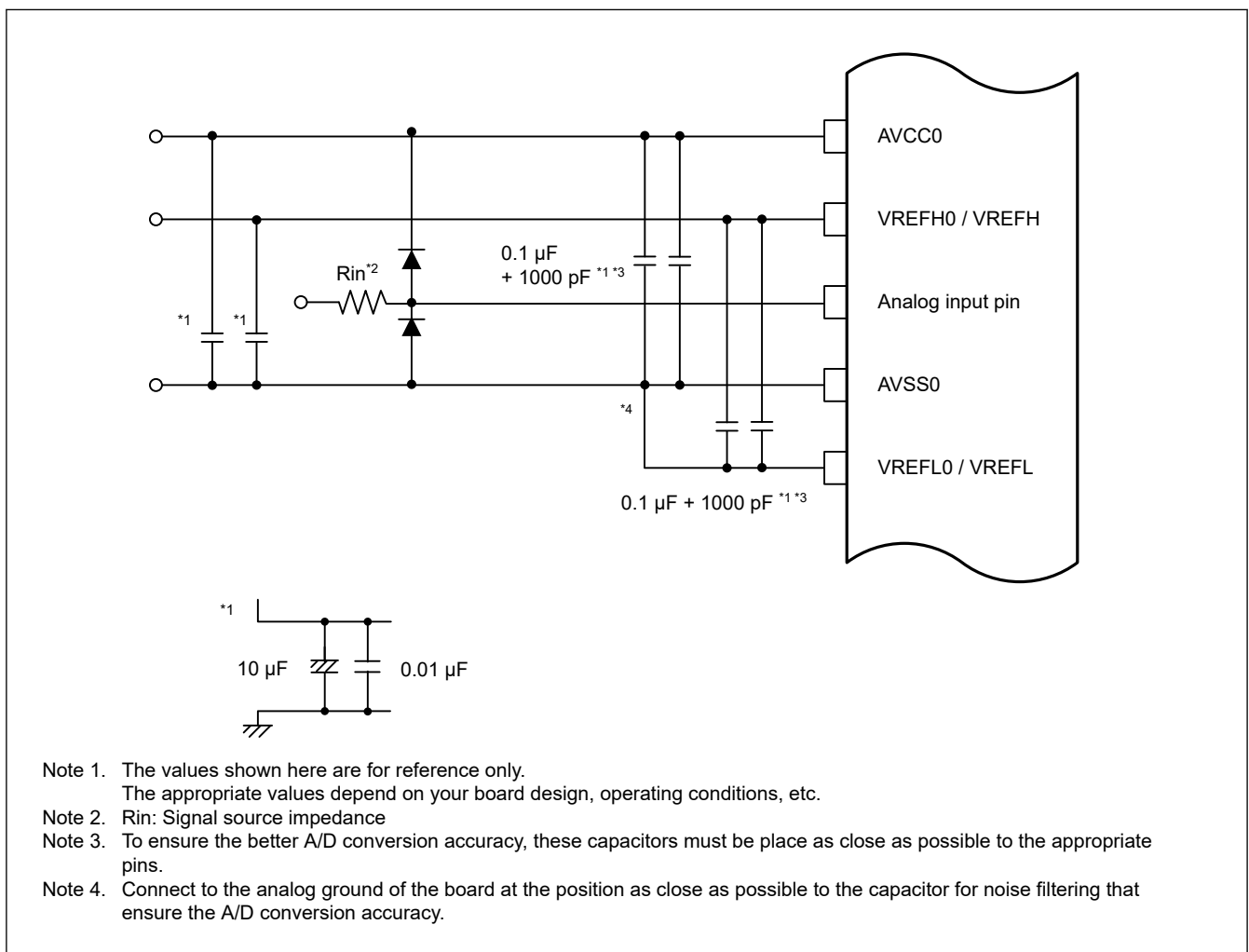


Figure 54.66 Example of protection circuit and noise prevention

54.11.8 Notes on Synchronous Operation

Synchronous operation function is disabled in the initial state after reset released. If using synchronous operation function, follow the restrictions described in [section 54.3.19. Synchronous Operation](#).

54.11.9 Notes on Channel-dedicated Sample-and-hold Circuit

When using channel-dedicated sample-and-hold circuit, follow the restrictions. For details, see [section 54.3.16. Channel-dedicated Sample-and-hold Circuit](#).

54.11.10 Restrictions on Analog Channel Shared Among Multiple A/D Converters (ADC16H) and High-Speed Analog Comparator (ACMPHS)

To avoid degradation of the accuracy of the A/D conversion result, A/D conversion of the same analog channel (same analog signal source) from both ADC0 and ADC1 is prohibited. Also, an analog source during A/D conversion should not be selected as an input for ACMPHS.

If this restriction is violated, the A/D conversion results are not guaranteed because the A/D conversion accuracy of the target analog channel may deteriorate significantly.

For the A/D conversion targets in [Table 54.64](#), unit 0 and unit 1 cannot perform A/D conversion at the same time.

Table 54.64 A/D conversion targets that are mutually exclusive with each other

A/D conversion target	
Unit 0	Unit 1
AN016	
AN017	
AN018	
AN019	
AN020	
AN021	
AN022	
Temperature sensor	
Internal reference voltage	
VBATT 1/6 voltage monitor output	
D/A converter channel 0	
D/A converter channel 1	

The A/D conversion targets in [Table 54.65](#) should not be selected as ACMPHS input during A/D conversion, because these pins are multiplexed with the ADC16H and ACMPHS.

Table 54.65 A/D conversion targets that are mutually exclusive with ACMPHS (1 of 2)

A/D conversion target		
Unit 0	Unit 1	ACMPHS
AN000	—	ACMPHS0.IVCMP2
AN001	—	ACMPHS0.IVCMP3
AN002	—	ACMPHS1.IVCMP2
AN003	—	ACMPHS1.IVCMP3
AN004	—	ACMPHS2.IVCMP2
AN005	—	ACMPHS2.IVCMP3
—	AN006	ACMPHS3.IVCMP2
—	AN007	ACMPHS3.IVCMP3
—	AN008	ACMPHS0 to ACMPHS3.IVREF0
—	AN009	ACMPHS0 to ACMPHS3.IVREF1
AN014	—	ACMPHS3.IVCMP0

Table 54.65 A/D conversion targets that are mutually exclusive with ACPHPS (2 of 2)

A/D conversion target		
Unit 0	Unit 1	ACPHPS
—	AN015	ACPHPS2.IVCMP0
AN016	AN016	ACPHPS1.IVCMP0
AN017	AN017	ACPHPS0.IVCMP0
Internal reference voltage	Internal reference voltage	ACPHPS0 to ACPHPS3.IVREF2
D/A converter channel 0	D/A converter channel 0	ACPHPS0 to ACPHPS3.IVREF3
D/A converter channel 1	D/A converter channel 1	ACPHPS0 to ACPHPS3.IVCMP1

54.11.11 Notes on A/D Conversion Start Trigger

(1) Restrictions on A/D conversion start triggers for the same scan group

A/D conversion start triggers for the same scan group are not accepted until scanning operation of the scan group is completed, except Background continuous scan mode and One-channel continuous scan mode (in this case, A/D converter start triggers are ignored).

An A/D conversion trigger for the same scan group should be input after the scanning operation of its scan group is completed (ADSCANENDSR.SCENDFn = 1 (n = 0 to 8)) and after 6 PCLKA clock cycles or more have elapsed.

(2) Prohibition on calibration and A/D conversion

It is prohibited to start calibration and A/D conversion in the ADC unit during sleep operation. In addition, it is prohibited to set the ADC unit to Sleep mode during calibration and A/D conversion operations.

54.11.12 Notes on Self-calibration

Observe the restrictions on self-calibration function. For details, see [section 54.3.8. Self-calibration](#).

54.11.13 Notes on Group Priority Operation

Observe the restrictions when using group priority operation. For details, see [section 54.3.18. Group Priority Operation](#).

54.11.14 Restrictions on SAR Mode

(1) Notes on resolution and accuracy

In SAR mode, the A/D converter functions as a 12-bit A/D converter. Although the data output from the A/D converter is internally extended to 16-bit data length for calculation processing, even if 14-bit or 16-bit length data format is selected, the resolution and accuracy exceeding 12 bits are not guaranteed for that A/D conversion data. Select 14-bit or 16-bit length data format in SAR mode if you want to increase the computational resolution, such as when using A/D-converted value addition/averaging function or User's Gain/User's Offset function.

(2) Restrictions on digital filter function

The digital filter is prohibited in SAR mode. Operation is not guaranteed when the digital filter function is enabled.

54.11.15 Restrictions on Oversampling Mode

(1) Restrictions on digital filter configuration

In Oversampling mode, the use of the digital filter function is mandatory. If A/D conversion is performed without using the digital filter function, the A/D conversion characteristics are not guaranteed.

(2) Restrictions on trigger interval in One-channel continuous scan mode

In Oversampling mode – One-channel continuous scan mode, A/D conversion start trigger should be input after the following intervals.

[Required intervals for A/D conversion start trigger input]

- When trigger delay function is not used, or software trigger is used: 8 ADCLK or more
- When trigger delay function is used: $(8 + (\text{the setting value of ADTRGDLRm.TRGDLYn}[7:0])) \times \text{ADCLK}$, or more

Note: $m = 0$ to 4, $n = 0$ to 8

If this restriction is violated, the A/D conversion start trigger is not accepted and is ignored.

54.11.16 Restrictions on Hybrid Mode

(1) Prohibition of scanning operation with only one channel

In Hybrid mode, be sure to assign two or more virtual channels to one scan group (up to 4 virtual channels). Scanning operation with only one virtual channel assigned to one scan group is prohibited. If this restriction is violated, operation is not guaranteed.

(2) Restrictions on virtual channel configuration

In Hybrid mode, do not assign the same analog channel to multiple virtual channels that are assigned to the same scan group. If this restriction is violated, A/D conversion result is not guaranteed.

(3) Restrictions on digital filter configuration

In Hybrid mode, the use of the digital filter function is mandatory. If A/D conversion in Hybrid mode is performed without using the digital filter function, the A/D conversion characteristics are not guaranteed.

In addition, since multiple digital filters are used simultaneously in Hybrid mode, the digital filters used for virtual channels in the same scan group must be set to be exclusive. The digital filters selected by the ADDOPCRAn.DFSEL[2:0] ($n = 0$ to 32) bits must be set to be exclusive among virtual channels in the same scan group. If the same digital filter is selected from multiple virtual channels in the same scan group, operation is not guaranteed.

(4) Restrictions on A/D-converted value addition/averaging function

When using A/D-converted value addition/averaging function in Hybrid mode, set the number of addition/averaging times for all virtual channels in the same scan group to be the same. If this restriction is violated, operation is not guaranteed.

(5) Restrictions on trigger interval in Background continuous scan mode

In Hybrid mode – Background continuous scan mode, A/D conversion start trigger should be input after the following interval or more.

[Required Interval for A/D Conversion Start Trigger Input]

- When trigger delay function is not used, or software trigger is used: 8 ADCLK, or more
- When trigger delay function is used: $(8 + (\text{the setting value of ADTRGDLRm.TRGDLYn}[7:0])) \times \text{ADCLK}$, or more

Note: $m = 0$ to 4, $n = 0$ to 8

If this restriction is violated, the A/D conversion start trigger is not accepted and is ignored.

(6) Restrictions on Fixed-channel continuous scan mode

Observe the restrictions on Hybrid mode – Fixed-channel continuous scan mode.

- Fixed channels in Fixed-channel continuous scan mode use the scan group and virtual channels determined for each unit. See [Table 54.66](#) for the scan group and virtual channels that can be assigned to fixed channels. The scan group setting for virtual channels that can be assigned to a fixed channel should be set to the no setting ($\text{ADCHCRn.SGSEL}[4:0] = 0x00$ ($n = 0$ to 32)).

Table 54.66 Virtual channel number of Fixed-channel continuous scan mode (1 of 2)

Unit	Unit 0	Unit 1
Scan group	Scan group 0	Scan group 1
1st Fixed channel	Virtual channel 0	Virtual channel 4
2nd Fixed channel	Virtual channel 1	Virtual channel 5

Table 54.66 Virtual channel number of Fixed-channel continuous scan mode (2 of 2)

Unit	Unit 0	Unit 1
3rd Fixed channel	Virtual channel 2	Virtual channel 6

- The scan period must be set to a number equal to or greater than the total number of conversions for fixed channels and non-fixed channels. If the scan period is set to less than the total number of conversions for fixed and non-fixed channels, conversion cycles beyond the scan period are not subject to A/D conversion.
- To prevent scan cycle variations due to non-fixed channels, set the same time for all virtual channel sampling states (ADSSTRx.SSTy[9:0]) used for non-fixed channels.
- Before starting scan in this mode, it is necessary to set ADMDR.ADMDm[3:0] = 0000 (m = 0, 1) and perform gain/offset calibration for SAR. The gain/offset correction coefficients for SAR are prepared separately from the correction coefficients for Hybrid that are calculated by the gain/offset calibration performed in this mode. These correction factors are used to correct the conversion results for fixed and non-fixed channels respectively.
- The use of fixed channels and non-fixed channels in fixed channel scanning in addition/average mode is prohibited.
- When using fixed-channel continuous scan mode, also perform gain/offset calibration for SAR mode (single scan mode).

For more details on Fixed-channel continuous mode, see [section 54.3.7.9. Hybrid Mode – Fixed-channel Continuous Scan Mode](#).

(7) Restrictions on High Accuracy Mode

In Hybrid mode, enable High accuracy mode.

54.11.17 Restrictions on Composite Compare Match

The use of a common compare table for multi units is prohibited for composite compare match function. If selected, the composite condition MATCH interrupt is asserted when the condition is satisfied at any unit. When the condition is not satisfied for all units, the compound condition UNMATCH interrupt is asserted.

54.11.18 Restrictions on High Accuracy Mode Setting Register

When the High accuracy mode is used (ADACMDR.ADHACMDn (n = 0, 1) = 1), observe the following setting restrictions.

- $ADSSTRn.SSTm[9:0] \geq ADCNVSTR.CSTn (n = 0, 1) [5:0] \times 1.6$
- $ADSSTRn.SSTm[9:0] \geq 10'h4$
- ADSSTRn.SSTm[9:0] is set to a multiple of 2
- When synchronous operation is used, the synchronous operation period should be set as follows.
 - $ADSYCR.ADSYCYC[7:0] \times i = ADSSTRn.SSTm[9:0] + ADCNVSTR.CSTn[5:0]$
(i: Any even number or equal to 1 (i = 1, 2, 4, 6...))
- High accuracy mode can only be used with SAR mode or Hybrid mode. The scan modes supported by high accuracy mode are shown in [Table 54.67](#).

Table 54.67 Scan modes supported by high accuracy mode

Operation mode of A/D converter	High accuracy mode	
	Disabled (ADACMDR.ADHACMDn = 0)	Enabled (ADACMDR.ADHACMDn = 1)
SAR mode	✓	✓
Oversampling mode	✓	—
Hybrid mode	—	✓

54.11.19 Restrictions on Measuring of VBATT Voltage Monitor Output

When measuring VBATT 1/6 voltage monitor output, perform two or more A/D conversions and ignore the first conversion result, or perform multiple A/D conversions and use the average of the conversion results.

54.11.20 Sampling Error and Sampling Time Estimation

This subsection describes how to estimate the sampling error and the sampling time.

The sampling time required in A/D conversion for externally input analog signal is determined by the charging time to the sampling capacitance in A/D converter. The simplified model of the circuit is shown in [Figure 54.67](#).

The sampling time can be roughly estimated by the following equation.

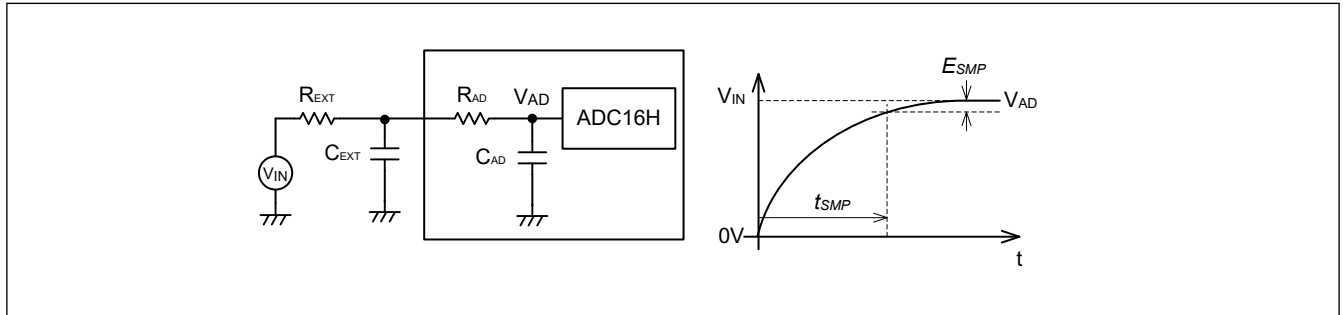


Figure 54.67 Simplified model of sample-and-hold circuit for A/D converter

You can estimate the sampling error (0 to 1, it is 1 when $V_{IN}=V_{REFH0}/V_{REFH}$). Set sampling time (t_{SMP}) so that sampling error (E_{SMP}) is within the expected range.

[Estimation Equation for Sampling Error]

$$E_{SMP}(t) = \frac{\frac{k_1 C_{AD}}{C_{EXT} + k_1 C_{AD}}}{1 - \frac{C_{EXT}}{C_{EXT} + k_1 C_{AD}} e^{-\frac{t_{SCAN} + t_{SMP}/k_2(1 - \frac{C_{EXT}}{C_{EXT} + k_1 C_{AD}})}{\tau}}} e^{-\frac{t_{SMP}/k_2}{\tau}}$$

$$\tau = R_{EXT}(C_{EXT} + k_1 C_{AD}) + R_{AD}k_1 C_{AD}$$

$$\tau_{EXT} = R_{EXT}C_{EXT}$$

E_{SMP} : Estimated sampling error

t_{SMP} : Sampling time

C_{EXT} : External capacitance (pin capacitance + PCB parasitic capacitance)

C_{AD} : Internal sampling capacitance

R_{EXT} : External input signal source Impedance

R_{AD} : Internal resistance

k_1, k_2 : Correction coefficients according to operation mode

t_{SCAN} : Channel scan interval

[Figure 54.68](#) shows definition of channel scan interval.

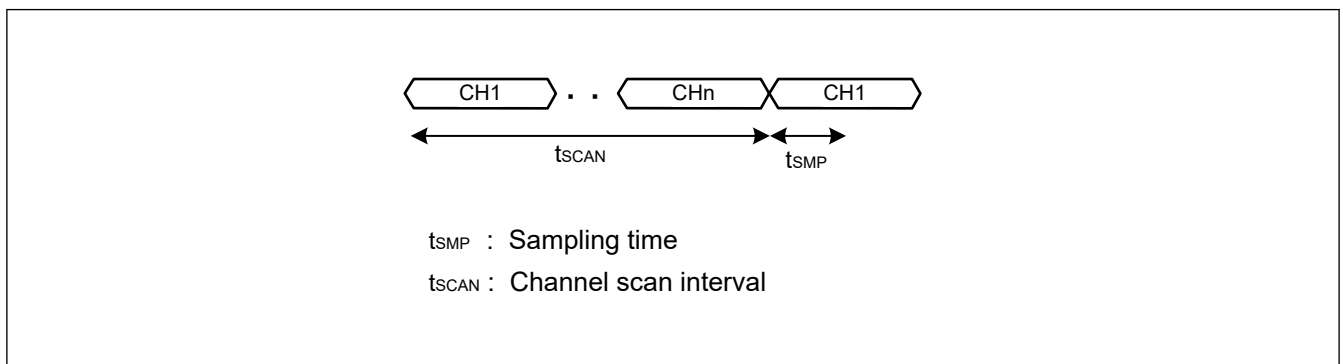


Figure 54.68 Definition of channel scan interval

Note: The typical values for each parameter are as follows.

These are reference value:

- Internal sampling capacitance (C_{AD}): 7 pF

Table 54.68 Reference values of k_1 , k_2

Power	Scan mode	k_1		k_2
		High-speed channels (AN000 to AN011) Low-speed channels (AN016 to AN022)	Middle-speed channels (AN012 to AN015)	
AVCC: 2.7 to 3.63 V VCC: 2.7 to 3.63 V VREFH0/VREFH: 2.7 V to AVCC	SAR mode (Normal mode)	1.3	1.3	1.0
	SAR mode (High accuracy mode)	1.3	1.3	1.8
	Oversampling mode	1.0	1.0	1.0
	Hybrid mode	1.3	1.3	1.8
AVCC: 1.62 to 2.7 V VCC: 1.62 to 2.7 V VREFH0/VREFH: 1.62 V to AVCC	SAR mode (Normal mode)	1.2	1.3	1.0
	SAR mode (High accuracy mode)	1.2	1.3	1.8
	Oversampling mode	1.0	1.0	1.0
	Hybrid mode	1.2	1.3	1.8

Table 54.69 Reference values of RAD

Power	RAD (Ω)		
	High-speed channels (AN000 to AN011)	Middle-speed channels (AN012 to AN015)	Low-speed channels (AN016 to AN022)
AVCC: 2.7 to 3.63 V VCC: 2.7 to 3.63 V VREFH0/VREFH: 2.7 V to AVCC	700	2000	3000
	1500	3000	5000

[Usage Notes]

- Sampling time must be greater than or equal to the min value of the sampling time specified on [section 70, Electrical Characteristics](#) and restrictions in this chapter.
- The equation described in this subsection is simplified in view of general use case. It does not guarantee the accurate sampling error and sampling time. You should only use it to estimate the roughly sampling error and sampling time.

54.11.21 Port Settings When Using the ADC16H Input

When using the high-speed and middle-speed channels (AN000 to AN015), do not use PORT0 as digital output ports. Renesas recommends that you do not use the digital output that is also used as the AD analog input if low-speed channel (AN016 to AN022) is used. If the digital output that is also used as the AD analog input is used for output signals, perform A/D conversion several times, eliminate the maximum and minimum values, and obtain the average of the other results.

54.11.22 Notes on Simultaneous Operation with Other Modules

The following combinations may degrade the ADC accuracy depending on the specific timing of operation.

- All channels of ADC unit 1 when DAC12 is operated.
- All channels of ADC unit 0 and 1 when there is access to the external bus.
- Specific channels of ADC unit 0 and 1 when ACMPHS is operated. For details, see [Table 54.65](#).

To mitigate deterioration, it is recommended to take following measures.

- Performing A/D conversion with average mode.
- For even better results, performing A/D conversion several times, eliminate the maximum and minimum values, and obtain the average of the other results.